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FITTED EMPIRICAL REFERENCE CROSS SECTIONS FOR *K*-SHELL IONIZATION BY PROTONS*

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On the basis of our recent review of experimental cross sections for *K*-shell ionization by light ions, we present a table of best-fitted cross sections for protons on all elements from $_4$Be to $_{92}$U. Experimental cross sections are first normalized by the ECPSSR theory of Brandt and Lapicki; then the normalized values are averaged, interpolated, and used to produce reference cross sections (19 values per energy decade). Particular attention is given to assigning a reasonable uncertainty to the reference values: the error consists of a calculated random contribution and an estimated systematic contribution which describes both the influence of a small deficiency of the theory and the influence of possible bias in the data. A complete list of experimental data is given. © 1989 Academic Press, Inc.

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INTRODUCTION

K-shell ionization cross sections are useful for the calibration of x-ray detectors, for applications like PIXE,[1] and as a testing ground for theoretical descriptions.[2,3] We have recently reviewed[4] the large number of experimental data available (about 4000 data points for protons; see the Appendix) and we have produced a small table of K-shell reference cross sections, σ_{ref}, based on weighted averages of these data. In this paper, we present a similar table for all targets from $_4$Be to $_{92}$U, and for many more energy values (19 per decade) to facilitate interpolation. Compared with our earlier estimates[4] it turns out that the error can be reduced considerably in many cases (that is, for target atomic numbers $Z_2 > 30$ and medium energy), but that it must be increased for certain low- and high-energy values.

Since the cross sections vary over many orders of magnitude, it is useful to normalize them first using a convenient theoretical function. We therefore define $s = \sigma_I/\sigma_{ECPSSR}$, where σ_I is the experimental ionization cross section deduced from Auger cross sections σ_A by $\sigma_I = \sigma_A/(1-\omega)$, and from x-ray cross sections σ_X by σ_X/ω. Here, ω is the single-hole fluorescence yield taken[5] from Krause's Table 3 for $Z_2 \geq 16$, and from his Table 1 (for solid targets) for $11 \leq Z_2 \leq 15$. X-ray measurements for $Z_2 < 11$ are not used for averaging.

ECPSSR refers to the theoretical approach of Brandt and Lapicki.[6] This approach is based on the plane-wave Born approximation with hydrogenic wave functions, corrected for energy loss of the projectile during the collision (E), for deceleration and deflection of the projectile in the Coulomb field (C), for the perturbation of the stationary target electron states by the passing projectile (PSS), and for relativistic electron motion (R). An extensive table of ECPSSR cross-section values has been presented recently by Cohen and Harrigan.[7] These authors, however, do not follow the Brandt–Lapicki procedure[8] literally but effectively apply a relativistic correction to the energy-loss function $f(z_S)$. This makes their low-energy values larger (for example, 1.916×10^{-5} b compared to our value of 1.580×10^{-5} b for 0.7-MeV protons on Au). We therefore also list our values of σ_{ECPSSR} in the main table (Table I). These were calculated in a manner essentially equivalent to the Brandt–Lapicki procedure using a computer program due to Kropf.[9] For 0.7-MeV protons on Au, the Kropf program differs by 1.3% from the Brandt–Lapicki procedure; the difference is smaller for other, less adiabatic collisions.[10]

The normalized cross sections s are considered as functions of the scaled velocity $\xi = (2/\theta)(v_1/v_{2K})$, where $\theta = I_K/(Z_2 - 0.3)^2$ Ry, v_1 is the laboratory-frame velocity of the projectile, $v_{2K} = (Z_2 - 0.3)(2\ \text{Ry}/m_e)^{0.5}$ is the hydrogenic Bohr velocity of the target K-shell electron, I_K is the experimental ionization energy,[11] Ry the Rydberg energy, and m_e the electron mass. The data are grouped into five ranges of target atomic numbers which are treated separately: Z_2 = 4–10, 11–30, 21–40, 41–60, and 61–92. These ranges overlap in one place, so that the data for $21 \leq Z_2 \leq 30$, which are most numerous and most accurate, serve to calculate σ_{ref} down to $_{11}$Na and up to $_{40}$Zr. We subdivide the entire ξ range into intervals of equal Δ log

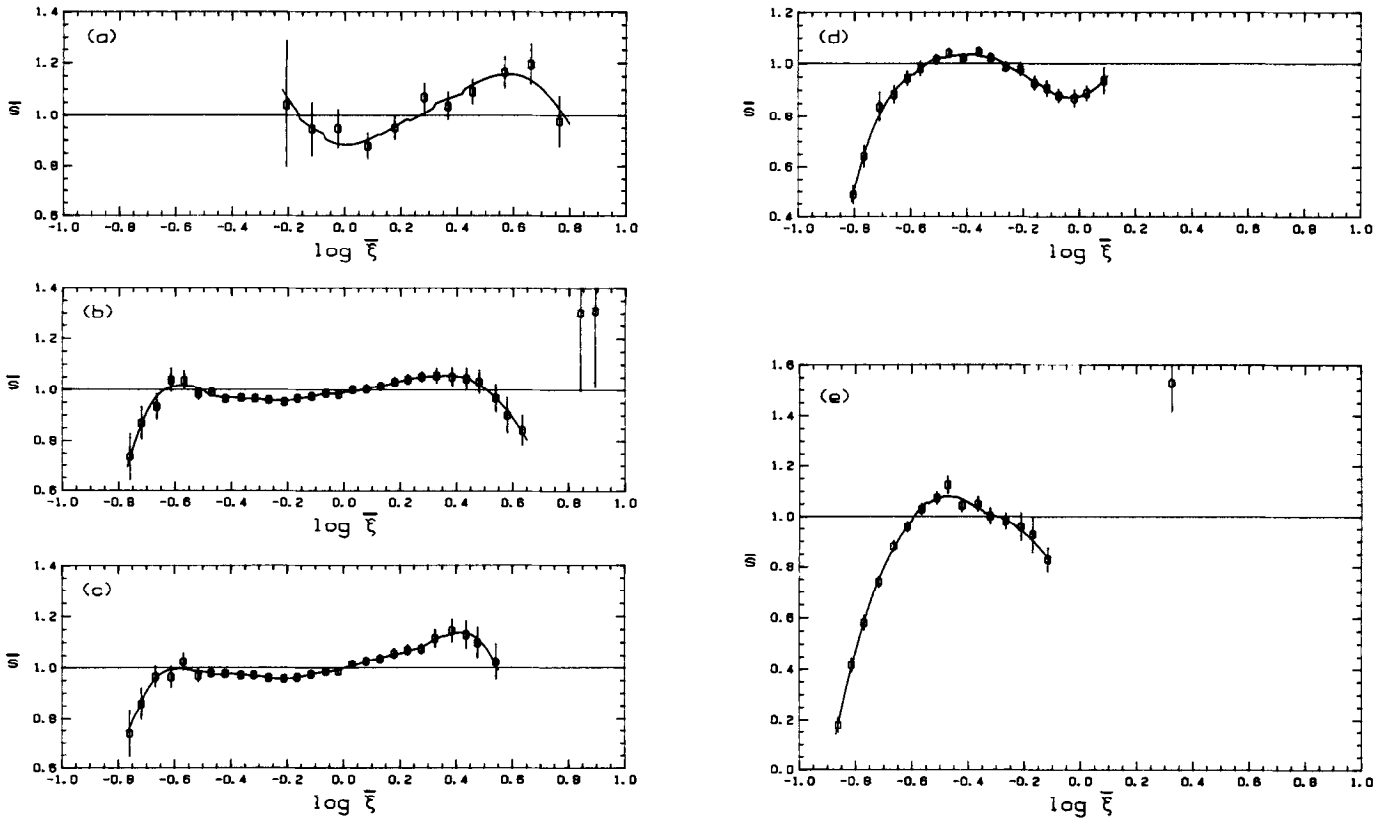


Figure 1. The averaged normalized cross section, $\bar{s} = \overline{\sigma_1/\sigma_{\mathrm{ECPSSR}}}$, versus the average logarithm of the scaled velocity, $\bar{\xi}$. The data were averaged in equal intervals $\Delta \log \xi = 0.1$ (for $Z_2 \leq 10$) and $\Delta \log \xi = 0.05$ (for $Z_2 > 10$). The curve shows the result obtained by carrying out the interpolation for 100 equidistant abscissae and connecting the resulting points by straight lines. The ranges of target atomic numbers are as follows: (a) $Z_2 = 4$–10. (b) $Z_2 = 11$–30. The two averages for $\log \xi > 0.8$, based on 160-MeV data by JA72 and on 24- to 30-MeV data by SE80 (cf. Ref. 4), are not used here. (c) $Z_2 = 21$–40. (d) $Z_2 = 41$–60. (e) $Z_2 = 61$–92. The average at $\log \xi = 0.33$, based on three 160-MeV data[4] by JA72, is not used here.

ξ, and we calculate[4] a weighted average $\bar{s} \pm \delta$ for every interval, and corresponding weighted averages $\log \bar{\xi}$ and $\bar{Z}_2$. An average is calculated only if there are data from at least two references or, if not, from at least two different targets within the interval.

Figure 1 shows five sets of points ($\bar{s}$, $\log \bar{\xi}$) corresponding to the five ranges of Z_2. To produce a particular value σ_{ref} for a certain laboratory-frame energy E_1 and a certain target atomic number Z_2, we calculate the corresponding scaled velocity ξ_1. We take the set of $\bar{s}$ values from Fig. 1 that is appropriate to the Z_2 (for $21 \leq Z_2 \leq 30$, the set corresponding to $11 \leq Z_2 \leq 30$). We then interpolate the function $\bar{s}$ versus $\log \bar{\xi}$ using a polynomial of the third degree; this yields a value $s_1 \pm \delta$. Here the error (standard deviation) δ is taken from the point ($\bar{s}$, $\log \bar{\xi}$) closest to $\log \xi_1$. The polynomial is fitted to 4 neighboring points if δ is small; the number of points is increased with increasing δ up to a maximum of 13. The curves in Fig. 1 show the result obtained by carrying out the interpolation for 100 equidistant points. The extreme ξ values for which the fit is carried out are chosen corresponding to the smallest and largest ξ of available experimental data, within each Z_2 range. The reference cross section is then obtained as $\sigma_{\mathrm{ref}}(E_1) = \sigma_{\mathrm{ECPSSR}}(E_1) \times (s_1 \pm \delta)$.

Although electron capture by the projectile is generally small (it contributes, e.g., 6% to ionization for 0.4 MeV protons on Ne and less for heavier targets), it should be noted that our reference cross sections include, in principle, the sum of direct ionization *and* electron capture.

Our data file, described in Ref. 4, contains 100 publications with proton data; of these, 20 were rejected using a statistical criterion.[4] Five recent references have been added: Geretschläger and Benka[12] (Al, Ti, and Cu targets), the Warsaw group[13–15] (Nd, Sm, Tb, Ho, Tm, Ta), and McElroy et al.[16] (C and O). The Warsaw group data are all based on a comparison with known Coulomb ionization cross sections of nuclear states. The McElroy data

are a welcome addition to the small number of absolute Auger measurements (among the various gases measured, we chose O_2 and CO for O, and C_2H_6 for C). The complete file is listed in the Appendix as Table II.

Several recent references have not been added: McDaniel et al.[17] because the data are not absolute; Braziewicz et al.[18] because the intensity calibration was based on other published K-shell cross sections; Saleh and Al-Saleh[19] because the data appear generally low compared with other available measurements; Olabanji and Calvert[20] because the Cr data appear to be too high and because the calibration involved other investigators' published cross sections.[21]

The present method has the advantage that reference cross sections can be produced automatically from a given set of data, whereas our earlier fitting procedure[4] is difficult and time consuming to apply.

Estimation of Errors

To obtain the error δ of $\bar{s}$ mentioned above, we have chosen[4] to take the larger of the internal error (based on the measurement errors stated by the various authors) and the external error (based on a comparison of the numbers averaged). Also, we have not used the laws of error propagation in polynomial fitting but have simply taken the error of the closest anchor point. In spite of this conservative procedure, this "random" error δ_r turns out to be unreasonably small in several cases, indicating that it should be increased.

One obvious possibility is to add a "systematic" error δ_s. The fact that the five curves in Fig. 1 are not identical may indicate a slightly incorrect Z_2 dependence in the ECPSSR theory. Since we can ascribe an average value $\bar{Z}_2$ to every point in Fig. 1, we can reinterpret these curves as $\bar{s}$ versus $\bar{Z}_2$ (with $\log \bar{\xi}$ as a parameter) and hence estimate $\Delta\bar{s}/\Delta\bar{Z}_2$ as a function of $\log \bar{\xi}$. The result is shown in Table A. In some cases, this may indeed indicate a certain defect of the theory[4]; in others, it may just mean biased data. Whichever the case, we calculate $\delta_s = (Z_2 - \bar{Z}_2)|\Delta\bar{s}/\Delta\bar{Z}_2|$ and add this systematic error linearly to obtain the total error $\delta_t = \delta_r + \delta_s$. This then automatically produces a large error if the atomic number of the element of interest is far from that of the elements actually measured.

Table B shows the result of this procedure for 7 selected elements and for a few energies. Inspection of this table can serve to test the procedure. The reference cross sections were calculated on the basis of the five sets of $\bar{s}$ values shown in Figs. 1a–1e, and also on the basis of using only measurements on the element in question ($\Delta Z_2 = 0$). The latter values usually have larger errors but they cannot be influenced by an erroneous Z_2 dependence of the theory. It should be noted that values for σ_{ref} ($\Delta Z_2 = 0$) are given only where measurements from at least two different references are available for averaging. The values for $\Delta Z_2 > 0$ represent a small selection from (the main) Table I.

TABLE A

Estimated Value of $|\Delta\bar{s}/\Delta\bar{Z}_2|$ as a Function of $\log \bar{\xi}$

Range of $\log \bar{\xi}$	$\lvert\Delta\bar{s}/\Delta\bar{Z}_2\rvert$
(−0.9, −0.8)	0.01
(−0.8, −0.7)	0.01
(−0.7, −0.6)	0.004
(−0.6, −0.5)	0.0025
(−0.5, −0.4)	0.0019
(−0.4, −0.3)	0.0018
(−0.3, −0.2)	0.0019
(−0.2, −0.1)	0.0021
(−0.1, 0)	0.0023
(0, 0.1)	0.0025
(0.1, 0.2)	0.003
(0.2, 0.3)	0.0045
(0.3, 0.4)	0.006
(0.4, 0.5)	0.0075
(0.5, 0.6)	0.01
(0.6, 0.7)	0.025
(0.7, 0.8)	0.03

Inspection of Table B shows that the values of σ_{ref} ($\Delta Z_2 = 0$) and σ_{ref} ($\Delta Z_2 > 0$) generally agree within the error of the former. This is satisfactory. The values for Al form an exception since here the values based on $\Delta Z_2 = 0$ are up to 20% smaller. But the low-energy Al values are almost entirely based on the thick target measurements of Khan and Potter[22] because the newer measurements of Basbas et al.[23] were normalized to the Khan data; the Khan data were in turn based on rather low stopping-power values. By contrast, the recent value of (16,806 ± 1681) b for 1-MeV protons on Al due to Geretschläger and Benka[12] agrees well with our σ_{ref} ($\Delta Z_2 = 19$). We think, therefore, that the values σ_{ref} ($\Delta Z_2 > 0$) are reliable; in the following, we use only these values.

Table B shows that the systematic error usually increases toward low and high energies. This is partly because $|\Delta\bar{s}/\Delta\bar{Z}_2|$ increases toward small and large ξ (Table A); in addition, the error is large for the highest-energy values for Ti and Cu because these are based on measurements for much lighter elements (cf. the value of $\bar{Z}_2$).

The influence of a slightly incorrect Z_2 dependence of the ECPSSR theory[4] upon the reference cross sections was checked further by recalculating some values using a recent modification of the ECPSSR theory.[24] Table C shows a few σ_{ref} values based on the ECPSSR theory as described above, compared with $\sigma_{ref}^{(M)}$ values calculated in the same manner, but using the MECPSSR theory,[24]

TABLE B

Reference Cross Sections (in Barns) for Selected Elements, Calculated (a) on the Basis of Data for the Element in Question ($\Delta Z_2 = 0$), and (b) Also Using Neighboring Elements ($\Delta Z_2 > 0$)

Target	Energy (MeV)	log ξ	σ_{ref} ($\Delta Z_2 = 0$)	Error (%)	σ_{ref}	$\bar{Z}_2$	Error (%) δ_r	δ_s	δ_t Total
					$(4 \leq Z_2 \leq 10)$				
$_{10}$Ne	0.1	−0.217			1.238E+3	8.0	23.6	0.4	24.0
	0.2	−0.067			8.114E+3	7.2	8.1	0.7	8.8
	0.5	0.133	4.237E+4	10.0	4.441E+4	8.0	4.9	0.7	5.6
	1	0.283	9.199E+4	9.2	8.619E+4	7.8	5.0	1.0	5.9
	2	0.434	1.137E+5	9.2	1.100E+5	7.9	4.3	1.4	5.8
	5	0.633	8.102E+4	11.0	8.868E+4	7.5	6.4	5.5	11.8
	10	0.783			5.147E+4	6.6	10.1	10.2	20.4
					$(11 \leq Z_2 \leq 30)$				
$_{13}$Al	0.02	−0.702			8.196E−3	27.6	7.4	16.1	23.5
	0.05	−0.503	2.502E+0	19.3	2.666E+0	27.3	2.5	3.6	6.0
	0.1	−0.353	4.147E+1	11.8	5.116E+1	26.2	1.5	2.5	3.9
	0.2	−0.202	4.750E+2	12.9	5.780E+2	25.9	1.0	2.6	3.6
	0.5	−0.003	5.491E+3	11.8	6.084E+3	24.9	1.1	2.8	3.8
	1	0.147	1.397E+4	7.1	1.738E+4	24.4	1.6	3.4	5.0
	2	0.298	2.408E+4	10.2	2.941E+4	25.7	1.8	5.4	7.2
	5	0.497	2.685E+4	8.5	2.979E+4	22.4	4.6	7.0	11.5
	10	0.647	1.870E+4	7.7	1.827E+4	13.9	7.2	2.7	9.9
					$(11 \leq Z_2 \leq 30)$				
$_{22}$Ti	0.1	−0.624	4.234E−2	15.2	4.610E−2	24.9	4.6	1.1	5.7
	0.2	−0.473	1.272E+0	5.0	1.235E+0	26.0	1.9	0.8	2.7
	0.5	−0.274	3.372E+1	2.9	3.398E+1	26.1	1.1	0.8	1.9
	1.0	−0.124	2.359E+2	2.1	2.362E+2	26.0	0.9	0.9	1.7
	2	0.027	9.735E+2	2.0	9.417E+2	22.8	1.4	0.2	1.6
	5	0.226	2.752E+3	7.2	2.580E+3	25.5	1.9	1.5	3.4
	10	0.376	3.494E+3	5.9	3.233E+3	21.1	3.5	0.5	4.0
	20	0.527	2.822E+3	8.7	2.655E+3	21.3	5.8	0.7	6.4
	30	0.615			1.951E+3	13.9	7.2	23.4	30.5
					$(11 \leq Z_2 \leq 30)$				
$_{29}$Cu	0.1	−0.760			2.906E−4	28.4	12.8	0.8	13.6
	0.2	−0.609	3.143E−2	7.4	3.326E−2	24.9	4.6	1.6	6.2
	0.5	−0.410	1.634E+0	2.9	1.618E+0	26.8	1.5	0.4	1.9
	1	−0.260	1.599E+1	1.9	1.584E+1	26.1	1.1	0.6	1.6
	2	−0.109	9.767E+1	1.9	9.666E+1	26.0	0.9	0.6	1.5
	5	0.090	4.646E+2	3.4	4.635E+2	23.0	1.7	1.5	3.2
	10	0.241	8.735E+2	3.4	8.459E+2	25.5	1.9	1.5	3.4
	20	0.391	1.034E+3	7.7	9.914E+2	21.1	3.5	4.5	8.0
	50	0.590			6.460E+2	16.0	7.9	14.4	22.3
					$(21 \leq Z_2 \leq 40)$				
$_{34}$Se	0.2	−0.688			2.781E−3	30.0	4.4	1.8	6.1
	0.5	−0.489			2.484E−1	27.7	1.9	1.2	3.1
	1	−0.339	3.034E+0	2.9	2.981E+0	28.2	1.2	1.1	2.2
	2	−0.188	2.213E+1	4.5	2.201E+1	27.4	1.0	1.5	2.4
	5	0.011	1.562E+2	7.7	1.487E+2	24.3	1.3	2.4	3.7
	10	0.161			3.461E+2	27.8	1.9	1.8	3.7
	20	0.312			5.106E+2	26.6	3.2	4.0	7.2
	50	0.511			4.474E+2	24.4	7.0	9.0	16.0
					$(41 \leq Z_2 \leq 60)$				
$_{47}$Ag	0.3	−0.763	1.338E−4	9.0	1.294E−4	49.1	6.8	3.2	10.0
	0.5	−0.652	3.629E−3	6.8	3.678E−3	51.7	3.8	2.1	5.9
	1	−0.502	8.288E−2	2.5	8.383E−2	49.8	1.8	0.7	2.5
	2	−0.351	8.994E−1	2.2	9.178E−1	50.2	1.5	0.6	2.1
	5	−0.152	9.497E+0	5.5	9.667E+0	47.8	2.8	0.2	3.0
	10	−0.002	3.374E+1	4.0	3.327E+1	46.6	3.8	0.1	4.0
					$(61 \leq Z_2 \leq 92)$				
$_{79}$Au	0.7	−0.852			3.659E−6	82.5	17.7	15.1	32.8
	1	−0.775	9.440E−5	7.9	1.013E−4	80.7	4.9	2.9	7.8
	2	−0.625	3.944E−3	5.9	4.040E−3	70.6	2.2	3.6	5.7
	3	−0.537	1.811E−2	6.0	1.848E−2	67.6	2.2	2.7	4.9
	5	−0.426			8.858E−2	69.9	2.4	1.6	4.0
	10	−0.275			5.039E−1	73.4	3.3	1.1	4.4
	20	−0.125			1.833E+0	67.4	5.7	2.9	8.6

TABLE C

Comparison of Reference Cross Sections σ_{ref} and $\sigma_{ref}^{(M)}$ (in Barns) Based on the ECPSSR and the MECPSSR Theories, and Using $\Delta Z_2 > 0$

Target	Energy (MeV)	σ_{ECPSSR}	σ_{ref}	$\sigma_{MECPSSR}$	$\sigma_{ref}^{(M)}$	δ_r (%)	δ_s (%)	Diff (%)
$_{13}$Al	0.016	1.150E−3	8.781E−4	1.093E−3	9.813E−4	13.0	17.2	−10.5
	0.02	9.043E−3	8.196E−3	8.734E−3	8.664E−3	7.4	14.7	−5.4
	0.05	2.666E+0	2.666E+0	2.657E+0	2.670E+0	2.5	3.6	−0.1
$_{22}$Ti	0.055	7.652E−4	5.773E−4	6.851E−4	6.109E−4	13.0	7.2	−5.5
	0.1	4.572E−2	4.610E−2	4.452E−2	4.648E−2	4.6	1.1	−0.8
	0.2	1.259E+0	1.235E+0	1.255E+0	1.235E+0	1.9	0.8	0
$_{29}$Cu	0.1	3.969E−4	2.906E−4	3.315E−4	2.906E−4	13.0	0.7	0
	0.2	3.282E−2	3.326E−2	3.182E−2	3.318E−2	4.6	1.6	0.2
	0.5	1.668E+0	1.618E+0	1.665E+0	1.617E+0	1.5	0.4	0.1
$_{34}$Se	0.14	2.528E−4	1.887E−4	1.981E−4	1.812E−4	12.8	5.9	4.1
	0.2	2.996E−3	2.781E−3	2.688E−3	2.723E−3	4.4	1.6	2.1
	0.5	2.527E−1	2.484E−1	2.512E−1	2.480E−1	1.9	1.2	0.2
$_{47}$Ag	0.26	6.709E−5	3.569E−5	4.015E−5	3.638E−5	7.4	1.5	−1.9
	0.5	4.072E−3	3.678E−3	3.670E−3	3.740E−3	3.8	1.8	−1.7
	1	8.209E−2	8.383E−2	8.123E−2	8.385E−2	1.8	0.7	0.0
	2	8.926E−1	9.178E−1	8.915E−1	9.174E−1	1.5	0.6	0.0
$_{79}$Au	0.7	1.580E−5	3.659E−6	3.385E−6	4.287E−6	17.7	2.8	−14.6
	1	1.795E−4	1.013E−4	8.286E−5	9.587E−5	4.8	1.4	5.7
	2	4.269E−3	4.040E−3	3.826E−3	4.010E−3	2.2	3.2	0.7
	5	8.278E−2	8.858E−2	8.238E−2	8.834E−2	2.4	1.6	0.3

Note. The random error δ_r and the systematic error δ_s refer to $\sigma_{ref}^{(M)}$. Diff (%) = $(\sigma_{ref} - \sigma_{ref}^{(M)}) \times 100/\sigma_{ref}^{(M)}$.

which describes the low-energy data better. Only low-energy values are shown, since the two approaches differ only for low ξ (where the differences between theory and data are also largest).

Inspection of the largest discrepancies in Table C gives the following result: For Au, the difference is due to the different ways in which interpolation works in the two cases; this difference is covered by the interpolation error, that is, δ_r. In the case of $_{13}$Al, the difference comes from the fact that the cross sections are effectively based on theoretical and experimental Ni values ($\bar{Z}_2 \simeq 28$; cf. Table B), and the two theories have different Z_2 dependences; the difference here is well covered by the estimated systematic error. The case of $_{22}$Ti is similar, except that $\bar{Z}_2 \simeq 25$ is much closer to $Z_2 = 22$. The values in Table A have been chosen large enough that here also the difference is covered by the systematic error.

In a few cases ($61 \leqslant Z_2 \leqslant 68$, low ξ), the total error becomes >100% (see Table I). This is because the experimental values are far away ($\bar{Z}_2 = 82.5$), and because s is small. Even so, we think that these σ_{ref} values should give the correct order of magnitude.

Discussion

When we first proposed empirical reference cross sections[25,26] for protons, we considered all targets simultaneously and obtained a single function s_c for the normalized cross section. We later divided the data into three groups[26,4] according to Z_2 ranges 4–30, 31–60, and 61–92. This proved more satisfactory.[2] It is of interest to compare the present analysis, based on five groups, with the previous version, that is, to compare the curves for $\bar{s}$ in Fig. 1 with the corresponding curves for s'_c in Figs. 5a–5c or Eq. (4) of Ref. 4. The result is as follows:

For $4 \leqslant Z_2 \leqslant 10$, the main differences are the reduction of the Z_2 range and the fact that we now calculate σ_{ref} only for $\log \xi \geqslant -0.22$; our earlier fitting function s'_c for $4 \leqslant Z_2 \leqslant 30$ was based only on data for $Z_2 > 10$ in the region $-0.76 \leqslant \log \xi \leqslant -0.22$. In addition, the new data[16] have raised the fitted function $\bar{s}$ by up to 3% (at $\log \bar{\xi} = 0.6$). The functions $\bar{s}$ and s'_c now differ by up to 15%. Hence, the present σ_{ref} values for Ne deviate by up to 15% from those in Ref. 4. This difference is comparable to our error δ_t.

For $11 \leqslant Z_2 \leqslant 30$, the curves for $\bar{s}$ and s'_c agree within 2% except above $\log \xi = 0.5$, where the difference increases up to 19%; the difference is covered by δ_t. Also, σ_{ref} is now calculated only up to $\log \xi = 0.65$.

For $31 \leqslant Z_2 \leqslant 40$, the two curves are rather different: the new $\bar{s}$ goes much higher in velocity and differs by up to 7% from the old one. The reason is that $\bar{s}$ now depends heavily on the $_{29}$Cu data, whereas the earlier s'_c depended essentially on $_{47}$Ag data. The new method is better than the earlier one, since the Cu data are closer in Z_2, and since they cover a larger range in ξ. The change is within the old[4] estimated error of 8%. Some of the new values of σ_{ref} have errors as small as 1.4%.

For $41 \leqslant Z_2 \leqslant 60$, the two curves are similar; the maximum difference is 4% around $\log \xi = 0$, well within the old error of 8%. Some values of σ_{ref} now have errors as small as 1.5%.

For $61 \leqslant Z_2 \leqslant 92$, the two curves were identical within 2% before the new data were added. This shows that the two methods of interpolation yield identical results if applied to the same data. The new data[13-15] have raised the fitted curve by as much as 6%; consequently, σ_{ref} for 10-MeV protons on Au was also raised by 6%. This is well within the old error estimate of 10%, but slightly larger than our present $\delta_t = 4.4\%$. Some errors of σ_{ref} are now as low as 2.1%.

A comparison of σ_{ref} from Table I with our[26] analytical cross section σ_X^a (based on a single function s_c) shows agreement roughly within the error of σ_X^a, for $11 \leqslant Z_2 \leqslant 40$. For heavier targets, the difference can be as large as 10%–15%. This difference occurs in the places where $\bar{s}$ (in Figs. 1d and 1e) differs most from unity and, hence, from the single function s_c (cf. Fig. 1 of Ref. 26). Evidently, the error of σ_X^a should be increased in these places.

Various analytical cross-section fits due to other authors have been compared with our reference cross sections in Table 8 of Ref. 4. A recent list of fitted cross sections for 2-MeV protons by Braziewicz et al.[18] agrees fairly well with our σ_{ref}; only the value for Al is 36% lower, probably due to the difficulties with Al mentioned above.

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EXPLANATION OF TABLES

TABLE I. Empirical Reference Cross Sections for K-Shell Ionization by Protons

This Table presents ionization cross sections (i.e., direct ionization plus electron capture by the projectile), obtained by averaging the available experimental data published before 1988. All targets from $_4$Be to $_{92}$U are covered. The energy values are approximately equidistant on a logarithmic scale; the extreme energy values are chosen corresponding to the smallest and largest values of the scaled variable ξ for which experimental data (suitable[4] for averaging) exist, within each of five ranges of Z_2 (see Figs. 1a–1e). Values calculated from the ECPSSR theory by Brandt and Lapicki[6] (which is used for normalizing the data) are also presented. In addition to the random error (based on the errors stated in the publications, and on the actual discrepancies of the data), we introduce an estimated systematic error. The total error is the sum of the two. The rather large changes of the error from one line to the next which occur at low- and high-energy values in the Table are a consequence of the discrete steps used in estimating the systematic error.

The uncertainty of the fluorescence yield, ω, is not included in our error estimate. Hence, for values based essentially on x-ray measurements, our relative error is really the uncertainty of the x-ray production cross sections. The error of ω can be found in Ref. 5.

If interpolation between energy values in the Table is necessary, we suggest using linear interpolation of log σ_{ref} vs log E_1. In the worst case (0.75-MeV p on $_{80}$Hg) this will give an interpolation error half as large as δ_t; in all other cases the interpolation error is smaller (relative to δ_t).

The table and a FORTRAN subroutine for interpolation are available on floppy disk to interested users.

TARGET	Atomic number and element symbol
ENERGY	Projectile energy in MeV
LOG XI	log ξ, where ξ is the scaled velocity of the projectile
SIGMA	
ECPSSR	Ionization cross section according to the ECPSSR theory,[6] in barns
REFERENCE	Empirical reference cross section σ_{ref}, in barns (1 b = 10^{-24} cm^2 = 100 fm^2, 1.132E+03 = $1.132 \times 10^{+3}$)
ERROR	
RND	Random error of σ_{ref}, in percent (standard deviation)
TOT	Total (random plus systematic) error of σ_{ref}, in percent (standard deviation)

EXPLANATION OF TABLES continued

TABLE II. Experimental Cross Sections for K-Shell X-Ray Production, Auger-Electron Production, and Ionization

Table II lists the data file which forms the basis of our fitted cross sections. It contains proton data from the compilations by Rutledge and Watson[A1] and by Gardner and Gray,[A2] and from more recent publications. Information on misprints discovered in the original tables can be found in the "footnotes" to Table 1 of Ref. 4.

As explained in Ref. 4, the data from 20 references (AV83, BD78, BO75, BS78, CK79, CR74, HO81, KH75, KL76, KN77, LE53, LE73, MA79, ME58, MI76, RS77, SH71, SI57, ST72, WL76) were rejected using a statistical criterion. For various reasons explained in the Introduction several recent references[17-21] have not been included in the data file.

In Ref. 4 and in an earlier publication,[A3] the normalized data were also compared graphically.

Z2	Atomic number of target
ENERGY	Projectile energy in MeV
SIGMA	Total *K*-Shell x-ray production, Auger-electron production, or ionization cross section as identified by the first letter of the REFERENCE (see below). The values are given in barns (1 b = 10^{-24} cm^2 = 100 fm^2, 1.132E+03 = $1.132 \times 10^{+3}$).
ERROR	Error (in percent) as given by the authors of the REFERENCE. Where no error was given, 80% has been arbitrarily assumed.
REFERENCE	Reference source, coded as follows:
First letter:	X x-ray cross section A Auger cross section I Ionization cross section. For (older) measurements on light elements, this is equivalent to the Auger cross section. For some recent measurements on heavy elements, the label I is used where the authors give only the ionization cross section
Second letter:	G Gaseous target N Thin solid target K Thick solid target (infinitely thick for the projectiles)
Remaining letters and numbers:	Reference code (see References for Table II)

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A2. R. K. Gardner and T. G. Gray, ATOMIC DATA AND NUCLEAR DATA TABLES **21,** 515 (1978), Table I; **24,** 281 (1979) (Erratum)

A3. H. Paul, ATOMIC DATA AND NUCLEAR DATA TABLES **24,** 243 (1979)

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
4-Be	0.012	-0.204	1.369E+04	1.468E+04	23.6	24.3
	0.014	-0.171	2.864E+04	2.924E+04	23.6	24.4
	0.016	-0.142	5.283E+04	5.119E+04	11.2	11.9
	0.018	-0.116	8.846E+04	8.396E+04	11.2	11.9
	0.020	-0.093	1.372E+05	1.282E+05	11.2	12.0
	0.023	-0.063	2.368E+05	2.138E+05	8.1	8.9
	0.026	-0.036	3.685E+05	3.278E+05	8.1	8.9
	0.030	-0.005	5.896E+05	5.205E+05	8.1	8.9
	0.035	0.028	9.247E+05	8.179E+05	8.1	9.0
	0.040	0.057	1.303E+06	1.163E+06	5.8	6.8
	0.045	0.083	1.704E+06	1.539E+06	5.8	6.8
	0.050	0.106	2.111E+06	1.931E+06	5.8	7.0
	0.055	0.126	2.510E+06	2.313E+06	5.8	7.0
	0.060	0.145	2.895E+06	2.699E+06	4.9	6.2
	0.070	0.179	3.602E+06	3.436E+06	4.9	6.1
	0.080	0.208	4.214E+06	4.105E+06	4.9	6.7
	0.090	0.233	4.732E+06	4.629E+06	5.0	6.7
	0.100	0.256	5.165E+06	5.123E+06	5.0	6.7
	0.110	0.277	5.522E+06	5.549E+06	5.0	6.6
	0.120	0.296	5.815E+06	5.913E+06	5.0	6.6
	0.140	0.329	6.246E+06	6.556E+06	5.2	7.2
	0.160	0.358	6.522E+06	6.922E+06	5.2	7.2
	0.180	0.384	6.688E+06	7.164E+06	5.2	7.2
	0.200	0.407	6.775E+06	7.315E+06	5.2	7.7
	0.230	0.437	6.808E+06	7.548E+06	4.3	6.9
	0.260	0.464	6.763E+06	7.600E+06	4.3	6.9
	0.300	0.495	6.634E+06	7.554E+06	4.3	6.9
	0.350	0.528	6.417E+06	7.383E+06	5.1	8.4
	0.400	0.557	6.174E+06	7.142E+06	5.1	8.4
	0.450	0.583	5.928E+06	6.867E+06	5.1	8.4
	0.500	0.606	5.689E+06	6.577E+06	5.1	13.4
	0.550	0.626	5.460E+06	6.286E+06	6.4	14.0
	0.600	0.645	5.245E+06	6.000E+06	6.4	14.0
	0.700	0.679	4.855E+06	5.456E+06	6.4	14.1
	0.800	0.708	4.516E+06	4.958E+06	6.4	15.9
	0.900	0.733	4.219E+06	4.504E+06	10.1	17.5
	1.000	0.756	3.960E+06	4.109E+06	10.1	17.7
	1.100	0.777	3.732E+06	3.755E+06	10.1	17.9
	1.200	0.796	3.530E+06	3.438E+06	10.1	18.2
5-B	0.023	-0.192	1.193E+04	1.255E+04	23.6	24.2
	0.026	-0.166	1.990E+04	2.017E+04	23.6	24.2
	0.030	-0.134	3.515E+04	3.384E+04	11.2	11.6
	0.035	-0.101	6.246E+04	5.866E+04	11.2	11.7
	0.040	-0.072	9.922E+04	9.169E+04	11.2	11.7
	0.045	-0.046	1.449E+05	1.295E+05	8.1	8.6
	0.050	-0.023	1.986E+05	1.759E+05	8.1	8.6
	0.055	-0.003	2.588E+05	2.285E+05	8.1	8.6
	0.060	0.016	3.242E+05	2.863E+05	8.1	8.7
	0.070	0.050	4.650E+05	4.137E+05	5.8	6.6
	0.080	0.079	6.118E+05	5.512E+05	5.8	6.6
	0.090	0.104	7.578E+05	6.927E+05	5.8	6.7
	0.100	0.127	8.986E+05	8.285E+05	5.8	6.7
	0.110	0.148	1.032E+06	9.635E+05	4.9	5.9
	0.120	0.167	1.156E+06	1.093E+06	4.9	5.8
	0.140	0.200	1.375E+06	1.332E+06	4.9	6.3
	0.160	0.229	1.557E+06	1.519E+06	4.9	6.3
	0.180	0.255	1.706E+06	1.690E+06	5.0	6.2
	0.200	0.278	1.826E+06	1.836E+06	5.0	6.2
	0.230	0.308	1.964E+06	2.012E+06	5.0	6.6
	0.260	0.335	2.060E+06	2.167E+06	5.2	6.7
	0.300	0.366	2.143E+06	2.281E+06	5.2	6.6
	0.350	0.399	2.196E+06	2.365E+06	5.2	6.6
	0.400	0.428	2.211E+06	2.439E+06	4.3	6.3
	0.450	0.454	2.202E+06	2.463E+06	4.3	6.3
	0.500	0.477	2.178E+06	2.462E+06	4.3	6.2
	0.550	0.497	2.145E+06	2.444E+06	4.3	6.2
	0.600	0.516	2.105E+06	2.415E+06	5.1	7.6
	0.700	0.550	2.018E+06	2.332E+06	5.1	7.6
	0.800	0.578	1.928E+06	2.233E+06	5.1	7.6
	0.900	0.604	1.840E+06	2.128E+06	5.1	11.2
	1.000	0.627	1.756E+06	2.022E+06	6.4	11.8
	1.100	0.648	1.678E+06	1.918E+06	6.4	11.8
	1.200	0.667	1.606E+06	1.818E+06	6.4	11.9
	1.400	0.700	1.477E+06	1.633E+06	6.4	13.1
	1.600	0.729	1.367E+06	1.466E+06	10.1	14.6
	1.800	0.755	1.272E+06	1.322E+06	10.1	14.8
	2.000	0.777	1.190E+06	1.196E+06	10.1	14.9
6-C	0.035	-0.191	7.756E+03	8.139E+03	23.6	24.0
	0.040	-0.162	1.296E+04	1.307E+04	23.6	24.0
	0.045	-0.136	1.995E+04	1.923E+04	11.2	11.4
	0.050	-0.113	2.881E+04	2.728E+04	11.2	11.4
	0.055	-0.092	3.954E+04	3.693E+04	11.2	11.5
	0.060	-0.073	5.205E+04	4.813E+04	11.2	11.5
	0.070	-0.040	8.188E+04	7.295E+04	8.1	8.4
	0.080	-0.011	1.169E+05	1.033E+05	8.1	8.4
	0.090	0.014	1.557E+05	1.375E+05	8.1	8.4
	0.100	0.037	1.969E+05	1.745E+05	5.8	6.3
	0.110	0.058	2.392E+05	2.135E+05	5.8	6.3
	0.120	0.077	2.817E+05	2.536E+05	5.8	6.3
	0.140	0.111	3.649E+05	3.349E+05	5.8	6.4
	0.160	0.140	4.428E+05	4.114E+05	4.9	5.5
	0.180	0.165	5.139E+05	4.856E+05	4.9	5.5
	0.200	0.188	5.776E+05	5.547E+05	4.9	5.5
	0.230	0.218	6.599E+05	6.478E+05	4.9	5.8
	0.260	0.245	7.277E+05	7.169E+05	5.0	5.8
	0.300	0.276	7.989E+05	8.024E+05	5.0	5.7
	0.350	0.310	8.635E+05	8.856E+05	5.0	6.0
	0.400	0.338	9.077E+05	9.562E+05	5.2	6.1
	0.450	0.364	9.368E+05	9.964E+05	5.2	6.1
	0.500	0.387	9.547E+05	1.024E+06	5.2	6.1
	0.550	0.408	9.644E+05	1.042E+06	5.2	6.3
	0.600	0.426	9.680E+05	1.067E+06	4.3	5.6
	0.700	0.460	9.626E+05	1.080E+06	4.3	5.6
	0.800	0.489	9.467E+05	1.076E+06	4.3	5.6
	0.900	0.515	9.252E+05	1.061E+06	5.1	6.7
	1.000	0.537	9.008E+05	1.039E+06	5.1	6.7
	1.100	0.558	8.751E+05	1.012E+06	5.1	6.7
	1.200	0.577	8.492E+05	9.836E+05	5.1	6.7
	1.400	0.610	7.988E+05	9.229E+05	5.1	9.1
	1.600	0.640	7.521E+05	8.621E+05	6.4	9.6
	1.800	0.665	7.095E+05	8.038E+05	6.4	9.7
	2.000	0.688	6.709E+05	7.489E+05	6.4	9.7
	2.300	0.718	6.199E+05	6.721E+05	10.1	11.8
	2.600	0.745	5.761E+05	6.065E+05	10.1	11.9
	3.000	0.776	5.268E+05	5.307E+05	10.1	12.0
7-N	0.050	-0.201	4.316E+03	4.603E+03	23.6	23.8
	0.055	-0.181	6.124E+03	6.335E+03	23.6	23.8
	0.060	-0.162	8.340E+03	8.411E+03	23.6	23.8
	0.070	-0.128	1.404E+04	1.344E+04	11.2	11.2
	0.080	-0.099	2.140E+04	2.007E+04	11.2	11.2
	0.090	-0.073	3.030E+04	2.802E+04	11.2	11.2
	0.100	-0.051	4.056E+04	3.633E+04	8.1	8.1
	0.110	-0.030	5.193E+04	4.609E+04	8.1	8.1
	0.120	-0.011	6.419E+04	5.671E+04	8.1	8.1
	0.140	0.022	9.050E+04	7.997E+04	8.1	8.1
	0.160	0.051	1.180E+05	1.050E+05	5.8	6.0
	0.180	0.077	1.455E+05	1.310E+05	5.8	6.0
	0.200	0.100	1.723E+05	1.571E+05	5.8	6.0
	0.230	0.130	2.103E+05	1.943E+05	5.8	6.0
	0.260	0.157	2.449E+05	2.301E+05	4.9	5.2
	0.300	0.188	2.854E+05	2.740E+05	4.9	5.2
	0.350	0.221	3.273E+05	3.220E+05	4.9	5.3
	0.400	0.250	3.609E+05	3.568E+05	5.0	5.3
	0.450	0.276	3.874E+05	3.891E+05	5.0	5.3
	0.500	0.299	4.081E+05	4.158E+05	5.0	5.3
	0.550	0.319	4.241E+05	4.434E+05	5.0	5.4
	0.600	0.338	4.363E+05	4.596E+05	5.2	5.5
	0.700	0.372	4.519E+05	4.820E+05	5.2	5.5
	0.800	0.401	4.591E+05	4.947E+05	5.2	5.6
	0.900	0.426	4.608E+05	5.080E+05	4.3	4.9
	1.000	0.449	4.587E+05	5.119E+05	4.3	4.9
	1.100	0.470	4.541E+05	5.118E+05	4.3	4.9
	1.200	0.489	4.478E+05	5.088E+05	4.3	4.9
	1.400	0.522	4.325E+05	4.969E+05	5.1	5.8
	1.600	0.551	4.155E+05	4.803E+05	5.1	5.8
	1.800	0.577	3.983E+05	4.614E+05	5.1	5.8
	2.000	0.600	3.816E+05	4.416E+05	5.1	5.8
	2.300	0.630	3.582E+05	4.118E+05	6.4	7.4
	2.600	0.657	3.369E+05	3.833E+05	6.4	7.5
	3.000	0.688	3.117E+05	3.480E+05	6.4	7.5
	3.500	0.721	2.849E+05	3.080E+05	10.1	11.2
	4.000	0.750	2.624E+05	2.743E+05	10.1	11.3
	4.500	0.776	2.432E+05	2.451E+05	10.1	11.3

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
7-N	5.000	0.799	2.268E+05	2.196E+05	10.1	11.3
8-O	0.070	-0.190	3.495E+03	3.664E+03	23.6	23.6
	0.080	-0.161	5.519E+03	5.444E+03	11.2	11.3
	0.090	-0.135	8.093E+03	7.798E+03	11.2	11.3
	0.100	-0.112	1.120E+04	1.061E+04	11.2	11.3
	0.110	-0.092	1.482E+04	1.384E+04	11.2	11.4
	0.120	-0.073	1.890E+04	1.747E+04	11.2	11.4
	0.140	-0.039	2.820E+04	2.512E+04	8.1	8.3
	0.160	-0.010	3.867E+04	3.416E+04	8.1	8.3
	0.180	0.015	4.987E+04	4.403E+04	8.1	8.3
	0.200	0.038	6.144E+04	5.447E+04	5.8	5.9
	0.230	0.068	7.892E+04	7.075E+04	5.8	5.9
	0.260	0.095	9.596E+04	8.724E+04	5.8	5.9
	0.300	0.126	1.174E+05	1.081E+05	5.8	5.9
	0.350	0.160	1.414E+05	1.331E+05	4.9	4.9
	0.400	0.189	1.623E+05	1.559E+05	4.9	4.9
	0.450	0.214	1.801E+05	1.763E+05	4.9	4.9
	0.500	0.237	1.952E+05	1.914E+05	5.0	5.1
	0.550	0.258	2.079E+05	2.065E+05	5.0	5.1
	0.600	0.277	2.186E+05	2.196E+05	5.0	5.1
	0.700	0.310	2.348E+05	2.409E+05	5.0	5.1
	0.800	0.339	2.457E+05	2.589E+05	5.2	5.5
	0.900	0.365	2.527E+05	2.688E+05	5.2	5.5
	1.000	0.387	2.568E+05	2.755E+05	5.2	5.5
	1.100	0.408	2.589E+05	2.796E+05	5.2	5.5
	1.200	0.427	2.593E+05	2.860E+05	4.3	4.4
	1.400	0.461	2.570E+05	2.884E+05	4.3	4.4
	1.600	0.490	2.521E+05	2.865E+05	4.3	4.4
	1.800	0.515	2.458E+05	2.819E+05	5.1	5.3
	2.000	0.538	2.389E+05	2.755E+05	5.1	5.3
	2.300	0.568	2.280E+05	2.641E+05	5.1	5.3
	2.600	0.595	2.174E+05	2.517E+05	5.1	5.3
	3.000	0.626	2.040E+05	2.349E+05	6.4	7.5
	3.500	0.660	1.890E+05	2.148E+05	6.4	7.5
	4.000	0.689	1.758E+05	1.962E+05	6.4	7.5
	4.500	0.714	1.643E+05	1.788E+05	10.1	14.0
	5.000	0.737	1.542E+05	1.638E+05	10.1	14.1
	5.500	0.758	1.452E+05	1.503E+05	10.1	14.2
	6.000	0.777	1.373E+05	1.382E+05	10.1	14.3
9-F	0.080	-0.216	1.594E+03	1.741E+03	23.6	23.8
	0.090	-0.191	2.397E+03	2.515E+03	23.6	23.8
	0.100	-0.168	3.401E+03	3.457E+03	23.6	23.8
	0.110	-0.147	4.609E+03	4.486E+03	11.2	11.5
	0.120	-0.128	6.016E+03	5.761E+03	11.2	11.6
	0.140	-0.094	9.389E+03	8.782E+03	11.2	11.6
	0.160	-0.065	1.341E+04	1.213E+04	8.1	8.5
	0.180	-0.040	1.796E+04	1.600E+04	8.1	8.5
	0.200	-0.017	2.290E+04	2.025E+04	8.1	8.5
	0.230	0.013	3.077E+04	2.716E+04	8.1	8.6
	0.260	0.040	3.892E+04	3.452E+04	5.8	6.2
	0.300	0.071	4.977E+04	4.467E+04	5.8	6.1
	0.350	0.104	6.281E+04	5.743E+04	5.8	6.2
	0.400	0.133	7.489E+04	6.931E+04	4.9	5.2
	0.450	0.159	8.584E+04	8.078E+04	4.9	5.2
	0.500	0.182	9.565E+04	9.145E+04	4.9	5.2
	0.550	0.203	1.043E+05	1.013E+05	4.9	5.4
	0.600	0.221	1.120E+05	1.102E+05	4.9	5.4
	0.700	0.255	1.246E+05	1.236E+05	5.0	5.5
	0.800	0.284	1.342E+05	1.355E+05	5.0	5.5
	0.900	0.310	1.414E+05	1.451E+05	5.0	5.7
	1.000	0.332	1.467E+05	1.542E+05	5.2	6.1
	1.100	0.353	1.506E+05	1.595E+05	5.2	6.1
	1.200	0.372	1.532E+05	1.634E+05	5.2	6.0
	1.400	0.405	1.558E+05	1.682E+05	5.2	6.2
	1.600	0.435	1.561E+05	1.729E+05	4.3	5.1
	1.800	0.460	1.549E+05	1.738E+05	4.3	5.1
	2.000	0.483	1.527E+05	1.731E+05	4.3	5.1
	2.300	0.513	1.484E+05	1.701E+05	5.1	6.2
	2.600	0.540	1.436E+05	1.656E+05	5.1	6.1
	3.000	0.571	1.368E+05	1.585E+05	5.1	6.1
	3.500	0.604	1.286E+05	1.488E+05	5.1	7.7
	4.000	0.633	1.210E+05	1.390E+05	6.4	9.7
	4.500	0.659	1.141E+05	1.297E+05	6.4	9.7
	5.000	0.682	1.079E+05	1.209E+05	6.4	9.7

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
9-F	5.500	0.703	1.022E+05	1.127E+05	6.4	10.5
	6.000	0.721	9.709E+04	1.049E+05	10.1	16.8
	7.000	0.755	8.829E+04	9.175E+04	10.1	17.0
	8.000	0.784	8.099E+04	8.054E+04	10.1	17.3
10-Ne	0.100	-0.217	1.132E+03	1.238E+03	23.6	24.0
	0.110	-0.196	1.562E+03	1.653E+03	23.6	24.0
	0.120	-0.177	2.076E+03	2.138E+03	23.6	24.0
	0.140	-0.144	3.353E+03	3.255E+03	11.2	11.8
	0.160	-0.115	4.950E+03	4.694E+03	11.2	11.8
	0.180	-0.089	6.834E+03	6.373E+03	11.2	11.8
	0.200	-0.067	8.964E+03	8.114E+03	8.1	8.8
	0.230	-0.036	1.252E+04	1.114E+04	8.1	8.8
	0.260	-0.009	1.640E+04	1.448E+04	8.1	8.8
	0.300	0.022	2.183E+04	1.929E+04	8.1	8.9
	0.350	0.055	2.874E+04	2.562E+04	5.8	6.4
	0.400	0.084	3.552E+04	3.209E+04	5.8	6.4
	0.450	0.110	4.197E+04	3.850E+04	5.8	6.5
	0.500	0.132	4.801E+04	4.441E+04	4.9	5.6
	0.550	0.153	5.359E+04	5.023E+04	4.9	5.5
	0.600	0.172	5.870E+04	5.574E+04	4.9	5.5
	0.700	0.206	6.759E+04	6.573E+04	4.9	5.8
	0.800	0.235	7.485E+04	7.328E+04	5.0	6.0
	0.900	0.260	8.073E+04	8.028E+04	5.0	6.0
	1.000	0.283	8.544E+04	8.619E+04	5.0	5.9
	1.100	0.304	8.920E+04	9.115E+04	5.0	6.3
	1.200	0.323	9.215E+04	9.647E+04	5.0	6.2
	1.400	0.356	9.619E+04	1.020E+05	5.2	6.6
	1.600	0.385	9.839E+04	1.054E+05	5.2	6.6
	1.800	0.411	9.933E+04	1.074E+05	5.2	6.9
	2.000	0.433	9.939E+04	1.100E+05	4.3	5.8
	2.300	0.464	9.839E+04	1.106E+05	4.3	5.7
	2.600	0.491	9.660E+04	1.098E+05	4.3	5.7
	3.000	0.522	9.357E+04	1.075E+05	5.1	7.0
	3.500	0.555	8.936E+04	1.033E+05	5.1	7.0
	4.000	0.584	8.510E+04	9.857E+04	5.1	7.0
	4.500	0.610	8.101E+04	9.361E+04	5.1	9.8
	5.000	0.632	7.718E+04	8.868E+04	6.4	11.8
	5.500	0.653	7.362E+04	8.392E+04	6.4	11.9
	6.000	0.672	7.034E+04	7.936E+04	6.4	11.9
	7.000	0.706	6.451E+04	7.097E+04	6.4	13.2
	8.000	0.734	5.956E+04	6.348E+04	10.1	19.7
	9.000	0.760	5.533E+04	5.709E+04	10.1	20.0
	10.000	0.783	5.168E+04	5.147E+04	10.1	20.3
11-Na	0.010	-0.765	4.738E-04	3.374E-04	12.8	37.2
	0.011	-0.744	1.395E-03	1.097E-03	12.8	34.9
	0.012	-0.725	3.421E-03	2.892E-03	7.4	27.0
	0.014	-0.692	1.410E-02	1.310E-02	5.5	12.0
	0.016	-0.663	4.161E-02	4.060E-02	5.5	11.7
	0.018	-0.637	9.881E-02	9.886E-02	4.6	10.1
	0.020	-0.614	2.020E-01	2.044E-01	4.6	10.1
	0.023	-0.584	4.856E-01	4.947E-01	3.7	7.2
	0.026	-0.557	9.916E-01	1.008E+00	3.7	7.2
	0.030	-0.526	2.162E+00	2.184E+00	2.5	6.5
	0.035	-0.493	4.745E+00	4.718E+00	2.5	5.6
	0.040	-0.464	9.031E+00	8.816E+00	1.9	4.8
	0.045	-0.438	1.555E+01	1.510E+01	1.5	4.6
	0.050	-0.415	2.485E+01	2.412E+01	1.5	4.6
	0.055	-0.395	3.750E+01	3.633E+01	1.5	4.4
	0.060	-0.376	5.408E+01	5.231E+01	1.5	4.3
	0.070	-0.342	1.012E+02	9.751E+01	1.5	4.3
	0.080	-0.313	1.704E+02	1.644E+02	1.3	4.2
	0.090	-0.288	2.658E+02	2.553E+02	1.1	4.1
	0.100	-0.265	3.905E+02	3.743E+02	1.1	4.1
	0.110	-0.244	5.475E+02	5.242E+02	1.1	4.1
	0.120	-0.225	7.386E+02	7.070E+02	1.0	4.0
	0.140	-0.192	1.228E+03	1.178E+03	1.0	4.3
	0.160	-0.163	1.864E+03	1.802E+03	0.9	4.3
	0.180	-0.137	2.642E+03	2.564E+03	0.8	4.1
	0.200	-0.114	3.552E+03	3.458E+03	0.8	4.1
	0.230	-0.084	5.137E+03	5.047E+03	0.9	4.5
	0.260	-0.057	6.940E+03	6.843E+03	0.9	4.5
	0.300	-0.026	9.593E+03	9.468E+03	1.1	4.3
	0.350	0.007	1.315E+04	1.305E+04	1.4	4.4
	0.400	0.036	1.683E+04	1.674E+04	1.4	4.4
	0.450	0.062	2.049E+04	2.051E+04	1.7	4.7
	0.500	0.085	2.406E+04	2.427E+04	1.7	4.7

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
11-Na	0.550	0.105	2.747E+04	2.781E+04	1.7	5.3
	0.600	0.124	3.071E+04	3.118E+04	1.6	5.6
	0.700	0.158	3.660E+04	3.744E+04	1.8	5.9
	0.800	0.187	4.170E+04	4.300E+04	1.8	5.9
	0.900	0.212	4.605E+04	4.782E+04	1.9	8.2
	1.000	0.235	4.974E+04	5.194E+04	1.9	8.2
	1.100	0.256	5.285E+04	5.541E+04	1.8	8.1
	1.200	0.275	5.545E+04	5.831E+04	1.8	8.0
	1.400	0.308	5.941E+04	6.264E+04	2.8	9.5
	1.600	0.337	6.208E+04	6.552E+04	2.8	9.5
	1.800	0.363	6.379E+04	6.749E+04	3.5	9.3
	2.000	0.386	6.479E+04	6.846E+04	3.5	9.3
	2.300	0.416	6.537E+04	6.876E+04	4.0	11.8
	2.600	0.443	6.517E+04	6.813E+04	4.0	11.8
	3.000	0.474	6.419E+04	6.582E+04	4.6	12.9
	3.500	0.507	6.233E+04	6.237E+04	4.6	15.9
	4.000	0.536	6.014E+04	5.852E+04	5.8	16.4
	4.500	0.562	5.785E+04	5.461E+04	7.9	13.2
	5.000	0.585	5.559E+04	5.079E+04	7.9	13.4
	5.500	0.605	5.340E+04	4.716E+04	7.9	22.1
	6.000	0.624	5.132E+04	4.374E+04	7.2	15.6
12-Mg	0.014	-0.738	2.028E-03	1.631E-03	7.4	26.8
	0.016	-0.709	7.130E-03	6.353E-03	7.4	24.9
	0.018	-0.683	1.920E-02	1.815E-02	5.5	11.5
	0.020	-0.660	4.308E-02	4.217E-02	5.5	11.3
	0.023	-0.630	1.147E-01	1.153E-01	4.6	9.7
	0.026	-0.603	2.518E-01	2.555E-01	4.6	9.6
	0.030	-0.572	5.883E-01	5.993E-01	3.7	7.0
	0.035	-0.539	1.370E+00	1.391E+00	2.5	6.2
	0.040	-0.510	2.719E+00	2.727E+00	2.5	6.3
	0.045	-0.484	4.829E+00	4.754E+00	1.9	4.6
	0.050	-0.461	7.905E+00	7.712E+00	1.9	4.6
	0.055	-0.441	1.216E+01	1.181E+01	1.5	4.4
	0.060	-0.422	1.782E+01	1.727E+01	1.5	4.4
	0.070	-0.388	3.422E+01	3.313E+01	1.5	4.1
	0.080	-0.359	5.885E+01	5.671E+01	1.5	4.1
	0.090	-0.334	9.338E+01	8.997E+01	1.3	4.0
	0.100	-0.311	1.392E+02	1.343E+02	1.3	4.0
	0.110	-0.290	1.980E+02	1.903E+02	1.1	3.9
	0.120	-0.271	2.707E+02	2.593E+02	1.1	3.9
	0.140	-0.238	4.616E+02	4.419E+02	1.0	3.8
	0.160	-0.209	7.173E+02	6.864E+02	1.0	3.8
	0.180	-0.183	1.040E+03	9.985E+02	0.9	4.1
	0.200	-0.160	1.428E+03	1.382E+03	0.9	4.1
	0.230	-0.130	2.129E+03	2.068E+03	0.8	3.9
	0.260	-0.103	2.959E+03	2.890E+03	0.8	3.9
	0.300	-0.072	4.231E+03	4.168E+03	0.9	4.2
	0.350	-0.039	6.026E+03	5.937E+03	1.1	4.1
	0.400	-0.010	7.967E+03	7.887E+03	1.1	4.1
	0.450	0.016	9.986E+03	9.921E+03	1.4	4.1
	0.500	0.039	1.203E+04	1.197E+04	1.4	4.1
	0.550	0.059	1.405E+04	1.405E+04	1.7	4.4
	0.600	0.078	1.603E+04	1.611E+04	1.7	4.4
	0.700	0.112	1.978E+04	2.004E+04	1.6	5.3
	0.800	0.141	2.318E+04	2.360E+04	1.6	5.3
	0.900	0.166	2.623E+04	2.689E+04	1.8	5.6
	1.000	0.189	2.892E+04	2.984E+04	1.8	5.6
	1.100	0.210	3.128E+04	3.246E+04	1.9	7.8
	1.200	0.229	3.335E+04	3.477E+04	1.9	7.7
	1.400	0.262	3.669E+04	3.850E+04	1.8	7.6
	1.600	0.291	3.917E+04	4.125E+04	1.8	7.6
	1.800	0.317	4.099E+04	4.324E+04	2.8	8.9
	2.000	0.340	4.228E+04	4.463E+04	2.8	8.9
	2.300	0.370	4.349E+04	4.600E+04	3.5	8.7
	2.600	0.397	4.406E+04	4.649E+04	3.5	8.7
	3.000	0.428	4.415E+04	4.634E+04	4.0	11.1
	3.500	0.461	4.361E+04	4.504E+04	4.6	12.1
	4.000	0.490	4.266E+04	4.327E+04	4.6	12.2
	4.500	0.516	4.150E+04	4.122E+04	5.8	15.1
	5.000	0.539	4.025E+04	3.906E+04	5.8	15.4
	5.500	0.559	3.896E+04	3.689E+04	5.8	15.6
	6.000	0.578	3.769E+04	3.478E+04	7.9	12.3
	7.000	0.612	3.527E+04	3.079E+04	7.2	12.6
	8.000	0.641	3.306E+04	2.722E+04	7.2	12.9
13-Al	0.016	-0.751	1.150E-03	8.781E-04	12.8	33.0
13-Al	0.018	-0.725	3.602E-03	3.044E-03	7.4	24.7
	0.020	-0.702	9.043E-03	8.196E-03	7.4	23.5
	0.023	-0.672	2.720E-02	2.620E-02	5.5	11.0
	0.026	-0.645	6.513E-02	6.474E-02	5.5	10.8
	0.030	-0.614	1.652E-01	1.671E-01	4.6	9.3
	0.035	-0.581	4.127E-01	4.205E-01	3.7	6.7
	0.040	-0.552	8.603E-01	8.735E-01	3.7	6.7
	0.045	-0.526	1.584E+00	1.600E+00	2.5	6.0
	0.050	-0.503	2.666E+00	2.666E+00	2.5	6.0
	0.055	-0.483	4.193E+00	4.126E+00	1.9	4.4
	0.060	-0.464	6.256E+00	6.107E+00	1.9	4.4
	0.070	-0.430	1.235E+01	1.199E+01	1.5	4.2
	0.080	-0.401	2.171E+01	2.105E+01	1.5	4.2
	0.090	-0.376	3.506E+01	3.391E+01	1.5	3.9
	0.100	-0.353	5.310E+01	5.116E+01	1.5	3.9
	0.110	-0.332	7.649E+01	7.370E+01	1.3	3.8
	0.120	-0.313	1.058E+02	1.021E+02	1.3	3.8
	0.140	-0.280	1.844E+02	1.769E+02	1.1	3.7
	0.160	-0.251	2.922E+02	2.799E+02	1.1	3.7
	0.180	-0.225	4.316E+02	4.131E+02	1.0	3.6
	0.200	-0.202	6.035E+02	5.779E+02	1.0	3.6
	0.230	-0.172	9.226E+02	8.895E+02	0.9	3.9
	0.260	-0.145	1.312E+03	1.272E+03	0.9	3.8
	0.300	-0.114	1.932E+03	1.881E+03	0.8	3.7
	0.350	-0.081	2.843E+03	2.796E+03	0.9	4.0
	0.400	-0.052	3.871E+03	3.814E+03	0.9	4.0
	0.450	-0.026	4.981E+03	4.916E+03	1.1	3.8
	0.500	-0.003	6.141E+03	6.084E+03	1.1	3.8
	0.550	0.017	7.326E+03	7.280E+03	1.4	3.9
	0.600	0.036	8.517E+03	8.473E+03	1.4	3.9
	0.700	0.070	1.086E+04	1.089E+04	1.7	4.2
	0.800	0.099	1.308E+04	1.323E+04	1.7	4.2
	0.900	0.124	1.515E+04	1.538E+04	1.6	5.0
	1.000	0.147	1.704E+04	1.738E+04	1.6	5.0
	1.100	0.168	1.876E+04	1.924E+04	1.8	5.3
	1.200	0.187	2.031E+04	2.094E+04	1.8	5.3
	1.400	0.220	2.293E+04	2.386E+04	1.9	7.3
	1.600	0.249	2.502E+04	2.619E+04	1.9	7.3
	1.800	0.275	2.665E+04	2.802E+04	1.8	7.2
	2.000	0.298	2.791E+04	2.941E+04	1.8	7.2
	2.300	0.328	2.926E+04	3.088E+04	2.8	8.3
	2.600	0.355	3.012E+04	3.179E+04	2.8	8.3
	3.000	0.386	3.072E+04	3.246E+04	3.5	8.1
	3.500	0.419	3.088E+04	3.247E+04	4.0	10.4
	4.000	0.448	3.063E+04	3.196E+04	4.0	10.4
	4.500	0.474	3.014E+04	3.091E+04	4.6	11.4
	5.000	0.497	2.951E+04	2.979E+04	4.6	11.5
	5.500	0.517	2.881E+04	2.857E+04	5.8	14.2
	6.000	0.536	2.806E+04	2.731E+04	5.8	14.3
	7.000	0.570	2.655E+04	2.480E+04	7.9	11.1
	8.000	0.599	2.510E+04	2.243E+04	7.9	11.3
	9.000	0.624	2.375E+04	2.025E+04	7.2	9.8
	10.000	0.647	2.252E+04	1.827E+04	7.2	9.9
14-Si	0.018	-0.764	6.206E-04	4.430E-04	12.8	33.0
	0.020	-0.741	1.787E-03	1.421E-03	12.8	30.9
	0.023	-0.711	6.229E-03	5.513E-03	7.4	22.8
	0.026	-0.684	1.656E-02	1.562E-02	5.5	10.7
	0.030	-0.653	4.635E-02	4.572E-02	5.5	10.4
	0.035	-0.620	1.260E-01	1.273E-01	4.6	8.9
	0.040	-0.591	2.785E-01	2.828E-01	4.6	7.2
	0.045	-0.565	5.352E-01	5.448E-01	3.7	6.5
	0.050	-0.542	9.307E-01	9.425E-01	3.7	6.5
	0.055	-0.522	1.502E+00	1.515E+00	2.5	5.8
	0.060	-0.503	2.288E+00	2.287E+00	2.5	5.8
	0.070	-0.469	4.665E+00	4.569E+00	1.9	4.2
	0.080	-0.440	8.394E+00	8.155E+00	1.5	4.0
	0.090	-0.415	1.381E+01	1.340E+01	1.5	4.0
	0.100	-0.392	2.124E+01	2.057E+01	1.5	3.9
	0.110	-0.371	3.101E+01	2.999E+01	1.5	3.7
	0.120	-0.352	4.341E+01	4.183E+01	1.5	3.8
	0.140	-0.319	7.722E+01	7.441E+01	1.3	3.6
	0.160	-0.290	1.245E+02	1.197E+02	1.1	3.5
	0.180	-0.264	1.870E+02	1.792E+02	1.1	3.5
	0.200	-0.241	2.655E+02	2.542E+02	1.1	3.5
	0.230	-0.211	4.147E+02	3.968E+02	1.0	3.4
	0.260	-0.184	6.019E+02	5.778E+02	0.9	3.6
	0.300	-0.153	9.087E+02	8.798E+02	0.9	3.6
	0.350	-0.120	1.376E+03	1.338E+03	0.8	3.4

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
14-Si	0.400	-0.091	1.923E+03	1.885E+03	0.8	3.7
	0.450	-0.065	2.533E+03	2.498E+03	0.9	3.8
	0.500	-0.042	3.190E+03	3.138E+03	0.9	3.8
	0.550	-0.022	3.880E+03	3.832E+03	1.1	3.6
	0.600	-0.003	4.591E+03	4.549E+03	1.1	3.6
	0.700	0.031	6.037E+03	6.011E+03	1.4	3.6
	0.800	0.060	7.467E+03	7.469E+03	1.7	3.9
	0.900	0.085	8.843E+03	8.923E+03	1.7	3.9
	1.000	0.108	1.014E+04	1.027E+04	1.6	4.7
	1.100	0.129	1.136E+04	1.154E+04	1.6	4.7
	1.200	0.148	1.248E+04	1.273E+04	1.6	4.7
	1.400	0.181	1.446E+04	1.489E+04	1.8	5.0
	1.600	0.210	1.611E+04	1.672E+04	1.9	6.9
	1.800	0.236	1.747E+04	1.824E+04	1.9	6.9
	2.000	0.259	1.857E+04	1.948E+04	1.8	6.8
	2.300	0.289	1.985E+04	2.090E+04	1.8	6.8
	2.600	0.316	2.076E+04	2.190E+04	2.8	7.8
	3.000	0.347	2.155E+04	2.274E+04	2.8	7.8
	3.500	0.380	2.204E+04	2.331E+04	3.5	7.6
	4.000	0.409	2.218E+04	2.336E+04	3.5	8.6
	4.500	0.435	2.208E+04	2.314E+04	4.0	9.7
	5.000	0.458	2.184E+04	2.260E+04	4.0	9.8
	5.500	0.478	2.149E+04	2.198E+04	4.6	10.7
	6.000	0.497	2.109E+04	2.128E+04	4.6	10.8
	7.000	0.531	2.019E+04	1.977E+04	5.8	13.2
	8.000	0.560	1.927E+04	1.824E+04	5.8	13.5
	9.000	0.585	1.836E+04	1.677E+04	7.9	10.1
	10.000	0.608	1.751E+04	1.539E+04	7.2	7.5
	11.000	0.629	1.671E+04	1.412E+04	7.2	7.5
	12.000	0.648	1.598E+04	1.294E+04	7.2	7.5
15-P	0.023	-0.748	1.348E-03	1.042E-03	12.8	30.1
	0.026	-0.721	4.054E-03	3.472E-03	7.4	22.1
	0.030	-0.690	1.274E-02	1.187E-02	5.5	10.3
	0.035	-0.657	3.822E-02	3.756E-02	5.5	10.1
	0.040	-0.628	9.044E-02	9.100E-02	4.6	8.5
	0.045	-0.602	1.827E-01	1.853E-01	4.6	8.5
	0.050	-0.580	3.299E-01	3.361E-01	3.7	6.2
	0.055	-0.559	5.483E-01	5.576E-01	3.7	6.2
	0.060	-0.540	8.553E-01	8.684E-01	2.5	5.5
	0.070	-0.506	1.808E+00	1.811E+00	2.5	5.5
	0.080	-0.477	3.340E+00	3.280E+00	1.9	4.0
	0.090	-0.452	5.607E+00	5.458E+00	1.9	4.1
	0.100	-0.429	8.767E+00	8.503E+00	1.5	3.8
	0.110	-0.408	1.298E+01	1.259E+01	1.5	3.8
	0.120	-0.389	1.839E+01	1.780E+01	1.5	3.6
	0.140	-0.356	3.337E+01	3.215E+01	1.5	3.6
	0.160	-0.327	5.478E+01	5.278E+01	1.3	3.4
	0.180	-0.301	8.343E+01	8.035E+01	1.3	3.4
	0.200	-0.278	1.201E+02	1.152E+02	1.1	3.3
	0.230	-0.248	1.913E+02	1.832E+02	1.1	3.3
	0.260	-0.221	2.826E+02	2.706E+02	1.0	3.2
	0.300	-0.190	4.363E+02	4.184E+02	1.0	3.4
	0.350	-0.157	6.776E+02	6.557E+02	0.9	3.4
	0.400	-0.128	9.691E+02	9.415E+02	0.8	3.2
	0.450	-0.102	1.304E+03	1.274E+03	0.8	3.2
	0.500	-0.079	1.675E+03	1.647E+03	0.9	3.5
	0.550	-0.059	2.074E+03	2.046E+03	0.9	3.5
	0.600	-0.040	2.495E+03	2.458E+03	1.1	3.4
	0.700	-0.006	3.379E+03	3.346E+03	1.1	3.4
	0.800	0.023	4.285E+03	4.262E+03	1.4	3.4
	0.900	0.048	5.186E+03	5.173E+03	1.4	3.4
	1.000	0.071	6.063E+03	6.083E+03	1.7	3.7
	1.100	0.092	6.904E+03	6.974E+03	1.7	3.7
	1.200	0.111	7.701E+03	7.802E+03	1.6	4.4
	1.400	0.144	9.153E+03	9.325E+03	1.6	4.4
	1.600	0.173	1.041E+04	1.070E+04	1.8	4.8
	1.800	0.199	1.149E+04	1.189E+04	1.8	4.7
	2.000	0.221	1.241E+04	1.291E+04	1.9	6.4
	2.300	0.252	1.352E+04	1.416E+04	1.9	6.4
	2.600	0.278	1.436E+04	1.510E+04	1.8	6.3
	3.000	0.310	1.517E+04	1.600E+04	2.8	7.2
	3.500	0.343	1.580E+04	1.668E+04	2.8	7.2
	4.000	0.372	1.613E+04	1.706E+04	3.5	7.0
	4.500	0.398	1.625E+04	1.715E+04	3.5	7.0
	5.000	0.421	1.624E+04	1.707E+04	4.0	9.0
	5.500	0.441	1.612E+04	1.686E+04	4.0	9.0
15-P	6.000	0.460	1.593E+04	1.647E+04	4.6	9.9
	7.000	0.494	1.545E+04	1.563E+04	4.6	10.0
	8.000	0.523	1.488E+04	1.468E+04	5.8	12.2
	9.000	0.548	1.429E+04	1.372E+04	5.8	12.3
	10.000	0.571	1.371E+04	1.279E+04	7.9	9.0
	11.000	0.592	1.316E+04	1.189E+04	7.9	9.0
	12.000	0.611	1.263E+04	1.105E+04	7.2	10.3
	14.000	0.644	1.167E+04	9.537E+03	7.2	10.6
16-S	0.026	-0.754	1.003E-03	7.558E-04	12.8	29.3
	0.030	-0.723	3.602E-03	3.073E-03	7.4	21.0
	0.035	-0.689	1.209E-02	1.130E-02	5.5	9.9
	0.040	-0.660	3.093E-02	3.027E-02	5.5	9.7
	0.045	-0.635	6.607E-02	6.622E-02	4.6	8.1
	0.050	-0.612	1.245E-01	1.261E-01	4.6	8.1
	0.055	-0.591	2.139E-01	2.172E-01	4.6	6.8
	0.060	-0.572	3.426E-01	3.489E-01	3.7	6.0
	0.070	-0.539	7.535E-01	7.647E-01	2.5	5.2
	0.080	-0.510	1.432E+00	1.436E+00	2.5	5.3
	0.090	-0.484	2.456E+00	2.418E+00	1.9	3.8
	0.100	-0.461	3.907E+00	3.811E+00	1.9	3.9
	0.110	-0.440	5.864E+00	5.697E+00	1.5	3.6
	0.120	-0.422	8.407E+00	8.148E+00	1.5	3.6
	0.140	-0.388	1.556E+01	1.507E+01	1.5	3.4
	0.160	-0.359	2.595E+01	2.500E+01	1.5	3.4
	0.180	-0.334	4.008E+01	3.862E+01	1.3	3.2
	0.200	-0.311	5.838E+01	5.631E+01	1.3	3.2
	0.230	-0.280	9.450E+01	9.066E+01	1.1	3.1
	0.260	-0.254	1.417E+02	1.357E+02	1.1	3.1
	0.300	-0.223	2.227E+02	2.132E+02	1.0	3.0
	0.350	-0.189	3.533E+02	3.389E+02	1.0	3.2
	0.400	-0.160	5.150E+02	4.981E+02	0.9	3.2
	0.450	-0.135	7.053E+02	6.847E+02	0.8	3.0
	0.500	-0.112	9.209E+02	8.972E+02	0.8	3.0
	0.550	-0.091	1.158E+03	1.135E+03	0.8	3.2
	0.600	-0.072	1.413E+03	1.392E+03	0.9	3.3
	0.700	-0.039	1.962E+03	1.933E+03	1.1	3.1
	0.800	-0.010	2.544E+03	2.519E+03	1.1	3.1
	0.900	0.016	3.139E+03	3.119E+03	1.4	3.1
	1.000	0.039	3.733E+03	3.716E+03	1.4	3.1
	1.100	0.060	4.316E+03	4.317E+03	1.7	3.4
	1.200	0.078	4.880E+03	4.905E+03	1.7	3.4
	1.400	0.112	5.937E+03	6.016E+03	1.6	4.1
	1.600	0.141	6.887E+03	7.010E+03	1.6	4.1
	1.800	0.166	7.726E+03	7.922E+03	1.8	4.5
	2.000	0.189	8.461E+03	8.731E+03	1.8	4.4
	2.300	0.220	9.385E+03	9.762E+03	1.9	6.0
	2.600	0.246	1.013E+04	1.060E+04	1.9	6.0
	3.000	0.277	1.088E+04	1.144E+04	1.8	5.9
	3.500	0.311	1.152E+04	1.215E+04	2.8	6.6
	4.000	0.340	1.193E+04	1.259E+04	2.8	6.6
	4.500	0.365	1.216E+04	1.287E+04	3.5	6.4
	5.000	0.388	1.227E+04	1.296E+04	3.5	6.4
	5.500	0.409	1.229E+04	1.294E+04	3.5	7.2
	6.000	0.428	1.224E+04	1.284E+04	4.0	8.2
	7.000	0.461	1.201E+04	1.240E+04	4.6	9.2
	8.000	0.490	1.168E+04	1.185E+04	4.6	9.3
	9.000	0.516	1.131E+04	1.123E+04	5.8	11.1
	10.000	0.539	1.093E+04	1.060E+04	5.8	11.2
	11.000	0.559	1.054E+04	9.979E+03	5.8	11.4
	12.000	0.578	1.016E+04	9.376E+03	7.9	8.0
	14.000	0.612	9.462E+03	8.260E+03	7.2	13.2
	16.000	0.641	8.834E+03	7.270E+03	7.2	13.6
17-Cl	0.030	-0.753	9.783E-04	7.378E-04	12.8	27.9
	0.035	-0.720	3.738E-03	3.217E-03	7.4	19.7
	0.040	-0.691	1.044E-02	9.719E-03	5.5	9.4
	0.045	-0.665	2.379E-02	2.313E-02	5.5	9.3
	0.050	-0.643	4.703E-02	4.686E-02	5.5	9.2
	0.055	-0.622	8.387E-02	8.464E-02	4.6	7.7
	0.060	-0.603	1.384E-01	1.404E-01	4.6	7.7
	0.070	-0.570	3.183E-01	3.241E-01	3.7	5.7
	0.080	-0.540	6.242E-01	6.339E-01	2.5	5.0
	0.090	-0.515	1.096E+00	1.102E+00	2.5	5.0
	0.100	-0.492	1.777E+00	1.755E+00	1.9	3.6
	0.110	-0.471	2.707E+00	2.653E+00	1.9	3.7
	0.120	-0.452	3.930E+00	3.827E+00	1.9	3.7

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
17-Cl	0.140	-0.419	7.423E+00	7.208E+00	1.5	3.4
	0.160	-0.390	1.258E+01	1.218E+01	1.5	3.2
	0.180	-0.364	1.969E+01	1.904E+01	1.5	3.2
	0.200	-0.342	2.904E+01	2.798E+01	1.5	3.2
	0.230	-0.311	4.770E+01	4.602E+01	1.3	3.0
	0.260	-0.285	7.253E+01	6.963E+01	1.1	2.9
	0.300	-0.254	1.159E+02	1.110E+02	1.1	2.9
	0.350	-0.220	1.874E+02	1.794E+02	1.0	2.8
	0.400	-0.191	2.779E+02	2.665E+02	1.0	3.0
	0.450	-0.166	3.868E+02	3.738E+02	0.9	3.0
	0.500	-0.143	5.126E+02	4.971E+02	0.9	3.0
	0.550	-0.122	6.537E+02	6.356E+02	0.8	2.8
	0.600	-0.103	8.080E+02	7.893E+02	0.8	2.8
	0.700	-0.069	1.149E+03	1.133E+03	0.9	3.1
	0.800	-0.041	1.521E+03	1.499E+03	1.1	2.9
	0.900	-0.015	1.912E+03	1.891E+03	1.1	2.9
	1.000	0.008	2.311E+03	2.294E+03	1.4	2.9
	1.100	0.029	2.711E+03	2.699E+03	1.4	2.9
	1.200	0.047	3.107E+03	3.098E+03	1.4	2.9
	1.400	0.081	3.867E+03	3.899E+03	1.7	3.2
	1.600	0.110	4.571E+03	4.631E+03	1.6	3.8
	1.800	0.136	5.213E+03	5.299E+03	1.6	3.8
	2.000	0.159	5.789E+03	5.922E+03	1.8	4.2
	2.300	0.189	6.538E+03	6.746E+03	1.8	4.2
	2.600	0.215	7.160E+03	7.440E+03	1.9	5.6
	3.000	0.247	7.822E+03	8.186E+03	1.9	5.6
	3.500	0.280	8.429E+03	8.865E+03	1.8	5.5
	4.000	0.309	8.849E+03	9.331E+03	2.8	6.1
	4.500	0.335	9.128E+03	9.635E+03	2.8	6.1
	5.000	0.357	9.301E+03	9.814E+03	3.5	5.9
	5.500	0.378	9.394E+03	9.932E+03	3.5	5.9
	6.000	0.397	9.426E+03	9.946E+03	3.5	5.9
	7.000	0.431	9.368E+03	9.826E+03	4.0	7.5
	8.000	0.460	9.206E+03	9.517E+03	4.6	8.5
	9.000	0.485	8.987E+03	9.148E+03	4.6	8.5
	10.000	0.508	8.739E+03	8.740E+03	4.6	9.9
	11.000	0.529	8.480E+03	8.318E+03	5.8	10.2
	12.000	0.548	8.218E+03	7.896E+03	5.8	10.3
	14.000	0.581	7.710E+03	7.085E+03	7.9	9.0
	16.000	0.610	7.240E+03	6.342E+03	7.2	16.0
	18.000	0.636	6.814E+03	5.673E+03	7.2	16.5
18-Ar	0.035	-0.750	1.104E-03	8.463E-04	12.8	26.4
	0.040	-0.721	3.417E-03	2.931E-03	7.4	18.6
	0.045	-0.695	8.376E-03	7.719E-03	7.4	11.6
	0.050	-0.673	1.750E-02	1.684E-02	5.5	8.9
	0.055	-0.652	3.257E-02	3.218E-02	5.5	8.8
	0.060	-0.633	5.560E-02	5.578E-02	4.6	7.3
	0.070	-0.600	1.345E-01	1.365E-01	4.6	6.3
	0.080	-0.571	2.732E-01	2.783E-01	3.7	5.5
	0.090	-0.545	4.928E-01	4.994E-01	3.7	5.5
	0.100	-0.522	8.151E-01	8.221E-01	2.5	4.8
	0.110	-0.501	1.263E+00	1.261E+00	2.5	4.8
	0.120	-0.482	1.859E+00	1.829E+00	1.9	3.5
	0.140	-0.449	3.586E+00	3.490E+00	1.9	3.5
	0.160	-0.420	6.180E+00	5.988E+00	1.5	3.2
	0.180	-0.394	9.809E+00	9.503E+00	1.5	3.1
	0.200	-0.371	1.463E+01	1.415E+01	1.5	3.0
	0.230	-0.341	2.442E+01	2.353E+01	1.5	3.0
	0.260	-0.315	3.761E+01	3.630E+01	1.3	2.9
	0.300	-0.283	6.106E+01	5.861E+01	1.1	2.7
	0.350	-0.250	1.004E+02	9.619E+01	1.1	2.7
	0.400	-0.221	1.514E+02	1.449E+02	1.0	2.6
	0.450	-0.195	2.139E+02	2.050E+02	1.0	2.7
	0.500	-0.173	2.874E+02	2.770E+02	0.9	2.8
	0.550	-0.152	3.713E+02	3.596E+02	0.9	2.8
	0.600	-0.133	4.646E+02	4.511E+02	0.8	2.6
	0.700	-0.100	6.755E+02	6.606E+02	0.8	2.7
	0.800	-0.071	9.122E+02	8.989E+02	0.9	2.8
	0.900	-0.045	1.167E+03	1.149E+03	0.9	2.8
	1.000	-0.022	1.433E+03	1.415E+03	1.1	2.7
	1.100	-0.001	1.705E+03	1.690E+03	1.1	2.7
	1.200	0.018	1.979E+03	1.967E+03	1.4	2.6
	1.400	0.051	2.520E+03	2.515E+03	1.4	2.6
	1.600	0.080	3.036E+03	3.053E+03	1.7	2.9
	1.800	0.106	3.518E+03	3.561E+03	1.7	3.2
	2.000	0.128	3.963E+03	4.026E+03	1.6	3.5
18-Ar	2.300	0.159	4.556E+03	4.661E+03	1.8	3.9
	2.600	0.185	5.066E+03	5.222E+03	1.8	3.9
	3.000	0.216	5.627E+03	5.849E+03	1.9	5.2
	3.500	0.250	6.169E+03	6.461E+03	1.9	5.1
	4.000	0.279	6.568E+03	6.907E+03	1.8	5.0
	4.500	0.305	6.856E+03	7.229E+03	2.8	5.5
	5.000	0.327	7.056E+03	7.447E+03	2.8	5.5
	5.500	0.348	7.190E+03	7.589E+03	2.8	5.5
	6.000	0.367	7.270E+03	7.692E+03	3.5	5.3
	7.000	0.400	7.318E+03	7.717E+03	3.5	5.7
	8.000	0.430	7.266E+03	7.623E+03	4.0	6.8
	9.000	0.455	7.153E+03	7.443E+03	4.0	6.8
	10.000	0.478	7.005E+03	7.164E+03	4.6	7.8
	11.000	0.499	6.837E+03	6.890E+03	4.6	7.8
	12.000	0.518	6.660E+03	6.604E+03	5.8	9.1
	14.000	0.551	6.301E+03	6.029E+03	5.8	9.2
	16.000	0.580	5.954E+03	5.479E+03	7.9	10.1
	18.000	0.606	5.631E+03	4.971E+03	7.9	13.6
	20.000	0.628	5.335E+03	4.509E+03	7.2	19.3
19-K	0.040	-0.748	1.105E-03	8.525E-04	12.8	25.0
	0.045	-0.723	2.947E-03	2.511E-03	7.4	17.5
	0.050	-0.700	6.557E-03	5.978E-03	7.4	16.8
	0.055	-0.679	1.281E-02	1.220E-02	5.5	8.5
	0.060	-0.660	2.273E-02	2.225E-02	5.5	8.4
	0.070	-0.627	5.819E-02	5.858E-02	4.6	6.9
	0.080	-0.598	1.231E-01	1.249E-01	4.6	6.0
	0.090	-0.572	2.286E-01	2.329E-01	3.7	5.3
	0.100	-0.549	3.868E-01	3.925E-01	3.7	5.3
	0.110	-0.529	6.100E-01	6.169E-01	2.5	4.5
	0.120	-0.510	9.112E-01	9.140E-01	2.5	4.5
	0.140	-0.477	1.799E+00	1.766E+00	1.9	3.3
	0.160	-0.447	3.154E+00	3.068E+00	1.9	3.3
	0.180	-0.422	5.076E+00	4.920E+00	1.5	3.0
	0.200	-0.399	7.662E+00	7.426E+00	1.5	2.9
	0.230	-0.369	1.297E+01	1.255E+01	1.5	2.8
	0.260	-0.342	2.024E+01	1.950E+01	1.5	2.8
	0.300	-0.311	3.332E+01	3.214E+01	1.3	2.7
	0.350	-0.278	5.570E+01	5.341E+01	1.1	2.5
	0.400	-0.249	8.517E+01	8.156E+01	1.1	2.5
	0.450	-0.223	1.219E+02	1.167E+02	1.0	2.4
	0.500	-0.200	1.658E+02	1.588E+02	1.0	2.4
	0.550	-0.179	2.167E+02	2.084E+02	0.9	2.5
	0.600	-0.161	2.741E+02	2.650E+02	0.9	2.5
	0.700	-0.127	4.065E+02	3.950E+02	0.8	2.4
	0.800	-0.098	5.586E+02	5.465E+02	0.8	2.5
	0.900	-0.072	7.261E+02	7.153E+02	0.9	2.6
	1.000	-0.049	9.047E+02	8.911E+02	0.9	2.6
	1.100	-0.029	1.091E+03	1.076E+03	1.1	2.4
	1.200	-0.010	1.281E+03	1.268E+03	1.1	2.4
	1.400	0.023	1.666E+03	1.657E+03	1.4	2.4
	1.600	0.052	2.043E+03	2.040E+03	1.4	2.3
	1.800	0.078	2.404E+03	2.416E+03	1.7	2.7
	2.000	0.101	2.744E+03	2.776E+03	1.7	2.9
	2.300	0.131	3.210E+03	3.262E+03	1.6	3.2
	2.600	0.158	3.620E+03	3.703E+03	1.8	3.6
	3.000	0.189	4.087E+03	4.218E+03	1.8	3.6
	3.500	0.222	4.556E+03	4.743E+03	1.9	4.7
	4.000	0.252	4.918E+03	5.152E+03	1.9	4.7
	4.500	0.277	5.193E+03	5.459E+03	1.8	4.6
	5.000	0.300	5.398E+03	5.689E+03	1.8	4.6
	5.500	0.321	5.547E+03	5.853E+03	2.8	4.9
	6.000	0.340	5.652E+03	5.966E+03	2.8	4.9
	7.000	0.373	5.762E+03	6.094E+03	3.5	4.7
	8.000	0.402	5.779E+03	6.093E+03	3.5	5.0
	9.000	0.428	5.738E+03	6.023E+03	4.0	6.1
	10.000	0.451	5.659E+03	5.900E+03	4.0	6.1
	11.000	0.471	5.558E+03	5.708E+03	4.6	7.0
	12.000	0.490	5.442E+03	5.520E+03	4.6	7.1
	14.000	0.524	5.192E+03	5.119E+03	5.8	8.1
	16.000	0.553	4.939E+03	4.717E+03	5.8	8.2
	18.000	0.578	4.695E+03	4.333E+03	7.9	11.2
	20.000	0.601	4.467E+03	3.975E+03	7.9	16.4
	23.000	0.631	4.156E+03	3.491E+03	7.2	22.4
20-Ca	0.045	-0.749	1.016E-03	7.815E-04	12.8	23.7
	0.050	-0.726	2.427E-03	2.043E-03	7.4	16.4

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
20-Ca	0.055	-0.706	5.007E-03	4.500E-03	7.4	15.9
	0.060	-0.687	9.274E-03	8.706E-03	5.5	8.1
	0.070	-0.653	2.529E-02	2.495E-02	5.5	8.0
	0.080	-0.624	5.590E-02	5.635E-02	4.6	6.5
	0.090	-0.599	1.073E-01	1.089E-01	4.6	5.8
	0.100	-0.576	1.860E-01	1.895E-01	3.7	5.0
	0.110	-0.555	2.992E-01	3.040E-01	3.7	5.0
	0.120	-0.536	4.541E-01	4.605E-01	2.5	4.3
	0.140	-0.503	9.188E-01	9.183E-01	2.5	4.3
	0.160	-0.474	1.640E+00	1.609E+00	1.9	3.1
	0.180	-0.448	2.678E+00	2.605E+00	1.9	3.1
	0.200	-0.425	4.090E+00	3.965E+00	1.5	2.8
	0.230	-0.395	7.028E+00	6.809E+00	1.5	2.8
	0.260	-0.368	1.110E+01	1.073E+01	1.5	2.6
	0.300	-0.337	1.854E+01	1.786E+01	1.3	2.5
	0.350	-0.304	3.143E+01	3.028E+01	1.3	2.5
	0.400	-0.275	4.871E+01	4.669E+01	1.1	2.3
	0.450	-0.249	7.057E+01	6.758E+01	1.1	2.3
	0.500	-0.226	9.707E+01	9.292E+01	1.0	2.2
	0.550	-0.206	1.282E+02	1.227E+02	1.0	2.2
	0.600	-0.187	1.637E+02	1.571E+02	0.9	2.3
	0.700	-0.153	2.473E+02	2.394E+02	0.9	2.3
	0.800	-0.124	3.455E+02	3.359E+02	0.8	2.1
	0.900	-0.099	4.558E+02	4.459E+02	0.8	2.3
	1.000	-0.076	5.757E+02	5.667E+02	0.9	2.4
	1.100	-0.055	7.028E+02	6.927E+02	0.9	2.4
	1.200	-0.036	8.349E+02	8.229E+02	1.1	2.2
	1.400	-0.003	1.107E+03	1.097E+03	1.1	2.2
	1.600	0.026	1.382E+03	1.375E+03	1.4	2.1
	1.800	0.052	1.650E+03	1.648E+03	1.4	2.1
	2.000	0.075	1.908E+03	1.916E+03	1.7	2.4
	2.300	0.105	2.270E+03	2.297E+03	1.7	2.6
	2.600	0.132	2.597E+03	2.637E+03	1.6	2.9
	3.000	0.163	2.979E+03	3.052E+03	1.8	3.3
	3.500	0.196	3.376E+03	3.490E+03	1.8	3.3
	4.000	0.225	3.694E+03	3.847E+03	1.9	4.3
	4.500	0.251	3.945E+03	4.132E+03	1.9	4.3
	5.000	0.274	4.141E+03	4.354E+03	1.8	4.2
	5.500	0.294	4.292E+03	4.522E+03	1.8	4.2
	6.000	0.313	4.407E+03	4.648E+03	2.8	4.4
	7.000	0.347	4.550E+03	4.802E+03	2.8	4.4
	8.000	0.376	4.611E+03	4.876E+03	3.5	4.2
	9.000	0.401	4.618E+03	4.869E+03	3.5	4.3
	10.000	0.424	4.587E+03	4.818E+03	4.0	5.4
	11.000	0.445	4.533E+03	4.735E+03	4.0	5.4
	12.000	0.464	4.462E+03	4.602E+03	4.6	6.3
	14.000	0.497	4.295E+03	4.332E+03	4.6	6.3
	16.000	0.526	4.113E+03	4.044E+03	5.8	7.1
	18.000	0.552	3.932E+03	3.758E+03	5.8	7.1
	20.000	0.575	3.757E+03	3.484E+03	7.9	12.3
	23.000	0.605	3.513E+03	3.105E+03	7.9	19.3
	26.000	0.632	3.294E+03	2.766E+03	7.2	25.4
21-Sc	0.050	-0.751	9.009E-04	6.880E-04	12.8	22.5
	0.055	-0.730	1.973E-03	1.638E-03	7.4	15.4
	0.060	-0.711	3.830E-03	3.389E-03	7.4	14.9
	0.070	-0.678	1.119E-02	1.068E-02	5.5	7.7
	0.080	-0.649	2.594E-02	2.571E-02	5.5	7.6
	0.090	-0.623	5.153E-02	5.198E-02	4.6	6.1
	0.100	-0.600	9.177E-02	9.314E-02	4.6	6.1
	0.110	-0.580	1.507E-01	1.536E-01	3.7	4.8
	0.120	-0.561	2.326E-01	2.366E-01	3.7	4.8
	0.140	-0.527	4.829E-01	4.881E-01	2.5	4.0
	0.160	-0.498	8.785E-01	8.760E-01	2.5	3.7
	0.180	-0.473	1.455E+00	1.427E+00	1.9	2.9
	0.200	-0.450	2.249E+00	2.189E+00	1.9	2.9
	0.230	-0.419	3.921E+00	3.807E+00	1.5	2.6
	0.260	-0.393	6.264E+00	6.068E+00	1.5	2.6
	0.300	-0.362	1.060E+01	1.022E+01	1.5	2.4
	0.350	-0.328	1.824E+01	1.757E+01	1.3	2.3
	0.400	-0.299	2.859E+01	2.752E+01	1.3	2.4
	0.450	-0.274	4.188E+01	4.013E+01	1.1	2.1
	0.500	-0.251	5.819E+01	5.573E+01	1.1	2.1
	0.550	-0.230	7.757E+01	7.425E+01	1.0	2.0
	0.600	-0.211	9.997E+01	9.564E+01	1.0	2.0
	0.700	-0.178	1.535E+02	1.477E+02	0.9	2.1
	0.800	-0.149	2.176E+02	2.108E+02	0.9	2.1
	0.900	-0.123	2.910E+02	2.829E+02	0.8	1.9
21-Sc	1.000	-0.100	3.721E+02	3.638E+02	0.8	1.9
	1.100	-0.080	4.594E+02	4.518E+02	0.9	2.1
	1.200	-0.061	5.515E+02	5.440E+02	0.9	2.1
	1.400	-0.027	7.451E+02	7.353E+02	1.1	2.0
	1.600	0.002	9.448E+02	9.368E+02	1.1	2.1
	1.800	0.027	1.144E+03	1.139E+03	1.4	1.8
	2.000	0.050	1.340E+03	1.337E+03	1.4	1.8
	2.300	0.081	1.619E+03	1.629E+03	1.7	2.2
	2.600	0.107	1.878E+03	1.902E+03	1.6	2.6
	3.000	0.138	2.188E+03	2.226E+03	1.6	2.6
	3.500	0.172	2.519E+03	2.587E+03	1.8	3.0
	4.000	0.201	2.793E+03	2.891E+03	1.8	3.6
	4.500	0.226	3.016E+03	3.143E+03	1.9	3.9
	5.000	0.249	3.197E+03	3.348E+03	1.9	3.8
	5.500	0.270	3.342E+03	3.511E+03	1.8	3.8
	6.000	0.289	3.456E+03	3.639E+03	1.8	3.8
	7.000	0.322	3.613E+03	3.813E+03	2.8	3.8
	8.000	0.351	3.700E+03	3.905E+03	2.8	3.8
	9.000	0.377	3.737E+03	3.951E+03	3.5	3.6
	10.000	0.400	3.739E+03	3.943E+03	3.5	3.6
	11.000	0.420	3.717E+03	3.908E+03	4.0	4.7
	12.000	0.439	3.679E+03	3.850E+03	4.0	4.7
	14.000	0.473	3.573E+03	3.666E+03	4.6	5.6
	16.000	0.502	3.446E+03	3.464E+03	4.6	5.9
	18.000	0.527	3.313E+03	3.253E+03	5.8	6.1
	20.000	0.550	3.180E+03	3.045E+03	5.8	6.1
	23.000	0.581	2.990E+03	2.749E+03	7.9	13.4
	26.000	0.607	2.815E+03	2.479E+03	7.9	22.2
	30.000	0.638	2.608E+03	2.158E+03	7.2	28.6
22-Ti	0.055	-0.753	7.652E-04	5.773E-04	12.8	21.3
	0.060	-0.734	1.566E-03	1.277E-03	7.4	14.3
	0.070	-0.701	4.940E-03	4.492E-03	7.4	13.6
	0.080	-0.672	1.208E-02	1.163E-02	5.5	7.2
	0.090	-0.646	2.493E-02	2.476E-02	5.5	7.2
	0.100	-0.624	4.572E-02	4.610E-02	4.6	5.7
	0.110	-0.603	7.682E-02	7.794E-02	4.6	5.7
	0.120	-0.584	1.208E-01	1.230E-01	3.7	4.5
	0.140	-0.550	2.578E-01	2.616E-01	3.7	4.5
	0.160	-0.521	4.784E-01	4.824E-01	2.5	3.8
	0.180	-0.496	8.049E-01	8.017E-01	2.5	3.5
	0.200	-0.473	1.259E+00	1.235E+00	1.9	2.7
	0.230	-0.443	2.228E+00	2.165E+00	1.5	2.4
	0.260	-0.416	3.602E+00	3.497E+00	1.5	2.4
	0.300	-0.385	6.176E+00	5.979E+00	1.5	2.3
	0.350	-0.352	1.077E+01	1.038E+01	1.5	2.3
	0.400	-0.323	1.709E+01	1.646E+01	1.3	2.1
	0.450	-0.297	2.527E+01	2.432E+01	1.3	2.2
	0.500	-0.274	3.545E+01	3.397E+01	1.1	1.9
	0.550	-0.253	4.766E+01	4.565E+01	1.1	1.9
	0.600	-0.235	6.193E+01	5.928E+01	1.0	1.8
	0.700	-0.201	9.651E+01	9.243E+01	1.0	1.8
	0.800	-0.172	1.387E+02	1.337E+02	0.9	1.9
	0.900	-0.147	1.878E+02	1.820E+02	0.9	1.9
	1.000	-0.124	2.429E+02	2.362E+02	0.8	1.7
	1.100	-0.103	3.031E+02	2.961E+02	0.8	1.7
	1.200	-0.084	3.675E+02	3.610E+02	0.9	1.9
	1.400	-0.051	5.053E+02	4.978E+02	0.9	1.9
	1.600	-0.022	6.506E+02	6.425E+02	1.1	1.7
	1.800	0.004	7.987E+02	7.922E+02	1.1	1.8
	2.000	0.027	9.463E+02	9.417E+02	1.4	1.6
	2.300	0.057	1.162E+03	1.161E+03	1.7	1.9
	2.600	0.084	1.365E+03	1.377E+03	1.7	1.9
	3.000	0.115	1.615E+03	1.637E+03	1.6	2.3
	3.500	0.148	1.888E+03	1.926E+03	1.6	2.3
	4.000	0.177	2.120E+03	2.181E+03	1.8	2.7
	4.500	0.203	2.315E+03	2.398E+03	1.8	3.2
	5.000	0.226	2.477E+03	2.580E+03	1.9	3.4
	5.500	0.247	2.611E+03	2.732E+03	1.9	3.4
	6.000	0.266	2.720E+03	2.856E+03	1.8	3.3
	7.000	0.299	2.879E+03	3.034E+03	1.8	3.3
	8.000	0.328	2.978E+03	3.143E+03	2.8	3.2
	9.000	0.354	3.033E+03	3.201E+03	2.8	3.2
	10.000	0.376	3.057E+03	3.233E+03	3.5	4.0
	11.000	0.397	3.059E+03	3.227E+03	3.5	4.0
	12.000	0.416	3.044E+03	3.202E+03	4.0	4.0
	14.000	0.449	2.982E+03	3.110E+03	4.0	4.0
	16.000	0.479	2.897E+03	2.962E+03	4.6	4.9

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
22-Ti	18.000	0.504	2.801E+03	2.810E+03	4.6	4.9
	20.000	0.527	2.702E+03	2.655E+03	5.8	6.4
	23.000	0.557	2.555E+03	2.426E+03	5.8	6.5
	26.000	0.584	2.417E+03	2.211E+03	7.9	14.5
	30.000	0.615	2.249E+03	1.951E+03	7.2	30.5
	35.000	0.648	2.065E+03	1.670E+03	7.2	32.2
23-V	0.060	-0.757	6.345E-04	4.714E-04	12.8	20.1
	0.070	-0.723	2.179E-03	1.856E-03	7.4	12.8
	0.080	-0.694	5.648E-03	5.221E-03	7.4	9.4
	0.090	-0.669	1.216E-02	1.178E-02	5.5	6.8
	0.100	-0.646	2.303E-02	2.289E-02	5.5	6.8
	0.110	-0.625	3.967E-02	3.997E-02	4.6	5.3
	0.120	-0.606	6.361E-02	6.451E-02	4.6	5.3
	0.140	-0.573	1.399E-01	1.425E-01	3.7	4.3
	0.160	-0.544	2.652E-01	2.687E-01	3.7	4.3
	0.180	-0.518	4.536E-01	4.566E-01	2.5	3.5
	0.200	-0.495	7.188E-01	7.156E-01	2.5	3.3
	0.230	-0.465	1.291E+00	1.261E+00	1.9	2.5
	0.260	-0.438	2.113E+00	2.052E+00	1.5	2.2
	0.300	-0.407	3.669E+00	3.559E+00	1.5	2.2
	0.350	-0.374	6.482E+00	6.270E+00	1.5	2.1
	0.400	-0.345	1.040E+01	1.002E+01	1.5	2.1
	0.450	-0.319	1.553E+01	1.497E+01	1.3	1.9
	0.500	-0.296	2.197E+01	2.114E+01	1.3	2.0
	0.550	-0.275	2.978E+01	2.855E+01	1.1	1.7
	0.600	-0.257	3.899E+01	3.735E+01	1.1	1.7
	0.700	-0.223	6.160E+01	5.897E+01	1.0	1.6
	0.800	-0.194	8.965E+01	8.594E+01	1.0	1.6
	0.900	-0.169	1.228E+02	1.185E+02	0.9	1.7
	1.000	-0.146	1.605E+02	1.556E+02	0.9	1.7
	1.100	-0.125	2.022E+02	1.966E+02	0.8	1.5
	1.200	-0.106	2.474E+02	2.414E+02	0.8	1.5
	1.400	-0.073	3.459E+02	3.407E+02	0.9	1.7
	1.600	-0.044	4.517E+02	4.445E+02	0.9	1.7
	1.800	-0.018	5.617E+02	5.553E+02	1.1	1.5
	2.000	0.005	6.730E+02	6.676E+02	1.1	1.6
	2.300	0.035	8.384E+02	8.339E+02	1.4	1.5
	2.600	0.062	9.981E+02	9.990E+02	1.7	1.7
	3.000	0.093	1.198E+03	1.210E+03	1.7	1.7
	3.500	0.126	1.422E+03	1.444E+03	1.6	2.0
	4.000	0.155	1.617E+03	1.653E+03	1.6	2.0
	4.500	0.181	1.785E+03	1.837E+03	1.8	2.4
	5.000	0.204	1.927E+03	1.996E+03	1.8	2.7
	5.500	0.225	2.048E+03	2.132E+03	1.9	3.0
	6.000	0.243	2.149E+03	2.247E+03	1.9	3.0
	7.000	0.277	2.303E+03	2.421E+03	1.8	2.9
	8.000	0.306	2.406E+03	2.537E+03	2.8	3.0
	9.000	0.331	2.471E+03	2.608E+03	2.8	3.0
	10.000	0.354	2.509E+03	2.648E+03	2.8	3.0
	11.000	0.375	2.526E+03	2.671E+03	3.5	4.6
	12.000	0.394	2.527E+03	2.668E+03	3.5	4.6
	14.000	0.427	2.498E+03	2.623E+03	4.0	4.7
	16.000	0.456	2.445E+03	2.531E+03	4.0	4.8
	18.000	0.482	2.378E+03	2.425E+03	4.6	5.1
	20.000	0.505	2.305E+03	2.311E+03	4.6	5.2
	23.000	0.535	2.193E+03	2.136E+03	5.8	7.5
	26.000	0.562	2.083E+03	1.966E+03	7.9	15.4
	30.000	0.593	1.948E+03	1.757E+03	7.9	15.7
	35.000	0.626	1.797E+03	1.525E+03	7.2	34.0
24-Cr	0.070	-0.744	9.526E-04	7.487E-04	12.8	18.4
	0.080	-0.715	2.633E-03	2.302E-03	7.4	11.5
	0.090	-0.690	5.943E-03	5.546E-03	5.5	6.4
	0.100	-0.667	1.165E-02	1.131E-02	5.5	6.4
	0.110	-0.646	2.062E-02	2.049E-02	5.5	6.4
	0.120	-0.627	3.380E-02	3.402E-02	4.6	4.9
	0.140	-0.594	7.676E-02	7.795E-02	4.6	4.8
	0.160	-0.565	1.489E-01	1.516E-01	3.7	4.0
	0.180	-0.539	2.591E-01	2.630E-01	2.5	3.3
	0.200	-0.516	4.162E-01	4.187E-01	2.5	3.3
	0.230	-0.486	7.596E-01	7.484E-01	1.9	2.3
	0.260	-0.459	1.258E+00	1.227E+00	1.9	2.3
	0.300	-0.428	2.214E+00	2.147E+00	1.5	2.0
	0.350	-0.395	3.962E+00	3.839E+00	1.5	2.0
	0.400	-0.366	6.424E+00	6.211E+00	1.5	1.9
	0.450	-0.340	9.688E+00	9.334E+00	1.5	1.9
	0.500	-0.317	1.382E+01	1.332E+01	1.3	1.7
24-Cr	0.550	-0.297	1.887E+01	1.816E+01	1.3	1.8
	0.600	-0.278	2.488E+01	2.385E+01	1.1	1.5
	0.700	-0.244	3.981E+01	3.812E+01	1.1	1.5
	0.800	-0.215	5.862E+01	5.612E+01	1.0	1.4
	0.900	-0.190	8.114E+01	7.782E+01	1.0	1.4
	1.000	-0.167	1.071E+02	1.034E+02	0.9	1.5
	1.100	-0.146	1.362E+02	1.320E+02	0.9	1.5
	1.200	-0.127	1.680E+02	1.633E+02	0.8	1.3
	1.400	-0.094	2.385E+02	2.337E+02	0.8	1.3
	1.600	-0.065	3.158E+02	3.115E+02	0.9	1.4
	1.800	-0.039	3.975E+02	3.916E+02	1.1	1.3
	2.000	-0.016	4.814E+02	4.761E+02	1.1	1.3
	2.300	0.014	6.084E+02	6.043E+02	1.4	1.7
	2.600	0.041	7.332E+02	7.301E+02	1.4	1.7
	3.000	0.072	8.923E+02	8.953E+02	1.7	2.0
	3.500	0.105	1.075E+03	1.088E+03	1.7	2.0
	4.000	0.134	1.238E+03	1.258E+03	1.6	1.7
	4.500	0.160	1.380E+03	1.413E+03	1.8	2.1
	5.000	0.183	1.504E+03	1.550E+03	1.8	2.1
	5.500	0.203	1.611E+03	1.669E+03	1.8	2.3
	6.000	0.222	1.703E+03	1.772E+03	1.9	2.6
	7.000	0.256	1.847E+03	1.936E+03	1.8	2.5
	8.000	0.285	1.949E+03	2.051E+03	1.8	2.5
	9.000	0.310	2.019E+03	2.129E+03	2.8	3.6
	10.000	0.333	2.064E+03	2.179E+03	2.8	3.6
	11.000	0.354	2.091E+03	2.207E+03	2.8	3.6
	12.000	0.373	2.104E+03	2.225E+03	3.5	5.1
	14.000	0.406	2.099E+03	2.211E+03	3.5	5.6
	16.000	0.435	2.069E+03	2.167E+03	4.0	5.5
	18.000	0.461	2.024E+03	2.091E+03	4.6	5.8
	20.000	0.484	1.972E+03	2.009E+03	4.6	5.8
	23.000	0.514	1.887E+03	1.877E+03	5.8	8.4
	26.000	0.541	1.802E+03	1.745E+03	5.8	8.5
	30.000	0.572	1.693E+03	1.577E+03	7.9	16.5
	35.000	0.605	1.569E+03	1.386E+03	7.9	30.6
	40.000	0.634	1.459E+03	1.219E+03	7.2	37.4
25-Mn	0.070	-0.764	4.148E-04	2.961E-04	12.8	17.6
	0.080	-0.735	1.230E-03	1.000E-03	7.4	10.6
	0.090	-0.710	2.920E-03	2.595E-03	7.4	10.3
	0.100	-0.687	5.945E-03	5.580E-03	5.5	6.0
	0.110	-0.666	1.084E-02	1.053E-02	5.5	6.0
	0.120	-0.647	1.818E-02	1.804E-02	5.5	6.0
	0.140	-0.614	4.274E-02	4.326E-02	4.6	4.6
	0.160	-0.585	8.497E-02	8.656E-02	3.7	3.8
	0.180	-0.559	1.505E-01	1.531E-01	3.7	3.8
	0.200	-0.536	2.453E-01	2.488E-01	2.5	3.0
	0.230	-0.506	4.550E-01	4.555E-01	2.5	3.0
	0.260	-0.479	7.632E-01	7.501E-01	1.9	2.1
	0.300	-0.448	1.361E+00	1.324E+00	1.9	2.1
	0.350	-0.415	2.466E+00	2.394E+00	1.5	1.9
	0.400	-0.386	4.040E+00	3.912E+00	1.5	1.7
	0.450	-0.360	6.147E+00	5.923E+00	1.5	1.7
	0.500	-0.337	8.838E+00	8.516E+00	1.3	1.6
	0.550	-0.317	1.216E+01	1.171E+01	1.3	1.6
	0.600	-0.298	1.613E+01	1.552E+01	1.3	1.6
	0.700	-0.264	2.611E+01	2.503E+01	1.1	1.3
	0.800	-0.235	3.886E+01	3.720E+01	1.0	1.2
	0.900	-0.210	5.432E+01	5.198E+01	1.0	1.2
	1.000	-0.187	7.236E+01	6.940E+01	0.9	1.2
	1.100	-0.166	9.278E+01	8.963E+01	0.9	1.2
	1.200	-0.147	1.154E+02	1.118E+02	0.9	1.2
	1.400	-0.114	1.662E+02	1.618E+02	0.8	1.1
	1.600	-0.085	2.228E+02	2.188E+02	0.9	1.2
	1.800	-0.059	2.836E+02	2.797E+02	0.9	1.2
	2.000	-0.036	3.470E+02	3.420E+02	1.1	1.1
	2.300	-0.006	4.445E+02	4.403E+02	1.1	1.1
	2.600	0.021	5.421E+02	5.389E+02	1.4	2.0
	3.000	0.052	6.686E+02	6.675E+02	1.4	2.0
	3.500	0.085	8.170E+02	8.244E+02	1.7	2.2
	4.000	0.114	9.520E+02	9.649E+02	1.6	1.8
	4.500	0.140	1.073E+03	1.092E+03	1.6	1.8
	5.000	0.163	1.179E+03	1.208E+03	1.8	1.8
	5.500	0.183	1.273E+03	1.312E+03	1.8	1.8
	6.000	0.202	1.355E+03	1.403E+03	1.8	1.9
	7.000	0.236	1.487E+03	1.553E+03	1.9	2.1
	8.000	0.265	1.585E+03	1.664E+03	1.8	2.1
	9.000	0.290	1.655E+03	1.743E+03	1.8	2.0

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
25-Mn	10.000	0.313	1.704E+03	1.798E+03	2.8	4.2
	11.000	0.334	1.737E+03	1.834E+03	2.8	4.2
	12.000	0.353	1.757E+03	1.854E+03	2.8	4.2
	14.000	0.386	1.769E+03	1.869E+03	3.5	5.7
	16.000	0.415	1.756E+03	1.848E+03	4.0	6.2
	18.000	0.441	1.729E+03	1.808E+03	4.0	6.2
	20.000	0.464	1.692E+03	1.746E+03	4.6	6.5
	23.000	0.494	1.630E+03	1.648E+03	4.6	6.6
	26.000	0.521	1.564E+03	1.546E+03	5.8	9.5
	30.000	0.552	1.477E+03	1.412E+03	5.8	9.6
	35.000	0.585	1.376E+03	1.256E+03	7.9	17.8
	40.000	0.614	1.284E+03	1.116E+03	7.2	39.1
	45.000	0.640	1.203E+03	9.924E+02	7.2	40.8
26-Fe	0.080	-0.754	5.688E-04	4.269E-04	12.8	16.0
	0.090	-0.729	1.427E-03	1.190E-03	7.4	9.3
	0.100	-0.706	3.029E-03	2.719E-03	7.4	9.2
	0.110	-0.685	5.701E-03	5.367E-03	5.5	5.6
	0.120	-0.666	9.809E-03	9.525E-03	5.5	5.6
	0.140	-0.633	2.395E-02	2.402E-02	4.6	5.0
	0.160	-0.604	4.888E-02	4.959E-02	4.6	5.0
	0.180	-0.578	8.831E-02	8.998E-02	3.7	3.8
	0.200	-0.556	1.461E-01	1.484E-01	3.7	3.8
	0.230	-0.525	2.757E-01	2.784E-01	2.5	2.8
	0.260	-0.499	4.685E-01	4.673E-01	2.5	2.7
	0.300	-0.468	8.465E-01	8.271E-01	1.9	1.9
	0.350	-0.434	1.554E+00	1.509E+00	1.5	1.7
	0.400	-0.405	2.573E+00	2.495E+00	1.5	1.7
	0.450	-0.380	3.948E+00	3.821E+00	1.5	1.5
	0.500	-0.357	5.720E+00	5.512E+00	1.5	1.5
	0.550	-0.336	7.922E+00	7.633E+00	1.3	1.4
	0.600	-0.317	1.058E+01	1.019E+01	1.3	1.4
	0.700	-0.283	1.732E+01	1.662E+01	1.1	1.1
	0.800	-0.255	2.603E+01	2.493E+01	1.1	1.1
	0.900	-0.229	3.672E+01	3.515E+01	1.0	1.0
	1.000	-0.206	4.932E+01	4.721E+01	1.0	1.0
	1.100	-0.185	6.374E+01	6.117E+01	0.9	1.0
	1.200	-0.167	7.986E+01	7.713E+01	0.9	1.0
	1.400	-0.133	1.166E+02	1.132E+02	0.8	0.9
	1.600	-0.104	1.582E+02	1.545E+02	0.8	0.9
	1.800	-0.078	2.035E+02	2.002E+02	0.9	1.0
	2.000	-0.056	2.515E+02	2.479E+02	0.9	1.0
	2.300	-0.025	3.263E+02	3.221E+02	1.1	1.3
	2.600	0.001	4.025E+02	3.991E+02	1.1	1.3
	3.000	0.032	5.030E+02	5.000E+02	1.4	2.2
	3.500	0.066	6.232E+02	6.243E+02	1.7	2.5
	4.000	0.095	7.346E+02	7.424E+02	1.7	2.5
	4.500	0.120	8.360E+02	8.482E+02	1.6	2.1
	5.000	0.143	9.272E+02	9.445E+02	1.6	2.1
	5.500	0.164	1.009E+03	1.033E+03	1.8	2.0
	6.000	0.183	1.081E+03	1.113E+03	1.8	2.0
	7.000	0.216	1.200E+03	1.247E+03	1.9	2.1
	8.000	0.245	1.291E+03	1.351E+03	1.9	2.1
	9.000	0.271	1.360E+03	1.429E+03	1.8	1.9
	10.000	0.294	1.410E+03	1.485E+03	1.8	1.9
	11.000	0.315	1.446E+03	1.525E+03	2.8	4.7
	12.000	0.333	1.471E+03	1.552E+03	2.8	4.7
	14.000	0.367	1.494E+03	1.581E+03	3.5	6.3
	16.000	0.396	1.494E+03	1.577E+03	3.5	6.3
	18.000	0.422	1.480E+03	1.555E+03	4.0	6.9
	20.000	0.444	1.456E+03	1.521E+03	4.0	6.9
	23.000	0.475	1.411E+03	1.446E+03	4.6	7.3
	26.000	0.501	1.361E+03	1.368E+03	4.6	8.2
	30.000	0.533	1.292E+03	1.263E+03	5.8	10.5
	35.000	0.566	1.209E+03	1.135E+03	7.9	18.6
	40.000	0.595	1.133E+03	1.019E+03	7.9	19.1
	45.000	0.621	1.065E+03	9.144E+02	7.2	42.4
	50.000	0.643	1.003E+03	8.210E+02	7.2	44.2
27-Co	0.090	-0.748	6.941E-04	5.378E-04	12.8	14.6
	0.100	-0.725	1.541E-03	1.305E-03	7.4	8.1
	0.110	-0.704	3.002E-03	2.710E-03	7.4	8.1
	0.120	-0.685	5.308E-03	5.000E-03	5.5	5.9
	0.140	-0.652	1.349E-02	1.333E-02	5.5	5.9
	0.160	-0.623	2.833E-02	2.858E-02	4.6	5.4
	0.180	-0.597	5.225E-02	5.305E-02	4.6	5.1
	0.200	-0.574	8.782E-02	8.946E-02	3.7	4.0
	0.230	-0.544	1.687E-01	1.709E-01	3.7	4.0
27-Co	0.260	-0.517	2.907E-01	2.926E-01	2.5	2.5
	0.300	-0.486	5.326E-01	5.248E-01	1.9	2.1
	0.350	-0.453	9.908E-01	9.648E-01	1.9	2.1
	0.400	-0.424	1.657E+00	1.606E+00	1.5	1.5
	0.450	-0.398	2.565E+00	2.485E+00	1.5	1.5
	0.500	-0.375	3.743E+00	3.621E+00	1.5	1.6
	0.550	-0.354	5.218E+00	5.027E+00	1.5	1.6
	0.600	-0.336	7.008E+00	6.753E+00	1.3	1.4
	0.700	-0.302	1.160E+01	1.117E+01	1.3	1.4
	0.800	-0.273	1.760E+01	1.686E+01	1.1	1.2
	0.900	-0.248	2.504E+01	2.397E+01	1.1	1.2
	1.000	-0.225	3.390E+01	3.245E+01	1.0	1.2
	1.100	-0.204	4.413E+01	4.226E+01	1.0	1.2
	1.200	-0.185	5.567E+01	5.343E+01	0.9	1.1
	1.400	-0.152	8.230E+01	7.970E+01	0.9	1.1
	1.600	-0.123	1.130E+02	1.098E+02	0.8	1.1
	1.800	-0.097	1.468E+02	1.437E+02	0.8	1.1
	2.000	-0.074	1.831E+02	1.803E+02	0.9	1.1
	2.300	-0.044	2.406E+02	2.368E+02	0.9	1.1
	2.600	-0.017	3.000E+02	2.967E+02	1.1	1.6
	3.000	0.014	3.797E+02	3.771E+02	1.4	2.5
	3.500	0.047	4.768E+02	4.755E+02	1.4	2.5
	4.000	0.076	5.684E+02	5.710E+02	1.7	2.7
	4.500	0.102	6.533E+02	6.609E+02	1.7	2.9
	5.000	0.125	7.308E+02	7.420E+02	1.6	2.4
	5.500	0.146	8.009E+02	8.163E+02	1.6	2.4
	6.000	0.164	8.639E+02	8.853E+02	1.8	2.3
	7.000	0.198	9.704E+02	1.004E+03	1.8	2.3
	8.000	0.227	1.054E+03	1.098E+03	1.9	2.6
	9.000	0.252	1.119E+03	1.173E+03	1.8	2.3
	10.000	0.275	1.169E+03	1.229E+03	1.8	2.3
	11.000	0.296	1.206E+03	1.270E+03	1.8	2.3
	12.000	0.315	1.233E+03	1.300E+03	2.8	5.3
	14.000	0.348	1.264E+03	1.334E+03	2.8	5.3
	16.000	0.377	1.273E+03	1.347E+03	3.5	6.8
	18.000	0.403	1.269E+03	1.337E+03	3.5	7.7
	20.000	0.426	1.255E+03	1.318E+03	4.0	7.6
	23.000	0.456	1.224E+03	1.267E+03	4.0	7.7
	26.000	0.483	1.186E+03	1.210E+03	4.6	8.0
	30.000	0.514	1.133E+03	1.127E+03	5.8	11.5
	35.000	0.547	1.066E+03	1.024E+03	5.8	11.7
	40.000	0.576	1.003E+03	9.277E+02	7.9	19.8
	45.000	0.602	9.449E+02	8.396E+02	7.9	38.9
	50.000	0.625	8.926E+02	7.600E+02	7.2	45.7
	55.000	0.646	8.452E+02	6.883E+02	7.2	47.4
28-Ni	0.090	-0.765	3.360E-04	2.386E-04	12.8	13.4
	0.100	-0.742	7.835E-04	6.206E-04	12.8	13.3
	0.110	-0.722	1.585E-03	1.357E-03	7.4	7.9
	0.120	-0.703	2.887E-03	2.614E-03	7.4	7.9
	0.140	-0.669	7.665E-03	7.414E-03	5.5	6.3
	0.160	-0.640	1.659E-02	1.656E-02	5.5	6.3
	0.180	-0.615	3.129E-02	3.166E-02	4.6	5.8
	0.200	-0.592	5.348E-02	5.431E-02	4.6	5.4
	0.230	-0.562	1.047E-01	1.066E-01	3.7	4.3
	0.260	-0.535	1.831E-01	1.855E-01	2.5	2.6
	0.300	-0.504	3.402E-01	3.402E-01	2.5	2.6
	0.350	-0.470	6.413E-01	6.283E-01	1.9	2.3
	0.400	-0.441	1.084E+00	1.053E+00	1.5	1.7
	0.450	-0.416	1.691E+00	1.642E+00	1.5	1.7
	0.500	-0.393	2.486E+00	2.408E+00	1.5	1.7
	0.550	-0.372	3.488E+00	3.373E+00	1.5	1.8
	0.600	-0.353	4.712E+00	4.540E+00	1.5	1.8
	0.700	-0.320	7.877E+00	7.590E+00	1.3	1.6
	0.800	-0.291	1.206E+01	1.159E+01	1.1	1.4
	0.900	-0.265	1.729E+01	1.657E+01	1.1	1.4
	1.000	-0.242	2.358E+01	2.258E+01	1.1	1.4
	1.100	-0.222	3.091E+01	2.959E+01	1.0	1.4
	1.200	-0.203	3.924E+01	3.758E+01	1.0	1.4
	1.400	-0.169	5.869E+01	5.664E+01	0.9	1.3
	1.600	-0.140	8.142E+01	7.898E+01	0.9	1.3
	1.800	-0.115	1.068E+02	1.040E+02	0.8	1.3
	2.000	-0.092	1.344E+02	1.318E+02	0.8	1.3
	2.300	-0.062	1.787E+02	1.763E+02	0.9	1.4
	2.600	-0.035	2.251E+02	2.219E+02	1.1	1.8
	3.000	-0.004	2.884E+02	2.857E+02	1.1	1.8
	3.500	0.030	3.668E+02	3.651E+02	1.4	2.7
	4.000	0.059	4.421E+02	4.421E+02	1.7	3.0

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
28-Ni	4.500	0.084	5.129E+02	5.175E+02	1.7	3.0
	5.000	0.107	5.785E+02	5.858E+02	1.6	2.7
	5.500	0.128	6.387E+02	6.488E+02	1.6	2.7
	6.000	0.147	6.934E+02	7.070E+02	1.6	2.7
	7.000	0.180	7.877E+02	8.108E+02	1.8	2.6
	8.000	0.209	8.637E+02	8.960E+02	1.9	3.0
	9.000	0.235	9.242E+02	9.649E+02	1.9	3.0
	10.000	0.258	9.717E+02	1.019E+03	1.8	2.8
	11.000	0.278	1.009E+03	1.061E+03	1.8	2.8
	12.000	0.297	1.037E+03	1.092E+03	1.8	2.8
	14.000	0.331	1.072E+03	1.132E+03	2.8	5.9
	16.000	0.360	1.088E+03	1.148E+03	3.5	7.4
	18.000	0.385	1.091E+03	1.153E+03	3.5	7.4
	20.000	0.408	1.085E+03	1.142E+03	3.5	8.4
	23.000	0.438	1.065E+03	1.114E+03	4.0	8.3
	26.000	0.465	1.037E+03	1.069E+03	4.6	8.7
	30.000	0.496	9.959E+02	1.006E+03	4.6	8.8
	35.000	0.530	9.421E+02	9.231E+02	5.8	12.6
	40.000	0.559	8.900E+02	8.435E+02	5.8	12.8
	45.000	0.584	8.414E+02	7.694E+02	7.9	21.1
	50.000	0.607	7.968E+02	7.016E+02	7.9	42.0
	55.000	0.628	7.562E+02	6.399E+02	7.2	48.9
	60.000	0.647	7.191E+02	5.840E+02	7.2	50.6
29-Cu	0.100	-0.760	3.969E-04	2.906E-04	12.8	13.6
	0.110	-0.739	8.360E-04	6.694E-04	7.4	9.2
	0.120	-0.720	1.572E-03	1.353E-03	7.4	9.1
	0.140	-0.686	4.372E-03	4.107E-03	5.5	6.7
	0.160	-0.657	9.770E-03	9.595E-03	5.5	6.7
	0.180	-0.632	1.886E-02	1.894E-02	4.6	6.2
	0.200	-0.609	3.282E-02	3.326E-02	4.6	6.2
	0.230	-0.579	6.558E-02	6.682E-02	3.7	4.5
	0.260	-0.552	1.163E-01	1.181E-01	3.7	4.5
	0.300	-0.521	2.193E-01	2.211E-01	2.5	2.9
	0.350	-0.487	4.192E-01	4.132E-01	1.9	2.5
	0.400	-0.458	7.156E-01	6.977E-01	1.9	2.5
	0.450	-0.433	1.126E+00	1.093E+00	1.5	1.9
	0.500	-0.410	1.668E+00	1.618E+00	1.5	1.9
	0.550	-0.389	2.353E+00	2.279E+00	1.5	2.0
	0.600	-0.370	3.197E+00	3.092E+00	1.5	2.0
	0.700	-0.337	5.396E+00	5.199E+00	1.3	1.8
	0.800	-0.308	8.328E+00	8.029E+00	1.3	1.8
	0.900	-0.282	1.203E+01	1.155E+01	1.1	1.6
	1.000	-0.259	1.653E+01	1.584E+01	1.1	1.6
	1.100	-0.239	2.181E+01	2.088E+01	1.0	1.6
	1.200	-0.220	2.785E+01	2.666E+01	1.0	1.6
	1.400	-0.186	4.212E+01	4.040E+01	0.9	1.5
	1.600	-0.157	5.901E+01	5.710E+01	0.9	1.5
	1.800	-0.132	7.815E+01	7.589E+01	0.8	1.5
	2.000	-0.109	9.914E+01	9.666E+01	0.8	1.5
	2.300	-0.079	1.333E+02	1.311E+02	0.9	1.6
	2.600	-0.052	1.696E+02	1.671E+02	0.9	1.6
	3.000	-0.021	2.197E+02	2.170E+02	1.1	2.0
	3.500	0.013	2.830E+02	2.810E+02	1.4	3.0
	4.000	0.041	3.448E+02	3.434E+02	1.4	3.0
	4.500	0.067	4.037E+02	4.046E+02	1.7	3.2
	5.000	0.090	4.590E+02	4.635E+02	1.7	3.2
	5.500	0.111	5.104E+02	5.170E+02	1.6	3.0
	6.000	0.130	5.577E+02	5.667E+02	1.6	2.9
	7.000	0.163	6.405E+02	6.561E+02	1.8	2.9
	8.000	0.192	7.088E+02	7.320E+02	1.8	2.9
	9.000	0.218	7.644E+02	7.947E+02	1.9	3.4
	10.000	0.241	8.092E+02	8.459E+02	1.9	3.4
	11.000	0.261	8.449E+02	8.866E+02	1.8	3.2
	12.000	0.280	8.730E+02	9.182E+02	1.8	3.2
	14.000	0.314	9.112E+02	9.610E+02	2.8	6.4
	16.000	0.343	9.315E+02	9.832E+02	2.8	6.4
	18.000	0.368	9.395E+02	9.939E+02	3.5	8.0
	20.000	0.391	9.388E+02	9.914E+02	3.5	8.0
	23.000	0.421	9.274E+02	9.747E+02	4.0	9.0
	26.000	0.448	9.084E+02	9.478E+02	4.0	9.1
	30.000	0.479	8.770E+02	8.963E+02	4.6	9.5
	35.000	0.513	8.342E+02	8.309E+02	5.8	13.5
	40.000	0.541	7.915E+02	7.657E+02	5.8	13.7
	45.000	0.567	7.507E+02	7.036E+02	7.9	21.8
	50.000	0.590	7.128E+02	6.460E+02	7.9	22.3
	55.000	0.611	6.779E+02	5.931E+02	7.2	50.4
	60.000	0.630	6.459E+02	5.446E+02	7.2	52.0

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
30-Zn	0.110	-0.756	4.351E-04	3.248E-04	12.8	14.9
	0.120	-0.737	8.470E-04	6.844E-04	7.4	10.4
	0.140	-0.703	2.478E-03	2.241E-03	7.4	10.1
	0.160	-0.674	5.736E-03	5.506E-03	5.5	7.1
	0.180	-0.649	1.136E-02	1.126E-02	5.5	7.1
	0.200	-0.626	2.015E-02	2.030E-02	4.6	6.6
	0.230	-0.595	4.114E-02	4.177E-02	4.6	5.8
	0.260	-0.569	7.412E-02	7.548E-02	3.7	4.8
	0.300	-0.538	1.420E-01	1.440E-01	2.5	3.1
	0.350	-0.504	2.752E-01	2.752E-01	2.5	3.1
	0.400	-0.475	4.748E-01	4.660E-01	1.9	2.7
	0.450	-0.450	7.538E-01	7.336E-01	1.9	2.7
	0.500	-0.427	1.124E+00	1.090E+00	1.5	2.1
	0.550	-0.406	1.596E+00	1.548E+00	1.5	2.1
	0.600	-0.387	2.180E+00	2.111E+00	1.5	2.2
	0.700	-0.354	3.715E+00	3.580E+00	1.5	2.2
	0.800	-0.325	5.782E+00	5.572E+00	1.3	2.0
	0.900	-0.299	8.414E+00	8.099E+00	1.3	2.0
	1.000	-0.276	1.163E+01	1.115E+01	1.1	1.8
	1.100	-0.256	1.544E+01	1.479E+01	1.1	1.8
	1.200	-0.237	1.984E+01	1.899E+01	1.0	1.8
	1.400	-0.203	3.032E+01	2.903E+01	1.0	1.8
	1.600	-0.174	4.289E+01	4.132E+01	0.9	1.7
	1.800	-0.149	5.730E+01	5.552E+01	0.9	1.7
	2.000	-0.126	7.327E+01	7.121E+01	0.8	1.7
	2.300	-0.095	9.957E+01	9.749E+01	0.8	1.8
	2.600	-0.069	1.279E+02	1.261E+02	0.9	1.8
	3.000	-0.038	1.677E+02	1.652E+02	1.1	2.3
	3.500	-0.004	2.186E+02	2.165E+02	1.1	2.2
	4.000	0.025	2.691E+02	2.677E+02	1.4	3.2
	4.500	0.050	3.180E+02	3.173E+02	1.4	3.2
	5.000	0.073	3.645E+02	3.658E+02	1.7	3.5
	5.500	0.094	4.081E+02	4.124E+02	1.7	3.5
	6.000	0.113	4.488E+02	4.548E+02	1.6	3.2
	7.000	0.146	5.211E+02	5.313E+02	1.6	3.2
	8.000	0.175	5.820E+02	5.983E+02	1.8	3.2
	9.000	0.201	6.326E+02	6.548E+02	1.8	3.9
	10.000	0.224	6.742E+02	7.019E+02	1.9	3.9
	11.000	0.244	7.081E+02	7.408E+02	1.9	3.8
	12.000	0.263	7.355E+02	7.720E+02	1.8	3.6
	14.000	0.297	7.745E+02	8.160E+02	1.8	3.6
	16.000	0.326	7.975E+02	8.417E+02	2.8	7.0
	18.000	0.351	8.093E+02	8.542E+02	2.8	7.0
	20.000	0.374	8.130E+02	8.598E+02	3.5	8.5
	23.000	0.405	8.083E+02	8.516E+02	3.5	9.8
	26.000	0.431	7.959E+02	8.347E+02	4.0	9.7
	30.000	0.462	7.728E+02	7.977E+02	4.6	10.2
	35.000	0.496	7.392E+02	7.466E+02	4.6	10.3
	40.000	0.525	7.044E+02	6.937E+02	5.8	14.6
	45.000	0.550	6.705E+02	6.121E+02	5.8	14.8
	50.000	0.573	6.383E+02	5.933E+02	7.9	23.0
	55.000	0.594	6.085E+02	5.480E+02	7.9	23.5
	60.000	0.613	5.809E+02	5.062E+02	7.2	53.4
	70.000	0.646	5.318E+02	4.324E+02	7.2	56.7
31-Ga	0.120	-0.753	4.559E-04	3.568E-04	12.6	15.7
	0.140	-0.720	1.408E-03	1.214E-03	7.2	10.1
	0.160	-0.691	3.383E-03	3.127E-03	4.4	4.8
	0.180	-0.665	6.883E-03	6.612E-03	4.4	4.8
	0.200	-0.642	1.246E-02	1.223E-02	4.4	4.8
	0.230	-0.612	2.602E-02	2.586E-02	4.5	5.2
	0.260	-0.585	4.766E-02	4.754E-02	3.4	3.9
	0.300	-0.554	9.277E-02	9.225E-02	3.4	3.9
	0.350	-0.521	1.825E-01	1.800E-01	2.7	3.2
	0.400	-0.491	3.183E-01	3.130E-01	1.9	2.5
	0.450	-0.466	5.097E-01	4.963E-01	1.9	2.5
	0.500	-0.443	7.654E-01	7.446E-01	1.3	1.4
	0.550	-0.422	1.094E+00	1.064E+00	1.3	1.4
	0.600	-0.403	1.502E+00	1.466E+00	1.3	1.4
	0.700	-0.370	2.582E+00	2.516E+00	1.4	2.0
	0.800	-0.341	4.050E+00	3.929E+00	1.2	1.7
	0.900	-0.315	5.937E+00	5.743E+00	1.2	1.7
	1.000	-0.293	8.258E+00	7.977E+00	0.9	1.4
	1.100	-0.272	1.103E+01	1.060E+01	0.9	1.4
	1.200	-0.253	1.424E+01	1.365E+01	0.9	1.4
	1.400	-0.220	2.199E+01	2.103E+01	1.0	1.7
	1.600	-0.191	3.138E+01	3.009E+01	1.0	1.7
	1.800	-0.165	4.227E+01	4.070E+01	0.9	1.7

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
31-Ga	2.000	-0.142	5.446E+01	5.271E+01	0.9	1.7
	2.300	-0.112	7.477E+01	7.291E+01	0.8	1.7
	2.600	-0.085	9.695E+01	9.518E+01	0.9	1.9
	3.000	-0.054	1.285E+02	1.266E+02	0.9	1.9
	3.500	-0.021	1.694E+02	1.681E+02	1.1	2.3
	4.000	0.009	2.107E+02	2.111E+02	1.3	3.0
	4.500	0.034	2.512E+02	2.532E+02	1.3	3.0
	5.000	0.057	2.902E+02	2.950E+02	1.7	3.1
	5.500	0.078	3.272E+02	3.353E+02	1.7	3.1
	6.000	0.097	3.621E+02	3.735E+02	1.7	3.1
	7.000	0.130	4.250E+02	4.414E+02	1.8	2.7
	8.000	0.159	4.789E+02	5.022E+02	1.9	2.8
	9.000	0.184	5.245E+02	5.527E+02	1.9	2.8
	10.000	0.207	5.627E+02	5.960E+02	2.0	3.3
	11.000	0.228	5.945E+02	6.348E+02	2.0	3.3
	12.000	0.247	6.207E+02	6.655E+02	2.0	3.3
	14.000	0.280	6.594E+02	7.151E+02	1.8	3.2
	16.000	0.310	6.840E+02	7.535E+02	3.2	5.5
	18.000	0.335	6.983E+02	7.793E+02	3.2	5.5
	20.000	0.358	7.051E+02	7.946E+02	3.9	7.0
	23.000	0.388	7.056E+02	8.000E+02	3.9	7.0
	26.000	0.415	6.985E+02	7.948E+02	4.7	8.0
	30.000	0.446	6.821E+02	7.726E+02	4.7	8.0
	35.000	0.479	6.562E+02	7.290E+02	5.4	9.2
	40.000	0.508	6.280E+02	6.753E+02	5.4	10.7
	45.000	0.534	5.999E+02	6.169E+02	7.0	13.5
32-Ge	0.120	-0.769	2.430E-04	1.791E-04	12.6	17.2
	0.140	-0.735	7.959E-04	6.504E-04	7.2	11.5
	0.160	-0.706	1.991E-03	1.778E-03	7.2	11.1
	0.180	-0.681	4.169E-03	3.919E-03	4.4	5.3
	0.200	-0.658	7.715E-03	7.469E-03	4.4	5.2
	0.230	-0.628	1.650E-02	1.632E-02	4.5	5.6
	0.260	-0.601	3.076E-02	3.063E-02	4.5	5.6
	0.300	-0.570	6.092E-02	6.072E-02	3.4	4.2
	0.350	-0.536	1.217E-01	1.202E-01	2.7	3.5
	0.400	-0.507	2.146E-01	2.114E-01	2.7	3.5
	0.450	-0.482	3.467E-01	3.404E-01	1.9	2.7
	0.500	-0.459	5.244E-01	5.104E-01	1.9	2.7
	0.550	-0.438	7.538E-01	7.334E-01	1.3	1.6
	0.600	-0.419	1.041E+00	1.016E+00	1.3	1.6
	0.700	-0.386	1.806E+00	1.761E+00	1.4	2.2
	0.800	-0.357	2.854E+00	2.774E+00	1.4	2.2
	0.900	-0.331	4.211E+00	4.081E+00	1.2	1.9
	1.000	-0.308	5.894E+00	5.695E+00	1.2	1.9
	1.100	-0.288	7.914E+00	7.635E+00	0.9	1.6
	1.200	-0.269	1.027E+01	9.868E+00	0.9	1.6
	1.400	-0.235	1.601E+01	1.532E+01	1.0	1.9
	1.600	-0.206	2.305E+01	2.206E+01	1.0	1.9
	1.800	-0.181	3.130E+01	3.005E+01	0.9	1.9
	2.000	-0.158	4.062E+01	3.917E+01	0.9	1.9
	2.300	-0.128	5.631E+01	5.470E+01	0.8	1.9
	2.600	-0.101	7.367E+01	7.204E+01	0.8	1.9
	3.000	-0.070	9.864E+01	9.713E+01	0.9	2.2
	3.500	-0.036	1.316E+02	1.301E+02	1.1	2.5
	4.000	-0.007	1.653E+02	1.648E+02	1.1	2.5
	4.500	0.018	1.988E+02	1.998E+02	1.3	3.2
	5.000	0.041	2.314E+02	2.338E+02	1.3	3.2
	5.500	0.062	2.627E+02	2.675E+02	1.7	3.3
	6.000	0.081	2.925E+02	3.006E+02	1.7	3.3
	7.000	0.114	3.469E+02	3.592E+02	1.8	3.0
	8.000	0.143	3.944E+02	4.115E+02	1.8	3.0
	9.000	0.169	4.353E+02	4.579E+02	1.9	3.1
	10.000	0.191	4.701E+02	4.961E+02	1.9	3.1
	11.000	0.212	4.995E+02	5.297E+02	2.0	3.7
	12.000	0.231	5.242E+02	5.601E+02	2.0	3.7
	14.000	0.265	5.618E+02	6.050E+02	1.8	3.7
	16.000	0.294	5.870E+02	6.408E+02	1.8	3.6
	18.000	0.319	6.029E+02	6.676E+02	3.2	6.1
	20.000	0.342	6.120E+02	6.852E+02	3.2	6.0
	23.000	0.372	6.165E+02	6.956E+02	3.9	7.6
	26.000	0.399	6.135E+02	6.971E+02	3.9	7.5
	30.000	0.430	6.026E+02	6.851E+02	4.7	8.6
	35.000	0.464	5.830E+02	6.551E+02	5.4	9.9
	40.000	0.493	5.605E+02	6.149E+02	5.4	10.0
	45.000	0.518	5.374E+02	5.693E+02	7.0	14.2
	50.000	0.541	5.147E+02	5.213E+02	7.0	14.5
33-As	0.140	-0.751	4.492E-04	3.542E-04	12.6	18.2
33-As	0.160	-0.722	1.173E-03	1.004E-03	7.2	12.5
	0.180	-0.696	2.535E-03	2.316E-03	7.2	9.2
	0.200	-0.673	4.800E-03	4.562E-03	4.4	5.7
	0.230	-0.643	1.053E-02	1.033E-02	4.4	5.6
	0.260	-0.616	2.000E-02	1.986E-02	4.5	6.0
	0.300	-0.585	4.033E-02	4.023E-02	3.4	4.4
	0.350	-0.552	8.182E-02	8.132E-02	3.4	4.4
	0.400	-0.523	1.460E-01	1.441E-01	2.7	3.7
	0.450	-0.497	2.380E-01	2.342E-01	2.7	3.5
	0.500	-0.474	3.626E-01	3.557E-01	1.9	2.9
	0.550	-0.454	5.245E-01	5.104E-01	1.9	2.9
	0.600	-0.435	7.278E-01	7.081E-01	1.3	1.8
	0.700	-0.401	1.274E+00	1.244E+00	1.3	1.8
	0.800	-0.372	2.029E+00	1.977E+00	1.4	2.4
	0.900	-0.347	3.013E+00	2.925E+00	1.4	2.4
	1.000	-0.324	4.241E+00	4.107E+00	1.2	2.0
	1.100	-0.303	5.724E+00	5.527E+00	1.2	2.1
	1.200	-0.284	7.469E+00	7.200E+00	0.9	1.8
	1.400	-0.251	1.174E+01	1.125E+01	0.9	1.8
	1.600	-0.222	1.705E+01	1.630E+01	1.0	2.1
	1.800	-0.196	2.331E+01	2.233E+01	1.0	2.2
	2.000	-0.173	3.046E+01	2.928E+01	0.9	2.1
	2.300	-0.143	4.263E+01	4.125E+01	0.9	2.1
	2.600	-0.116	5.624E+01	5.478E+01	0.8	2.1
	3.000	-0.085	7.606E+01	7.467E+01	0.9	2.4
	3.500	-0.052	1.026E+02	1.011E+02	0.9	2.4
	4.000	-0.023	1.301E+02	1.290E+02	1.1	2.7
	4.500	0.003	1.578E+02	1.578E+02	1.1	2.9
	5.000	0.026	1.850E+02	1.864E+02	1.3	3.5
	5.500	0.046	2.114E+02	2.141E+02	1.3	3.5
	6.000	0.065	2.368E+02	2.415E+02	1.7	3.6
	7.000	0.099	2.839E+02	2.930E+02	1.7	3.6
	8.000	0.128	3.255E+02	3.380E+02	1.8	3.3
	9.000	0.153	3.620E+02	3.789E+02	1.8	3.3
	10.000	0.176	3.935E+02	4.149E+02	1.9	3.4
	11.000	0.197	4.205E+02	4.443E+02	1.9	3.4
	12.000	0.216	4.435E+02	4.707E+02	2.0	4.1
	14.000	0.249	4.795E+02	5.143E+02	2.0	4.1
	16.000	0.278	5.045E+02	5.465E+02	1.8	4.1
	18.000	0.304	5.214E+02	5.726E+02	3.2	6.6
	20.000	0.327	5.320E+02	5.913E+02	3.2	6.6
	23.000	0.357	5.393E+02	6.076E+02	3.9	8.1
	26.000	0.384	5.397E+02	6.111E+02	3.9	8.1
	30.000	0.415	5.332E+02	6.067E+02	4.7	9.3
	35.000	0.448	5.188E+02	5.872E+02	4.7	9.3
	40.000	0.477	5.011E+02	5.577E+02	5.4	10.6
	45.000	0.503	4.822E+02	5.225E+02	5.4	12.5
	50.000	0.526	4.632E+02	4.843E+02	7.0	15.3
	55.000	0.546	4.448E+02	4.448E+02	7.0	15.6
34-Se	0.140	-0.766	2.528E-04	1.887E-04	12.6	19.8
	0.160	-0.737	6.911E-04	5.619E-04	7.2	14.0
	0.180	-0.711	1.543E-03	1.362E-03	7.2	13.5
	0.200	-0.688	2.996E-03	2.781E-03	4.4	6.1
	0.230	-0.658	6.756E-03	6.540E-03	4.4	6.1
	0.260	-0.631	1.308E-02	1.292E-02	4.5	6.5
	0.300	-0.600	2.687E-02	2.675E-02	4.5	6.4
	0.350	-0.567	5.543E-02	5.523E-02	3.4	4.7
	0.400	-0.538	1.001E-01	9.889E-02	2.7	4.0
	0.450	-0.512	1.647E-01	1.623E-01	2.7	4.0
	0.500	-0.489	2.527E-01	2.484E-01	1.9	3.1
	0.550	-0.469	3.677E-01	3.582E-01	1.9	3.1
	0.600	-0.450	5.130E-01	4.991E-01	1.9	3.1
	0.700	-0.416	9.060E-01	8.846E-01	1.3	2.0
	0.800	-0.387	1.453E+00	1.417E+00	1.4	2.6
	0.900	-0.362	2.171E+00	2.111E+00	1.4	2.6
	1.000	-0.339	3.073E+00	2.981E+00	1.2	2.2
	1.100	-0.318	4.169E+00	4.035E+00	1.2	2.2
	1.200	-0.299	5.464E+00	5.273E+00	1.2	2.3
	1.400	-0.266	8.666E+00	8.317E+00	0.9	2.0
	1.600	-0.237	1.267E+01	1.213E+01	1.0	2.3
	1.800	-0.211	1.745E+01	1.670E+01	1.0	2.3
	2.000	-0.188	2.296E+01	2.201E+01	1.0	2.4
	2.300	-0.158	3.241E+01	3.126E+01	0.9	2.3
	2.600	-0.131	4.310E+01	4.183E+01	0.8	2.4
	3.000	-0.100	5.886E+01	5.757E+01	0.8	2.3
	3.500	-0.067	8.024E+01	7.904E+01	0.9	2.6

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
34-Se	4.000	-0.038	1.027E+02	1.015E+02	1.1	3.0
	4.500	-0.012	1.255E+02	1.249E+02	1.1	3.0
	5.000	0.011	1.483E+02	1.487E+02	1.3	3.7
	5.500	0.031	1.706E+02	1.718E+02	1.3	3.7
	6.000	0.050	1.922E+02	1.949E+02	1.3	3.7
	7.000	0.084	2.327E+02	2.394E+02	1.7	3.8
	8.000	0.113	2.691E+02	2.786E+02	1.8	3.6
	9.000	0.138	3.015E+02	3.140E+02	1.8	3.6
	10.000	0.161	3.298E+02	3.461E+02	1.9	3.6
	11.000	0.182	3.544E+02	3.733E+02	1.9	3.6
	12.000	0.201	3.757E+02	3.974E+02	1.9	4.5
	14.000	0.234	4.097E+02	4.380E+02	2.0	4.5
	16.000	0.263	4.342E+02	4.674E+02	1.8	4.5
	18.000	0.289	4.514E+02	4.915E+02	1.8	4.5
	20.000	0.312	4.629E+02	5.106E+02	3.2	7.2
	23.000	0.342	4.724E+02	5.289E+02	3.2	7.1
	26.000	0.369	4.752E+02	5.355E+02	3.9	8.6
	30.000	0.400	4.723E+02	5.367E+02	3.9	8.6
	35.000	0.433	4.622E+02	5.252E+02	4.7	9.9
	40.000	0.462	4.485E+02	5.043E+02	5.4	11.2
	45.000	0.488	4.332E+02	4.776E+02	5.4	11.3
	50.000	0.511	4.175E+02	4.474E+02	7.0	16.0
	55.000	0.531	4.019E+02	4.155E+02	7.0	16.3
35-Br	0.160	-0.751	4.068E-04	3.204E-04	12.6	20.7
	0.180	-0.726	9.413E-04	7.962E-04	7.2	14.9
	0.200	-0.703	1.876E-03	1.691E-03	7.2	14.4
	0.230	-0.672	4.355E-03	4.144E-03	4.4	6.5
	0.260	-0.646	8.607E-03	8.422E-03	4.4	6.5
	0.300	-0.615	1.803E-02	1.791E-02	4.5	6.8
	0.350	-0.581	3.785E-02	3.775E-02	3.4	4.9
	0.400	-0.552	6.921E-02	6.879E-02	3.4	4.9
	0.450	-0.527	1.149E-01	1.134E-01	2.7	4.2
	0.500	-0.504	1.777E-01	1.750E-01	2.7	4.2
	0.550	-0.483	2.602E-01	2.555E-01	1.9	3.3
	0.600	-0.464	3.648E-01	3.552E-01	1.9	3.3
	0.700	-0.431	6.499E-01	6.323E-01	1.3	2.2
	0.800	-0.402	1.049E+00	1.024E+00	1.3	2.2
	0.900	-0.376	1.577E+00	1.538E+00	1.4	2.8
	1.000	-0.353	2.245E+00	2.181E+00	1.4	2.8
	1.100	-0.333	3.060E+00	2.966E+00	1.2	2.4
	1.200	-0.314	4.029E+00	3.896E+00	1.2	2.4
	1.400	-0.280	6.440E+00	6.202E+00	0.9	2.2
	1.600	-0.251	9.487E+00	9.088E+00	0.9	2.2
	1.800	-0.226	1.315E+01	1.258E+01	1.0	2.5
	2.000	-0.203	1.740E+01	1.666E+01	1.0	2.5
	2.300	-0.172	2.478E+01	2.383E+01	0.9	2.5
	2.600	-0.146	3.320E+01	3.211E+01	0.9	2.5
	3.000	-0.115	4.576E+01	4.458E+01	0.8	2.6
	3.500	-0.081	6.300E+01	6.190E+01	0.9	2.9
	4.000	-0.052	8.134E+01	8.014E+01	0.9	2.9
	4.500	-0.027	1.002E+02	9.927E+01	1.1	3.2
	5.000	-0.004	1.192E+02	1.190E+02	1.1	3.2
	5.500	0.017	1.380E+02	1.387E+02	1.3	4.0
	6.000	0.036	1.564E+02	1.577E+02	1.3	4.0
	7.000	0.069	1.912E+02	1.953E+02	1.7	4.1
	8.000	0.098	2.230E+02	2.301E+02	1.7	4.1
	9.000	0.124	2.516E+02	2.610E+02	1.8	3.9
	10.000	0.147	2.770E+02	2.893E+02	1.8	3.9
	11.000	0.168	2.993E+02	3.148E+02	1.9	3.9
	12.000	0.186	3.189E+02	3.362E+02	1.9	3.9
	14.000	0.220	3.507E+02	3.725E+02	2.0	5.0
	16.000	0.249	3.743E+02	4.015E+02	2.0	4.9
	18.000	0.274	3.914E+02	4.226E+02	1.8	4.9
	20.000	0.297	4.035E+02	4.413E+02	1.8	4.9
	23.000	0.328	4.144E+02	4.608E+02	3.2	7.7
	26.000	0.354	4.191E+02	4.716E+02	3.2	7.6
	30.000	0.385	4.189E+02	4.746E+02	3.9	9.1
	35.000	0.419	4.124E+02	4.693E+02	4.7	10.6
	40.000	0.448	4.020E+02	4.551E+02	4.7	10.6
	45.000	0.473	3.898E+02	4.352E+02	5.4	11.9
	50.000	0.496	3.768E+02	4.116E+02	5.4	12.1
	55.000	0.517	3.637E+02	3.860E+02	7.0	17.0
	60.000	0.536	3.509E+02	3.594E+02	7.0	17.4
36-Kr	0.160	-0.766	2.377E-04	1.777E-04	12.6	22.5

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
36-Kr	0.180	-0.740	5.713E-04	4.662E-04	12.6	21.7
	0.200	-0.717	1.171E-03	1.017E-03	7.2	15.9
	0.230	-0.687	2.803E-03	2.609E-03	4.4	7.0
	0.260	-0.660	5.663E-03	5.470E-03	4.4	6.9
	0.300	-0.629	1.211E-02	1.197E-02	4.5	7.3
	0.350	-0.595	2.589E-02	2.578E-02	4.5	6.2
	0.400	-0.567	4.796E-02	4.779E-02	3.4	5.2
	0.450	-0.541	8.042E-02	7.945E-02	2.7	4.5
	0.500	-0.518	1.253E-01	1.236E-01	2.7	4.5
	0.550	-0.497	1.846E-01	1.817E-01	2.7	4.0
	0.600	-0.478	2.603E-01	2.554E-01	1.9	3.5
	0.700	-0.445	4.676E-01	4.550E-01	1.3	2.4
	0.800	-0.416	7.605E-01	7.425E-01	1.3	2.4
	0.900	-0.390	1.150E+00	1.122E+00	1.4	3.0
	1.000	-0.368	1.645E+00	1.602E+00	1.4	3.0
	1.100	-0.347	2.253E+00	2.187E+00	1.4	3.0
	1.200	-0.328	2.979E+00	2.885E+00	1.2	2.6
	1.400	-0.294	4.798E+00	4.627E+00	1.2	2.7
	1.600	-0.266	7.117E+00	6.830E+00	0.9	2.4
	1.800	-0.240	9.930E+00	9.503E+00	0.9	2.4
	2.000	-0.217	1.322E+01	1.264E+01	1.0	2.7
	2.300	-0.187	1.897E+01	1.820E+01	0.9	2.8
	2.600	-0.160	2.561E+01	2.469E+01	0.9	2.7
	3.000	-0.129	3.560E+01	3.457E+01	0.8	2.8
	3.500	-0.095	4.951E+01	4.848E+01	0.8	3.0
	4.000	-0.067	6.447E+01	6.352E+01	0.9	3.1
	4.500	-0.041	8.005E+01	7.905E+01	1.1	3.4
	5.000	-0.018	9.587E+01	9.524E+01	1.1	3.4
	5.500	0.003	1.117E+02	1.117E+02	1.1	3.6
	6.000	0.022	1.272E+02	1.281E+02	1.3	4.2
	7.000	0.055	1.571E+02	1.596E+02	1.3	4.2
	8.000	0.084	1.848E+02	1.901E+02	1.7	4.3
	9.000	0.110	2.100E+02	2.172E+02	1.8	4.2
	10.000	0.132	2.326E+02	2.418E+02	1.8	4.2
	11.000	0.153	2.528E+02	2.646E+02	1.8	4.1
	12.000	0.172	2.706E+02	2.850E+02	1.9	4.2
	14.000	0.206	3.001E+02	3.177E+02	2.0	5.4
	16.000	0.235	3.226E+02	3.449E+02	2.0	5.4
	18.000	0.260	3.394E+02	3.650E+02	1.8	5.3
	20.000	0.283	3.516E+02	3.818E+02	1.8	5.3
	23.000	0.313	3.635E+02	4.012E+02	3.2	8.3
	26.000	0.340	3.696E+02	4.134E+02	3.2	8.2
	30.000	0.371	3.716E+02	4.191E+02	3.9	9.7
	35.000	0.405	3.680E+02	4.185E+02	3.9	11.1
	40.000	0.433	3.605E+02	4.096E+02	4.7	11.3
	45.000	0.459	3.508E+02	3.952E+02	5.4	12.5
	50.000	0.482	3.402E+02	3.772E+02	5.4	12.6
	55.000	0.503	3.293E+02	3.569E+02	5.4	15.3
	60.000	0.522	3.184E+02	3.353E+02	7.0	18.0
37-Rb	0.180	-0.754	3.475E-04	2.714E-04	12.6	23.4
	0.200	-0.731	7.342E-04	6.103E-04	7.2	17.5
	0.230	-0.700	1.815E-03	1.644E-03	7.2	16.6
	0.260	-0.674	3.752E-03	3.564E-03	4.4	7.4
	0.300	-0.643	8.206E-03	8.047E-03	4.4	7.3
	0.350	-0.609	1.787E-02	1.777E-02	4.5	7.6
	0.400	-0.580	3.356E-02	3.347E-02	3.4	5.4
	0.450	-0.555	5.683E-02	5.651E-02	3.4	5.4
	0.500	-0.532	8.925E-02	8.813E-02	2.7	4.7
	0.550	-0.511	1.323E-01	1.304E-01	2.7	4.7
	0.600	-0.492	1.875E-01	1.844E-01	1.9	3.7
	0.700	-0.459	3.399E-01	3.308E-01	1.9	3.7
	0.800	-0.430	5.564E-01	5.415E-01	1.3	2.6
	0.900	-0.404	8.461E-01	8.260E-01	1.3	2.6
	1.000	-0.381	1.216E+00	1.186E+00	1.4	3.2
	1.100	-0.361	1.673E+00	1.627E+00	1.4	3.2
	1.200	-0.342	2.222E+00	2.155E+00	1.2	2.8
	1.400	-0.308	3.605E+00	3.484E+00	1.2	2.8
	1.600	-0.279	5.382E+00	5.182E+00	0.9	2.6
	1.800	-0.254	7.554E+00	7.238E+00	0.9	2.6
	2.000	-0.231	1.011E+01	9.670E+00	1.0	2.9
	2.300	-0.200	1.462E+01	1.400E+01	1.0	2.9
	2.600	-0.174	1.988E+01	1.910E+01	0.9	3.0
	3.000	-0.143	2.786E+01	2.696E+01	0.9	3.0
	3.500	-0.109	3.910E+01	3.816E+01	0.8	3.0
	4.000	-0.080	5.134E+01	5.046E+01	0.9	3.3
	4.500	-0.055	6.421E+01	6.327E+01	0.9	3.3
	5.000	-0.032	7.741E+01	7.660E+01	1.1	3.7

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
37-Rb	5.500	-0.011	9.071E+01	9.031E+01	1.1	3.7
	6.000	0.008	1.039E+02	1.041E+02	1.3	4.5
	7.000	0.041	1.295E+02	1.309E+02	1.3	4.5
	8.000	0.070	1.536E+02	1.570E+02	1.7	4.6
	9.000	0.096	1.758E+02	1.813E+02	1.7	4.5
	10.000	0.119	1.959E+02	2.030E+02	1.8	4.5
	11.000	0.140	2.140E+02	2.230E+02	1.8	4.4
	12.000	0.158	2.302E+02	2.414E+02	1.9	4.5
	14.000	0.192	2.574E+02	2.717E+02	1.9	4.5
	16.000	0.221	2.786E+02	2.960E+02	2.0	5.8
	18.000	0.246	2.948E+02	3.161E+02	2.0	5.8
	20.000	0.269	3.070E+02	3.310E+02	1.8	5.7
	23.000	0.300	3.194E+02	3.498E+02	1.8	5.7
	26.000	0.326	3.265E+02	3.628E+02	3.2	8.8
	30.000	0.357	3.303E+02	3.721E+02	3.9	10.2
	35.000	0.391	3.290E+02	3.732E+02	3.9	10.2
	40.000	0.420	3.238E+02	3.684E+02	4.7	11.9
	45.000	0.445	3.163E+02	3.584E+02	4.7	11.9
	50.000	0.468	3.078E+02	3.448E+02	5.4	13.2
	55.000	0.489	2.987E+02	3.289E+02	5.4	13.4
	60.000	0.508	2.895E+02	3.115E+02	5.4	16.3
	70.000	0.541	2.714E+02	2.747E+02	7.0	19.5
38-Sr	0.180	-0.767	2.107E-04	1.564E-04	12.6	25.3
	0.200	-0.744	4.594E-04	3.701E-04	12.6	24.3
	0.230	-0.714	1.176E-03	1.030E-03	7.2	18.1
	0.260	-0.687	2.491E-03	2.316E-03	4.4	7.9
	0.300	-0.656	5.575E-03	5.406E-03	4.4	7.7
	0.350	-0.623	1.239E-02	1.228E-02	4.5	8.1
	0.400	-0.594	2.358E-02	2.348E-02	4.5	6.7
	0.450	-0.568	4.036E-02	4.022E-02	3.4	5.7
	0.500	-0.545	6.389E-02	6.341E-02	3.4	5.7
	0.550	-0.525	9.535E-02	9.410E-02	2.7	5.0
	0.600	-0.506	1.359E-01	1.338E-01	2.7	5.0
	0.700	-0.472	2.483E-01	2.435E-01	1.9	3.9
	0.800	-0.443	4.094E-01	3.983E-01	1.3	2.8
	0.900	-0.418	6.260E-01	6.113E-01	1.3	2.8
	1.000	-0.395	9.043E-01	8.824E-01	1.3	2.7
	1.100	-0.374	1.249E+00	1.217E+00	1.4	3.3
	1.200	-0.355	1.665E+00	1.618E+00	1.4	3.3
	1.400	-0.322	2.721E+00	2.634E+00	1.2	3.0
	1.600	-0.293	4.088E+00	3.940E+00	1.2	3.1
	1.800	-0.267	5.770E+00	5.538E+00	0.9	2.8
	2.000	-0.244	7.762E+00	7.431E+00	0.9	2.8
	2.300	-0.214	1.131E+01	1.082E+01	1.0	3.1
	2.600	-0.187	1.547E+01	1.484E+01	0.9	3.2
	3.000	-0.156	2.186E+01	2.109E+01	0.9	3.2
	3.500	-0.123	3.096E+01	3.011E+01	0.8	3.2
	4.000	-0.094	4.097E+01	4.014E+01	0.8	3.4
	4.500	-0.068	5.161E+01	5.083E+01	0.9	3.6
	5.000	-0.045	6.262E+01	6.167E+01	0.9	3.6
	5.500	-0.024	7.380E+01	7.314E+01	1.1	3.9
	6.000	-0.006	8.499E+01	8.476E+01	1.1	3.9
	7.000	0.028	1.069E+02	1.079E+02	1.3	4.7
	8.000	0.057	1.278E+02	1.299E+02	1.7	4.8
	9.000	0.082	1.472E+02	1.514E+02	1.7	4.8
	10.000	0.105	1.651E+02	1.706E+02	1.8	4.8
	11.000	0.126	1.814E+02	1.882E+02	1.8	4.7
	12.000	0.145	1.960E+02	2.046E+02	1.8	4.7
	14.000	0.178	2.210E+02	2.332E+02	1.9	4.8
	16.000	0.207	2.408E+02	2.550E+02	2.0	6.2
	18.000	0.233	2.563E+02	2.740E+02	2.0	6.2
	20.000	0.256	2.682E+02	2.882E+02	1.8	6.2
	23.000	0.286	2.809E+02	3.054E+02	1.8	6.1
	26.000	0.313	2.886E+02	3.185E+02	3.2	9.3
	30.000	0.344	2.937E+02	3.291E+02	3.2	9.2
	35.000	0.377	2.943E+02	3.326E+02	3.9	10.7
	40.000	0.406	2.910E+02	3.309E+02	3.9	12.4
	45.000	0.432	2.854E+02	3.244E+02	4.7	12.6
	50.000	0.455	2.786E+02	3.145E+02	4.7	12.6
	55.000	0.475	2.711E+02	3.021E+02	5.4	13.9
	60.000	0.494	2.634E+02	2.883E+02	5.4	14.1
	70.000	0.528	2.479E+02	2.581E+02	7.0	20.1
39-Y	0.200	-0.757	2.871E-04	2.213E-04	12.6	26.1
	0.230	-0.727	7.622E-04	6.419E-04	7.2	19.7
	0.260	-0.700	1.657E-03	1.501E-03	7.2	18.8
	0.300	-0.669	3.801E-03	3.632E-03	4.4	8.2
39-Y	0.350	-0.636	8.624E-03	8.494E-03	4.5	8.5
	0.400	-0.607	1.666E-02	1.657E-02	4.5	8.5
	0.450	-0.581	2.882E-02	2.875E-02	3.4	5.9
	0.500	-0.558	4.601E-02	4.579E-02	3.4	5.9
	0.550	-0.538	6.912E-02	6.829E-02	2.7	5.2
	0.600	-0.519	9.904E-02	9.769E-02	2.7	5.2
	0.700	-0.485	1.826E-01	1.794E-01	1.9	4.1
	0.800	-0.456	3.030E-01	2.949E-01	1.9	4.1
	0.900	-0.431	4.659E-01	4.534E-01	1.3	3.0
	1.000	-0.408	6.763E-01	6.602E-01	1.3	3.0
	1.100	-0.387	9.382E-01	9.151E-01	1.4	3.5
	1.200	-0.368	1.255E+00	1.223E+00	1.4	3.5
	1.400	-0.335	2.065E+00	2.002E+00	1.2	3.2
	1.600	-0.306	3.121E+00	3.014E+00	1.2	3.2
	1.800	-0.280	4.428E+00	4.264E+00	0.9	3.0
	2.000	-0.257	5.987E+00	5.739E+00	0.9	3.0
	2.300	-0.227	8.783E+00	8.400E+00	1.0	3.3
	2.600	-0.200	1.209E+01	1.158E+01	1.0	3.3
	3.000	-0.169	1.722E+01	1.657E+01	0.9	3.4
	3.500	-0.136	2.459E+01	2.383E+01	0.8	3.4
	4.000	-0.107	3.279E+01	3.202E+01	0.8	3.4
	4.500	-0.081	4.158E+01	4.086E+01	0.9	3.8
	5.000	-0.058	5.077E+01	5.003E+01	0.9	3.8
	5.500	-0.038	6.017E+01	5.946E+01	1.1	4.1
	6.000	-0.019	6.965E+01	6.918E+01	1.1	4.1
	7.000	0.015	8.843E+01	8.880E+01	1.3	5.0
	8.000	0.044	1.065E+02	1.078E+02	1.3	5.0
	9.000	0.069	1.235E+02	1.262E+02	1.7	5.1
	10.000	0.092	1.394E+02	1.436E+02	1.7	5.0
	11.000	0.113	1.539E+02	1.593E+02	1.8	5.0
	12.000	0.132	1.671E+02	1.737E+02	1.8	5.0
	14.000	0.165	1.899E+02	1.996E+02	1.9	5.1
	16.000	0.194	2.083E+02	2.200E+02	1.9	5.1
	18.000	0.220	2.230E+02	2.369E+02	2.0	6.7
	20.000	0.243	2.346E+02	2.513E+02	2.0	6.6
	23.000	0.273	2.472E+02	2.668E+02	1.8	6.6
	26.000	0.300	2.554E+02	2.796E+02	1.8	6.5
	30.000	0.331	2.614E+02	2.911E+02	3.2	9.8
	35.000	0.364	2.635E+02	2.964E+02	3.9	11.3
	40.000	0.393	2.618E+02	2.971E+02	3.9	11.2
	45.000	0.419	2.578E+02	2.933E+02	4.7	13.2
	50.000	0.442	2.524E+02	2.863E+02	4.7	13.3
	55.000	0.462	2.463E+02	2.770E+02	5.4	14.5
	60.000	0.481	2.399E+02	2.660E+02	5.4	14.7
	70.000	0.515	2.267E+02	2.414E+02	7.0	20.7
	80.000	0.544	2.139E+02	2.153E+02	7.0	21.5
40-Zr	0.230	-0.740	4.951E-04	4.042E-04	12.6	26.6
	0.260	-0.713	1.106E-03	9.714E-04	7.2	20.3
	0.300	-0.682	2.603E-03	2.442E-03	4.4	8.7
	0.350	-0.649	6.037E-03	5.894E-03	4.4	8.5
	0.400	-0.619	1.184E-02	1.175E-02	4.5	8.9
	0.450	-0.594	2.071E-02	2.063E-02	4.5	7.2
	0.500	-0.571	3.335E-02	3.325E-02	3.4	6.2
	0.550	-0.550	5.045E-02	5.013E-02	3.4	6.2
	0.600	-0.531	7.269E-02	7.178E-02	2.7	5.5
	0.700	-0.498	1.352E-01	1.330E-01	2.7	4.8
	0.800	-0.469	2.259E-01	2.200E-01	1.9	4.3
	0.900	-0.443	3.492E-01	3.397E-01	1.3	3.2
	1.000	-0.421	5.091E-01	4.955E-01	1.3	3.2
	1.100	-0.400	7.092E-01	6.921E-01	1.3	3.1
	1.200	-0.381	9.523E-01	9.285E-01	1.4	3.7
	1.400	-0.347	1.577E+00	1.531E+00	1.4	3.7
	1.600	-0.319	2.396E+00	2.319E+00	1.2	3.4
	1.800	-0.293	3.417E+00	3.294E+00	1.2	3.5
	2.000	-0.270	4.642E+00	4.459E+00	0.9	3.2
	2.300	-0.240	6.854E+00	6.559E+00	0.9	3.2
	2.600	-0.213	9.493E+00	9.080E+00	1.0	3.5
	3.000	-0.182	1.361E+01	1.307E+01	0.9	3.6
	3.500	-0.148	1.960E+01	1.894E+01	0.9	3.6
	4.000	-0.119	2.633E+01	2.562E+01	0.8	3.6
	4.500	-0.094	3.361E+01	3.293E+01	0.8	3.9
	5.000	-0.071	4.128E+01	4.063E+01	0.9	4.0
	5.500	-0.050	4.919E+01	4.845E+01	0.9	4.0
	6.000	-0.032	5.722E+01	5.662E+01	1.1	4.4
	7.000	0.002	7.329E+01	7.328E+01	1.1	4.6
	8.000	0.031	8.895E+01	8.955E+01	1.3	5.2
	9.000	0.057	1.038E+02	1.056E+02	1.7	5.3

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
40-Zr	10.000	0.079	1.178E+02	1.208E+02	1.7	5.3
	11.000	0.100	1.308E+02	1.350E+02	1.7	6.0
	12.000	0.119	1.426E+02	1.478E+02	1.8	5.3
	14.000	0.153	1.634E+02	1.710E+02	1.8	5.3
	16.000	0.182	1.805E+02	1.901E+02	1.9	5.4
	18.000	0.207	1.943E+02	2.058E+02	2.0	7.1
	20.000	0.230	2.054E+02	2.194E+02	2.0	7.1
	23.000	0.260	2.178E+02	2.342E+02	1.8	7.0
	26.000	0.287	2.262E+02	2.461E+02	1.8	6.9
	30.000	0.318	2.328E+02	2.577E+02	3.2	10.4
	35.000	0.352	2.361E+02	2.654E+02	3.2	10.3
	40.000	0.380	2.357E+02	2.666E+02	3.9	11.8
	45.000	0.406	2.330E+02	2.650E+02	3.9	13.7
	50.000	0.429	2.289E+02	2.603E+02	4.7	13.9
	55.000	0.450	2.240E+02	2.534E+02	4.7	13.9
	60.000	0.468	2.187E+02	2.449E+02	5.4	15.2
	70.000	0.502	2.075E+02	2.251E+02	5.4	19.0
	80.000	0.531	1.964E+02	2.033E+02	7.0	22.1
41-Nb	0.180	-0.805	4.616E-05	2.216E-05	7.4	22.8
	0.200	-0.782	1.122E-04	6.530E-05	6.7	20.7
	0.230	-0.752	3.226E-04	2.234E-04	6.7	18.4
	0.260	-0.725	7.420E-04	5.706E-04	6.8	19.4
	0.300	-0.694	1.794E-03	1.503E-03	6.8	11.4
	0.350	-0.661	4.256E-03	3.796E-03	3.8	8.6
	0.400	-0.632	8.481E-03	7.852E-03	3.0	7.4
	0.450	-0.606	1.501E-02	1.428E-02	3.0	7.3
	0.500	-0.583	2.438E-02	2.354E-02	2.8	5.6
	0.550	-0.563	3.714E-02	3.655E-02	2.8	5.6
	0.600	-0.544	5.381E-02	5.378E-02	2.8	5.5
	0.700	-0.510	1.010E-01	1.031E-01	1.8	4.0
	0.800	-0.481	1.698E-01	1.738E-01	1.8	3.6
	0.900	-0.456	2.639E-01	2.718E-01	1.8	3.6
	1.000	-0.433	3.865E-01	3.990E-01	1.4	3.2
	1.100	-0.412	5.404E-01	5.581E-01	1.4	3.2
	1.200	-0.393	7.282E-01	7.537E-01	1.4	3.1
	1.400	-0.360	1.213E+00	1.256E+00	1.5	3.1
	1.600	-0.331	1.853E+00	1.899E+00	1.8	3.2
	1.800	-0.305	2.656E+00	2.715E+00	1.8	3.2
	2.000	-0.282	3.624E+00	3.658E+00	2.0	3.4
	2.300	-0.252	5.382E+00	5.317E+00	2.0	3.5
	2.600	-0.225	7.496E+00	7.292E+00	2.5	4.0
	3.000	-0.194	1.082E+01	1.034E+01	2.5	4.2
	3.500	-0.161	1.570E+01	1.463E+01	2.8	4.3
	4.000	-0.132	2.124E+01	1.942E+01	3.2	4.6
	4.500	-0.106	2.728E+01	2.445E+01	3.2	4.7
	5.000	-0.083	3.369E+01	2.975E+01	2.9	4.3
	5.500	-0.063	4.035E+01	3.537E+01	2.9	4.3
	6.000	-0.044	4.717E+01	4.119E+01	3.8	5.3
	7.000	-0.010	6.093E+01	5.327E+01	3.8	5.3
	8.000	0.019	7.449E+01	6.577E+01	3.1	4.8
	9.000	0.044	8.752E+01	7.853E+01	3.1	4.8
	10.000	0.067	9.984E+01	9.144E+01	5.2	6.7
	11.000	0.088	1.114E+02	1.044E+02	5.2	6.7
42-Mo	0.200	-0.794	6.979E-05	3.700E-05	7.4	19.4
	0.230	-0.764	2.096E-04	1.366E-04	6.7	17.6
	0.260	-0.738	4.971E-04	3.660E-04	6.8	18.6
	0.300	-0.706	1.236E-03	1.006E-03	6.8	17.5
	0.350	-0.673	3.005E-03	2.628E-03	3.8	8.3
	0.400	-0.644	6.088E-03	5.559E-03	3.8	8.1
	0.450	-0.618	1.091E-02	1.025E-02	3.0	6.9
	0.500	-0.595	1.788E-02	1.717E-02	3.0	5.4
	0.550	-0.575	2.743E-02	2.670E-02	2.8	5.3
	0.600	-0.556	3.998E-02	3.957E-02	2.8	5.3
	0.700	-0.522	7.568E-02	7.675E-02	1.8	3.7
	0.800	-0.493	1.281E-01	1.310E-01	1.8	3.3
	0.900	-0.468	2.002E-01	2.052E-01	1.8	3.4
	1.000	-0.445	2.945E-01	3.037E-01	1.8	3.4
	1.100	-0.424	4.133E-01	4.270E-01	1.4	3.0
	1.200	-0.405	5.589E-01	5.776E-01	1.4	3.0
	1.400	-0.372	9.365E-01	9.701E-01	1.5	3.0
	1.600	-0.343	1.438E+00	1.477E+00	1.5	3.0
	1.800	-0.317	2.070E+00	2.115E+00	1.8	3.0
	2.000	-0.294	2.836E+00	2.883E+00	1.8	3.1
	2.300	-0.264	4.238E+00	4.228E+00	2.0	3.3
	2.600	-0.237	5.934E+00	5.817E+00	2.0	3.3
	3.000	-0.206	8.624E+00	8.298E+00	2.5	3.8
42-Mo	3.500	-0.173	1.260E+01	1.184E+01	2.8	4.1
	4.000	-0.144	1.716E+01	1.585E+01	2.8	4.1
	4.500	-0.118	2.217E+01	2.006E+01	3.2	4.4
	5.000	-0.095	2.754E+01	2.449E+01	3.2	4.5
	5.500	-0.075	3.315E+01	2.917E+01	2.9	4.0
	6.000	-0.056	3.893E+01	3.406E+01	2.9	4.0
	7.000	-0.022	5.070E+01	4.425E+01	3.8	5.0
	8.000	0.007	6.242E+01	5.483E+01	3.1	4.5
	9.000	0.032	7.381E+01	6.567E+01	3.1	4.5
	10.000	0.055	8.467E+01	7.665E+01	3.1	4.5
	11.000	0.076	9.491E+01	8.772E+01	5.2	6.4
	12.000	0.095	1.045E+02	9.879E+01	5.2	6.4
43-Tc	0.200	-0.806	4.329E-05	2.056E-05	7.4	18.7
	0.230	-0.776	1.362E-04	8.274E-05	6.7	16.8
	0.260	-0.749	3.335E-04	2.338E-04	6.7	15.4
	0.300	-0.718	8.548E-04	6.724E-04	6.8	16.6
	0.350	-0.685	2.131E-03	1.822E-03	6.8	10.4
	0.400	-0.656	4.392E-03	3.947E-03	3.8	7.7
	0.450	-0.630	7.966E-03	7.389E-03	3.0	6.5
	0.500	-0.607	1.318E-02	1.253E-02	3.0	6.4
	0.550	-0.587	2.038E-02	1.961E-02	2.8	5.1
	0.600	-0.568	2.988E-02	2.928E-02	2.8	5.1
	0.700	-0.534	5.708E-02	5.745E-02	1.8	3.5
	0.800	-0.505	9.728E-02	9.927E-02	1.8	3.5
	0.900	-0.480	1.528E-01	1.565E-01	1.8	3.2
	1.000	-0.457	2.258E-01	2.324E-01	1.8	3.2
	1.100	-0.436	3.181E-01	3.283E-01	1.4	2.8
	1.200	-0.417	4.315E-01	4.459E-01	1.4	2.8
	1.400	-0.384	7.271E-01	7.530E-01	1.5	2.8
	1.600	-0.355	1.122E+00	1.154E+00	1.5	2.8
	1.800	-0.329	1.622E+00	1.662E+00	1.8	2.8
	2.000	-0.306	2.232E+00	2.282E+00	1.8	2.8
	2.300	-0.276	3.352E+00	3.371E+00	2.0	3.1
	2.600	-0.249	4.718E+00	4.654E+00	2.0	3.1
	3.000	-0.218	6.899E+00	6.684E+00	2.5	3.6
	3.500	-0.185	1.015E+01	9.620E+00	2.8	3.9
	4.000	-0.156	1.391E+01	1.292E+01	2.8	3.9
	4.500	-0.130	1.808E+01	1.651E+01	3.2	4.2
	5.000	-0.107	2.257E+01	2.025E+01	3.2	4.2
	5.500	-0.087	2.730E+01	2.415E+01	2.9	3.8
	6.000	-0.068	3.221E+01	2.827E+01	2.9	3.8
	7.000	-0.034	4.228E+01	3.689E+01	3.8	4.8
	8.000	-0.005	5.242E+01	4.588E+01	3.8	4.8
	9.000	0.020	6.236E+01	5.511E+01	3.1	4.2
	10.000	0.043	7.192E+01	6.449E+01	3.1	4.2
	11.000	0.064	8.101E+01	7.396E+01	5.2	6.2
	12.000	0.083	8.957E+01	8.346E+01	5.2	6.1
44-Ru	0.230	-0.787	8.827E-05	4.949E-05	7.4	15.2
	0.260	-0.761	2.237E-04	1.483E-04	6.7	14.4
	0.300	-0.730	5.918E-04	4.483E-04	6.8	15.6
	0.350	-0.696	1.515E-03	1.263E-03	6.8	10.0
	0.400	-0.667	3.180E-03	2.808E-03	3.8	7.3
	0.450	-0.642	5.844E-03	5.350E-03	3.8	7.2
	0.500	-0.619	9.767E-03	9.176E-03	3.0	6.1
	0.550	-0.598	1.521E-02	1.458E-02	3.0	4.9
	0.600	-0.579	2.244E-02	2.176E-02	2.8	4.8
	0.700	-0.546	4.328E-02	4.319E-02	2.8	4.8
	0.800	-0.517	7.426E-02	7.555E-02	1.8	3.3
	0.900	-0.491	1.173E-01	1.199E-01	1.8	2.9
	1.000	-0.468	1.740E-01	1.783E-01	1.8	3.0
	1.100	-0.448	2.461E-01	2.537E-01	1.8	3.0
	1.200	-0.429	3.348E-01	3.458E-01	1.4	2.6
	1.400	-0.395	5.673E-01	5.871E-01	1.4	2.5
	1.600	-0.366	8.796E-01	9.111E-01	1.5	2.6
	1.800	-0.341	1.277E+00	1.311E+00	1.5	2.6
	2.000	-0.318	1.764E+00	1.802E+00	1.8	2.7
	2.300	-0.287	2.663E+00	2.696E+00	2.0	2.9
	2.600	-0.261	3.765E+00	3.736E+00	2.0	2.9
	3.000	-0.230	5.539E+00	5.402E+00	2.5	3.4
	3.500	-0.196	8.205E+00	7.847E+00	2.5	3.5
	4.000	-0.167	1.131E+01	1.059E+01	2.8	3.7
	4.500	-0.142	1.478E+01	1.363E+01	2.8	3.7
	5.000	-0.119	1.855E+01	1.678E+01	3.2	3.9
	5.500	-0.098	2.254E+01	2.008E+01	3.2	4.0
	6.000	-0.079	2.670E+01	2.354E+01	2.9	3.5

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
44-Ru	7.000	-0.046	3.533E+01	3.085E+01	2.9	3.5
	8.000	-0.017	4.410E+01	3.851E+01	3.8	4.5
	9.000	0.009	5.277E+01	4.639E+01	3.1	3.9
	10.000	0.032	6.118E+01	5.442E+01	3.1	3.9
	11.000	0.052	6.924E+01	6.254E+01	3.1	3.9
	12.000	0.071	7.688E+01	7.071E+01	5.2	5.9
45-Rh	0.230	-0.799	5.706E-05	2.913E-05	7.4	14.0
	0.260	-0.772	1.499E-04	9.323E-05	6.7	13.3
	0.300	-0.741	4.101E-04	2.977E-04	6.7	12.4
	0.350	-0.708	1.079E-03	8.753E-04	6.8	13.8
	0.400	-0.679	2.310E-03	1.999E-03	3.8	6.9
	0.450	-0.653	4.303E-03	3.882E-03	3.8	6.8
	0.500	-0.630	7.266E-03	6.740E-03	3.0	5.7
	0.550	-0.609	1.141E-02	1.082E-02	3.0	5.6
	0.600	-0.590	1.694E-02	1.633E-02	3.0	4.6
	0.700	-0.557	3.297E-02	3.261E-02	2.8	4.5
	0.800	-0.528	5.697E-02	5.757E-02	1.8	3.0
	0.900	-0.502	9.044E-02	9.233E-02	1.8	3.0
	1.000	-0.480	1.348E-01	1.380E-01	1.8	2.9
	1.100	-0.459	1.913E-01	1.969E-01	1.8	2.9
	1.200	-0.440	2.611E-01	2.694E-01	1.8	2.9
	1.400	-0.407	4.448E-01	4.597E-01	1.4	2.4
	1.600	-0.378	6.927E-01	7.175E-01	1.5	2.4
	1.800	-0.352	1.010E+00	1.038E+00	1.5	2.4
	2.000	-0.329	1.400E+00	1.434E+00	1.8	2.5
	2.300	-0.299	2.124E+00	2.164E+00	1.8	2.5
	2.600	-0.272	3.016E+00	3.025E+00	2.0	2.7
	3.000	-0.241	4.462E+00	4.382E+00	2.0	2.7
	3.500	-0.208	6.652E+00	6.404E+00	2.5	3.2
	4.000	-0.179	9.221E+00	8.699E+00	2.8	3.4
	4.500	-0.153	1.211E+01	1.123E+01	2.8	3.4
	5.000	-0.130	1.527E+01	1.395E+01	3.2	3.7
	5.500	-0.109	1.865E+01	1.675E+01	3.2	3.7
	6.000	-0.090	2.218E+01	1.966E+01	2.9	3.3
	7.000	-0.057	2.957E+01	2.588E+01	2.9	3.3
	8.000	-0.028	3.715E+01	3.241E+01	3.8	4.3
	9.000	-0.002	4.471E+01	3.916E+01	3.8	4.3
	10.000	0.020	5.211E+01	4.606E+01	3.1	3.6
	11.000	0.041	5.925E+01	5.304E+01	3.1	3.6
	12.000	0.060	6.607E+01	6.008E+01	5.2	5.6
	14.000	0.093	7.860E+01	7.420E+01	5.2	5.6
46-Pd	0.230	-0.810	3.682E-05	1.685E-05	7.4	12.6
	0.260	-0.783	1.005E-04	5.813E-05	6.7	12.1
	0.300	-0.752	2.847E-04	1.971E-04	6.7	11.2
	0.350	-0.719	7.719E-04	6.064E-04	6.8	12.8
	0.400	-0.690	1.685E-03	1.426E-03	6.8	9.0
	0.450	-0.664	3.183E-03	2.825E-03	3.8	6.4
	0.500	-0.641	5.433E-03	4.977E-03	3.8	6.3
	0.550	-0.621	8.600E-03	8.066E-03	3.0	5.2
	0.600	-0.602	1.285E-02	1.228E-02	3.0	5.2
	0.700	-0.568	2.527E-02	2.475E-02	2.8	4.3
	0.800	-0.539	4.396E-02	4.409E-02	2.8	4.3
	0.900	-0.513	7.016E-02	7.149E-02	1.8	2.8
	1.000	-0.491	1.050E-01	1.074E-01	1.8	2.5
	1.100	-0.470	1.496E-01	1.532E-01	1.8	2.7
	1.200	-0.451	2.048E-01	2.110E-01	1.8	2.7
	1.400	-0.418	3.506E-01	3.623E-01	1.4	2.2
	1.600	-0.389	5.484E-01	5.677E-01	1.4	2.2
	1.800	-0.363	8.025E-01	8.311E-01	1.5	2.3
	2.000	-0.340	1.116E+00	1.145E+00	1.5	2.3
	2.300	-0.310	1.701E+00	1.743E+00	1.8	2.3
	2.600	-0.283	2.427E+00	2.451E+00	2.0	2.5
	3.000	-0.252	3.609E+00	3.565E+00	2.0	2.5
	3.500	-0.219	5.412E+00	5.244E+00	2.5	3.0
	4.000	-0.190	7.543E+00	7.185E+00	2.5	3.1
	4.500	-0.164	9.960E+00	9.301E+00	2.8	3.2
	5.000	-0.141	1.262E+01	1.163E+01	2.8	3.2
	5.500	-0.120	1.547E+01	1.402E+01	3.2	3.5
	6.000	-0.102	1.847E+01	1.650E+01	3.2	3.5
	7.000	-0.068	2.481E+01	2.178E+01	2.9	3.0
	8.000	-0.039	3.136E+01	2.737E+01	3.8	4.0
	9.000	-0.014	3.796E+01	3.317E+01	3.8	4.0
	10.000	0.009	4.447E+01	3.910E+01	3.1	3.4
	11.000	0.030	5.079E+01	4.513E+01	3.1	3.4
	12.000	0.049	5.686E+01	5.121E+01	3.1	3.4
46-Pd	14.000	0.082	6.813E+01	6.345E+01	5.2	5.3
47-Ag	0.260	-0.794	6.709E-05	3.569E-05	7.4	10.0
	0.300	-0.763	1.974E-04	1.294E-04	6.7	9.9
	0.350	-0.729	5.518E-04	4.184E-04	6.8	11.7
	0.400	-0.701	1.230E-03	1.016E-03	6.8	11.3
	0.450	-0.675	2.359E-03	2.056E-03	3.8	6.0
	0.500	-0.652	4.072E-03	3.678E-03	3.8	5.9
	0.550	-0.631	6.501E-03	6.022E-03	3.0	4.8
	0.600	-0.612	9.781E-03	9.249E-03	3.0	4.8
	0.700	-0.579	1.942E-02	1.884E-02	2.8	4.1
	0.800	-0.550	3.404E-02	3.386E-02	2.8	4.0
	0.900	-0.524	5.462E-02	5.532E-02	1.8	2.5
	1.000	-0.502	8.209E-02	8.383E-02	1.8	2.5
	1.100	-0.481	1.173E-01	1.201E-01	1.8	2.5
	1.200	-0.462	1.611E-01	1.658E-01	1.8	2.5
	1.400	-0.428	2.773E-01	2.864E-01	1.4	2.0
	1.600	-0.400	4.356E-01	4.505E-01	1.4	2.0
	1.800	-0.374	6.397E-01	6.627E-01	1.5	2.1
	2.000	-0.351	8.926E-01	9.178E-01	1.5	2.1
	2.300	-0.321	1.367E+00	1.398E+00	1.8	2.1
	2.600	-0.294	1.958E+00	1.990E+00	1.8	2.1
	3.000	-0.263	2.926E+00	2.917E+00	2.0	2.3
	3.500	-0.229	4.413E+00	4.303E+00	2.5	2.8
	4.000	-0.200	6.184E+00	5.929E+00	2.5	2.8
	4.500	-0.175	8.204E+00	7.720E+00	2.8	3.0
	5.000	-0.152	1.044E+01	9.667E+00	2.8	3.0
	5.500	-0.131	1.285E+01	1.175E+01	3.2	3.3
	6.000	-0.112	1.541E+01	1.388E+01	3.2	3.3
	7.000	-0.079	2.083E+01	1.836E+01	2.9	3.0
	8.000	-0.050	2.651E+01	2.317E+01	2.9	3.0
	9.000	-0.024	3.226E+01	2.815E+01	3.8	3.9
	10.000	-0.002	3.797E+01	3.327E+01	3.8	3.9
	11.000	0.019	4.356E+01	3.847E+01	3.1	3.2
	12.000	0.038	4.897E+01	4.374E+01	3.1	3.2
	14.000	0.072	5.909E+01	5.436E+01	5.2	5.4
48-Cd	0.260	-0.805	4.469E-05	2.155E-05	7.4	8.2
	0.300	-0.774	1.367E-04	8.424E-05	6.7	8.5
	0.350	-0.740	3.949E-04	2.877E-04	6.7	8.3
	0.400	-0.711	8.997E-04	7.223E-04	6.8	10.1
	0.450	-0.686	1.753E-03	1.496E-03	6.8	8.1
	0.500	-0.663	3.060E-03	2.722E-03	3.8	5.5
	0.550	-0.642	4.930E-03	4.512E-03	3.8	5.4
	0.600	-0.623	7.470E-03	6.985E-03	3.0	4.4
	0.700	-0.590	1.499E-02	1.438E-02	3.0	3.8
	0.800	-0.561	2.646E-02	2.609E-02	2.8	3.8
	0.900	-0.535	4.269E-02	4.294E-02	1.8	2.3
	1.000	-0.512	6.444E-02	6.571E-02	1.8	2.3
	1.100	-0.491	9.244E-02	9.454E-02	1.8	2.2
	1.200	-0.473	1.273E-01	1.305E-01	1.8	2.3
	1.400	-0.439	2.202E-01	2.272E-01	1.4	1.9
	1.600	-0.410	3.473E-01	3.587E-01	1.4	1.9
	1.800	-0.385	5.119E-01	5.301E-01	1.5	1.9
	2.000	-0.362	7.164E-01	7.420E-01	1.5	1.9
	2.300	-0.331	1.102E+00	1.129E+00	1.8	1.9
	2.600	-0.305	1.584E+00	1.619E+00	1.8	1.9
	3.000	-0.274	2.379E+00	2.388E+00	2.0	2.1
	3.500	-0.240	3.608E+00	3.542E+00	2.0	2.1
	4.000	-0.211	5.081E+00	4.902E+00	2.5	2.6
	4.500	-0.186	6.772E+00	6.420E+00	2.5	2.6
	5.000	-0.163	8.653E+00	8.074E+00	2.8	2.9
	5.500	-0.142	1.070E+01	9.866E+00	2.8	2.9
	6.000	-0.123	1.287E+01	1.169E+01	3.2	3.5
	7.000	-0.090	1.752E+01	1.553E+01	2.9	3.2
	8.000	-0.061	2.243E+01	1.965E+01	2.9	3.2
	9.000	-0.035	2.744E+01	2.395E+01	3.8	4.2
	10.000	-0.012	3.246E+01	2.837E+01	3.8	4.2
	11.000	0.009	3.740E+01	3.288E+01	3.1	3.5
	12.000	0.027	4.221E+01	3.745E+01	3.1	3.5
	14.000	0.061	5.129E+01	4.668E+01	5.2	5.7
	16.000	0.090	5.953E+01	5.594E+01	5.2	5.7
49-In	0.300	-0.784	9.454E-05	5.429E-05	6.7	6.9
	0.350	-0.751	2.825E-04	1.958E-04	6.7	6.9
	0.400	-0.722	6.590E-04	5.128E-04	6.8	9.0
	0.450	-0.696	1.305E-03	1.089E-03	6.8	7.6

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
49-In	0.500	-0.673	2.307E-03	2.016E-03	3.8	5.1
	0.550	-0.653	3.751E-03	3.386E-03	3.8	5.0
	0.600	-0.634	5.725E-03	5.288E-03	3.0	3.9
	0.700	-0.600	1.161E-02	1.111E-02	3.0	3.9
	0.800	-0.571	2.065E-02	2.017E-02	2.8	3.5
	0.900	-0.546	3.351E-02	3.344E-02	2.8	3.5
	1.000	-0.523	5.080E-02	5.150E-02	1.8	2.0
	1.100	-0.502	7.313E-02	7.467E-02	1.8	2.0
	1.200	-0.483	1.010E-01	1.034E-01	1.8	2.1
	1.400	-0.450	1.756E-01	1.810E-01	1.8	2.1
	1.600	-0.421	2.780E-01	2.872E-01	1.4	1.7
	1.800	-0.395	4.111E-01	4.254E-01	1.4	1.7
	2.000	-0.372	5.772E-01	5.979E-01	1.5	1.7
	2.300	-0.342	8.914E-01	9.153E-01	1.5	1.7
	2.600	-0.315	1.286E+00	1.321E+00	1.8	1.8
	3.000	-0.284	1.940E+00	1.961E+00	2.0	2.0
	3.500	-0.251	2.958E+00	2.920E+00	2.0	2.0
	4.000	-0.222	4.185E+00	4.063E+00	2.5	2.6
	4.500	-0.196	5.603E+00	5.358E+00	2.5	2.6
	5.000	-0.173	7.189E+00	6.757E+00	2.8	3.1
	5.500	-0.153	8.920E+00	8.264E+00	2.8	3.1
	6.000	-0.134	1.077E+01	9.869E+00	3.2	3.7
	7.000	-0.100	1.476E+01	1.318E+01	3.2	3.7
	8.000	-0.071	1.901E+01	1.670E+01	2.9	3.5
	9.000	-0.046	2.338E+01	2.042E+01	2.9	3.5
	10.000	-0.023	2.778E+01	2.425E+01	3.8	4.5
	11.000	-0.002	3.215E+01	2.816E+01	3.8	4.5
	12.000	0.017	3.642E+01	3.213E+01	3.1	3.7
	14.000	0.050	4.456E+01	4.018E+01	3.1	3.7
	16.000	0.079	5.202E+01	4.827E+01	5.2	5.9
50-Sn	0.300	-0.794	6.533E-05	3.461E-05	7.4	10.4
	0.350	-0.761	2.025E-04	1.341E-04	6.7	8.1
	0.400	-0.732	4.837E-04	3.634E-04	6.8	7.7
	0.450	-0.706	9.746E-04	7.925E-04	6.8	7.6
	0.500	-0.684	1.744E-03	1.495E-03	3.8	4.6
	0.550	-0.663	2.864E-03	2.546E-03	3.8	4.6
	0.600	-0.644	4.405E-03	4.021E-03	3.8	4.6
	0.700	-0.610	9.033E-03	8.557E-03	3.0	3.5
	0.800	-0.581	1.619E-02	1.566E-02	2.8	3.3
	0.900	-0.556	2.641E-02	2.615E-02	2.8	3.3
	1.000	-0.533	4.023E-02	4.052E-02	1.8	1.9
	1.100	-0.512	5.811E-02	5.925E-02	1.8	1.9
	1.200	-0.493	8.052E-02	8.234E-02	1.8	1.9
	1.400	-0.460	1.406E-01	1.447E-01	1.8	1.9
	1.600	-0.431	2.235E-01	2.308E-01	1.4	1.5
	1.800	-0.405	3.316E-01	3.427E-01	1.4	1.5
	2.000	-0.383	4.668E-01	4.835E-01	1.5	1.6
	2.300	-0.352	7.238E-01	7.443E-01	1.5	1.6
	2.600	-0.326	1.048E+00	1.073E+00	1.8	1.9
	3.000	-0.294	1.588E+00	1.614E+00	1.8	1.9
	3.500	-0.261	2.432E+00	2.414E+00	2.0	2.2
	4.000	-0.232	3.457E+00	3.376E+00	2.5	2.8
	4.500	-0.206	4.648E+00	4.472E+00	2.5	2.8
	5.000	-0.184	5.987E+00	5.668E+00	2.8	3.3
	5.500	-0.163	7.456E+00	6.958E+00	2.8	3.3
	6.000	-0.144	9.037E+00	8.349E+00	2.8	3.3
	7.000	-0.111	1.246E+01	1.121E+01	3.2	3.9
	8.000	-0.082	1.613E+01	1.424E+01	2.9	3.8
	9.000	-0.056	1.994E+01	1.745E+01	2.9	3.8
	10.000	-0.033	2.381E+01	2.077E+01	3.8	4.7
	11.000	-0.012	2.767E+01	2.418E+01	3.8	4.7
	12.000	0.007	3.146E+01	2.764E+01	3.1	4.0
	14.000	0.040	3.875E+01	3.466E+01	3.1	4.0
	16.000	0.069	4.550E+01	4.174E+01	5.2	6.2
	18.000	0.094	5.163E+01	4.881E+01	5.2	6.2
51-Sb	0.300	-0.805	4.506E-05	2.175E-05	7.4	12.8
	0.350	-0.771	1.451E-04	9.076E-05	6.7	9.8
	0.400	-0.742	3.555E-04	2.569E-04	6.7	9.4
	0.450	-0.717	7.293E-04	5.764E-04	6.8	7.3
	0.500	-0.694	1.323E-03	1.110E-03	6.8	7.0
	0.550	-0.673	2.194E-03	1.918E-03	3.8	4.1
	0.600	-0.654	3.401E-03	3.063E-03	3.8	4.1
	0.700	-0.621	7.055E-03	6.615E-03	3.0	3.1
	0.800	-0.592	1.275E-02	1.228E-02	3.0	3.0
	0.900	-0.566	2.091E-02	2.052E-02	2.8	3.0
	1.000	-0.543	3.199E-02	3.199E-02	2.8	3.0
51-Sb	1.100	-0.522	4.638E-02	4.703E-02	1.8	2.1
	1.200	-0.504	6.446E-02	6.579E-02	1.8	2.1
	1.400	-0.470	1.131E-01	1.159E-01	1.8	1.8
	1.600	-0.441	1.804E-01	1.861E-01	1.8	1.8
	1.800	-0.416	2.685E-01	2.775E-01	1.4	1.5
	2.000	-0.393	3.791E-01	3.923E-01	1.4	1.5
	2.300	-0.362	5.899E-01	6.109E-01	1.5	1.7
	2.600	-0.336	8.572E-01	8.792E-01	1.8	2.1
	3.000	-0.305	1.304E+00	1.332E+00	1.8	2.1
	3.500	-0.271	2.007E+00	2.011E+00	2.0	2.4
	4.000	-0.242	2.864E+00	2.815E+00	2.0	2.4
	4.500	-0.217	3.866E+00	3.742E+00	2.5	3.0
	5.000	-0.194	4.998E+00	4.773E+00	2.5	3.0
	5.500	-0.173	6.247E+00	5.871E+00	2.8	3.5
	6.000	-0.154	7.596E+00	7.045E+00	2.8	3.5
	7.000	-0.121	1.054E+01	9.552E+00	3.2	4.2
	8.000	-0.092	1.372E+01	1.217E+01	2.9	4.0
	9.000	-0.066	1.704E+01	1.495E+01	2.9	4.0
	10.000	-0.043	2.043E+01	1.784E+01	3.8	5.0
	11.000	-0.023	2.384E+01	2.081E+01	3.8	5.0
	12.000	-0.004	2.721E+01	2.383E+01	3.8	5.0
	14.000	0.030	3.373E+01	2.997E+01	3.1	4.3
	16.000	0.059	3.983E+01	3.618E+01	5.2	6.5
	18.000	0.084	4.542E+01	4.240E+01	5.2	6.5
52-Te	0.350	-0.781	1.040E-04	6.100E-05	6.7	11.7
	0.400	-0.752	2.617E-04	1.810E-04	6.7	10.9
	0.450	-0.727	5.470E-04	4.189E-04	6.8	8.6
	0.500	-0.704	1.006E-03	8.238E-04	6.8	8.5
	0.550	-0.683	1.686E-03	1.447E-03	3.8	4.0
	0.600	-0.664	2.635E-03	2.338E-03	3.8	4.0
	0.700	-0.631	5.531E-03	5.127E-03	3.0	3.3
	0.800	-0.602	1.007E-02	9.622E-03	3.0	3.3
	0.900	-0.576	1.663E-02	1.617E-02	2.8	2.9
	1.000	-0.553	2.555E-02	2.535E-02	2.8	2.9
	1.100	-0.533	3.717E-02	3.746E-02	1.8	2.4
	1.200	-0.514	5.181E-02	5.280E-02	1.8	2.4
	1.400	-0.480	9.132E-02	9.352E-02	1.8	2.0
	1.600	-0.451	1.462E-01	1.507E-01	1.8	2.0
	1.800	-0.426	2.183E-01	2.255E-01	1.4	1.7
	2.000	-0.403	3.090E-01	3.195E-01	1.4	1.7
	2.300	-0.372	4.825E-01	4.999E-01	1.5	1.8
	2.600	-0.346	7.034E-01	7.226E-01	1.5	1.8
	3.000	-0.315	1.074E+00	1.102E+00	1.8	2.3
	3.500	-0.281	1.660E+00	1.674E+00	2.0	2.6
	4.000	-0.252	2.379E+00	2.350E+00	2.0	2.6
	4.500	-0.227	3.224E+00	3.138E+00	2.5	3.2
	5.000	-0.204	4.183E+00	4.018E+00	2.5	3.2
	5.500	-0.183	5.245E+00	4.964E+00	2.8	3.7
	6.000	-0.164	6.398E+00	5.976E+00	2.8	3.8
	7.000	-0.131	8.929E+00	8.158E+00	3.2	4.4
	8.000	-0.102	1.168E+01	1.044E+01	3.2	4.4
	9.000	-0.076	1.458E+01	1.284E+01	2.9	4.3
	10.000	-0.053	1.756E+01	1.536E+01	2.9	4.3
	11.000	-0.032	2.057E+01	1.795E+01	3.8	5.3
	12.000	-0.014	2.356E+01	2.059E+01	3.8	5.3
	14.000	0.020	2.940E+01	2.597E+01	3.1	4.6
	16.000	0.049	3.490E+01	3.143E+01	3.1	4.6
	18.000	0.074	3.999E+01	3.691E+01	5.2	6.8
	20.000	0.097	4.464E+01	4.236E+01	5.2	6.7
53-I	0.350	-0.791	7.435E-05	4.056E-05	7.4	15.8
	0.400	-0.762	1.925E-04	1.268E-04	6.7	12.7
	0.450	-0.736	4.104E-04	3.035E-04	6.8	10.0
	0.500	-0.714	7.659E-04	6.109E-04	6.8	9.8
	0.550	-0.693	1.298E-03	1.091E-03	6.8	7.9
	0.600	-0.674	2.046E-03	1.786E-03	3.8	4.4
	0.700	-0.640	4.349E-03	3.987E-03	3.8	4.4
	0.800	-0.612	7.987E-03	7.560E-03	3.0	3.7
	0.900	-0.586	1.327E-02	1.278E-02	2.8	3.2
	1.000	-0.563	2.048E-02	2.015E-02	2.8	3.2
	1.100	-0.542	2.990E-02	2.992E-02	2.8	3.2
	1.200	-0.523	4.180E-02	4.237E-02	1.8	2.6
	1.400	-0.490	7.402E-02	7.572E-02	1.8	2.4
	1.600	-0.461	1.189E-01	1.224E-01	1.8	2.2
	1.800	-0.435	1.781E-01	1.838E-01	1.4	1.9
	2.000	-0.412	2.527E-01	2.610E-01	1.4	1.9

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
53-I	2.300	-0.382	3.960E-01	4.101E-01	1.5	2.0
	2.600	-0.356	5.790E-01	5.957E-01	1.5	2.0
	3.000	-0.324	8.872E-01	9.079E-01	1.8	2.5
	3.500	-0.291	1.377E+00	1.397E+00	1.8	2.5
	4.000	-0.262	1.981E+00	1.967E+00	2.0	2.8
	4.500	-0.236	2.694E+00	2.640E+00	2.0	2.8
	5.000	-0.213	3.508E+00	3.389E+00	2.5	3.4
	5.500	-0.193	4.413E+00	4.212E+00	2.5	3.5
	6.000	-0.174	5.400E+00	5.078E+00	2.8	4.0
	7.000	-0.140	7.578E+00	6.980E+00	2.8	4.0
	8.000	-0.112	9.966E+00	8.967E+00	3.2	4.6
	9.000	-0.086	1.250E+01	1.105E+01	2.9	4.5
	10.000	-0.063	1.511E+01	1.325E+01	2.9	4.6
	11.000	-0.042	1.777E+01	1.551E+01	3.8	5.5
	12.000	-0.023	2.042E+01	1.783E+01	3.8	5.5
	14.000	0.010	2.564E+01	2.255E+01	3.1	4.9
	16.000	0.039	3.061E+01	2.736E+01	3.1	4.9
	18.000	0.065	3.524E+01	3.219E+01	5.2	7.1
	20.000	0.087	3.949E+01	3.701E+01	5.2	7.0
54-Xe	0.350	-0.801	5.305E-05	2.660E-05	7.4	18.6
	0.400	-0.772	1.415E-04	8.823E-05	6.7	14.6
	0.450	-0.746	3.080E-04	2.189E-04	6.7	13.6
	0.500	-0.723	5.835E-04	4.518E-04	6.8	11.1
	0.550	-0.703	1.000E-03	8.217E-04	6.8	10.9
	0.600	-0.684	1.591E-03	1.364E-03	3.8	4.9
	0.700	-0.650	3.426E-03	3.102E-03	3.8	4.8
	0.800	-0.621	6.347E-03	5.948E-03	3.0	4.2
	0.900	-0.596	1.061E-02	1.019E-02	3.0	3.7
	1.000	-0.573	1.645E-02	1.605E-02	2.8	3.4
	1.100	-0.552	2.411E-02	2.395E-02	2.8	3.4
	1.200	-0.533	3.381E-02	3.406E-02	1.8	2.9
	1.400	-0.500	6.015E-02	6.144E-02	1.8	2.6
	1.600	-0.471	9.699E-02	9.938E-02	1.8	2.4
	1.800	-0.445	1.456E-01	1.502E-01	1.8	2.3
	2.000	-0.422	2.072E-01	2.141E-01	1.4	2.0
	2.300	-0.392	3.257E-01	3.371E-01	1.4	2.0
	2.600	-0.365	4.776E-01	4.947E-01	1.5	2.2
	3.000	-0.334	7.344E-01	7.530E-01	1.8	2.6
	3.500	-0.301	1.145E+00	1.167E+00	1.8	2.6
	4.000	-0.272	1.653E+00	1.657E+00	2.0	3.0
	4.500	-0.246	2.255E+00	2.221E+00	2.0	3.0
	5.000	-0.223	2.946E+00	2.862E+00	2.5	3.6
	5.500	-0.203	3.717E+00	3.568E+00	2.5	3.6
	6.000	-0.184	4.562E+00	4.319E+00	2.8	4.2
	7.000	-0.150	6.437E+00	5.953E+00	2.8	4.2
	8.000	-0.121	8.507E+00	7.713E+00	3.2	4.9
	9.000	-0.096	1.071E+01	9.530E+00	3.2	5.1
	10.000	-0.073	1.301E+01	1.144E+01	2.9	4.8
	11.000	-0.052	1.535E+01	1.342E+01	2.9	4.8
	12.000	-0.033	1.771E+01	1.545E+01	3.8	5.8
	14.000	0.000	2.237E+01	1.961E+01	3.8	6.0
	16.000	0.029	2.685E+01	2.384E+01	3.1	5.1
	18.000	0.055	3.105E+01	2.811E+01	3.1	5.1
	20.000	0.078	3.495E+01	3.237E+01	5.2	7.3
55-Cs	0.400	-0.781	1.042E-04	6.113E-05	6.7	16.8
	0.450	-0.756	2.319E-04	1.578E-04	6.7	15.4
	0.500	-0.733	4.464E-04	3.346E-04	6.8	12.6
	0.550	-0.712	7.744E-04	6.203E-04	6.8	12.2
	0.600	-0.693	1.243E-03	1.044E-03	6.8	8.9
	0.700	-0.660	2.713E-03	2.424E-03	3.8	5.3
	0.800	-0.631	5.072E-03	4.703E-03	3.0	4.6
	0.900	-0.605	8.532E-03	8.125E-03	3.0	4.6
	1.000	-0.582	1.330E-02	1.285E-02	2.8	3.7
	1.100	-0.562	1.956E-02	1.927E-02	2.8	3.7
	1.200	-0.543	2.751E-02	2.752E-02	2.8	3.7
	1.400	-0.509	4.916E-02	5.021E-02	1.8	3.1
	1.600	-0.480	7.954E-02	8.145E-02	1.8	2.5
	1.800	-0.455	1.198E-01	1.234E-01	1.8	2.5
	2.000	-0.432	1.708E-01	1.764E-01	1.4	2.2
	2.300	-0.401	2.693E-01	2.785E-01	1.4	2.2
	2.600	-0.375	3.959E-01	4.102E-01	1.5	2.4
	3.000	-0.344	6.108E-01	6.274E-01	1.5	2.4
	3.500	-0.310	9.554E-01	9.789E-01	1.8	2.8
	4.000	-0.281	1.385E+00	1.396E+00	2.0	3.2
	4.500	-0.256	1.895E+00	1.876E+00	2.0	3.2
	5.000	-0.233	2.483E+00	2.426E+00	2.5	3.8
55-Cs	5.500	-0.212	3.142E+00	3.033E+00	2.5	3.8
	6.000	-0.193	3.867E+00	3.691E+00	2.5	3.9
	7.000	-0.160	5.484E+00	5.106E+00	2.8	4.4
	8.000	-0.131	7.282E+00	6.654E+00	3.2	5.1
	9.000	-0.105	9.211E+00	8.249E+00	3.2	5.1
	10.000	-0.082	1.123E+01	9.911E+00	2.9	5.1
	11.000	-0.062	1.330E+01	1.165E+01	2.9	5.1
	12.000	-0.043	1.539E+01	1.344E+01	3.8	6.1
	14.000	-0.009	1.956E+01	1.710E+01	3.8	6.1
	16.000	0.020	2.359E+01	2.084E+01	3.1	5.4
	18.000	0.045	2.741E+01	2.462E+01	3.1	5.4
	20.000	0.068	3.097E+01	2.840E+01	5.2	7.6
	23.000	0.099	3.580E+01	3.403E+01	5.2	7.5
56-Ba	0.400	-0.790	7.669E-05	4.198E-05	7.4	21.3
	0.450	-0.765	1.746E-04	1.132E-04	6.7	17.4
	0.500	-0.742	3.417E-04	2.471E-04	6.7	16.3
	0.550	-0.721	6.004E-04	4.677E-04	6.8	13.7
	0.600	-0.702	9.733E-04	7.997E-04	6.8	13.3
	0.700	-0.669	2.154E-03	1.896E-03	3.8	5.8
	0.800	-0.640	4.064E-03	3.728E-03	3.8	5.7
	0.900	-0.614	6.882E-03	6.495E-03	3.0	5.0
	1.000	-0.592	1.078E-02	1.038E-02	3.0	4.2
	1.100	-0.571	1.592E-02	1.555E-02	2.8	3.9
	1.200	-0.552	2.245E-02	2.230E-02	2.8	3.9
	1.400	-0.518	4.031E-02	4.096E-02	1.8	3.4
	1.600	-0.489	6.543E-02	6.694E-02	1.8	3.0
	1.800	-0.464	9.879E-02	1.016E-01	1.8	2.7
	2.000	-0.441	1.412E-01	1.457E-01	1.8	2.7
	2.300	-0.411	2.233E-01	2.306E-01	1.4	2.4
	2.600	-0.384	3.291E-01	3.408E-01	1.5	2.5
	3.000	-0.353	5.093E-01	5.238E-01	1.5	2.5
	3.500	-0.319	7.995E-01	8.172E-01	1.8	3.0
	4.000	-0.290	1.162E+00	1.179E+00	1.8	3.1
	4.500	-0.265	1.596E+00	1.593E+00	2.0	3.4
	5.000	-0.242	2.097E+00	2.061E+00	2.0	3.4
	5.500	-0.221	2.661E+00	2.583E+00	2.5	4.0
	6.000	-0.203	3.283E+00	3.152E+00	2.5	4.0
	7.000	-0.169	4.679E+00	4.385E+00	2.8	4.6
	8.000	-0.140	6.241E+00	5.747E+00	2.8	4.7
	9.000	-0.114	7.927E+00	7.150E+00	3.2	5.3
	10.000	-0.092	9.700E+00	8.605E+00	2.9	5.3
	11.000	-0.071	1.153E+01	1.013E+01	2.9	5.3
	12.000	-0.052	1.339E+01	1.170E+01	2.9	5.3
	14.000	-0.019	1.711E+01	1.494E+01	3.8	6.3
	16.000	0.010	2.074E+01	1.825E+01	3.1	5.7
	18.000	0.036	2.421E+01	2.159E+01	3.1	5.7
	20.000	0.059	2.746E+01	2.495E+01	5.2	7.9
	23.000	0.089	3.191E+01	2.997E+01	5.2	7.8
57-La	0.400	-0.800	5.645E-05	2.858E-05	7.4	24.4
	0.450	-0.774	1.316E-04	8.089E-05	6.7	19.6
	0.500	-0.751	2.622E-04	1.822E-04	6.7	18.1
	0.550	-0.730	4.667E-04	3.526E-04	6.8	15.2
	0.600	-0.712	7.644E-04	6.129E-04	6.8	14.7
	0.700	-0.678	1.716E-03	1.486E-03	3.8	6.3
	0.800	-0.649	3.270E-03	2.965E-03	3.8	6.2
	0.900	-0.624	5.575E-03	5.211E-03	3.0	5.5
	1.000	-0.601	8.775E-03	8.389E-03	3.0	5.4
	1.100	-0.580	1.301E-02	1.260E-02	2.8	4.2
	1.200	-0.561	1.841E-02	1.814E-02	2.8	4.2
	1.400	-0.528	3.320E-02	3.356E-02	1.8	3.6
	1.600	-0.499	5.407E-02	5.524E-02	1.8	3.2
	1.800	-0.473	8.185E-02	8.386E-02	1.8	2.9
	2.000	-0.450	1.173E-01	1.208E-01	1.8	2.9
	2.300	-0.420	1.859E-01	1.921E-01	1.4	2.6
	2.600	-0.393	2.747E-01	2.843E-01	1.4	2.5
	3.000	-0.362	4.263E-01	4.415E-01	1.5	2.7
	3.500	-0.329	6.715E-01	6.877E-01	1.8	3.2
	4.000	-0.300	9.790E-01	9.979E-01	1.8	3.2
	4.500	-0.274	1.348E+00	1.354E+00	2.0	3.6
	5.000	-0.251	1.776E+00	1.754E+00	2.0	3.6
	5.500	-0.230	2.260E+00	2.205E+00	2.5	4.1
	6.000	-0.212	2.795E+00	2.698E+00	2.5	4.2
	7.000	-0.178	4.003E+00	3.775E+00	2.8	4.9
	8.000	-0.149	5.361E+00	4.955E+00	2.8	4.9

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
57-La	9.000	-0.124	6.837E+00	6.211E+00	3.2	5.6
	10.000	-0.101	8.396E+00	7.495E+00	3.2	5.6
	11.000	-0.080	1.001E+01	8.830E+00	2.9	5.6
	12.000	-0.061	1.166E+01	1.022E+01	2.9	5.6
	14.000	-0.028	1.499E+01	1.308E+01	3.8	6.6
	16.000	0.001	1.826E+01	1.601E+01	3.8	6.8
	18.000	0.027	2.141E+01	1.898E+01	3.1	6.0
	20.000	0.050	2.438E+01	2.197E+01	3.1	5.9
	23.000	0.080	2.847E+01	2.644E+01	5.2	8.1
58-Ce	0.400	-0.809	4.149E-05	1.921E-05	7.4	28.1
	0.450	-0.783	9.922E-05	5.741E-05	6.7	22.1
	0.500	-0.760	2.013E-04	1.338E-04	6.7	20.1
	0.550	-0.740	3.633E-04	2.653E-04	6.7	18.9
	0.600	-0.721	6.013E-04	4.694E-04	6.8	16.2
	0.700	-0.687	1.370E-03	1.166E-03	6.8	10.3
	0.800	-0.658	2.638E-03	2.362E-03	3.8	6.6
	0.900	-0.633	4.530E-03	4.189E-03	3.0	5.9
	1.000	-0.610	7.167E-03	6.795E-03	3.0	5.9
	1.100	-0.589	1.067E-02	1.024E-02	3.0	4.8
	1.200	-0.570	1.514E-02	1.480E-02	2.8	4.5
	1.400	-0.537	2.743E-02	2.756E-02	2.8	4.4
	1.600	-0.508	4.483E-02	4.572E-02	1.8	3.9
	1.800	-0.482	6.804E-02	6.966E-02	1.8	3.1
	2.000	-0.459	9.767E-02	1.005E-01	1.8	3.1
	2.300	-0.429	1.553E-01	1.604E-01	1.4	2.8
	2.600	-0.402	2.300E-01	2.378E-01	1.4	2.8
	3.000	-0.371	3.579E-01	3.708E-01	1.5	2.9
	3.500	-0.338	5.654E-01	5.801E-01	1.5	2.9
	4.000	-0.309	8.267E-01	8.465E-01	1.8	3.3
	4.500	-0.283	1.141E+00	1.153E+00	2.0	3.7
	5.000	-0.260	1.508E+00	1.496E+00	2.0	3.8
	5.500	-0.240	1.923E+00	1.888E+00	2.0	3.8
	6.000	-0.221	2.385E+00	2.313E+00	2.5	4.3
	7.000	-0.187	3.430E+00	3.255E+00	2.5	4.6
	8.000	-0.158	4.613E+00	4.290E+00	2.8	5.1
	9.000	-0.133	5.904E+00	5.404E+00	3.2	5.8
	10.000	-0.110	7.277E+00	6.540E+00	3.2	5.8
	11.000	-0.089	8.706E+00	7.710E+00	2.9	5.8
	12.000	-0.070	1.017E+01	8.936E+00	2.9	5.9
	14.000	-0.037	1.315E+01	1.147E+01	3.8	6.8
	16.000	-0.008	1.609E+01	1.407E+01	3.8	6.8
	18.000	0.018	1.894E+01	1.672E+01	3.1	6.3
	20.000	0.041	2.165E+01	1.938E+01	3.1	6.2
	23.000	0.071	2.541E+01	2.337E+01	5.2	8.4
	26.000	0.098	2.878E+01	2.732E+01	5.2	8.3
59-Pr	0.450	-0.792	7.486E-05	4.050E-05	7.4	27.0
	0.500	-0.769	1.548E-04	9.805E-05	6.7	22.4
	0.550	-0.748	2.835E-04	1.995E-04	6.7	20.8
	0.600	-0.729	4.744E-04	3.597E-04	6.8	17.8
	0.700	-0.696	1.098E-03	9.167E-04	6.8	10.8
	0.800	-0.667	2.137E-03	1.887E-03	3.8	7.1
	0.900	-0.641	3.696E-03	3.384E-03	3.8	7.0
	1.000	-0.619	5.879E-03	5.524E-03	3.0	6.3
	1.100	-0.598	8.786E-03	8.420E-03	3.0	5.0
	1.200	-0.579	1.251E-02	1.213E-02	2.8	4.7
	1.400	-0.545	2.277E-02	2.273E-02	2.8	4.7
	1.600	-0.516	3.734E-02	3.799E-02	1.8	4.1
	1.800	-0.491	5.681E-02	5.811E-02	1.8	3.6
	2.000	-0.468	8.172E-02	8.375E-02	1.8	3.3
	2.300	-0.438	1.303E-01	1.344E-01	1.4	3.0
	2.600	-0.411	1.933E-01	1.997E-01	1.4	3.0
	3.000	-0.380	3.016E-01	3.124E-01	1.5	3.1
	3.500	-0.347	4.778E-01	4.910E-01	1.5	3.1
	4.000	-0.318	7.006E-01	7.157E-01	1.8	3.5
	4.500	-0.292	9.697E-01	9.844E-01	1.8	3.6
	5.000	-0.269	1.284E+00	1.285E+00	2.0	3.9
	5.500	-0.248	1.642E+00	1.619E+00	2.0	4.0
	6.000	-0.229	2.040E+00	1.989E+00	2.5	4.6
	7.000	-0.196	2.946E+00	2.817E+00	2.5	4.8
	8.000	-0.167	3.978E+00	3.723E+00	2.8	5.3
	9.000	-0.141	5.110E+00	4.711E+00	2.8	5.4
	10.000	-0.119	6.319E+00	5.718E+00	3.2	6.0
	11.000	-0.098	7.584E+00	6.757E+00	3.2	6.3
	12.000	-0.079	8.887E+00	7.833E+00	2.9	6.1
	14.000	-0.045	1.155E+01	1.008E+01	3.8	7.1
59-Pr	16.000	-0.016	1.420E+01	1.240E+01	3.8	7.1
	18.000	0.009	1.678E+01	1.476E+01	3.1	6.6
	20.000	0.032	1.925E+01	1.713E+01	3.1	6.5
	23.000	0.062	2.271E+01	2.070E+01	5.2	8.7
	26.000	0.089	2.583E+01	2.424E+01	5.2	8.6
60-Nd	0.450	-0.801	5.651E-05	2.835E-05	7.4	30.5
	0.500	-0.778	1.192E-04	7.156E-05	6.7	24.9
	0.550	-0.757	2.216E-04	1.497E-04	6.7	22.9
	0.600	-0.738	3.752E-04	2.755E-04	6.7	21.6
	0.700	-0.705	8.831E-04	7.214E-04	6.8	18.3
	0.800	-0.676	1.736E-03	1.511E-03	3.8	7.6
	0.900	-0.650	3.026E-03	2.741E-03	3.8	7.5
	1.000	-0.627	4.841E-03	4.506E-03	3.0	6.8
	1.100	-0.607	7.264E-03	6.908E-03	3.0	6.7
	1.200	-0.588	1.038E-02	9.977E-03	2.8	5.0
	1.400	-0.554	1.898E-02	1.881E-02	2.8	4.9
	1.600	-0.525	3.122E-02	3.161E-02	1.8	4.4
	1.800	-0.500	4.762E-02	4.864E-02	1.8	3.7
	2.000	-0.477	6.863E-02	7.030E-02	1.8	3.5
	2.300	-0.446	1.097E-01	1.131E-01	1.8	3.5
	2.600	-0.420	1.631E-01	1.685E-01	1.4	3.1
	3.000	-0.389	2.550E-01	2.640E-01	1.4	3.1
	3.500	-0.355	4.051E-01	4.168E-01	1.5	3.2
	4.000	-0.326	5.955E-01	6.095E-01	1.8	3.7
	4.500	-0.301	8.261E-01	8.426E-01	1.8	3.7
	5.000	-0.278	1.096E+00	1.104E+00	2.0	4.1
	5.500	-0.257	1.405E+00	1.391E+00	2.0	4.2
	6.000	-0.238	1.749E+00	1.715E+00	2.0	4.2
	7.000	-0.205	2.536E+00	2.438E+00	2.5	4.8
	8.000	-0.176	3.437E+00	3.236E+00	2.8	5.5
	9.000	-0.150	4.431E+00	4.098E+00	2.8	5.6
	10.000	-0.127	5.497E+00	5.009E+00	3.2	6.2
	11.000	-0.107	6.617E+00	5.932E+00	3.2	6.3
	12.000	-0.088	7.775E+00	6.880E+00	2.9	6.4
	14.000	-0.054	1.015E+01	8.880E+00	2.9	6.4
	16.000	-0.025	1.254E+01	1.094E+01	3.8	7.4
	18.000	0.000	1.489E+01	1.305E+01	3.8	7.7
	20.000	0.023	1.714E+01	1.517E+01	3.1	6.8
	23.000	0.054	2.031E+01	1.836E+01	3.1	6.8
	26.000	0.080	2.319E+01	2.155E+01	5.2	8.9
61-Pm	0.350	-0.864	4.770E-06	8.243E-07	17.7	142.1
	0.400	-0.835	1.639E-05	5.213E-06	6.9	67.1
	0.450	-0.809	4.259E-05	1.836E-05	6.9	51.3
	0.500	-0.786	9.177E-05	4.795E-05	4.9	42.5
	0.550	-0.766	1.733E-04	1.035E-04	4.9	37.8
	0.600	-0.747	2.970E-04	1.960E-04	4.9	34.6
	0.700	-0.713	7.110E-04	5.388E-04	2.9	16.3
	0.800	-0.684	1.414E-03	1.171E-03	2.4	7.9
	0.900	-0.659	2.484E-03	2.198E-03	2.4	7.6
	1.000	-0.636	3.996E-03	3.708E-03	2.2	6.3
	1.100	-0.615	6.022E-03	5.788E-03	2.2	6.2
	1.200	-0.596	8.632E-03	8.528E-03	2.2	4.6
	1.400	-0.563	1.586E-02	1.650E-02	2.1	4.2
	1.600	-0.534	2.618E-02	2.767E-02	2.2	3.8
	1.800	-0.508	4.003E-02	4.287E-02	2.2	3.7
	2.000	-0.485	5.780E-02	6.247E-02	3.0	4.7
	2.300	-0.455	9.257E-02	9.983E-02	3.0	4.7
	2.600	-0.428	1.379E-01	1.478E-01	2.4	3.9
	3.000	-0.397	2.162E-01	2.278E-01	2.4	3.9
	3.500	-0.364	3.443E-01	3.559E-01	2.8	5.0
	4.000	-0.335	5.072E-01	5.137E-01	3.4	5.8
	4.500	-0.309	7.052E-01	7.082E-01	3.4	5.8
	5.000	-0.286	9.380E-01	9.376E-01	3.3	5.6
	5.500	-0.266	1.204E+00	1.188E+00	3.3	5.6
	6.000	-0.247	1.502E+00	1.463E+00	3.3	5.7
	7.000	-0.213	2.187E+00	2.069E+00	5.9	8.1
	8.000	-0.184	2.974E+00	2.731E+00	7.5	9.7
	9.000	-0.159	3.846E+00	3.423E+00	7.5	9.8
	10.000	-0.136	4.787E+00	4.123E+00	5.7	7.3
	11.000	-0.115	5.779E+00	4.814E+00	5.7	7.3
62-Sm	0.400	-0.843	1.196E-05	3.320E-06	17.7	91.5
	0.450	-0.818	3.205E-05	1.266E-05	6.9	52.8

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
62-Sm	0.500	-0.795	7.060E-05	3.457E-05	6.9	44.0
	0.550	-0.774	1.355E-04	7.693E-05	4.9	37.7
	0.600	-0.755	2.353E-04	1.488E-04	4.9	34.4
	0.700	-0.722	5.735E-04	4.212E-04	2.9	15.4
	0.800	-0.693	1.154E-03	9.348E-04	2.9	7.4
	0.900	-0.667	2.044E-03	1.772E-03	2.4	7.2
	1.000	-0.644	3.308E-03	3.022E-03	2.4	7.0
	1.100	-0.624	5.007E-03	4.747E-03	2.2	5.8
	1.200	-0.605	7.202E-03	7.031E-03	2.2	5.7
	1.400	-0.571	1.330E-02	1.370E-02	2.1	4.0
	1.600	-0.542	2.202E-02	2.336E-02	2.1	4.0
	1.800	-0.517	3.376E-02	3.602E-02	2.2	3.5
	2.000	-0.494	4.884E-02	5.256E-02	2.2	3.2
	2.300	-0.463	7.839E-02	8.466E-02	3.0	4.5
	2.600	-0.437	1.170E-01	1.259E-01	2.4	3.8
	3.000	-0.406	1.838E-01	1.946E-01	2.4	3.8
	3.500	-0.372	2.934E-01	3.049E-01	2.8	4.9
	4.000	-0.343	4.332E-01	4.423E-01	2.8	4.9
	4.500	-0.318	6.036E-01	6.079E-01	3.4	5.6
	5.000	-0.295	8.043E-01	8.038E-01	3.4	5.8
	5.500	-0.274	1.035E+00	1.027E+00	3.3	5.4
	6.000	-0.255	1.293E+00	1.267E+00	3.3	5.5
	7.000	-0.222	1.889E+00	1.801E+00	5.9	7.9
	8.000	-0.193	2.578E+00	2.389E+00	5.9	8.2
	9.000	-0.167	3.345E+00	3.009E+00	7.5	9.5
	10.000	-0.144	4.174E+00	3.641E+00	7.5	9.6
	11.000	-0.124	5.054E+00	4.271E+00	5.7	7.1
63-Eu	0.400	-0.852	8.707E-06	2.054E-06	17.7	100.3
	0.450	-0.826	2.409E-05	8.624E-06	6.9	54.8
	0.500	-0.803	5.430E-05	2.477E-05	6.9	44.5
	0.550	-0.783	1.061E-04	5.696E-05	4.9	37.7
	0.600	-0.764	1.866E-04	1.128E-04	4.9	34.1
	0.700	-0.730	4.634E-04	3.293E-04	2.9	14.4
	0.800	-0.701	9.442E-04	7.453E-04	2.9	13.3
	0.900	-0.676	1.687E-03	1.431E-03	2.4	6.8
	1.000	-0.653	2.746E-03	2.464E-03	2.4	6.6
	1.100	-0.632	4.177E-03	3.903E-03	2.2	5.4
	1.200	-0.613	6.028E-03	5.812E-03	2.2	5.3
	1.400	-0.580	1.119E-02	1.140E-02	2.1	3.8
	1.600	-0.551	1.859E-02	1.957E-02	2.1	3.7
	1.800	-0.525	2.856E-02	3.035E-02	2.2	3.3
	2.000	-0.502	4.140E-02	4.444E-02	2.2	3.3
	2.300	-0.472	6.660E-02	7.198E-02	3.0	4.3
	2.600	-0.445	9.960E-02	1.072E-01	2.4	3.6
	3.000	-0.414	1.567E-01	1.668E-01	2.4	3.6
	3.500	-0.381	2.508E-01	2.618E-01	2.8	4.7
	4.000	-0.352	3.710E-01	3.807E-01	2.8	4.7
	4.500	-0.326	5.179E-01	5.230E-01	3.4	5.4
	5.000	-0.303	6.914E-01	6.929E-01	3.4	5.5
	5.500	-0.282	8.910E-01	8.886E-01	3.3	5.2
	6.000	-0.264	1.116E+00	1.099E+00	3.3	5.3
	7.000	-0.230	1.635E+00	1.571E+00	5.9	7.7
	8.000	-0.201	2.238E+00	2.093E+00	5.9	7.8
	9.000	-0.176	2.913E+00	2.648E+00	7.5	9.3
	10.000	-0.153	3.646E+00	3.218E+00	7.5	9.3
	11.000	-0.132	4.426E+00	3.790E+00	5.7	6.8
	12.000	-0.113	5.242E+00	4.352E+00	5.7	6.8
64-Gd	0.400	-0.860	6.318E-06	1.222E-06	17.7	113.3
	0.450	-0.834	1.807E-05	5.788E-06	6.9	57.3
	0.500	-0.812	4.175E-05	1.761E-05	6.9	45.2
	0.550	-0.791	8.303E-05	4.198E-05	4.9	37.8
	0.600	-0.772	1.481E-04	8.526E-05	4.9	33.8
	0.700	-0.738	3.751E-04	2.573E-04	2.9	13.4
	0.800	-0.709	7.742E-04	5.946E-04	2.9	12.3
	0.900	-0.684	1.395E-03	1.157E-03	2.4	6.5
	1.000	-0.661	2.287E-03	2.013E-03	2.4	6.2
	1.100	-0.640	3.494E-03	3.215E-03	2.2	5.0
	1.200	-0.621	5.061E-03	4.816E-03	2.2	4.9
	1.400	-0.588	9.442E-03	9.515E-03	2.1	3.6
	1.600	-0.559	1.574E-02	1.644E-02	2.1	3.5
	1.800	-0.533	2.424E-02	2.564E-02	2.2	3.1
	2.000	-0.510	3.521E-02	3.768E-02	2.2	3.1
	2.300	-0.480	5.676E-02	6.135E-02	3.0	4.2
	2.600	-0.454	8.502E-02	9.166E-02	3.0	4.2
64-Gd	3.000	-0.422	1.340E-01	1.432E-01	2.4	3.4
	3.500	-0.389	2.149E-01	2.252E-01	2.8	4.5
	4.000	-0.360	3.185E-01	3.285E-01	2.8	4.5
	4.500	-0.334	4.455E-01	4.511E-01	3.4	5.3
	5.000	-0.312	5.957E-01	5.986E-01	3.4	5.3
	5.500	-0.291	7.689E-01	7.705E-01	3.3	5.0
	6.000	-0.272	9.644E-01	9.555E-01	3.3	5.1
	7.000	-0.238	1.418E+00	1.371E+00	5.9	7.5
	8.000	-0.209	1.947E+00	1.835E+00	5.9	7.5
	9.000	-0.184	2.541E+00	2.332E+00	7.5	9.0
	10.000	-0.161	3.189E+00	2.846E+00	7.5	9.1
	11.000	-0.140	3.882E+00	3.366E+00	5.7	6.5
	12.000	-0.121	4.608E+00	3.879E+00	5.7	6.6
65-Tb	0.400	-0.868	4.561E-06	6.838E-07	17.7	134.4
	0.450	-0.842	1.352E-05	3.804E-06	17.7	79.9
	0.500	-0.820	3.203E-05	1.239E-05	6.9	46.1
	0.550	-0.799	6.493E-05	3.074E-05	6.9	38.9
	0.600	-0.780	1.175E-04	6.418E-05	4.9	33.5
	0.700	-0.747	3.039E-04	2.008E-04	4.9	28.6
	0.800	-0.718	6.357E-04	4.743E-04	2.9	11.2
	0.900	-0.692	1.156E-03	9.389E-04	2.9	5.9
	1.000	-0.669	1.908E-03	1.646E-03	2.4	5.8
	1.100	-0.648	2.930E-03	2.653E-03	2.4	5.7
	1.200	-0.629	4.260E-03	3.998E-03	2.2	4.6
	1.400	-0.596	7.989E-03	7.896E-03	2.2	3.6
	1.600	-0.567	1.337E-02	1.384E-02	2.1	3.3
	1.800	-0.541	2.064E-02	2.190E-02	2.1	3.2
	2.000	-0.519	3.002E-02	3.201E-02	2.2	2.8
	2.300	-0.488	4.850E-02	5.241E-02	3.0	4.0
	2.600	-0.462	7.277E-02	7.857E-02	3.0	4.0
	3.000	-0.431	1.150E-01	1.233E-01	2.4	3.2
	3.500	-0.397	1.846E-01	1.945E-01	2.4	3.2
	4.000	-0.368	2.741E-01	2.841E-01	2.8	4.3
	4.500	-0.343	3.840E-01	3.919E-01	3.4	5.1
	5.000	-0.320	5.144E-01	5.183E-01	3.4	5.1
	5.500	-0.299	6.650E-01	6.655E-01	3.4	5.2
	6.000	-0.280	8.353E-01	8.318E-01	3.3	4.9
	7.000	-0.247	1.232E+00	1.199E+00	3.3	4.9
	8.000	-0.218	1.696E+00	1.611E+00	5.9	7.3
	9.000	-0.192	2.220E+00	2.056E+00	5.9	7.5
	10.000	-0.169	2.793E+00	2.519E+00	7.5	8.8
	11.000	-0.148	3.408E+00	2.990E+00	7.5	8.8
	12.000	-0.130	4.056E+00	3.460E+00	5.7	6.3
66-Dy	0.450	-0.851	1.009E-05	2.436E-06	17.7	86.0
	0.500	-0.828	2.455E-05	8.614E-06	6.9	47.2
	0.550	-0.807	5.078E-05	2.237E-05	6.9	39.0
	0.600	-0.788	9.331E-05	4.814E-05	4.9	33.3
	0.700	-0.755	2.465E-04	1.565E-04	4.9	28.0
	0.800	-0.726	5.230E-04	3.785E-04	2.9	10.1
	0.900	-0.700	9.607E-04	7.611E-04	2.9	9.4
	1.000	-0.677	1.596E-03	1.349E-03	2.4	5.4
	1.100	-0.656	2.464E-03	2.192E-03	2.4	5.3
	1.200	-0.638	3.596E-03	3.327E-03	2.2	4.2
	1.400	-0.604	6.782E-03	6.627E-03	2.2	4.1
	1.600	-0.575	1.139E-02	1.167E-02	2.1	3.0
	1.800	-0.549	1.762E-02	1.857E-02	2.1	3.0
	2.000	-0.527	2.569E-02	2.727E-02	2.2	2.6
	2.300	-0.496	4.158E-02	4.472E-02	2.2	2.5
	2.600	-0.470	6.248E-02	6.752E-02	3.0	3.8
	3.000	-0.439	9.886E-02	1.064E-01	2.4	3.1
	3.500	-0.405	1.591E-01	1.684E-01	2.4	3.1
	4.000	-0.376	2.366E-01	2.464E-01	2.8	4.2
	4.500	-0.351	3.319E-01	3.404E-01	2.8	4.2
	5.000	-0.328	4.452E-01	4.498E-01	3.4	4.9
	5.500	-0.307	5.764E-01	5.784E-01	3.4	4.9
	6.000	-0.288	7.251E-01	7.255E-01	3.3	4.7
	7.000	-0.255	1.072E+00	1.050E+00	3.3	4.7
	8.000	-0.226	1.480E+00	1.417E+00	5.9	7.1
	9.000	-0.200	1.942E+00	1.814E+00	5.9	7.2
	10.000	-0.177	2.450E+00	2.231E+00	7.5	8.6
	11.000	-0.156	2.997E+00	2.658E+00	7.5	8.6
	12.000	-0.138	3.574E+00	3.087E+00	5.7	6.1
67-Ho	0.450	-0.859	7.502E-06	1.505E-06	17.7	94.9

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
67-Ho	0.500	-0.836	1.878E-05	5.899E-06	6.9	48.8
	0.550	-0.815	3.965E-05	1.614E-05	6.9	39.2
	0.600	-0.796	7.405E-05	3.590E-05	6.9	34.0
	0.700	-0.763	2.001E-04	1.217E-04	4.9	27.3
	0.800	-0.734	4.308E-04	3.018E-04	2.9	8.9
	0.900	-0.708	7.995E-04	6.170E-04	2.9	8.3
	1.000	-0.685	1.338E-03	1.106E-03	2.4	5.0
	1.100	-0.664	2.076E-03	1.813E-03	2.4	4.9
	1.200	-0.646	3.044E-03	2.773E-03	2.4	4.8
	1.400	-0.612	5.772E-03	5.574E-03	2.2	3.7
	1.600	-0.583	9.726E-03	9.868E-03	2.1	2.8
	1.800	-0.558	1.509E-02	1.578E-02	2.1	2.8
	2.000	-0.535	2.204E-02	2.329E-02	2.2	2.4
	2.300	-0.504	3.575E-02	3.835E-02	2.2	2.3
	2.600	-0.478	5.379E-02	5.815E-02	3.0	3.6
	3.000	-0.447	8.525E-02	9.176E-02	2.4	2.9
	3.500	-0.413	1.374E-01	1.461E-01	2.4	2.9
	4.000	-0.384	2.046E-01	2.141E-01	2.8	4.0
	4.500	-0.359	2.875E-01	2.963E-01	2.8	4.0
	5.000	-0.336	3.862E-01	3.913E-01	3.4	4.7
	5.500	-0.315	5.006E-01	5.037E-01	3.4	4.7
	6.000	-0.296	6.306E-01	6.305E-01	3.4	4.8
	7.000	-0.263	9.350E-01	9.207E-01	3.3	4.5
	8.000	-0.234	1.294E+00	1.247E+00	5.9	6.9
	9.000	-0.208	1.702E+00	1.603E+00	5.9	6.9
	10.000	-0.185	2.152E+00	1.978E+00	7.5	8.3
	11.000	-0.164	2.638E+00	2.365E+00	7.5	8.3
	12.000	-0.146	3.154E+00	2.756E+00	7.5	8.4
	14.000	-0.112	4.249E+00	3.521E+00	5.7	5.8
68-Er	0.450	-0.866	5.557E-06	8.831E-07	17.7	108.9
	0.500	-0.844	1.432E-05	3.959E-06	17.7	70.1
	0.550	-0.823	3.092E-05	1.152E-05	6.9	39.5
	0.600	-0.804	5.873E-05	2.661E-05	6.9	33.7
	0.700	-0.771	1.625E-04	9.437E-05	4.9	26.7
	0.800	-0.742	3.553E-04	2.404E-04	2.9	7.6
	0.900	-0.716	6.665E-04	5.004E-04	2.9	7.1
	1.000	-0.693	1.124E-03	9.097E-04	2.9	4.5
	1.100	-0.672	1.754E-03	1.501E-03	2.4	4.5
	1.200	-0.653	2.582E-03	2.313E-03	2.4	4.4
	1.400	-0.620	4.925E-03	4.697E-03	2.2	3.3
	1.600	-0.591	8.331E-03	8.358E-03	2.1	2.6
	1.800	-0.565	1.296E-02	1.344E-02	2.1	2.5
	2.000	-0.543	1.896E-02	2.010E-02	2.1	2.5
	2.300	-0.512	3.081E-02	3.295E-02	2.2	2.3
	2.600	-0.486	4.644E-02	5.019E-02	3.0	3.5
	3.000	-0.454	7.370E-02	7.948E-02	3.0	3.5
	3.500	-0.421	1.190E-01	1.270E-01	2.4	2.7
	4.000	-0.392	1.774E-01	1.863E-01	2.8	3.8
	4.500	-0.366	2.496E-01	2.585E-01	2.8	3.8
	5.000	-0.344	3.357E-01	3.428E-01	2.8	3.8
	5.500	-0.323	4.358E-01	4.396E-01	3.4	4.5
	6.000	-0.304	5.496E-01	5.509E-01	3.4	4.6
	7.000	-0.271	8.168E-01	8.086E-01	3.3	4.3
	8.000	-0.242	1.133E+00	1.099E+00	3.3	4.3
	9.000	-0.216	1.494E+00	1.417E+00	5.9	6.7
	10.000	-0.193	1.893E+00	1.755E+00	5.9	6.8
	11.000	-0.172	2.326E+00	2.106E+00	7.5	8.1
	12.000	-0.153	2.786E+00	2.461E+00	7.5	8.1
	14.000	-0.120	3.768E+00	3.165E+00	5.7	5.8
69-Tm	0.500	-0.851	1.090E-05	2.591E-06	17.7	74.5
	0.550	-0.831	2.408E-05	8.127E-06	6.9	39.9
	0.600	-0.812	4.655E-05	1.958E-05	6.9	33.4
	0.700	-0.778	1.321E-04	7.298E-05	4.9	26.0
	0.800	-0.749	2.934E-04	1.914E-04	4.9	22.7
	0.900	-0.724	5.565E-04	4.058E-04	2.9	5.9
	1.000	-0.701	9.460E-04	7.476E-04	2.9	5.7
	1.100	-0.680	1.485E-03	1.245E-03	2.4	4.0
	1.200	-0.661	2.196E-03	1.932E-03	2.4	4.0
	1.400	-0.628	4.214E-03	3.967E-03	2.2	2.8
	1.600	-0.599	7.156E-03	7.046E-03	2.2	2.6
	1.800	-0.573	1.116E-02	1.147E-02	2.1	2.3
	2.000	-0.550	1.636E-02	1.723E-02	2.1	2.3
	2.300	-0.520	2.664E-02	2.839E-02	2.2	2.5
	2.600	-0.493	4.021E-02	4.342E-02	2.2	2.4
	3.000	-0.462	6.391E-02	6.900E-02	3.0	3.3
	3.500	-0.429	1.033E-01	1.107E-01	2.4	2.5
69-Tm	4.000	-0.400	1.542E-01	1.628E-01	2.4	2.5
	4.500	-0.374	2.173E-01	2.260E-01	2.8	3.6
	5.000	-0.351	2.926E-01	3.002E-01	2.8	3.7
	5.500	-0.331	3.802E-01	3.845E-01	3.4	4.4
	6.000	-0.312	4.800E-01	4.824E-01	3.4	4.4
	7.000	-0.278	7.150E-01	7.113E-01	3.3	4.1
	8.000	-0.249	9.943E-01	9.699E-01	3.3	4.1
	9.000	-0.224	1.314E+00	1.255E+00	5.9	6.5
	10.000	-0.201	1.668E+00	1.559E+00	5.9	6.5
	11.000	-0.180	2.063E+00	1.876E+00	7.5	7.9
	12.000	-0.161	2.464E+00	2.200E+00	7.5	7.9
	14.000	-0.128	3.346E+00	2.846E+00	5.7	6.1
70-Yb	0.500	-0.859	8.275E-06	1.640E-06	17.7	80.8
	0.550	-0.838	1.873E-05	5.649E-06	6.9	40.6
	0.600	-0.819	3.687E-05	1.429E-05	6.9	33.1
	0.700	-0.786	1.073E-04	5.625E-05	4.9	25.2
	0.800	-0.757	2.426E-04	1.521E-04	4.9	21.9
	0.900	-0.731	4.655E-04	3.291E-04	2.9	4.6
	1.000	-0.709	7.978E-04	6.147E-04	2.9	4.4
	1.100	-0.688	1.260E-03	1.036E-03	2.4	3.6
	1.200	-0.669	1.872E-03	1.616E-03	2.4	3.5
	1.400	-0.636	3.615E-03	3.357E-03	2.2	2.4
	1.600	-0.607	6.164E-03	6.003E-03	2.2	2.4
	1.800	-0.581	9.641E-03	9.810E-03	2.1	2.1
	2.000	-0.558	1.416E-02	1.480E-02	2.1	2.1
	2.300	-0.528	2.310E-02	2.451E-02	2.2	2.8
	2.600	-0.501	3.491E-02	3.749E-02	2.2	2.8
	3.000	-0.470	5.556E-02	6.004E-02	3.0	3.1
	3.500	-0.437	8.992E-02	9.669E-02	2.4	2.4
	4.000	-0.407	1.344E-01	1.425E-01	2.4	2.4
	4.500	-0.382	1.896E-01	1.981E-01	2.8	3.5
	5.000	-0.359	2.555E-01	2.634E-01	2.8	3.5
	5.500	-0.338	3.324E-01	3.371E-01	3.4	4.2
	6.000	-0.319	4.201E-01	4.233E-01	3.4	4.2
	7.000	-0.286	6.271E-01	6.267E-01	3.3	3.9
	8.000	-0.257	8.739E-01	8.572E-01	3.3	3.9
	9.000	-0.231	1.157E+00	1.112E+00	5.9	6.3
	10.000	-0.208	1.472E+00	1.387E+00	5.9	6.3
	11.000	-0.188	1.816E+00	1.673E+00	7.5	7.6
	12.000	-0.169	2.183E+00	1.968E+00	7.5	7.6
	14.000	-0.135	2.974E+00	2.560E+00	5.7	6.3
71-Lu	0.500	-0.867	6.257E-06	9.866E-07	17.7	90.6
	0.550	-0.846	1.452E-05	3.841E-06	17.7	61.2
	0.600	-0.827	2.915E-05	1.031E-05	6.9	32.8
	0.700	-0.794	8.720E-05	4.314E-05	6.9	25.4
	0.800	-0.765	2.006E-04	1.206E-04	4.9	20.9
	0.900	-0.739	3.897E-04	2.666E-04	2.9	3.2
	1.000	-0.716	6.738E-04	5.054E-04	2.9	3.1
	1.100	-0.695	1.071E-03	8.607E-04	2.9	3.0
	1.200	-0.677	1.599E-03	1.353E-03	2.4	3.1
	1.400	-0.643	3.108E-03	2.845E-03	2.4	3.0
	1.600	-0.614	5.322E-03	5.124E-03	2.2	2.3
	1.800	-0.589	8.347E-03	8.405E-03	2.1	2.3
	2.000	-0.566	1.228E-02	1.273E-02	2.1	2.3
	2.300	-0.535	2.008E-02	2.121E-02	2.2	3.0
	2.600	-0.509	3.039E-02	3.255E-02	2.2	3.0
	3.000	-0.478	4.843E-02	5.235E-02	3.0	3.1
	3.500	-0.444	7.847E-02	8.441E-02	2.4	2.6
	4.000	-0.415	1.174E-01	1.250E-01	2.4	2.6
	4.500	-0.389	1.657E-01	1.738E-01	2.8	3.3
	5.000	-0.367	2.236E-01	2.316E-01	2.8	3.3
	5.500	-0.346	2.912E-01	2.978E-01	2.8	3.3
	6.000	-0.327	3.684E-01	3.722E-01	3.4	4.0
	7.000	-0.294	5.510E-01	5.530E-01	3.3	3.7
	8.000	-0.265	7.693E-01	7.586E-01	3.3	3.7
	9.000	-0.239	1.020E+00	9.874E-01	5.9	6.1
	10.000	-0.216	1.301E+00	1.234E+00	5.9	6.1
	11.000	-0.195	1.607E+00	1.494E+00	5.9	6.2
	12.000	-0.177	1.936E+00	1.762E+00	7.5	7.5
	14.000	-0.143	2.647E+00	2.304E+00	7.5	7.5
	16.000	-0.114	3.408E+00	2.834E+00	5.7	6.6
72-Hf	0.550	-0.854	1.121E-05	2.534E-06	17.7	64.1
	0.600	-0.835	2.296E-05	7.315E-06	6.9	32.5
	0.700	-0.801	7.068E-05	3.281E-05	6.9	24.4

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
72-Hf	0.800	-0.772	1.657E-04	9.518E-05	4.9	19.9
	0.900	-0.747	3.260E-04	2.153E-04	4.9	18.0
	1.000	-0.724	5.688E-04	4.146E-04	2.9	4.0
	1.100	-0.703	9.103E-04	7.142E-04	2.9	3.9
	1.200	-0.684	1.366E-03	1.132E-03	2.4	2.6
	1.400	-0.651	2.674E-03	2.410E-03	2.4	2.6
	1.600	-0.622	4.599E-03	4.374E-03	2.2	2.8
	1.800	-0.596	7.234E-03	7.148E-03	2.2	2.5
	2.000	-0.573	1.067E-02	1.096E-02	2.1	2.6
	2.300	-0.543	1.747E-02	1.852E-02	2.1	2.6
	2.600	-0.516	2.648E-02	2.827E-02	2.2	3.2
	3.000	-0.485	4.225E-02	4.567E-02	3.0	3.3
	3.500	-0.452	6.855E-02	7.388E-02	3.0	3.3
	4.000	-0.423	1.027E-01	1.097E-01	2.4	2.7
	4.500	-0.397	1.451E-01	1.528E-01	2.4	2.7
	5.000	-0.374	1.959E-01	2.038E-01	2.8	3.1
	5.500	-0.354	2.553E-01	2.624E-01	2.8	3.1
	6.000	-0.335	3.233E-01	3.274E-01	3.4	3.8
	7.000	-0.301	4.844E-01	4.851E-01	3.4	3.8
	8.000	-0.272	6.775E-01	6.715E-01	3.3	3.5
	9.000	-0.247	9.003E-01	8.765E-01	3.3	3.5
	10.000	-0.224	1.150E+00	1.099E+00	5.9	6.0
	11.000	-0.203	1.423E+00	1.333E+00	5.9	6.0
	12.000	-0.184	1.717E+00	1.577E+00	7.5	7.8
	14.000	-0.151	2.355E+00	2.073E+00	7.5	7.8
	16.000	-0.122	3.043E+00	2.563E+00	5.7	6.8
73-Ta	0.550	-0.861	8.642E-06	1.619E-06	17.7	68.4
	0.600	-0.842	1.808E-05	5.118E-06	17.7	51.2
	0.700	-0.809	5.736E-05	2.486E-05	6.9	23.4
	0.800	-0.780	1.371E-04	7.506E-05	4.9	18.9
	0.900	-0.754	2.734E-04	1.740E-04	4.9	16.9
	1.000	-0.731	4.817E-04	3.407E-04	2.9	5.4
	1.100	-0.711	7.763E-04	5.940E-04	2.9	5.2
	1.200	-0.692	1.171E-03	9.516E-04	2.9	3.8
	1.400	-0.658	2.309E-03	2.046E-03	2.4	2.7
	1.600	-0.629	3.990E-03	3.746E-03	2.2	3.2
	1.800	-0.604	6.295E-03	6.155E-03	2.2	3.2
	2.000	-0.581	9.300E-03	9.465E-03	2.1	2.8
	2.300	-0.550	1.527E-02	1.608E-02	2.1	2.8
	2.600	-0.524	2.317E-02	2.464E-02	2.2	3.5
	3.000	-0.493	3.701E-02	3.997E-02	2.2	3.1
	3.500	-0.459	6.012E-02	6.488E-02	3.0	3.4
	4.000	-0.430	9.014E-02	9.666E-02	2.4	2.9
	4.500	-0.405	1.274E-01	1.348E-01	2.4	2.9
	5.000	-0.382	1.722E-01	1.800E-01	2.8	2.9
	5.500	-0.361	2.246E-01	2.319E-01	2.8	3.0
	6.000	-0.342	2.847E-01	2.904E-01	3.4	3.7
	7.000	-0.309	4.272E-01	4.289E-01	3.4	3.7
	8.000	-0.280	5.985E-01	5.959E-01	3.3	3.3
	9.000	-0.254	7.966E-01	7.798E-01	3.3	3.3
	10.000	-0.231	1.019E+00	9.799E-01	5.9	6.2
	11.000	-0.211	1.263E+00	1.192E+00	5.9	6.2
	12.000	-0.192	1.527E+00	1.413E+00	5.9	6.2
	14.000	-0.158	2.101E+00	1.868E+00	7.5	8.0
	16.000	-0.129	2.721E+00	2.320E+00	5.7	7.1
74-W	0.550	-0.868	6.634E-06	9.786E-07	17.7	75.3
	0.600	-0.850	1.420E-05	3.495E-06	17.7	52.2
	0.700	-0.816	4.647E-05	1.867E-05	6.9	22.2
	0.800	-0.787	1.134E-04	5.889E-05	4.9	17.7
	0.900	-0.762	2.293E-04	1.402E-04	4.9	15.8
	1.000	-0.739	4.080E-04	2.795E-04	2.9	7.0
	1.100	-0.718	6.625E-04	4.935E-04	2.9	6.6
	1.200	-0.699	1.005E-03	7.983E-04	2.9	4.3
	1.400	-0.666	1.997E-03	1.738E-03	2.4	3.2
	1.600	-0.637	3.466E-03	3.212E-03	2.2	3.6
	1.800	-0.611	5.486E-03	5.306E-03	2.2	3.6
	2.000	-0.588	8.123E-03	8.183E-03	2.1	3.1
	2.300	-0.558	1.336E-02	1.397E-02	2.1	3.0
	2.600	-0.531	2.031E-02	2.151E-02	2.2	3.7
	3.000	-0.500	3.249E-02	3.490E-02	2.2	3.7
	3.500	-0.467	5.281E-02	5.706E-02	3.0	3.6
	4.000	-0.438	7.925E-02	8.526E-02	2.4	3.1
	4.500	-0.412	1.121E-01	1.191E-01	2.4	3.1
	5.000	-0.389	1.516E-01	1.590E-01	2.8	2.8
	5.500	-0.368	1.979E-01	2.052E-01	2.8	2.8
74-W	6.000	-0.350	2.510E-01	2.573E-01	2.8	2.8
	7.000	-0.316	3.772E-01	3.797E-01	3.4	3.5
	8.000	-0.287	5.293E-01	5.293E-01	3.3	3.4
	9.000	-0.262	7.056E-01	6.943E-01	3.3	3.4
	10.000	-0.239	9.040E-01	8.746E-01	5.9	6.4
	11.000	-0.218	1.122E+00	1.067E+00	5.9	6.4
	12.000	-0.199	1.359E+00	1.268E+00	5.9	6.4
	14.000	-0.166	1.875E+00	1.683E+00	7.5	8.2
	16.000	-0.137	2.436E+00	2.101E+00	5.7	7.3
	18.000	-0.111	3.026E+00	2.503E+00	5.7	7.4
75-Re	0.600	-0.857	1.111E-05	2.317E-06	17.7	53.6
	0.700	-0.824	3.759E-05	1.389E-05	6.9	20.8
	0.800	-0.795	9.372E-05	4.600E-05	6.9	17.4
	0.900	-0.769	1.923E-04	1.127E-04	4.9	14.5
	1.000	-0.746	3.459E-04	2.290E-04	4.9	13.4
	1.100	-0.725	5.660E-04	4.100E-04	2.9	8.1
	1.200	-0.706	8.635E-04	6.698E-04	2.9	7.8
	1.400	-0.673	1.730E-03	1.478E-03	2.4	3.7
	1.600	-0.644	3.018E-03	2.759E-03	2.4	3.6
	1.800	-0.618	4.793E-03	4.582E-03	2.2	4.0
	2.000	-0.596	7.112E-03	7.086E-03	2.2	3.3
	2.300	-0.565	1.173E-02	1.216E-02	2.1	3.3
	2.600	-0.539	1.784E-02	1.881E-02	2.1	3.3
	3.000	-0.507	2.858E-02	3.062E-02	2.2	3.9
	3.500	-0.474	4.651E-02	5.027E-02	3.0	3.8
	4.000	-0.445	6.984E-02	7.514E-02	2.4	3.3
	4.500	-0.419	9.887E-02	1.055E-01	2.4	3.3
	5.000	-0.397	1.338E-01	1.409E-01	2.4	3.2
	5.500	-0.376	1.747E-01	1.820E-01	2.8	3.0
	6.000	-0.357	2.217E-01	2.283E-01	2.8	3.0
	7.000	-0.324	3.337E-01	3.367E-01	3.4	3.5
	8.000	-0.295	4.688E-01	4.685E-01	3.4	3.5
	9.000	-0.269	6.259E-01	6.190E-01	3.3	3.6
	10.000	-0.246	8.031E-01	7.815E-01	3.3	3.6
	11.000	-0.225	9.986E-01	9.554E-01	5.9	6.6
	12.000	-0.206	1.210E+00	1.138E+00	5.9	6.6
	14.000	-0.173	1.675E+00	1.518E+00	7.5	8.5
	16.000	-0.144	2.182E+00	1.903E+00	7.5	8.5
	18.000	-0.119	2.718E+00	2.277E+00	5.7	7.6
76-Os	0.600	-0.864	8.673E-06	1.474E-06	17.7	55.9
	0.700	-0.831	3.036E-05	1.021E-05	6.9	19.2
	0.800	-0.802	7.742E-05	3.573E-05	6.9	15.9
	0.900	-0.776	1.614E-04	9.034E-05	4.9	13.2
	1.000	-0.753	2.934E-04	1.874E-04	4.9	12.2
	1.100	-0.733	4.841E-04	3.403E-04	2.9	9.7
	1.200	-0.714	7.430E-04	5.620E-04	2.9	9.2
	1.400	-0.680	1.501E-03	1.257E-03	2.4	4.2
	1.600	-0.651	2.633E-03	2.369E-03	2.4	4.1
	1.800	-0.626	4.195E-03	3.963E-03	2.2	4.5
	2.000	-0.603	6.240E-03	6.108E-03	2.2	4.4
	2.300	-0.573	1.031E-02	1.061E-02	2.1	3.5
	2.600	-0.546	1.572E-02	1.662E-02	2.1	3.5
	3.000	-0.515	2.520E-02	2.692E-02	2.2	4.2
	3.500	-0.481	4.105E-02	4.437E-02	3.0	4.0
	4.000	-0.452	6.168E-02	6.648E-02	3.0	4.0
	4.500	-0.427	8.737E-02	9.355E-02	2.4	3.4
	5.000	-0.404	1.183E-01	1.251E-01	2.4	3.5
	5.500	-0.383	1.546E-01	1.617E-01	2.8	3.1
	6.000	-0.364	1.963E-01	2.030E-01	2.8	3.1
	7.000	-0.331	2.957E-01	2.991E-01	3.4	3.7
	8.000	-0.302	4.160E-01	4.167E-01	3.4	3.7
	9.000	-0.276	5.561E-01	5.525E-01	3.3	3.8
	10.000	-0.253	7.145E-01	6.991E-01	3.3	3.8
	11.000	-0.233	8.896E-01	8.565E-01	5.9	6.7
	12.000	-0.214	1.080E+00	1.022E+00	5.9	6.8
	14.000	-0.180	1.498E+00	1.369E+00	7.5	8.7
	16.000	-0.151	1.957E+00	1.724E+00	7.5	8.7
	18.000	-0.126	2.443E+00	2.072E+00	5.7	7.8
77-Ir	0.700	-0.838	2.447E-05	7.402E-06	6.9	17.3
	0.800	-0.809	6.389E-05	2.757E-05	6.9	14.2
	0.900	-0.784	1.354E-04	7.215E-05	4.9	11.7
	1.000	-0.761	2.490E-04	1.530E-04	4.9	10.8
	1.100	-0.740	4.143E-04	2.822E-04	2.9	11.4
	1.200	-0.721	6.400E-04	4.713E-04	2.9	10.8

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
77-Ir	1.400	-0.688	1.305E-03	1.073E-03	2.4	4.7
	1.600	-0.659	2.301E-03	2.037E-03	2.4	4.6
	1.800	-0.633	3.680E-03	3.432E-03	2.2	4.9
	2.000	-0.610	5.487E-03	5.314E-03	2.2	4.8
	2.300	-0.580	9.089E-03	9.262E-03	2.1	3.8
	2.600	-0.553	1.387E-02	1.457E-02	2.1	3.8
	3.000	-0.522	2.227E-02	2.370E-02	2.2	4.4
	3.500	-0.489	3.631E-02	3.923E-02	3.0	4.1
	4.000	-0.460	5.459E-02	5.892E-02	3.0	4.1
	4.500	-0.434	7.736E-02	8.310E-02	2.4	3.6
	5.000	-0.411	1.048E-01	1.113E-01	2.4	3.6
	5.500	-0.391	1.370E-01	1.437E-01	2.8	3.3
	6.000	-0.372	1.740E-01	1.808E-01	2.8	3.3
	7.000	-0.338	2.625E-01	2.662E-01	3.4	3.9
	8.000	-0.309	3.697E-01	3.713E-01	3.4	3.9
	9.000	-0.284	4.948E-01	4.938E-01	3.3	4.0
	10.000	-0.261	6.365E-01	6.260E-01	3.3	4.0
	11.000	-0.240	7.936E-01	7.685E-01	3.3	4.0
	12.000	-0.221	9.644E-01	9.191E-01	5.9	7.0
	14.000	-0.188	1.341E+00	1.236E+00	7.5	8.9
	16.000	-0.159	1.756E+00	1.562E+00	7.5	9.0
	18.000	-0.133	2.198E+00	1.885E+00	5.7	8.0
	20.000	-0.110	2.658E+00	2.195E+00	5.7	8.1
78-Pt	0.700	-0.845	1.969E-05	5.264E-06	17.7	34.5
	0.800	-0.816	5.268E-05	2.112E-05	6.9	12.3
	0.900	-0.791	1.135E-04	5.738E-05	4.9	10.1
	1.000	-0.768	2.114E-04	1.247E-04	4.9	9.4
	1.100	-0.747	3.549E-04	2.337E-04	4.9	8.9
	1.200	-0.728	5.520E-04	3.952E-04	2.9	12.4
	1.400	-0.695	1.136E-03	9.142E-04	2.9	6.3
	1.600	-0.666	2.014E-03	1.752E-03	2.4	5.0
	1.800	-0.640	3.234E-03	2.976E-03	2.2	5.4
	2.000	-0.617	4.834E-03	4.630E-03	2.2	5.3
	2.300	-0.587	8.029E-03	8.101E-03	2.1	4.1
	2.600	-0.560	1.227E-02	1.280E-02	2.1	4.0
	3.000	-0.529	1.972E-02	2.091E-02	2.2	4.6
	3.500	-0.496	3.219E-02	3.462E-02	2.2	4.0
	4.000	-0.467	4.842E-02	5.231E-02	3.0	4.3
	4.500	-0.441	6.865E-02	7.396E-02	2.4	3.8
	5.000	-0.418	9.304E-02	9.921E-02	2.4	3.8
	5.500	-0.398	1.217E-01	1.282E-01	2.4	3.7
	6.000	-0.379	1.546E-01	1.613E-01	2.8	3.5
	7.000	-0.345	2.334E-01	2.386E-01	2.8	3.5
	8.000	-0.316	3.291E-01	3.313E-01	3.4	4.1
	9.000	-0.291	4.410E-01	4.419E-01	3.3	4.2
	10.000	-0.268	5.679E-01	5.613E-01	3.3	4.2
	11.000	-0.247	7.088E-01	6.903E-01	3.3	4.2
	12.000	-0.228	8.624E-01	8.272E-01	5.9	7.1
	14.000	-0.195	1.202E+00	1.117E+00	5.9	7.3
	16.000	-0.166	1.578E+00	1.417E+00	7.5	9.2
	18.000	-0.140	1.979E+00	1.716E+00	5.7	8.3
	20.000	-0.117	2.399E+00	2.006E+00	5.7	8.4
79-Au	0.700	-0.853	1.580E-05	3.659E-06	17.7	32.8
	0.800	-0.823	4.338E-05	1.603E-05	6.9	10.0
	0.900	-0.798	9.516E-05	4.543E-05	6.9	9.3
	1.000	-0.775	1.795E-04	1.013E-04	4.9	7.8
	1.100	-0.754	3.042E-04	1.933E-04	4.9	7.5
	1.200	-0.735	4.765E-04	3.311E-04	2.9	14.1
	1.400	-0.702	9.899E-04	7.793E-04	2.9	12.8
	1.600	-0.673	1.767E-03	1.509E-03	2.4	5.6
	1.800	-0.647	2.848E-03	2.584E-03	2.4	5.4
	2.000	-0.625	4.269E-03	4.041E-03	2.2	5.7
	2.300	-0.594	7.108E-03	7.097E-03	2.2	4.3
	2.600	-0.568	1.088E-02	1.126E-02	2.1	4.3
	3.000	-0.536	1.751E-02	1.848E-02	2.2	4.9
	3.500	-0.503	2.860E-02	3.069E-02	2.2	4.8
	4.000	-0.474	4.305E-02	4.653E-02	3.0	4.5
	4.500	-0.449	6.106E-02	6.576E-02	3.0	4.5
	5.000	-0.426	8.278E-02	8.858E-02	2.4	4.0
	5.500	-0.405	1.083E-01	1.146E-01	2.4	4.0
	6.000	-0.386	1.377E-01	1.442E-01	2.8	3.6
	7.000	-0.352	2.080E-01	2.136E-01	2.8	3.7
	8.000	-0.324	2.935E-01	2.962E-01	3.4	4.2
	9.000	-0.298	3.936E-01	3.938E-01	3.4	4.3
	10.000	-0.275	5.075E-01	5.039E-01	3.3	4.4
79-Au	11.000	-0.254	6.341E-01	6.209E-01	3.3	4.4
	12.000	-0.235	7.724E-01	7.453E-01	5.9	7.3
	14.000	-0.202	1.079E+00	1.010E+00	5.9	7.4
	16.000	-0.173	1.419E+00	1.286E+00	7.5	9.4
	18.000	-0.147	1.784E+00	1.563E+00	7.5	9.5
	20.000	-0.125	2.167E+00	1.833E+00	5.7	8.6
80-Hg	0.700	-0.860	1.263E-05	2.461E-06	17.7	30.5
	0.800	-0.831	3.564E-05	1.202E-05	6.9	7.3
	0.900	-0.805	7.966E-05	3.573E-05	6.9	7.2
	1.000	-0.782	1.523E-04	8.194E-05	4.9	6.1
	1.100	-0.762	2.606E-04	1.594E-04	4.9	5.9
	1.200	-0.743	4.113E-04	2.769E-04	2.9	16.0
	1.400	-0.709	8.634E-04	6.638E-04	2.9	14.3
	1.600	-0.680	1.551E-03	1.300E-03	2.4	6.1
	1.800	-0.655	2.510E-03	2.243E-03	2.4	5.9
	2.000	-0.632	3.774E-03	3.528E-03	2.2	6.2
	2.300	-0.601	6.302E-03	6.182E-03	2.2	6.0
	2.600	-0.575	9.663E-03	9.911E-03	2.1	4.5
	3.000	-0.544	1.557E-02	1.650E-02	2.1	4.4
	3.500	-0.510	2.545E-02	2.724E-02	2.2	5.1
	4.000	-0.481	3.833E-02	4.144E-02	3.0	4.7
	4.500	-0.456	5.439E-02	5.866E-02	3.0	4.7
	5.000	-0.433	7.376E-02	7.918E-02	2.4	4.1
	5.500	-0.412	9.652E-02	1.026E-01	2.4	4.2
	6.000	-0.393	1.227E-01	1.290E-01	2.8	3.8
	7.000	-0.360	1.856E-01	1.914E-01	2.8	3.8
	8.000	-0.331	2.621E-01	2.651E-01	3.4	4.4
	9.000	-0.305	3.518E-01	3.528E-01	3.4	4.4
	10.000	-0.282	4.539E-01	4.527E-01	3.3	4.5
	11.000	-0.262	5.678E-01	5.587E-01	3.3	4.6
	12.000	-0.243	6.923E-01	6.718E-01	3.3	4.6
	14.000	-0.209	9.694E-01	9.136E-01	5.9	7.6
	16.000	-0.180	1.277E+00	1.167E+00	7.5	9.6
	18.000	-0.155	1.609E+00	1.424E+00	7.5	9.7
	20.000	-0.132	1.958E+00	1.676E+00	5.7	8.8
81-Tl	0.700	-0.867	1.007E-05	1.583E-06	17.7	27.2
	0.800	-0.838	2.924E-05	8.899E-06	6.9	9.6
	0.900	-0.812	6.664E-05	2.791E-05	6.9	8.9
	1.000	-0.789	1.292E-04	6.605E-05	4.9	5.6
	1.100	-0.769	2.234E-04	1.312E-04	4.9	5.5
	1.200	-0.750	3.553E-04	2.312E-04	4.9	5.4
	1.400	-0.716	7.539E-04	5.654E-04	2.9	15.9
	1.600	-0.687	1.364E-03	1.123E-03	2.4	6.7
	1.800	-0.662	2.217E-03	1.949E-03	2.4	6.4
	2.000	-0.639	3.343E-03	3.085E-03	2.2	6.7
	2.300	-0.608	5.598E-03	5.436E-03	2.2	6.5
	2.600	-0.582	8.600E-03	8.740E-03	2.1	4.8
	3.000	-0.551	1.388E-02	1.461E-02	2.1	4.7
	3.500	-0.517	2.270E-02	2.423E-02	2.2	5.3
	4.000	-0.488	3.421E-02	3.696E-02	3.0	4.8
	4.500	-0.463	4.855E-02	5.243E-02	3.0	4.8
	5.000	-0.440	6.586E-02	7.091E-02	2.4	4.3
	5.500	-0.419	8.620E-02	9.195E-02	2.4	4.3
	6.000	-0.400	1.097E-01	1.157E-01	2.4	4.4
	7.000	-0.367	1.659E-01	1.718E-01	2.8	4.0
	8.000	-0.338	2.344E-01	2.377E-01	3.4	4.6
	9.000	-0.312	3.149E-01	3.165E-01	3.4	4.6
	10.000	-0.289	4.067E-01	4.072E-01	3.3	4.7
	11.000	-0.269	5.091E-01	5.034E-01	3.3	4.7
	12.000	-0.250	6.213E-01	6.063E-01	3.3	4.8
	14.000	-0.216	8.717E-01	8.271E-01	5.9	7.8
	16.000	-0.187	1.151E+00	1.060E+00	7.5	9.8
	18.000	-0.162	1.453E+00	1.298E+00	7.5	9.9
	20.000	-0.139	1.771E+00	1.532E+00	5.7	9.0
82-Pb	0.800	-0.845	2.393E-05	6.466E-06	17.7	19.5
	0.900	-0.819	5.566E-05	2.162E-05	6.9	11.6
	1.000	-0.796	1.095E-04	5.297E-05	6.9	10.7
	1.100	-0.776	1.914E-04	1.076E-04	4.9	7.3
	1.200	-0.757	3.070E-04	1.927E-04	4.9	7.0
	1.400	-0.723	6.588E-04	4.812E-04	2.9	17.7
	1.600	-0.694	1.200E-03	9.679E-04	2.9	8.2
	1.800	-0.669	1.961E-03	1.694E-03	2.4	6.9
	2.000	-0.646	2.966E-03	2.701E-03	2.4	6.7
	2.300	-0.615	4.982E-03	4.786E-03	2.2	6.9

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
82-Pb	2.600	-0.589	7.668E-03	7.717E-03	2.1	5.1
	3.000	-0.558	1.239E-02	1.295E-02	2.1	5.0
	3.500	-0.524	2.029E-02	2.157E-02	2.2	5.6
	4.000	-0.495	3.059E-02	3.290E-02	2.2	4.7
	4.500	-0.470	4.342E-02	4.693E-02	3.0	5.0
	5.000	-0.447	5.891E-02	6.342E-02	2.4	4.5
	5.500	-0.426	7.713E-02	8.256E-02	2.4	4.5
	6.000	-0.407	9.812E-02	1.040E-01	2.4	4.5
	7.000	-0.374	1.485E-01	1.544E-01	2.8	4.2
	8.000	-0.345	2.100E-01	2.146E-01	2.8	4.2
	9.000	-0.319	2.823E-01	2.844E-01	3.4	4.8
	10.000	-0.296	3.648E-01	3.648E-01	3.4	4.9
	11.000	-0.276	4.571E-01	4.540E-01	3.3	4.9
	12.000	-0.257	5.583E-01	5.476E-01	3.3	4.9
	14.000	-0.223	7.847E-01	7.494E-01	5.9	7.9
	16.000	-0.194	1.038E+00	9.636E-01	5.9	8.2
	18.000	-0.169	1.313E+00	1.183E+00	7.5	10.1
	20.000	-0.146	1.603E+00	1.401E+00	7.5	10.2
	23.000	-0.116	2.058E+00	1.715E+00	5.7	9.4
83-Bi	0.800	-0.852	1.954E-05	4.599E-06	17.7	19.8
	0.900	-0.826	4.644E-05	1.660E-05	6.9	14.8
	1.000	-0.803	9.276E-05	4.227E-05	6.9	13.1
	1.100	-0.783	1.640E-04	8.802E-05	4.9	9.3
	1.200	-0.764	2.654E-04	1.603E-04	4.9	8.8
	1.400	-0.730	5.763E-04	4.094E-04	2.9	19.5
	1.600	-0.701	1.058E-03	8.348E-04	2.9	17.8
	1.800	-0.676	1.737E-03	1.473E-03	2.4	7.5
	2.000	-0.653	2.636E-03	2.365E-03	2.4	7.2
	2.300	-0.623	4.443E-03	4.220E-03	2.2	7.4
	2.600	-0.596	6.851E-03	6.824E-03	2.2	5.3
	3.000	-0.565	1.109E-02	1.151E-02	2.1	5.2
	3.500	-0.531	1.818E-02	1.924E-02	2.2	5.8
	4.000	-0.502	2.741E-02	2.942E-02	2.2	5.8
	4.500	-0.477	3.892E-02	4.208E-02	3.0	5.2
	5.000	-0.454	5.281E-02	5.694E-02	3.0	5.2
	5.500	-0.433	6.915E-02	7.425E-02	2.4	4.7
	6.000	-0.414	8.798E-02	9.361E-02	2.4	4.7
	7.000	-0.381	1.332E-01	1.391E-01	2.8	4.3
	8.000	-0.352	1.884E-01	1.934E-01	2.8	4.4
	9.000	-0.326	2.535E-01	2.560E-01	3.4	4.9
	10.000	-0.303	3.278E-01	3.286E-01	3.4	5.0
	11.000	-0.283	4.110E-01	4.100E-01	3.3	5.1
	12.000	-0.264	5.025E-01	4.952E-01	3.3	5.1
	14.000	-0.230	7.074E-01	6.796E-01	5.9	8.1
	16.000	-0.201	9.373E-01	8.765E-01	5.9	8.2
	18.000	-0.176	1.187E+00	1.079E+00	7.5	10.3
	20.000	-0.153	1.452E+00	1.282E+00	7.5	10.4
	23.000	-0.123	1.869E+00	1.576E+00	5.7	9.6
84-Po	0.800	-0.859	1.590E-05	3.171E-06	17.7	25.2
	0.900	-0.833	3.864E-05	1.259E-05	6.9	18.7
	1.000	-0.810	7.845E-05	3.348E-05	6.9	15.9
	1.100	-0.790	1.404E-04	7.162E-05	4.9	11.4
	1.200	-0.771	2.292E-04	1.329E-04	4.9	10.7
	1.400	-0.737	5.041E-04	3.476E-04	2.9	21.4
	1.600	-0.708	9.327E-04	7.193E-04	2.9	19.5
	1.800	-0.683	1.539E-03	1.281E-03	2.4	8.0
	2.000	-0.660	2.345E-03	2.070E-03	2.4	7.7
	2.300	-0.629	3.966E-03	3.723E-03	2.2	7.9
	2.600	-0.603	6.129E-03	6.000E-03	2.2	7.6
	3.000	-0.572	9.934E-03	1.023E-02	2.1	5.5
	3.500	-0.538	1.630E-02	1.718E-02	2.2	6.1
	4.000	-0.509	2.460E-02	2.633E-02	2.2	6.0
	4.500	-0.484	3.494E-02	3.776E-02	3.0	5.4
	5.000	-0.461	4.741E-02	5.118E-02	3.0	5.4
	5.500	-0.440	6.208E-02	6.685E-02	2.4	4.8
	6.000	-0.421	7.900E-02	8.436E-02	2.4	4.9
	7.000	-0.388	1.196E-01	1.254E-01	2.8	4.5
	8.000	-0.359	1.693E-01	1.745E-01	2.8	4.5
	9.000	-0.333	2.279E-01	2.306E-01	3.4	5.1
	10.000	-0.310	2.949E-01	2.962E-01	3.4	5.1
	11.000	-0.290	3.699E-01	3.705E-01	3.3	5.3
	12.000	-0.271	4.526E-01	4.481E-01	3.3	5.3
	14.000	-0.237	6.381E-01	6.166E-01	5.9	8.3
	16.000	-0.208	8.469E-01	7.975E-01	5.9	8.4
84-Po	18.000	-0.183	1.074E+00	9.847E-01	7.5	10.5
	20.000	-0.160	1.316E+00	1.173E+00	7.5	10.6
	23.000	-0.129	1.698E+00	1.448E+00	5.7	9.8
85-At	0.800	-0.866	1.290E-05	2.102E-06	17.7	33.0
	0.900	-0.840	3.211E-05	9.411E-06	17.7	26.2
	1.000	-0.817	6.629E-05	2.633E-05	6.9	19.1
	1.100	-0.797	1.201E-04	5.804E-05	6.9	16.9
	1.200	-0.778	1.980E-04	1.099E-04	4.9	12.7
	1.400	-0.744	4.413E-04	2.949E-04	4.9	11.4
	1.600	-0.715	8.233E-04	6.198E-04	2.9	21.2
	1.800	-0.690	1.367E-03	1.118E-03	2.4	8.6
	2.000	-0.667	2.090E-03	1.814E-03	2.4	8.3
	2.300	-0.636	3.547E-03	3.289E-03	2.2	8.4
	2.600	-0.610	5.494E-03	5.324E-03	2.2	8.1
	3.000	-0.579	8.921E-03	9.104E-03	2.1	5.8
	3.500	-0.545	1.465E-02	1.550E-02	2.1	5.6
	4.000	-0.516	2.212E-02	2.361E-02	2.2	6.3
	4.500	-0.491	3.143E-02	3.395E-02	3.0	5.5
	5.000	-0.468	4.265E-02	4.609E-02	3.0	5.5
	5.500	-0.447	5.586E-02	6.013E-02	2.4	5.0
	6.000	-0.428	7.108E-02	7.616E-02	2.4	5.0
	7.000	-0.395	1.077E-01	1.133E-01	2.4	4.9
	8.000	-0.366	1.524E-01	1.578E-01	2.8	4.7
	9.000	-0.340	2.052E-01	2.082E-01	3.4	5.3
	10.000	-0.317	2.657E-01	2.675E-01	3.4	5.3
	11.000	-0.296	3.335E-01	3.335E-01	3.4	5.4
	12.000	-0.278	4.083E-01	4.060E-01	3.3	5.5
	14.000	-0.244	5.764E-01	5.601E-01	3.3	5.5
	16.000	-0.215	7.662E-01	7.263E-01	5.9	8.6
	18.000	-0.190	9.735E-01	8.991E-01	7.5	10.7
	20.000	-0.167	1.194E+00	1.074E+00	7.5	10.8
	23.000	-0.136	1.544E+00	1.331E+00	5.7	10.0
86-Rn	0.900	-0.847	2.666E-05	6.921E-06	17.7	31.2
	1.000	-0.824	5.601E-05	2.056E-05	6.9	22.8
	1.100	-0.803	1.028E-04	4.684E-05	6.9	19.7
	1.200	-0.785	1.712E-04	9.066E-05	4.9	15.0
	1.400	-0.751	3.868E-04	2.501E-04	4.9	13.2
	1.600	-0.722	7.280E-04	5.342E-04	2.9	23.0
	1.800	-0.696	1.215E-03	9.737E-04	2.9	10.3
	2.000	-0.674	1.866E-03	1.592E-03	2.4	8.8
	2.300	-0.643	3.179E-03	2.909E-03	2.4	8.4
	2.600	-0.617	4.936E-03	4.733E-03	2.2	8.6
	3.000	-0.586	8.029E-03	8.118E-03	2.1	6.0
	3.500	-0.552	1.320E-02	1.388E-02	2.1	5.9
	4.000	-0.523	1.994E-02	2.121E-02	2.2	6.5
	4.500	-0.498	2.834E-02	3.046E-02	2.2	5.4
	5.000	-0.475	3.846E-02	4.157E-02	3.0	5.7
	5.500	-0.454	5.037E-02	5.431E-02	3.0	5.7
	6.000	-0.435	6.410E-02	6.888E-02	2.4	5.2
	7.000	-0.401	9.709E-02	1.026E-01	2.4	5.3
	8.000	-0.373	1.375E-01	1.429E-01	2.8	4.9
	9.000	-0.347	1.851E-01	1.894E-01	2.8	4.9
	10.000	-0.324	2.398E-01	2.420E-01	3.4	5.5
	11.000	-0.303	3.012E-01	3.019E-01	3.4	5.5
	12.000	-0.285	3.689E-01	3.683E-01	3.3	5.7
	14.000	-0.251	5.215E-01	5.093E-01	3.3	5.7
	16.000	-0.222	6.941E-01	6.621E-01	5.9	8.7
	18.000	-0.196	8.831E-01	8.217E-01	5.9	9.1
	20.000	-0.174	1.085E+00	9.839E-01	7.5	11.0
	23.000	-0.143	1.406E+00	1.224E+00	7.5	11.2
	26.000	-0.117	1.741E+00	1.454E+00	5.7	10.4
87-Fr	0.900	-0.854	2.203E-05	4.954E-06	17.7	37.7
	1.000	-0.831	4.715E-05	1.584E-05	6.9	27.2
	1.100	-0.810	8.777E-05	3.747E-05	6.9	22.9
	1.200	-0.791	1.476E-04	7.430E-05	4.9	17.5
	1.400	-0.758	3.384E-04	2.112E-04	4.9	15.1
	1.600	-0.729	6.430E-04	4.593E-04	2.9	25.0
	1.800	-0.703	1.080E-03	8.468E-04	2.9	23.0
	2.000	-0.680	1.665E-03	1.395E-03	2.4	9.4
	2.300	-0.650	2.850E-03	2.572E-03	2.4	8.9
	2.600	-0.623	4.436E-03	4.206E-03	2.2	9.1
	3.000	-0.592	7.230E-03	7.238E-03	2.2	6.3
	3.500	-0.559	1.190E-02	1.243E-02	2.1	6.2
	4.000	-0.530	1.799E-02	1.906E-02	2.2	6.8

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
87-Fr	4.500	-0.504	2.557E-02	2.743E-02	2.2	6.7
	5.000	-0.481	3.471E-02	3.752E-02	3.0	5.9
	5.500	-0.461	4.546E-02	4.907E-02	3.0	5.9
	6.000	-0.442	5.784E-02	6.218E-02	2.4	5.4
	7.000	-0.408	8.762E-02	9.292E-02	2.4	5.4
	8.000	-0.379	1.241E-01	1.295E-01	2.8	5.0
	9.000	-0.354	1.671E-01	1.718E-01	2.8	5.1
	10.000	-0.331	2.166E-01	2.191E-01	3.4	5.7
	11.000	-0.310	2.721E-01	2.734E-01	3.4	5.7
	12.000	-0.291	3.334E-01	3.343E-01	3.3	5.9
	14.000	-0.258	4.720E-01	4.633E-01	3.3	5.9
	16.000	-0.229	6.290E-01	6.036E-01	5.9	8.9
	18.000	-0.203	8.013E-01	7.509E-01	5.9	9.0
	20.000	-0.181	9.860E-01	9.014E-01	7.5	11.2
	23.000	-0.150	1.280E+00	1.126E+00	7.5	11.4
	26.000	-0.123	1.588E+00	1.341E+00	5.7	10.6
88-Ra	0.900	-0.861	1.818E-05	3.448E-06	17.7	46.7
	1.000	-0.838	3.967E-05	1.207E-05	6.9	32.6
	1.100	-0.817	7.490E-05	2.980E-05	6.9	26.6
	1.200	-0.798	1.274E-04	6.070E-05	6.9	23.3
	1.400	-0.765	2.965E-04	1.782E-04	4.9	17.1
	1.600	-0.736	5.688E-04	3.949E-04	2.9	27.1
	1.800	-0.710	9.618E-04	7.370E-04	2.9	24.8
	2.000	-0.687	1.490E-03	1.226E-03	2.4	10.1
	2.300	-0.657	2.560E-03	2.275E-03	2.4	9.5
	2.600	-0.630	3.996E-03	3.745E-03	2.2	9.6
	3.000	-0.599	6.526E-03	6.422E-03	2.2	6.6
	3.500	-0.566	1.076E-02	1.115E-02	2.1	6.4
	4.000	-0.537	1.627E-02	1.717E-02	2.2	7.0
	4.500	-0.511	2.313E-02	2.475E-02	2.2	7.0
	5.000	-0.488	3.140E-02	3.393E-02	3.0	6.1
	5.500	-0.468	4.112E-02	4.443E-02	3.0	6.1
	6.000	-0.449	5.233E-02	5.636E-02	3.0	6.1
	7.000	-0.415	7.926E-02	8.438E-02	2.4	5.6
	8.000	-0.386	1.123E-01	1.176E-01	2.8	5.2
	9.000	-0.361	1.512E-01	1.560E-01	2.8	5.2
	10.000	-0.338	1.960E-01	1.987E-01	3.4	5.8
	11.000	-0.317	2.464E-01	2.481E-01	3.4	5.8
	12.000	-0.298	3.020E-01	3.022E-01	3.4	6.0
	14.000	-0.265	4.279E-01	4.220E-01	3.3	6.1
	16.000	-0.236	5.709E-01	5.510E-01	5.9	9.1
	18.000	-0.210	7.282E-01	6.870E-01	5.9	9.2
	20.000	-0.187	8.972E-01	8.265E-01	7.5	11.4
	23.000	-0.157	1.167E+00	1.036E+00	7.5	11.6
	26.000	-0.130	1.450E+00	1.239E+00	5.7	10.8
89-Ac	0.900	-0.867	1.495E-05	2.295E-06	17.7	60.0
	1.000	-0.845	3.328E-05	9.036E-06	17.7	41.6
	1.100	-0.824	6.379E-05	2.348E-05	6.9	30.9
	1.200	-0.805	1.097E-04	4.928E-05	6.9	26.6
	1.400	-0.771	2.596E-04	1.498E-04	4.9	19.3
	1.600	-0.743	5.031E-04	3.388E-04	2.9	29.3
	1.800	-0.717	8.566E-04	6.407E-04	2.9	26.7
	2.000	-0.694	1.333E-03	1.076E-03	2.9	11.7
	2.300	-0.664	2.302E-03	2.013E-03	2.4	10.1
	2.600	-0.637	3.603E-03	3.336E-03	2.2	10.1
	3.000	-0.606	5.899E-03	5.748E-03	2.2	9.7
	3.500	-0.572	9.740E-03	1.002E-02	2.1	6.7
	4.000	-0.544	1.474E-02	1.561E-02	2.1	6.6
	4.500	-0.518	2.096E-02	2.235E-02	2.2	7.2
	5.000	-0.495	2.845E-02	3.061E-02	2.2	6.0
	5.500	-0.474	3.726E-02	4.027E-02	3.0	6.2
	6.000	-0.456	4.741E-02	5.113E-02	3.0	6.2
	7.000	-0.422	7.180E-02	7.670E-02	2.4	5.8
	8.000	-0.393	1.017E-01	1.068E-01	2.8	5.3
	9.000	-0.367	1.370E-01	1.419E-01	2.8	5.4
	10.000	-0.345	1.776E-01	1.815E-01	2.8	5.4
	11.000	-0.324	2.233E-01	2.253E-01	3.4	6.0
	12.000	-0.305	2.738E-01	2.746E-01	3.4	6.0
	14.000	-0.271	3.883E-01	3.846E-01	3.3	6.3
	16.000	-0.243	5.186E-01	5.033E-01	3.3	6.3
	18.000	-0.217	6.623E-01	6.288E-01	5.9	9.4
	20.000	-0.194	8.169E-01	7.582E-01	5.9	9.8
	23.000	-0.164	1.065E+00	9.535E-01	7.5	11.8
	26.000	-0.137	1.325E+00	1.144E+00	5.7	11.0
	1.000	-0.851	2.788E-05	6.634E-06	17.7	49.2
90-Th	1.100	-0.831	5.428E-05	1.833E-05	6.9	36.0
	1.200	-0.812	9.450E-05	3.978E-05	6.9	30.2
	1.400	-0.778	2.273E-04	1.257E-04	4.9	21.8
	1.600	-0.749	4.453E-04	2.905E-04	4.9	19.2
	1.800	-0.724	7.636E-04	5.569E-04	2.9	28.7
	2.000	-0.701	1.195E-03	9.441E-04	2.9	25.7
	2.300	-0.670	2.073E-03	1.783E-03	2.4	10.7
	2.600	-0.644	3.255E-03	2.976E-03	2.4	10.2
	3.000	-0.613	5.341E-03	5.153E-03	2.2	10.2
	3.500	-0.579	8.833E-03	9.007E-03	2.1	7.0
	4.000	-0.550	1.337E-02	1.409E-02	2.1	6.8
	4.500	-0.525	1.902E-02	2.022E-02	2.2	7.5
	5.000	-0.502	2.583E-02	2.773E-02	2.2	7.4
	5.500	-0.481	3.382E-02	3.656E-02	3.0	6.4
	6.000	-0.462	4.303E-02	4.647E-02	3.0	6.4
	7.000	-0.429	6.517E-02	6.984E-02	2.4	5.9
	8.000	-0.400	9.228E-02	9.737E-02	2.4	5.8
	9.000	-0.374	1.243E-01	1.293E-01	2.8	5.5
	10.000	-0.351	1.612E-01	1.654E-01	2.8	5.6
	11.000	-0.331	2.027E-01	2.050E-01	3.4	6.2
	12.000	-0.312	2.487E-01	2.499E-01	3.4	6.2
	14.000	-0.278	3.529E-01	3.510E-01	3.3	6.4
	16.000	-0.249	4.717E-01	4.601E-01	3.3	6.5
	18.000	-0.224	6.031E-01	5.761E-01	5.9	9.5
	20.000	-0.201	7.447E-01	6.960E-01	5.9	9.6
	23.000	-0.170	9.721E-01	8.780E-01	7.5	12.0
	26.000	-0.144	1.212E+00	1.056E+00	7.5	12.1
	30.000	-0.113	1.543E+00	1.280E+00	5.7	11.4
91-Pa	1.000	-0.858	2.328E-05	4.739E-06	17.7	59.4
	1.100	-0.837	4.608E-05	1.412E-05	6.9	42.2
	1.200	-0.818	8.123E-05	3.185E-05	6.9	34.5
	1.400	-0.785	1.988E-04	1.050E-04	4.9	24.5
	1.600	-0.756	3.940E-04	2.484E-04	4.9	21.3
	1.800	-0.730	6.808E-04	4.834E-04	2.9	30.8
	2.000	-0.707	1.071E-03	8.279E-04	2.9	28.5
	2.300	-0.677	1.868E-03	1.579E-03	2.4	11.3
	2.600	-0.650	2.942E-03	2.653E-03	2.4	10.7
	3.000	-0.619	4.842E-03	4.622E-03	2.2	10.7
	3.500	-0.586	8.021E-03	8.105E-03	2.1	7.3
	4.000	-0.557	1.216E-02	1.272E-02	2.1	7.1
	4.500	-0.531	1.729E-02	1.831E-02	2.2	7.7
	5.000	-0.508	2.348E-02	2.515E-02	2.2	7.7
	5.500	-0.488	3.075E-02	3.323E-02	3.0	6.6
	6.000	-0.469	3.912E-02	4.227E-02	3.0	6.6
	7.000	-0.435	5.923E-02	6.366E-02	2.4	6.1
	8.000	-0.406	8.386E-02	8.884E-02	2.4	6.1
	9.000	-0.381	1.130E-01	1.180E-01	2.8	5.7
	10.000	-0.358	1.465E-01	1.509E-01	2.8	5.8
	11.000	-0.337	1.842E-01	1.868E-01	3.4	6.4
	12.000	-0.318	2.261E-01	2.277E-01	3.4	6.4
	14.000	-0.285	3.210E-01	3.206E-01	3.3	6.6
	16.000	-0.256	4.295E-01	4.210E-01	3.3	6.7
	18.000	-0.230	5.496E-01	5.281E-01	5.9	9.7
	20.000	-0.207	6.793E-01	6.392E-01	5.9	9.8
	23.000	-0.177	8.882E-01	8.088E-01	7.5	12.2
	26.000	-0.150	1.109E+00	9.762E-01	7.5	12.3
	30.000	-0.119	1.416E+00	1.188E+00	5.7	11.6
92-U	1.000	-0.865	1.940E-05	3.263E-06	17.7	74.1
	1.100	-0.844	3.905E-05	1.071E-05	17.7	52.3
	1.200	-0.825	6.973E-05	2.528E-05	6.9	39.5
	1.400	-0.792	1.738E-04	8.732E-05	4.9	27.5
	1.600	-0.763	3.486E-04	2.119E-04	4.9	23.5
	1.800	-0.737	6.072E-04	4.191E-04	2.9	33.0
	2.000	-0.714	9.603E-04	7.255E-04	2.9	30.4
	2.300	-0.684	1.685E-03	1.398E-03	2.4	11.9
	2.600	-0.657	2.663E-03	2.365E-03	2.4	11.3
	3.000	-0.626	4.395E-03	4.149E-03	2.2	11.2
	3.500	-0.593	7.294E-03	7.300E-03	2.2	7.5
	4.000	-0.564	1.106E-02	1.149E-02	2.1	7.4
	4.500	-0.538	1.575E-02	1.660E-02	2.2	8.0
	5.000	-0.515	2.138E-02	2.284E-02	2.2	7.9
	5.500	-0.495	2.800E-02	3.013E-02	2.2	6.5
	6.000	-0.476	3.562E-02	3.850E-02	3.0	6.8
	7.000	-0.442	5.392E-02	5.796E-02	2.4	6.3
	8.000	-0.413	7.632E-02	8.115E-02	2.4	6.3
	9.000	-0.388	1.028E-01	1.078E-01	2.8	5.9

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by Protons

See page 112 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] ECPSSR	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
92-U	10.000	-0.365	1.333E-01	1.379E-01	2.8	5.9
	11.000	-0.344	1.677E-01	1.713E-01	2.8	6.0
	12.000	-0.325	2.058E-01	2.077E-01	3.4	6.5
	14.000	-0.292	2.924E-01	2.931E-01	3.3	6.8
	16.000	-0.263	3.914E-01	3.855E-01	3.3	6.9
	18.000	-0.237	5.013E-01	4.843E-01	5.9	9.9
	20.000	-0.214	6.202E-01	5.874E-01	5.9	10.0
	23.000	-0.184	8.122E-01	7.453E-01	7.5	12.3
	26.000	-0.157	1.016E+00	9.021E-01	7.5	12.5
	30.000	-0.126	1.299E+00	1.102E+00	5.7	11.8

APPENDIX: TABLE II. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger-Electron Production, and Ionization

See page 112 for Explanation of Tables

Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
4	.008	2.500E-01	12.0	XNSH80	6	.097	4.401E+02	12.0	XGHA73H	6	1.096	2.180E+03	30.3	XKKH65
4	.009	7.380E-01	12.0	XNSH80	6	.097	2.108E+02	12.0	XGHA73O	6	1.100	1.890E+03	24.9	XNLA76
4	.012	2.500E+00	12.0	XNSH80	6	.100	2.080E+02	29.8	XKKH65	6	1.200	1.250E+06	15.6	AGME87
4	.015	8.816E+00	80.0	XKTE72	6	.100	4.700E+02	80.0	XKTE71	6	1.270	2.100E+03	30.5	XKKH65
4	.017	1.080E+01	12.0	XNSH80	6	.100	1.830E+05	16.4	AGST72	6	1.360	2.100E+03	30.5	XKKH65
4	.020	3.010E+01	80.0	XKTE72	6	.100	1.520E+05	29.6	AGST72	6	1.400	9.600E+05	20.8	IGKB76
4	.021	3.160E+01	12.0	XNSH80	6	.100	1.610E+05	14.9	IGST75	6	1.400	1.290E+06	14.7	AGME87
4	.025	7.600E+01	80.0	XKTE72	6	.100	4.150E+02	25.1	XGLA76	6	1.500	3.000E+03	80.0	XKOG64
4	.026	7.290E+01	12.0	XNSH80	6	.100	1.690E+05	15.0	IGST73	6	1.500	8.500E+05	21.2	AGTO72B
4	.030	1.611E+02	80.0	XKTE72	6	.106	5.427E+02	12.0	XGHA73H	6	1.505	2.020E+03	29.7	XKKH65
4	.035	2.200E+02	12.0	XNSH80	6	.106	2.737E+02	12.0	XGHA73O	6	1.600	1.210E+06	15.0	AGME87
4	.040	3.952E+02	80.0	XKTE72	6	.110	2.740E+02	29.9	XKKH65	6	1.658	1.940E+03	29.9	XKKH65
4	.045	4.450E+02	12.0	XNSH80	6	.116	6.669E+02	12.0	XGHA73H	6	1.800	1.140E+06	21.9	IGKB76
4	.050	6.384E+02	80.0	XKTE72	6	.116	3.468E+02	12.0	XGHA73O	6	1.800	1.110E+06	15.4	AGME87
4	.054	7.320E+02	12.0	XNSH80	6	.120	6.600E+02	80.0	XKTE71	6	1.908	1.900E+03	29.5	XKKH65
4	.060	1.034E+03	80.0	XKTE72	6	.125	2.400E+05	15.0	IGST75	6	2.000	7.500E+05	18.7	AGTO72B
4	.070	1.307E+03	80.0	XKTE72	6	.125	6.500E+02	25.1	XGLA76	6	2.000	2.010E+03	19.9	XNBI76
4	.074	1.280E+03	12.0	XNSH80	6	.125	2.590E+05	15.0	IGST73	6	2.000	9.800E+05	17.1	AGME87
4	.080	1.611E+03	80.0	XKTE72	6	.126	7.830E+02	12.0	XGHA73H	6	2.210	8.800E+05	21.6	IGKB76
4	.085	1.590E+03	12.0	XNSH80	6	.126	4.046E+02	12.0	XGHA73O	6	2.600	5.700E+05	22.8	IGKB76
4	.090	1.794E+03	80.0	XKTE72	6	.135	8.991E+02	12.0	XGHA73H	6	3.000	1.160E+03	19.9	XNBI76
4	.094	1.840E+03	12.0	XNSH80	6	.135	4.794E+02	12.0	XGHA73O	6	4.000	1.230E+03	20.3	XNBI76
4	.100	2.006E+03	80.0	XKTE72	6	.140	9.100E+02	80.0	XKTE71	6	5.000	1.080E+03	19.4	XNBI76
4	.114	2.170E+03	12.0	XNSH80	6	.145	1.029E+03	12.0	XGHA73H	6	6.000	9.300E+02	20.4	XNBI76
4	.120	2.310E+03	80.0	XKTE72	6	.145	5.525E+02	12.0	XGHA73O	6	7.000	8.300E+02	20.5	XNBI76
4	.140	2.675E+03	80.0	XKTE72	6	.150	3.310E+05	18.1	AGST72	6	8.000	8.100E+02	19.8	XNBI76
4	.160	3.010E+03	80.0	XKTE72	6	.150	2.840E+05	21.1	AGST72	6	9.000	7.200E+02	19.4	XNBI76
4	.160	6.000E+06	25.0	IKPO77	6	.150	3.230E+05	15.2	IGST75	6	10.000	6.700E+02	19.4	XNBI76
4	.180	3.192E+03	80.0	XKTE72	6	.150	9.100E+02	24.9	XGLA76	6	11.000	6.500E+02	20.0	XNBI76
4	.200	3.405E+03	80.0	XKTE72	6	.150	8.400E+02	25.0	XNLA76	6	12.000	6.300E+02	20.6	XNBI76
4	.300	2.530E+03	80.0	XKKW77	6	.150	3.510E+05	15.0	IGST73	6	13.000	5.500E+02	20.0	XNBI76
4	.500	2.260E+03	80.0	XKKW77	6	.160	1.200E+03	80.0	XKTE71	6	14.000	5.300E+02	20.8	XNBI76
4	.700	2.000E+03	80.0	XKKW77	6	.175	1.150E+03	25.2	XGLA76	6	15.000	5.300E+02	24.5	XNBI76
4	1.000	1.680E+03	80.0	XKKW77	6	.175	1.120E+03	25.0	XNLA76	6	16.000	5.800E+02	25.9	XNBI76
4	1.200	1.610E+03	80.0	XKKW77	6	.180	1.300E+03	80.0	XKTE71	6	18.000	4.280E+02	20.1	XNBI77
4	1.500	1.520E+03	80.0	XKKW77	6	.200	1.500E+03	80.0	XKTE71	6	19.000	3.790E+02	20.1	XNBI77
4	1.800	1.420E+03	80.0	XKKW77	6	.200	5.240E+05	14.3	AGST72	6	20.000	3.690E+02	20.0	XNBI77
5	.300	1.800E+06	33.3	AGTO72B	6	.200	4.300E+05	25.6	AGST72	6	21.000	3.590E+02	19.9	XNBI77
5	.500	1.800E+06	27.8	AGTO72B	6	.200	5.010E+05	15.0	IGST75	6	22.000	3.390E+02	20.3	XNBI77
5	.500	1.510E+03	80.0	XKKW77	6	.200	1.360E+03	25.0	XGLA76	6	23.000	3.300E+02	20.1	XNBI77
5	.750	1.800E+06	23.0	AGKO79	6	.200	1.330E+03	24.8	XNLA76	6	24.000	3.100E+02	19.8	XNBI77
5	.750	1.430E+03	80.0	XKKW77	6	.200	5.480E+05	15.0	IGST73	6	25.000	3.100E+02	19.8	XNBI77
5	1.000	1.800E+06	27.8	AGTO72B	6	.250	1.770E+03	24.9	XGLA76	6	26.000	3.100E+02	19.8	XNBI77
5	1.000	1.800E+06	23.0	AGKO79	6	.250	1.640E+03	25.0	XNLA76	7	.028	2.688E+00	12.1	XGHA73
5	1.000	1.320E+03	80.0	XKKW77	6	.250	6.300E+05	15.0	IGST73	7	.037	7.191E+00	11.8	XGHA73
5	1.200	1.230E+03	80.0	XKKW77	6	.290	2.210E+03	19.9	XNBI76	7	.048	1.809E+01	11.9	XGHA73
5	1.400	1.140E+03	80.0	XKKW77	6	.300	7.500E+05	16.0	AGST72	7	.050	4.220E+03	14.2	IGST75
5	1.500	1.600E+06	23.0	AGKO79	6	.300	9.200E+05	27.2	AGST72	7	.050	6.500E+03	15.0	IGST73
5	1.600	1.070E+03	80.0	XKKW77	6	.300	9.400E+05	19.1	AGTO72B	7	.058	3.379E+01	12.0	XGHA73
5	1.800	1.010E+03	80.0	XKKW77	6	.300	7.460E+05	15.0	IGST75	7	.060	7.200E+03	15.3	IGST75
5	2.000	9.000E+05	20.0	AGTO72B	6	.300	2.040E+03	25.0	XGLA76	7	.063	8.500E+03	15.3	IGST75
5	2.000	9.780E+02	80.0	XKKW77	6	.300	1.860E+03	24.7	XNLA76	7	.068	5.922E+01	11.9	XGHA73
6	.015	5.760E-02	29.9	XKKH65	6	.300	7.800E+05	15.0	IGST73	7	.075	1.460E+04	15.8	IGST75
6	.020	2.000E-01	30.0	XKKH65	6	.400	7.800E+05	15.4	AGST72	7	.075	2.200E+04	15.0	IGST73
6	.020	1.700E+00	80.0	XKTE71	6	.400	9.900E+05	25.3	AGST72	7	.077	9.212E+01	11.7	XGHA73
6	.025	5.580E-01	30.1	XKKH65	6	.400	8.420E+05	15.0	IGST75	7	.087	1.260E+02	11.9	XGHA73
6	.025	5.000E+00	80.0	XKTE71	6	.400	2.200E+03	25.0	XGLA76	7	.097	1.678E+02	12.0	XGHA73
6	.028	7.560E+00	12.0	XGHA73H	6	.400	2.120E+03	25.0	XNLA76	7	.100	3.540E+04	15.0	IGST75
6	.030	1.840E+00	30.4	XKKH65	6	.400	9.200E+05	15.0	IGST73	7	.100	5.000E+04	15.0	IGST73
6	.030	1.200E+01	80.0	XKTE71	6	.499	2.180E+03	30.3	XKKH65	7	.106	2.200E+02	12.0	XGHA73
6	.037	2.020E+01	12.0	XGHA73H	6	.500	9.300E+05	16.1	AGST72	7	.116	2.731E+02	12.0	XGHA73
6	.037	9.724E+00	12.0	XGHA73O	6	.500	1.090E+06	22.9	AGST72	7	.125	6.050E+04	15.0	IGST75
6	.040	6.340E+00	30.0	XKKH65	6	.500	1.110E+06	22.5	AGTO72B	7	.125	3.170E+02	25.0	XGLA76
6	.040	3.600E+01	80.0	XKTE71	6	.500	9.240E+05	15.0	IGST75	7	.126	3.262E+02	12.0	XGHA73
6	.048	5.535E+01	12.0	XGHA73H	6	.500	2.330E+03	24.9	XGLA76	7	.135	3.840E+02	12.0	XGHA73
6	.048	2.227E+01	12.0	XGHA73O	6	.500	2.230E+03	25.1	XNLA76	7	.145	4.277E+02	12.1	XGHA73
6	.050	1.540E+01	29.9	XKKH65	6	.500	1.010E+06	15.0	IGST73	7	.150	9.000E+04	15.6	IGST75
6	.050	7.600E+01	80.0	XKTE71	6	.520	2.800E+03	20.0	XNBI76	7	.150	4.700E+02	24.9	XGLA76
6	.050	2.370E+04	15.2	IGST75	6	.595	2.240E+03	30.4	XKKH65	7	.150	1.190E+05	15.0	IGST73
6	.050	3.200E+04	15.0	IGST73	6	.600	9.540E+05	15.0	IGST75	7	.175	6.420E+02	24.9	XGLA76
6	.058	9.936E+01	12.0	XGHA73H	6	.600	2.350E+03	25.1	XGLA76	7	.200	1.470E+05	15.0	IGST75
6	.058	4.556E+01	12.0	XGHA73O	6	.600	2.250E+03	24.9	XNLA76	7	.200	8.210E+02	25.0	XGLA76
6	.060	3.400E+01	30.0	XKKH65	6	.600	9.200E+05	21.7	IGKB76	7	.200	1.750E+05	15.0	IGST73
6	.060	1.300E+02	80.0	XKTE71	6	.600	1.020E+06	15.0	IGST73	7	.250	1.090E+03	24.8	XGLA76
6	.060	3.950E+04	14.9	IGST75	6	.600	1.270E+06	17.9	AGME87	7	.300	3.000E+05	20.0	AGTO72B
6	.060	4.600E+04	15.0	IGST73	6	.698	2.280E+03	29.8	XKKH65	7	.300	2.590E+05	15.1	IGST75
6	.063	4.900E+04	14.3	IGST75	6	.700	2.280E+03	25.0	XGLA76	7	.300	1.360E+03	25.0	XGLA76
6	.068	1.550E+02	12.0	XGHA73H	6	.700	2.210E+03	24.9	XNLA76	7	.300	3.000E+05	15.0	IGST73
6	.068	7.276E+01	12.0	XGHA73O	6	.720	2.770E+03	20.2	XNBI76	7	.400	3.620E+05	14.9	IGST75
6	.070	5.940E+01	30.0	XKKH65	6	.775	2.280E+03	29.8	XKKH65	7	.400	1.710E+03	25.1	XGLA76
6	.070	2.000E+02	80.0	XKTE71	6	.800	2.090E+03	24.9	XGLA76	7	.400	4.000E+05	15.0	IGST73
6	.075	1.130E+05	17.7	AGST72	6	.800	2.160E+03	25.0	XNLA76	7	.500	5.200E+05	21.2	AGTO72B
6	.075	8.300E+04	15.7	IGST75	6	.800	1.350E+06	15.2	AGME87	7	.500	3.890E+05	14.9	IGST75
6	.075	9.500E+04	15.0	IGST73	6	.900	2.080E+03	25.0	XNLA76	7	.500	2.050E+03	24.9	XGLA76
6	.077	2.430E+02	12.0	XGHA73H	6	.910	2.300E+03	30.4	XKKH65	7	.500	4.500E+05	15.0	IGST73
6	.077	1.149E+02	12.0	XGHA73O	6	1.000	1.050E+06	20.0	AGTO72B	7	.600	4.200E+05	19.0	IGKB76
6	.080	1.180E+02	30.5	XKKH65	6	1.000	2.100E+03	24.8	XGLA76	7	1.000	5.800E+05	20.7	AGTO72B
6	.080	2.800E+02	80.0	XKTE71	6	1.000	1.990E+03	25.1	XNLA76	7	1.000	6.000E+05	30.0	IGMK76
6	.087	3.321E+02	12.0	XGHA73H	6	1.000	1.310E+06	15.3	AGME87	7	1.010	5.200E+05	19.2	IGKB76
6	.087	1.649E+02	12.0	XGHA73O	6	1.010	1.250E+06	22.4	IGKB76	7	1.400	5.500E+05	20.0	IGKB76
6	.090	1.640E+02	30.5	XKKH65	6	1.020	2.450E+03	20.0	XNBI76	7	1.500	5.100E+05	20.4	AGTO72B
6	.090	3.700E+02	80.0	XKTE71	6	1.022	2.240E+03	30.4	XKKH65	7	1.800	6.000E+05	20.0	IGKB76

APPENDIX: TABLE II. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger-Electron Production, and Ionization

See page 112 for Explanation of Tables

Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
7	2.000	4.500E+05	20.0	AGTO72B	9	2.000	2.000E+05	50.0	IGKB76	12	.090	2.411E+00	29.7	XKKH65
7	2.000	4.600E+05	30.0	IGMK76	9	2.000	1.000E+05	30.0	IGMK76	12	.100	3.471E+00	29.8	XKKH65
7	2.200	5.300E+05	20.8	IGKB76	9	2.000	1.830E+03	15.0	XKKW80	12	.100	2.809E+00	15.1	XKKH64
7	2.600	4.600E+05	19.6	IGKB76	9	2.400	1.800E+05	50.0	IGKB76	12	.150	1.193E+01	15.1	XKKH64
7	3.000	4.360E+05	30.0	IGMK76	9	3.000	9.900E+04	30.0	IGMK76	12	.200	2.809E+01	15.1	XKKH64
7	4.000	3.510E+05	29.9	IGMK76	9	4.000	9.200E+04	30.0	IGMK76	12	.300	9.540E+01	14.7	XKKH64
7	5.000	2.810E+05	29.9	IGMK76	9	5.000	8.460E+04	30.0	IGMK76	12	.400	1.802E+02	14.7	XKKH64
8	.020	1.856E-01	50.0	XKHA69	10	.048	1.411E+00	12.1	XGHA73	12	.500	2.306E+02	14.9	XKKH64
8	.025	4.582E-01	50.6	XKHA69	10	.058	2.666E+00	12.2	XGHA73	12	.602	4.240E+02	30.0	XKKH65
8	.030	9.396E-01	50.0	XKHA69	10	.068	4.681E+00	11.9	XGHA73	12	.686	4.479E+02	30.2	XKKH65
8	.035	1.740E+00	50.0	XKHA69	10	.077	7.827E+00	12.1	XGHA73	12	.800	5.061E+02	29.8	XKKH65
8	.040	2.900E+00	50.0	XKHA69	10	.097	1.783E+01	12.2	XGHA73	12	.900	5.777E+02	29.8	XKKH65
8	.045	4.698E+00	50.6	XKHA69	10	.100	7.300E+02	15.1	IGST75	12	1.000	6.413E+02	30.2	XKKH65
8	.050	7.018E+00	50.4	XKHA69	10	.116	3.286E+01	11.8	XGHA73	12	1.100	7.261E+02	29.9	XKKH65
8	.055	1.009E+01	50.0	XKHA69	10	.125	1.610E+03	14.9	IGST75	12	1.200	7.738E+02	30.1	XKKH65
8	.060	1.392E+01	50.0	XKHA69	10	.125	3.200E+01	25.0	XGLA76	12	1.310	8.745E+02	30.3	XKKH65
8	.065	1.856E+01	50.0	XKHA69	10	.135	5.394E+01	12.1	XGHA73	12	1.400	9.540E+02	30.6	XKKH65
8	.070	2.436E+01	50.0	XKHA69	10	.150	2.830E+03	11.8	IGST75	12	1.500	1.034E+03	30.8	XKKH65
8	.075	7.500E+03	13.3	AGST72	10	.150	5.700E+01	24.9	XGLA76	12	1.600	1.060E+03	30.0	XKKH65
8	.075	3.132E+01	50.0	XKHA69	10	.200	6.240E+03	15.1	IGST75	12	1.700	1.140E+03	30.2	XKKH65
8	.080	3.886E+01	49.3	XKHA69	10	.200	1.250E+02	24.8	XGLA76	12	3.000	1.099E+03	8.0	XNBS78
8	.085	4.756E+01	50.0	XKHA69	10	.250	1.250E+04	15.2	IGST75	13	.025	1.428E-03	30.0	XKKH65
8	.090	5.684E+01	50.0	XKHA69	10	.250	2.090E+02	24.9	XGLA76	13	.025	2.020E-03	24.8	XKBB73
8	.095	6.728E+01	50.0	XKHA69	10	.300	1.800E+04	20.0	AGTO72B	13	.030	4.998E-03	30.0	XKKH65
8	.100	1.600E+04	12.5	AGST72	10	.300	1.660E+04	15.1	IGST75	13	.030	6.540E-03	24.9	XKBB73
8	.100	7.888E+01	50.0	XKHA69	10	.300	3.180E+02	24.8	XGLA76	13	.035	1.610E-02	24.8	XKBB73
8	.150	4.300E+04	16.3	AGST72	10	.400	2.610E+04	14.9	IGST75	13	.040	2.999E-02	29.8	XKKH65
8	.200	6.300E+04	15.9	AGST72	10	.400	5.630E+02	25.0	XGLA76	13	.040	3.260E-02	24.8	XKBB73
8	.300	1.340E+05	14.9	AGST72	10	.500	5.200E+04	20.0	AGTO72B	13	.050	9.175E-02	30.0	XKKH65
8	.300	1.200E+05	25.0	AGTO72B	10	.500	3.590E+04	15.0	IGST75	13	.050	1.070E-01	25.2	XKBB73
8	.300	7.600E+04	28.4	AGME87	10	.500	7.040E+02	25.0	XGLA76	13	.060	2.106E-01	30.5	XKKH65
8	.400	1.600E+05	12.5	AGST72	10	.500	3.900E+04	20.5	AGWO76	13	.060	1.714E-01	14.8	XKKH64
8	.400	1.470E+05	23.6	AGME87	10	.600	4.540E+04	15.0	IGST75	13	.060	2.270E-01	25.1	XKBB73
8	.450	1.630E+03	80.0	XKKW77	10	.600	8.370E+02	25.0	XGLA76	13	.070	4.105E-01	30.4	XKKH65
8	.500	2.610E+05	11.5	AGST72	10	.600	4.200E+04	38.1	IGKB76	13	.070	4.330E-01	24.9	XKBB73
8	.500	1.730E+05	21.6	AGME87	10	.600	4.800E+04	20.8	AGWO76	13	.080	6.962E-01	30.3	XKKH65
8	.600	2.100E+05	23.8	IGKB76	10	.700	9.600E+02	25.0	XGLA76	13	.080	7.330E-01	25.0	XKBB73
8	.600	1.600E+03	80.0	XKKW77	10	.750	6.700E+04	19.4	AGWO76	13	.090	1.060E+00	30.0	XKKH65
8	.600	1.930E+05	19.8	AGME87	10	.800	1.050E+03	24.8	XGLA76	13	.090	1.050E+00	5.1	XKSH71
8	.800	1.790E+03	80.0	XKKW77	10	.900	1.200E+03	25.0	XGLA76	13	.090	1.140E+00	24.6	XKBB73
8	.800	2.330E+05	16.4	AGME87	10	.970	1.110E+05	15.3	AGWO74	13	.100	1.535E+00	30.2	XKKH65
8	1.000	2.800E+05	21.4	AGTO72B	10	1.000	8.800E+04	20.0	AGTO72B	13	.100	1.499E+00	15.0	XKKH64
8	1.000	1.770E+03	80.0	XKKW77	10	1.000	1.300E+03	24.6	XGLA76	13	.100	1.489E+00	5.0	XKSH71
8	1.000	2.490E+05	15.6	AGME87	10	1.000	1.300E+05	30.0	IGMK76	13	.100	1.680E+00	25.0	XKBB73
8	1.010	3.500E+05	22.9	IGKB76	10	1.000	8.500E+04	20.0	AGWO76	13	.110	2.350E+00	25.1	XKBB73
8	1.200	2.510E+05	15.6	AGME87	10	1.010	8.700E+04	26.4	IGKB76	13	.120	3.460E+00	24.9	XKBB73
8	1.400	2.900E+05	24.1	IGKB76	10	1.100	1.420E+03	24.6	XGLA76	13	.132	3.891E+00	14.7	XKKH64
8	1.400	1.670E+03	80.0	XKKW77	10	1.200	1.500E+03	25.3	XGLA76	13	.140	4.677E+00	5.3	XKSH71
8	1.400	2.560E+05	15.5	AGME87	10	1.250	8.600E+04	19.8	AGWO76	13	.140	6.440E+00	25.0	XKBB73
8	1.500	2.800E+05	21.4	AGTO72B	10	1.400	1.030E+05	26.2	IGKB76	13	.150	5.605E+00	15.3	XKKH64
8	1.600	2.470E+05	15.7	AGME87	10	1.500	1.040E+05	20.2	AGWO76	13	.153	7.420E+00	24.9	XKBB73
8	1.800	3.100E+05	22.6	IGKB76	10	1.750	1.070E+05	19.6	AGWO76	13	.160	9.140E+00	24.9	XKBB73
8	1.800	1.580E+03	80.0	XKKW77	10	1.800	9.900E+04	26.3	IGKB76	13	.179	1.220E+01	24.6	XKBB73
8	1.800	2.530E+05	16.3	AGME87	10	2.000	1.000E+05	20.0	AGTO72B	13	.180	1.060E+01	5.1	XKSH71
8	2.000	2.700E+05	14.8	AGTO72B	10	2.000	1.420E+05	30.3	IGMK76	13	.200	1.499E+01	15.0	XKKH64
8	2.000	2.330E+05	17.3	AGME87	10	2.000	1.000E+05	20.0	AGWO76	13	.200	1.790E+01	25.1	XKBB73
8	2.210	2.600E+05	23.1	IGKB76	10	2.000	1.180E+05	15.3	AGWO74	13	.204	1.810E+01	25.0	XKBB73
8	2.610	2.400E+05	25.0	IGKB76	10	2.210	9.800E+04	26.5	IGKB76	13	.250	3.110E+01	25.1	XKBB73
9	.300	5.000E+04	20.0	AGTO72B	10	2.500	9.300E+04	20.4	AGWO76	13	.255	3.280E+01	25.0	XKBB73
9	.300	5.000E+04	20.0	AGTO77	10	2.600	1.010E+05	25.7	IGKB76	13	.300	5.498E+01	14.9	XKKH64
9	.300	5.000E+04	20.0	AGTO77	10	3.000	1.290E+05	30.2	IGMK76	13	.300	6.850E+01	10.0	XNGB86
9	.300	3.140E+02	15.0	XKKW80	10	3.000	8.700E+04	19.5	AGWO76	13	.306	5.340E+01	24.9	XKBB73
9	.350	4.150E+02	15.0	XKKW80	10	3.000	1.110E+05	15.3	AGWO74	13	.357	7.810E+01	25.0	XKBB73
9	.400	5.140E+02	15.0	XKKW80	10	3.000	9.800E+04	20.0	AGTO83	13	.400	1.214E+02	14.7	XKKH64
9	.500	1.000E+05	20.0	AGTO72B	10	3.500	8.400E+04	20.2	AGWO76	13	.408	9.810E+01	25.0	XKBB73
9	.500	9.000E+04	22.0	AGTO77	10	3.670	8.900E+04	20.0	AGTO83	13	.459	1.380E+02	24.6	XKBB73
9	.500	1.100E+05	18.0	AGTO77	10	4.000	1.060E+05	30.2	IGMK76	13	.484	1.928E+02	29.6	XKKH65
9	.500	7.700E+04	20.0	IGCH77	10	4.000	7.900E+04	20.3	AGWO76	13	.500	2.285E+02	15.2	XKKH64
9	.500	7.540E+02	15.0	XKKW80	10	4.000	9.400E+04	15.0	AGWO74	13	.510	1.700E+02	24.7	XKBB73
9	.600	1.020E+03	15.0	XKKW80	10	4.200	8.600E+04	20.0	AGTO83	13	.598	2.713E+02	30.3	XKKH65
9	.700	1.270E+03	15.0	XKKW80	10	4.500	6.900E+04	20.3	AGWO76	13	.600	3.000E+02	10.0	XNGB86
9	.750	1.300E+05	23.0	AGKO79	10	5.000	9.500E+04	30.0	IGMK76	13	.612	2.340E+02	24.8	XKBB73
9	.800	1.490E+03	15.0	XKKW80	10	5.000	7.000E+04	20.0	AGWO76	13	.700	3.427E+02	30.2	XKKH65
9	.900	1.680E+03	15.0	XKKW80	10	5.000	8.400E+04	15.0	AGWO74	13	.714	2.950E+02	25.1	XKBB73
9	1.000	1.600E+05	18.8	AGTO72B	10	5.500	6.400E+04	20.3	AGWO76	13	.750	2.960E+02	12.2	XNTA76
9	1.000	9.400E+04	30.0	IGMK76	10	6.000	6.400E+04	20.3	AGWO76	13	.800	4.177E+02	29.9	XKKH65
9	1.000	1.300E+05	23.0	AGTO77	10	6.500	5.700E+04	19.3	AGWO76	13	.816	3.500E+02	25.1	XKBB73
9	1.000	1.400E+05	21.0	AGTO77	10	7.000	5.800E+04	20.7	AGWO76	13	.900	4.855E+02	30.1	XKKH65
9	1.000	1.870E+03	15.0	XKKW80	10	7.500	5.500E+04	20.0	AGWO76	13	.918	4.010E+02	24.9	XKBB73
9	1.000	1.500E+05	23.0	AGKO79	10	8.000	5.500E+04	20.0	AGWO76	13	1.000	5.462E+02	30.1	XKKH65
9	1.100	1.940E+03	15.0	XKKW80	10	8.500	5.300E+04	20.8	AGWO76	13	1.000	4.780E+02	11.9	XNTA76
9	1.200	2.400E+05	50.0	IGKB76	10	9.000	5.200E+04	19.2	AGWO76	13	1.000	6.050E+02	10.0	XNGB86
9	1.200	2.020E+03	15.0	XKKW80	10	9.500	5.000E+04	20.0	AGWO76	13	1.020	4.460E+02	24.9	XKBB73
9	1.400	2.060E+03	15.0	XKKW80	10	10.000	4.800E+04	20.8	AGWO76	13	1.100	6.176E+02	30.1	XKKH65
9	1.500	1.560E+05	20.0	IGCH77	12	.025	4.320E-03	30.1	XKKH65	13	1.120	4.840E+02	25.0	XKBB73
9	1.500	1.800E+05	23.0	AGKO79	12	.030	1.537E-02	29.3	XKKH65	13	1.200	6.676E+02	29.9	XKKH65
9	1.600	2.600E+05	50.0	IGKB76	12	.040	7.340E-02	30.0	XKKH65	13	1.220	5.210E+02	25.0	XKBB73
9	1.600	2.020E+03	15.0	XKKW80	12	.050	2.279E-01	30.2	XKKH65	13	1.250	6.160E+02	12.0	XNTA76
9	1.800	1.300E+05	23.0	AGTO77	12	.060	4.452E-01	29.8	XKKH65	13	1.310	7.176E+02	29.9	XKKH65
9	1.800	1.400E+05	21.0	AGTO77	12	.060	3.498E-01	15.2	XKKH64	13	1.330	5.480E+02	25.0	XKBB73
9	1.800	1.920E+03	15.0	XKKW80	12	.070	8.533E-01	30.1	XKKH65	13	1.400	7.675E+02	30.2	XKKH65
9	2.000	1.600E+05	18.8	AGTO72B	12	.080	1.855E+00	30.0	XKKH65	13	1.430	5.760E+02	25.0	XKBB73

APPENDIX: TABLE II. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger-Electron Production, and Ionization

See page 112 for Explanation of Tables

Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
13	1.500	9.996E+02	80.0	XKOG64
13	1.500	8.282E+02	30.2	XKKH65
13	1.500	6.980E+02	12.0	XNTA76
13	1.530	6.040E+02	25.0	XKBB73
13	1.600	8.782E+02	30.1	XKKH65
13	1.630	6.270E+02	25.0	XKBB73
13	1.700	9.246E+02	30.1	XKKH65
13	1.730	6.500E+02	25.1	XKBB73
13	1.840	6.700E+02	24.9	XKBB73
13	1.940	6.890E+02	25.0	XKBB73
13	2.000	8.100E+02	12.0	XNTA76
13	2.040	7.050E+02	25.0	XKBB73
13	2.140	7.230E+02	25.0	XKBB73
13	2.240	7.390E+02	25.0	XKBB73
13	2.350	7.540E+02	24.9	XKBB73
13	2.450	7.680E+02	25.0	XKBB73
13	2.500	9.470E+02	12.0	XNTA76
13	2.500	9.550E+02	10.0	XNBB78
13	2.550	7.810E+02	25.0	XKBB73
13	2.650	7.960E+02	25.0	XKBB73
13	2.750	8.090E+02	25.0	XKBB73
13	2.850	8.230E+02	25.0	XKBB73
13	2.960	8.370E+02	25.0	XKBB73
13	2.985	9.690E+02	12.0	XNSE80
13	3.000	9.930E+02	12.0	XNTA76
13	3.000	9.880E+02	10.0	XNBB78
13	3.000	1.088E+03	8.0	XNBS78
13	3.500	9.980E+02	12.0	XNTA76
13	3.940	9.920E+02	12.0	XNSE80
13	4.000	9.730E+02	12.0	XNTA76
13	5.000	9.160E+02	10.0	XNBB78
13	6.070	8.790E+02	12.0	XNSE80
13	7.500	7.800E+02	10.0	XNBB78
13	9.000	7.190E+02	10.0	XNBB78
13	9.130	7.330E+02	12.0	XNSE80
13	9.750	6.830E+02	10.0	XNBB78
13	12.210	6.130E+02	12.0	XNSE80
13	18.290	4.690E+02	12.0	XNSE80
13	23.980	3.940E+02	12.0	XNSE80
13	30.130	3.420E+02	12.0	XNSE80
13	35.610	2.970E+02	12.0	XNSE80
13	39.560	2.770E+02	12.0	XNSE80
14	.750	2.470E+02	12.1	XNTA76
14	1.000	3.950E+02	11.9	XNTA76
14	1.250	4.850E+02	12.0	XNTA76
14	1.500	6.350E+02	12.0	XNTA76
14	2.000	7.680E+02	12.0	XNTA76
14	2.500	8.610E+02	12.0	XNTA76
14	3.000	9.340E+02	12.0	XNTA76
14	3.000	1.000E+03	8.0	XNBS78
14	3.500	9.550E+02	12.0	XNTA76
14	4.000	9.600E+02	12.0	XNTA76
15	.600	1.590E+02	15.0	XNAV81
15	.800	3.486E+02	15.0	XNAV81
15	1.000	4.560E+02	15.0	XNAV81
15	1.200	5.592E+02	15.0	XNAV81
15	1.400	6.720E+02	15.0	XNAV81
15	1.400	5.270E+02	20.0	XNSN81
15	1.500	5.510E+02	20.0	XNSN81
15	1.600	7.920E+02	15.0	XNAV81
15	1.600	6.120E+02	20.0	XNSN81
15	1.700	6.330E+02	20.0	XNSN81
15	1.800	9.120E+02	15.0	XNAV81
15	1.800	6.700E+02	20.0	XNSN81
15	1.900	6.950E+02	20.0	XNSN81
15	2.000	9.240E+02	15.0	XNAV81
15	2.000	7.250E+02	20.0	XNSN81
15	2.200	9.720E+02	15.0	XNAV81
15	2.400	9.660E+02	15.0	XNAV81
15	2.600	9.480E+02	15.0	XNAV81
16	.600	1.550E+02	15.0	XNAV81
16	.800	2.952E+02	15.0	XNAV81
16	.830	1.800E+02	16.7	XNHO75B
16	1.000	4.043E+02	15.0	XNAV81
16	1.200	5.297E+02	15.0	XNAV81
16	1.400	6.429E+02	15.0	XNAV81
16	1.400	4.620E+02	20.0	XNSN81
16	1.500	4.560E+02	20.0	XNSN81
16	1.580	5.400E+02	14.8	XNHO75B
16	1.600	7.446E+02	15.0	XNAV81
16	1.600	4.910E+02	20.0	XNSN81
16	1.700	5.260E+02	20.0	XNSN81
16	1.800	8.093E+02	15.0	XNAV81
16	1.800	5.580E+02	20.0	XNSN81
16	1.900	5.650E+02	20.0	XNSN81
16	2.000	8.774E+02	15.0	XNAV81
16	2.000	5.970E+02	20.0	XNSN81
16	2.200	9.676E+02	15.0	XNAV81
16	2.400	9.840E+02	15.0	XNAV81
16	2.560	7.500E+02	14.7	XNHO75B
16	2.600	1.033E+03	15.0	XNAV81
16	3.000	8.460E+02	7.0	XNBS78
16	3.280	1.080E+03	14.8	XNHO75B
16	4.000	8.420E+02	10.0	XNBD78
17	.800	1.633E+02	15.0	XNAV81
17	.950	2.600E+02	15.0	XNMI76
17	1.000	2.636E+02	15.0	XNAV81
17	1.200	2.798E+02	15.0	XNAV81
17	1.400	3.314E+02	15.0	XNAV81
17	1.400	3.630E+02	15.0	XNSN81
17	1.500	3.740E+02	15.0	XNSN81
17	1.600	4.183E+02	15.0	XNAV81
17	1.600	4.020E+02	15.0	XNSN81
17	1.700	4.230E+02	15.0	XNSN81
17	1.800	4.804E+02	15.0	XNAV81
17	1.800	5.250E+02	15.0	XNSN81
17	1.900	5.150E+02	15.0	XNSN81
17	2.000	5.319E+02	15.0	XNAV81
17	2.000	5.340E+02	15.0	XNSN81
17	2.200	5.463E+02	15.0	XNAV81
17	2.400	5.329E+02	15.0	XNAV81
17	2.600	5.711E+02	15.0	XNAV81
17	2.800	5.529E+02	15.0	XNAV81
17	3.000	6.950E+02	7.2	XNBS78
18	.068	1.852E-02	11.8	XGHA73
18	.077	3.013E-02	11.8	XGHA73
18	.087	5.543E-02	12.0	XGHA73
18	.097	1.035E-01	12.2	XGHA73
18	.106	1.484E-01	11.6	XGHA73
18	.116	2.265E-01	12.2	XGHA73
18	.125	3.120E-01	25.0	XGLA76
18	.126	3.047E-01	12.1	XGHA73
18	.135	4.013E-01	12.0	XGHA73
18	.135	4.160E-01	35.0	XNLO80
18	.150	5.920E-01	25.0	XGLA76
18	.200	1.590E+00	25.2	XGLA76
18	.200	1.574E+00	15.0	XNLO80
18	.250	3.570E+00	24.9	XGLA76
18	.250	3.221E+00	15.0	XNLO80
18	.300	6.510E+00	25.0	XGLA76
18	.300	5.795E+00	15.0	XNLO80
18	.350	9.309E+00	10.0	XNLO80
18	.400	1.580E+01	24.7	XGLA76
18	.400	1.537E+01	10.0	XNLO80
18	.450	2.245E+01	10.0	XNLO80
18	.500	3.000E+01	25.0	XGLA76
18	.500	3.001E+01	10.0	XNLO80
18	.600	4.790E+01	25.1	XGLA76
18	.600	4.807E+01	10.0	XNLO80
18	.700	6.850E+01	25.0	XGLA76
18	.700	6.991E+01	10.0	XNLO80
18	.800	9.350E+01	25.0	XGLA76
18	.800	9.955E+01	10.0	XNLO80
18	.900	1.180E+02	21.6	XGLA76
18	.900	1.269E+02	10.0	XNLO80
18	1.000	1.400E+02	25.0	XGLA76
18	1.000	1.476E+02	10.0	XNLO80
18	1.100	1.696E+02	10.0	XNLO80
18	1.200	2.013E+02	10.0	XNLO80
18	1.300	2.306E+02	10.0	XNLO80
18	1.400	2.574E+02	10.0	XNLO80
18	1.500	3.190E+02	11.0	XGWN73A
18	1.500	2.891E+02	10.0	XNLO80
18	1.600	3.379E+02	10.0	XNLO80
18	1.700	3.660E+02	10.0	XNLO80
18	1.800	4.063E+02	10.0	XNLO80
18	1.890	4.720E+02	20.3	XGCZ75
18	1.900	4.355E+02	10.0	XNLO80
18	2.000	4.570E+02	14.0	XGWN73A
18	2.000	5.230E+02	15.3	XGRN76
18	2.000	4.160E+02	10.0	XNLO80
18	2.500	5.530E+02	14.0	XGWN73A
18	2.530	6.150E+02	20.0	XGCZ75
18	3.000	6.380E+02	14.0	XGWN73A
18	3.000	7.030E+02	20.1	XGCZ75
18	3.500	7.250E+02	14.0	XGWN73A
18	4.000	7.610E+02	14.0	XGWN73A
18	4.500	7.990E+02	14.0	XGWN73A
18	5.000	8.680E+02	14.0	XGWN73A
19	.600	4.043E+01	15.0	XNAV81
19	.800	8.653E+01	15.0	XNAV81
19	.950	1.210E+02	14.9	XNMI76
19	1.000	1.228E+02	15.0	XNAV81
19	1.200	1.794E+02	15.0	XNAV81
19	1.400	2.456E+02	15.0	XNAV81
19	1.400	2.200E+02	15.0	XNSN81
19	1.500	2.270E+02	15.0	XNSN81
19	1.600	2.981E+02	15.0	XNAV81
19	1.600	2.480E+02	15.0	XNSN81
19	1.700	2.700E+02	15.0	XNSN81
19	1.800	3.409E+02	15.0	XNAV81
19	1.800	3.430E+02	15.0	XNSN81
19	1.900	3.550E+02	15.0	XNSN81
19	2.000	4.306E+02	15.0	XNAV81
19	2.000	3.720E+02	15.0	XNSN81
19	2.200	5.216E+02	15.0	XNAV81
19	2.400	5.561E+02	15.0	XNAV81
19	2.600	5.727E+02	15.0	XNAV81
19	2.800	6.044E+02	15.0	XNAV81
19	3.000	5.350E+02	6.5	XNBS78
19	4.000	7.210E+02	10.0	XNBD78
20	.200	6.300E-01	10.0	XNLO79A
20	.225	8.100E-01	10.0	XNLO79A
20	.250	1.320E+00	10.0	XNLO79A
20	.275	1.710E+00	10.0	XNLO79A
20	.300	2.380E+00	10.0	XNLO79A
20	.325	3.160E+00	10.0	XNLO79A
20	.350	3.720E+00	10.0	XNLO79A
20	.375	5.000E+00	10.0	XNLO79A
20	.400	5.880E+00	10.0	XNLO79A
20	.425	7.620E+00	10.0	XNLO79A
20	.450	8.860E+00	10.0	XNLO79A
20	.475	1.100E+01	10.0	XNLO79A
20	.500	1.280E+01	10.0	XNLO79A
20	.500	1.090E+01	10.0	XNLO78
20	.550	1.580E+01	10.0	XNLO78
20	.600	1.950E+01	10.0	XNLO78
20	.600	2.641E+01	15.0	XNAV81
20	.650	2.550E+01	10.0	XNLO78
20	.700	3.160E+01	10.0	XNLO78
20	.750	3.770E+01	10.0	XNLO78
20	.800	4.360E+01	10.0	XNLO78
20	.800	5.656E+01	15.0	XNAV81
20	.850	5.070E+01	10.0	XNLO78
20	.900	5.850E+01	10.0	XNLO78
20	.950	6.610E+01	10.0	XNLO78
20	1.000	7.500E+01	10.0	XNLO78
20	1.000	9.161E+01	15.0	XNAV81
20	1.050	8.410E+01	10.0	XNLO78
20	1.100	9.470E+01	10.0	XNLO78
20	1.150	1.010E+02	10.0	XNLO78
20	1.200	1.100E+02	10.0	XNLO78
20	1.200	1.399E+02	15.0	XNAV81
20	1.250	1.170E+02	10.0	XNLO78
20	1.300	1.290E+02	10.0	XNLO78
20	1.350	1.380E+02	10.0	XNLO78
20	1.400	1.440E+02	10.0	XNLO78
20	1.400	1.809E+02	15.0	XNAV81
20	1.450	1.550E+02	10.0	XNLO78
20	1.500	1.670E+02	10.0	XNLO78
20	1.550	1.750E+02	10.0	XNLO78
20	1.600	1.890E+02	10.0	XNLO78
20	1.600	2.233E+02	15.0	XNAV81
20	1.650	1.970E+02	10.0	XNLO78
20	1.700	2.000E+02	10.0	XNLO78
20	1.750	2.160E+02	10.0	XNLO78
20	1.800	2.230E+02	10.0	XNLO78
20	1.800	2.608E+02	15.0	XNAV81
20	1.850	2.250E+02	10.0	XNLO78
20	1.900	2.460E+02	10.0	XNLO78
20	1.950	2.510E+02	10.0	XNLO78
20	2.000	2.347E+02	17.4	XNBI70
20	2.000	2.660E+02	10.0	XNLO78
20	2.000	2.934E+02	15.0	XNAV81
20	2.200	3.733E+02	15.0	XNAV81
20	2.400	4.238E+02	15.0	XNAV81
20	2.600	1.238E+02	15.0	XNAV81
20	2.800	4.564E+02	15.0	XNAV81
20	3.000	3.945E+02	17.1	XNBI70
20	3.000	4.850E+02	6.6	XNBS78
20	4.000	4.939E+02	17.5	XNBI70
20	5.000	5.754E+02	17.3	XNBI70
20	6.000	5.933E+02	17.3	XNBI70
20	7.000	6.324E+02	17.3	XNBI70
20	8.000	6.292E+02	17.4	XNBI70
20	9.000	6.357E+02	18.2	XNBI70
20	10.000	6.129E+02	18.1	XNBI70
20	11.000	6.161E+02	18.3	XNBI70
20	12.000	6.178E+02	18.2	XNBI70
20	13.000	6.080E+02	18.0	XNBI70
20	16.500	5.395E+02	18.1	XNBI70
20	20.000	5.460E+02	19.7	XNBI70
20	25.000	4.483E+02	20.0	XNBI70
21	.200	3.600E-01	10.0	XNLO79A
21	.200	3.762E-01	8.0	XNJL80
21	.225	5.700E-01	10.0	XNLO79A
21	.250	9.700E-01	10.0	XNLO79A
21	.275	1.320E+00	10.0	XNLO79A
21	.300	1.870E+00	10.0	XNLO79A
21	.300	1.820E+00	7.0	XNJL80
21	.325	2.150E+00	10.0	XNLO79A
21	.350	3.200E+00	10.0	XNLO79A
21	.375	4.170E+00	10.0	XNLO79A
21	.400	4.070E+00	10.0	XNLO79A
21	.400	4.978E+00	7.0	XNJL80
21	.425	5.130E+00	10.0	XNLO79A

APPENDIX: TABLE II. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger-Electron Production, and Ionization

See page 112 for Explanation of Tables

ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
.450	6.240E+00	10.0	XNLO79A	22	.150	1.230E-01	30.0	XKZA79	22	1.450	1.120E+02	10.0	XNLO78
.475	7.330E+00	10.0	XNLO79A	22	.160	9.500E-02	20.0	XNBG80	22	1.500	1.099E+02	80.0	XKOG64
.500	9.820E+00	10.0	XNLO79A	22	.170	1.649E-01	5.0	XKSH71	22	1.500	1.360E+02	10.3	XKRS76
.500	9.880E+00	10.0	XNLO78	22	.185	1.770E-01	20.0	XNBG80	22	1.500	1.300E+02	7.7	XNAK74
.500	9.956E+00	6.0	XNJL80	22	.200	1.099E-01	80.0	XKJO62	22	1.500	1.180E+02	10.0	XNLO78
.500	9.550E+00	5.0	XNBR84	22	.200	2.700E-01	10.0	XNLO79A	22	1.500	1.570E+02	12.0	XNBO75
.550	1.320E+01	10.0	XNLO78	22	.200	2.740E-01	6.0	XNLA79	22	1.500	1.441E+02	15.0	XNBA80
.600	1.720E+01	10.0	XNLO78	22	.225	3.900E-01	10.0	XNLO79A	22	1.550	1.270E+02	10.0	XNLO78
.600	1.740E+01	6.0	XNJL80	22	.250	6.200E-01	10.0	XNLO79A	22	1.600	1.550E+02	10.3	XKRS76
.600	1.711E+01	5.0	XNBR84	22	.275	8.200E-01	10.0	XNLO79A	22	1.600	1.340E+02	10.0	XNLO78
.650	2.140E+01	10.0	XNLO78	22	.285	9.570E-01	20.0	XNBG80	22	1.600	1.430E+02	5.0	XNLA79
.667	2.370E+01	15.2	XNHO75A	22	.300	4.708E-01	80.0	XKJO62	22	1.600	1.820E+02	12.0	XNBO75
.700	2.620E+01	10.0	XNLO78	22	.300	1.320E+00	10.0	XNLO79A	22	1.600	1.382E+02	4.8	XNBR82
.700	2.622E+01	6.0	XNJL80	22	.300	1.480E+00	6.0	XNLA79	22	1.600	1.318E+02	5.0	XNBR84
.700	2.707E+01	5.0	XNBR84	22	.300	1.200E+00	5.5	XNBR82	22	1.650	1.410E+02	10.0	XNLO78
.750	3.130E+01	10.0	XNLO78	22	.325	1.430E+00	10.0	XNLO79A	22	1.700	1.740E+02	9.8	XKRS76
.800	3.720E+01	10.0	XNLO78	22	.350	2.010E+00	10.0	XNLO79A	22	1.700	1.490E+02	10.0	XNLO78
.800	3.534E+01	6.0	XNJL80	22	.375	2.480E+00	10.0	XNLO79A	22	1.700	1.940E+02	12.0	XNBO75
.800	3.910E+01	5.0	XNBR84	22	.390	2.890E+00	20.0	XNBG80	22	1.750	1.550E+02	10.0	XNLO78
.830	4.200E+01	14.3	XNHO75B	22	.400	1.200E+00	80.0	XKJO62	22	1.750	1.866E+02	15.0	XNBA80
.850	4.330E+01	10.0	XNLO78	22	.400	3.170E+00	10.0	XNLO79A	22	1.800	1.930E+02	9.8	XKRS76
.900	4.970E+01	10.0	XNLO78	22	.400	3.990E+00	6.0	XNLA79	22	1.800	1.610E+02	10.0	XNLO78
.900	4.807E+01	6.0	XNJL80	22	.400	4.600E+00	12.0	XNBO75	22	1.800	2.220E+02	12.0	XNBO75
.950	5.700E+01	10.0	XNLO78	22	.400	3.373E+00	5.4	XNBR82	22	1.800	1.726E+02	4.8	XNBR82
1.000	6.830E+01	14.9	XNHO75A	22	.425	3.630E+00	10.0	XNLO79A	22	1.800	1.647E+02	5.0	XNBR84
1.000	6.560E+01	10.0	XNLO78	22	.441	1.629E+00	80.0	XKJO62	22	1.850	1.720E+02	10.0	XNLO78
1.000	6.862E+01	5.0	XNBR84	22	.450	4.560E+00	10.0	XNLO79A	22	1.900	2.080E+02	10.1	XKRS76
1.050	7.330E+01	10.0	XNLO78	22	.475	5.570E+00	10.0	XNLO79A	22	1.900	1.780E+02	10.0	XNLO78
1.100	8.270E+01	10.0	XNLO78	22	.490	6.080E+00	20.0	XNBG80	22	1.900	2.380E+02	12.0	XNBO75
1.150	8.980E+01	10.0	XNLO78	22	.500	2.100E+00	80.0	XKJO62	22	1.950	1.900E+02	10.0	XNLO78
1.200	9.710E+01	10.0	XNLO78	22	.500	6.640E+00	10.0	XNLO79A	22	2.000	2.212E+02	15.8	XNBI70
1.200	1.025E+02	5.0	XNBR84	22	.500	6.650E+00	10.0	XNLO78	22	2.000	2.300E+02	10.0	XKRS76
1.250	1.060E+02	10.0	XNLO78	22	.500	7.990E+00	6.0	XNLA79	22	2.000	2.150E+02	7.9	XNAK74
1.300	1.150E+02	10.0	XNLO78	22	.500	9.400E+00	12.0	XNBO75	22	2.000	1.531E+02	9.0	XNKN77
1.333	1.290E+02	14.7	XNHO75A	22	.500	7.096E+00	5.3	XNBR82	22	2.000	1.960E+02	10.0	XNLO78
1.350	1.220E+02	10.0	XNLO78	22	.500	6.877E+00	5.0	XNBR84	22	2.000	2.150E+02	3.0	XNLA79
1.400	1.320E+02	10.0	XNLO78	22	.550	9.800E+00	10.0	XNLO78	22	2.000	2.590E+02	12.0	XNBO75
1.400	1.404E+02	5.0	XNBR84	22	.590	1.080E+01	20.0	XNBG80	22	2.000	2.343E+02	15.0	XNBA80
1.450	1.400E+02	10.0	XNLO78	22	.600	1.320E+01	10.0	XNLO78	22	2.000	2.078E+02	4.7	XNBR82
1.500	1.480E+02	10.0	XNLO78	22	.600	1.320E+01	6.0	XNLA79	22	2.000	1.969E+02	5.0	XNBR84
1.550	1.570E+02	10.0	XNLO78	22	.600	1.600E+01	12.0	XNBO75	22	2.100	2.500E+02	10.0	XKRS76
1.580	1.790E+02	15.1	XNHO75B	22	.600	1.259E+01	5.2	XNBR82	22	2.100	2.740E+02	12.0	XNBO75
1.600	1.650E+02	10.0	XNLO78	22	.600	1.235E+01	5.0	XNBR84	22	2.200	2.670E+02	10.1	XKRS76
1.600	1.750E+02	5.0	XNBR84	22	.650	1.630E+01	10.0	XNLO78	22	2.200	3.030E+02	12.0	XNBO75
1.650	1.690E+02	10.0	XNLO78	22	.690	1.700E+01	20.0	XNBG80	22	2.200	2.365E+02	4.7	XNBR82
1.667	2.140E+02	15.0	XNHO75A	22	.700	1.990E+01	10.0	XNLO78	22	2.200	2.256E+02	5.0	XNBR84
1.700	1.800E+02	10.0	XNLO78	22	.700	2.500E+01	12.0	XNBO75	22	2.250	1.852E+02	14.7	XNBE73
1.750	1.890E+02	10.0	XNLO78	22	.700	1.903E+01	5.1	XNBR82	22	2.250	2.759E+02	15.0	XNBA80
1.800	1.990E+02	10.0	XNLO78	22	.700	1.936E+01	5.0	XNBR84	22	2.300	2.850E+02	10.2	XKRS76
1.800	2.049E+02	5.0	XNBR84	22	.750	2.320E+01	10.0	XNLO78	22	2.300	3.200E+02	12.0	XNBO75
1.850	2.060E+02	10.0	XNLO78	22	.790	2.500E+01	20.0	XNBG80	22	2.300	2.497E+02	5.0	XNBR84
1.900	2.150E+02	10.0	XNLO78	22	.800	2.820E+01	10.0	XNLO78	22	2.400	3.000E+02	10.0	XKRS76
1.950	2.260E+02	10.0	XNLO78	22	.800	3.000E+01	6.0	XNLA79	22	2.400	2.720E+02	6.0	XNLA79
2.000	2.680E+02	14.9	XNHO75A	22	.800	3.700E+01	12.0	XNBO75	22	2.400	2.694E+02	4.7	XNBR82
2.000	2.300E+02	10.0	XNLO78	22	.800	2.869E+01	5.0	XNBR82	22	2.400	2.606E+02	5.0	XNBR84
2.000	2.425E+02	5.0	XNBR84	22	.800	2.803E+01	5.0	XNBR84	22	2.500	3.180E+02	10.1	XKRS76
2.200	2.839E+02	5.0	XNBR84	22	.850	3.300E+01	10.0	XNLO78	22	2.500	3.120E+02	8.0	XNAK74
2.300	3.064E+02	5.0	XNBR84	22	.900	3.780E+01	10.0	XNLO78	22	2.500	3.197E+02	15.0	XNBA80
2.333	3.170E+02	15.1	XNHO75A	22	.900	4.800E+01	12.0	XNBO75	22	2.500	2.759E+02	5.0	XNBR84
2.400	3.140E+02	5.0	XNBR84	22	.950	4.410E+01	10.0	XNLO78	22	2.600	3.310E+02	10.0	XKRS76
2.500	3.252E+02	5.0	XNBR84	22	1.000	5.450E+01	10.1	XKRS76	22	2.700	3.460E+02	10.1	XKRS76
2.560	3.210E+02	15.0	XNHO75B	22	1.000	3.749E+01	15.2	XNBE73	22	2.750	3.701E+02	15.0	XNBA80
2.667	3.790E+02	15.0	XNHO75A	22	1.000	3.679E+01	8.9	XNKN77	22	2.800	3.590E+02	10.0	XKRS76
3.000	4.440E+02	15.1	XNHO75A	22	1.000	4.890E+01	10.0	XNLO78	22	2.900	3.720E+02	9.9	XKRS76
3.280	4.710E+02	15.1	XNHO75B	22	1.000	5.150E+01	3.0	XNLA79	22	3.000	3.701E+02	16.0	XNBI70
3.333	4.850E+02	15.1	XNHO75A	22	1.000	6.400E+01	12.0	XNBO75	22	3.000	3.850E+02	10.1	XKRS76
3.667	5.970E+02	15.1	XNHO75A	22	1.000	5.869E+01	15.0	XNBA80	22	3.000	2.931E+02	15.1	XNBE73
4.000	5.760E+02	14.9	XNHO75A	22	1.000	4.884E+01	5.0	XNBR82	22	3.000	3.830E+02	8.1	XNAK74
4.333	6.050E+02	15.0	XNHO75A	22	1.000	4.884E+01	5.0	XNBR84	22	3.000	4.117E+02	15.0	XNBA80
4.667	6.370E+02	15.1	XNHO75A	22	1.000	4.940E+01	10.0	XNGB86	22	3.000	3.700E+02	5.4	XNBS78
5.000	6.690E+02	14.9	XNHO75A	22	1.050	5.430E+01	10.0	XNLO78	22	4.000	4.906E+02	16.1	XNBI70
5.333	6.810E+02	15.0	XNHO75A	22	1.100	6.830E+01	10.0	XKRS76	22	4.000	4.800E+02	10.0	XNPE79
5.667	6.990E+02	15.0	XNHO75A	22	1.100	6.110E+01	10.0	XNLO78	22	4.960	5.820E+02	8.1	XNAK74
.060	2.090E-04	30.0	XKZA79	22	1.100	8.000E+01	12.0	XNBO75	22	5.000	5.519E+02	16.3	XNBI70
.070	6.160E-04	30.0	XKZA79	22	1.150	6.770E+01	10.0	XNLO78	22	5.000	5.173E+02	9.0	XNKN77
.080	1.570E-03	30.0	XKZA79	22	1.200	8.370E+01	10.0	XKRS76	22	5.960	6.500E+02	8.0	XNAK74
.085	3.770E-03	22.0	XNBG80	22	1.200	7.340E+01	10.0	XNLO78	22	6.000	6.263E+02	16.1	XNBI70
.090	5.650E-03	5.0	XKSH71	22	1.200	8.120E+01	5.0	XNLA79	22	6.000	6.810E+02	10.0	XNPE79
.090	3.570E-03	30.0	XKZA79	22	1.200	9.700E+01	12.0	XNBO75	22	6.960	7.080E+02	8.1	XNAK74
.100	8.979E-03	29.3	XKDU72	22	1.200	7.665E+01	4.9	XNBR82	22	7.000	6.636E+02	16.2	XNBI70
.100	7.450E-03	30.0	XKZA79	22	1.200	7.293E+01	5.0	XNBR84	22	8.000	6.964E+02	16.0	XNBI70
.110	1.879E-02	5.0	XKSH71	22	1.250	8.060E+01	10.0	XNLO78	22	8.000	7.160E+02	10.0	XNPE79
.110	1.450E-02	30.0	XKZA79	22	1.250	9.483E+01	15.0	XNBA80	22	8.940	7.480E+02	8.0	XNAK74
.110	1.560E-02	22.0	XNBG80	22	1.300	1.020E+02	9.8	XKRS76	22	9.000	6.964E+02	16.4	XNBI70
.120	2.640E-02	30.0	XKZA79	22	1.300	8.970E+01	10.0	XNLO78	22	10.000	7.183E+02	16.2	XNBI70
.125	2.650E-02	29.8	XKDU72	22	1.300	1.130E+02	12.0	XNBO75	22	10.000	7.560E+02	10.0	XNPE79
.130	4.533E-02	4.8	XKSH71	22	1.350	9.610E+01	10.0	XNLO78	22	10.930	7.580E+02	8.0	XNAK74
.130	4.580E-02	30.0	XKZA79	22	1.400	1.180E+02	10.2	XKRS76	22	11.000	6.920E+02	16.8	XNBI70
.135	4.340E-02	21.0	XNBG80	22	1.400	1.050E+02	10.0	XNLO78	22	12.000	6.745E+02	16.9	XNBI70
.140	7.630E-02	30.0	XKZA79	22	1.400	1.330E+02	12.0	XNBO75	22	12.000	7.400E+02	10.0	XNPE79
.150	9.001E-02	5.1	XKSH71	22	1.400	1.075E+02	4.9	XNBR82	22	13.000	6.942E+02	16.7	XNBI70
.150	6.417E-02	30.0	XKDU72	22	1.400	1.021E+02	5.0	XNBR84	22	14.000	7.160E+02	10.0	XNPE79

APPENDIX: TABLE II. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger-Electron Production, and Ionization

See page 112 for Explanation of Tables

Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
22	16.000	6.700E+02	10.0	XNPE79
22	16.500	6.942E+02	16.7	XNBI70
22	18.000	6.130E+02	10.0	XNPE79
22	20.000	6.132E+02	17.5	XNBI70
22	20.000	5.930E+02	10.0	XNPE79
22	22.000	5.650E+02	10.0	XNPE79
22	25.000	5.760E+02	17.5	XNBI70
23	.100	5.575E-03	30.0	XKDU72
23	.125	1.750E-02	30.0	XKDU72
23	.150	4.300E-02	30.2	XKDU72
23	.200	1.450E-01	10.0	XNLO79A
23	.225	2.420E-01	10.0	XNLO79A
23	.250	3.610E-01	10.0	XNLO79A
23	.275	4.710E-01	10.0	XNLO79A
23	.300	7.800E-01	10.0	XNLO79A
23	.325	1.090E+00	10.0	XNLO79A
23	.350	1.420E+00	10.0	XNLO79A
23	.375	1.900E+00	10.0	XNLO79A
23	.400	2.230E+00	10.0	XNLO79A
23	.400	3.000E+00	12.0	XNBO75
23	.425	2.820E+00	10.0	XNLO79A
23	.450	3.310E+00	10.0	XNLO79A
23	.475	4.300E+00	10.0	XNLO79A
23	.500	5.170E+00	10.0	XNLO79A
23	.500	4.800E+00	10.0	XNLO78
23	.500	5.630E+00	10.0	XNLA79
23	.500	6.300E+00	12.0	XNBO75
23	.550	6.920E+00	10.0	XNLO78
23	.600	8.750E+00	10.0	XNLO78
23	.600	1.110E+01	10.0	XNLA79
23	.600	1.110E+01	12.0	XNBO75
23	.650	1.130E+01	10.0	XNLO78
23	.700	1.400E+01	10.0	XNLO78
23	.700	1.730E+01	12.0	XNBO75
23	.750	1.660E+01	10.0	XNLO78
23	.800	2.060E+01	10.0	XNLO78
23	.800	2.040E+01	10.0	XNLA79
23	.800	2.600E+01	12.0	XNBO75
23	.850	2.520E+01	10.0	XNLO78
23	.900	2.920E+01	10.0	XNLO78
23	.900	3.500E+01	12.0	XNBO75
23	.950	2.270E+01	10.1	XNMI76
23	.950	3.390E+01	10.0	XNLO78
23	1.000	3.520E+01	5.1	XKRS77
23	1.000	3.840E+01	10.0	XNLO78
23	1.000	4.000E+01	10.0	XNLA79
23	1.000	4.500E+01	12.0	XNBO75
23	1.000	4.125E+01	10.0	XNBR84
23	1.050	4.290E+01	10.0	XNLO78
23	1.100	4.380E+01	5.0	XKRS77
23	1.100	4.720E+01	10.0	XNLO78
23	1.100	5.800E+01	12.0	XNBO75
23	1.150	5.430E+01	10.0	XNLO78
23	1.200	5.910E+01	10.0	XNLO78
23	1.200	5.750E+01	10.0	XNLA79
23	1.200	7.000E+01	12.0	XNBO75
23	1.200	6.200E+01	10.0	XNBR84
23	1.250	6.570E+01	10.0	XNLO78
23	1.300	6.350E+01	5.8	XKRS77
23	1.300	7.170E+01	10.0	XNLO78
23	1.300	8.700E+01	12.0	XNBO75
23	1.350	7.790E+01	10.0	XNLO78
23	1.400	8.350E+01	10.0	XNLO78
23	1.400	1.000E+02	12.0	XNBO75
23	1.400	8.400E+01	10.0	XNBR84
23	1.450	9.050E+01	10.0	XNLO78
23	1.500	8.600E+01	5.0	XKRS77
23	1.500	9.480E+01	10.0	XNLO78
23	1.500	1.180E+02	12.0	XNBO75
23	1.550	1.050E+02	10.0	XNLO78
23	1.600	1.100E+02	10.0	XNLO78
23	1.600	1.100E+02	10.0	XNLA79
23	1.600	1.340E+02	12.0	XNBO75
23	1.600	1.085E+02	10.0	XNBR84
23	1.650	1.160E+02	10.0	XNLO78
23	1.700	1.110E+02	5.4	XKRS77
23	1.700	1.230E+02	10.0	XNLO78
23	1.700	1.470E+02	12.0	XNBO75
23	1.750	1.280E+02	10.0	XNLO78
23	1.800	1.340E+02	10.0	XNLO78
23	1.800	1.690E+02	12.0	XNBO75
23	1.800	1.348E+02	10.0	XNBR84
23	1.850	1.400E+02	10.0	XNLO78
23	1.900	1.370E+02	5.1	XKRS77
23	1.900	1.450E+02	10.0	XNLO78
23	1.900	1.810E+02	12.0	XNBO75
23	1.950	1.560E+02	10.0	XNLO78
23	2.000	1.510E+02	5.3	XKRS77
23	2.000	1.560E+02	10.0	XNLO78
23	2.000	1.540E+02	10.0	XNLA79
23	2.000	1.970E+02	12.0	XNBO75
23	2.000	1.710E+02	10.0	XNBR84
23	2.100	1.640E+02	4.9	XKRS77
23	2.100	2.130E+02	12.0	XNBO75
23	2.200	2.190E+02	12.0	XNBO75
23	2.200	1.885E+02	10.0	XNBR84
23	2.250	1.433E+02	15.1	XNBE73
23	2.300	1.900E+02	5.3	XKRS77
23	2.300	2.440E+02	12.0	XNBO75
23	2.400	2.210E+02	10.0	XNLA79
23	2.500	2.140E+02	5.1	XKRS77
23	2.700	2.350E+02	5.1	XKRS77
23	2.800	2.450E+02	4.9	XKRS77
23	2.900	2.530E+02	5.1	XKRS77
23	3.000	2.630E+02	4.9	XKRS77
24	.100	2.820E-03	30.0	XKDU72
24	.125	1.043E-02	29.7	XKDU72
24	.150	2.707E-02	30.2	XKDU72
24	.200	1.070E-01	10.0	XNLO79A
24	.225	1.690E-01	10.0	XNLO79A
24	.250	2.770E-01	10.0	XNLO79A
24	.275	4.050E-01	10.0	XNLO79A
24	.300	5.800E-01	10.0	XNLO79A
24	.300	6.373E-01	5.5	XNBR82
24	.325	7.800E-01	10.0	XNLO79A
24	.350	1.020E+00	10.0	XNLO79A
24	.375	1.310E+00	10.0	XNLO79A
24	.400	1.680E+00	10.0	XNLO79A
24	.400	1.715E+00	5.4	XNBR82
24	.425	2.020E+00	10.0	XNLO79A
24	.450	2.470E+00	10.0	XNLO79A
24	.475	2.980E+00	10.0	XNLO79A
24	.500	3.660E+00	10.0	XNLO79A
24	.500	3.520E+00	10.0	XNLO78
24	.500	4.900E+00	12.0	XNBO75
24	.500	3.863E+00	5.3	XNBR82
24	.550	4.840E+00	10.0	XNLO78
24	.600	6.570E+00	10.0	XNLO78
24	.600	8.700E+00	12.0	XNBO75
24	.600	6.965E+00	5.2	XNBR82
24	.650	8.190E+00	10.0	XNLO78
24	.700	1.020E+01	10.0	XNLO78
24	.700	1.370E+01	12.0	XNBO75
24	.700	1.055E+01	5.1	XNBR82
24	.750	1.240E+01	10.0	XNLO78
24	.800	1.490E+01	10.0	XNLO78
24	.800	2.090E+01	12.0	XNBO75
24	.800	1.588E+01	5.0	XNBR82
24	.800	1.503E+01	5.0	XNBR84
24	.850	1.790E+01	10.0	XNLO78
24	.900	1.600E+01	12.5	XNKL76
24	.900	2.070E+01	10.0	XNLO78
24	.900	2.700E+01	12.0	XNBO75
24	.950	1.730E+01	9.8	XNMI76
24	.950	2.380E+01	10.0	XNLO78
24	1.000	2.500E+01	13.2	XNKL76
24	1.000	2.800E+01	10.0	XNLO78
24	1.000	3.600E+01	12.0	XNBO75
24	1.000	2.848E+01	5.0	XNBR82
24	1.000	2.764E+01	5.0	XNBR84
24	1.050	3.130E+01	10.0	XNLO78
24	1.100	2.800E+01	12.9	XNKL76
24	1.100	3.600E+01	10.0	XNLO78
24	1.100	4.700E+01	12.0	XNBO75
24	1.150	3.870E+01	10.0	XNLO78
24	1.200	3.600E+01	13.1	XNKL76
24	1.200	4.400E+01	10.0	XNLO78
24	1.200	5.500E+01	12.0	XNBO75
24	1.200	4.484E+01	4.9	XNBR82
24	1.200	4.315E+01	5.0	XNBR84
24	1.250	4.920E+01	10.0	XNLO78
24	1.300	4.200E+01	12.9	XNKL76
24	1.300	5.290E+01	10.0	XNLO78
24	1.300	6.800E+01	12.0	XNBO75
24	1.350	5.770E+01	10.0	XNLO78
24	1.400	5.200E+01	12.9	XNKL76
24	1.400	6.310E+01	10.0	XNLO78
24	1.400	8.200E+01	12.0	XNBO75
24	1.400	6.373E+01	4.9	XNBR82
24	1.400	6.176E+01	5.0	XNBR84
24	1.450	6.740E+01	10.0	XNLO78
24	1.500	7.260E+01	10.0	XNLO78
24	1.500	9.400E+01	12.0	XNBO75
24	1.550	7.820E+01	10.0	XNLO78
24	1.600	6.100E+01	13.0	XNKL76
24	1.600	8.490E+01	10.0	XNLO78
24	1.600	1.080E+02	12.0	XNBO75
24	1.600	8.460E+01	4.8	XNBR82
24	1.600	8.291E+01	5.0	XNBR84
24	1.650	8.930E+01	10.0	XNLO78
24	1.700	7.200E+01	12.9	XNKL76
24	1.700	9.460E+01	10.0	XNLO78
24	1.700	1.240E+02	12.0	XNBO75
24	1.750	1.000E+02	10.0	XNLO78
24	1.800	8.500E+01	12.9	XNKL76
24	1.800	1.060E+02	10.0	XNLO78
24	1.800	1.350E+02	12.0	XNBO75
24	1.800	1.094E+02	4.8	XNBR82
24	1.800	1.080E+02	5.0	XNBR84
24	1.850	1.130E+02	10.0	XNLO78
24	1.900	8.200E+01	13.4	XNKL76
24	1.900	1.170E+02	10.0	XNLO78
24	1.900	1.490E+02	12.0	XNBO75
24	1.950	1.220E+02	10.0	XNLO78
24	2.000	8.600E+01	12.8	XNKL76
24	2.000	1.280E+02	10.0	XNLO78
24	2.000	1.670E+02	12.0	XNBO75
24	2.000	1.269E+02	4.7	XNBR82
24	2.000	1.292E+02	5.0	XNBR84
24	2.100	1.200E+02	13.3	XNKL76
24	2.100	1.770E+02	12.0	XNBO75
24	2.200	1.140E+02	13.2	XNKL76
24	2.200	1.980E+02	12.0	XNBO75
24	2.200	1.520E+02	4.7	XNBR82
24	2.200	1.483E+02	5.0	XNBR84
24	2.300	1.290E+02	13.2	XNKL76
24	2.300	1.610E+02	5.0	XNBR84
24	2.400	1.370E+02	13.1	XNKL76
24	2.400	1.779E+02	4.7	XNBR82
24	2.400	1.717E+02	5.0	XNBR84
24	2.500	1.710E+02	12.9	XNKL76
24	2.500	1.785E+02	5.0	XNBR84
24	3.000	2.810E+02	5.0	XNBS78
24	7.000	3.976E+02	14.2	XNDO81A
25	.100	1.130E-03	30.6	XKDU72
25	.125	4.239E-03	29.6	XKDU72
25	.150	1.287E-02	29.3	XKDU72
25	.200	4.800E-02	10.0	XNLO79A
25	.225	8.900E-02	10.0	XNLO79A
25	.250	1.550E-01	10.0	XNLO79A
25	.275	2.210E-01	10.0	XNLO79A
25	.300	3.450E-01	10.0	XNLO79A
25	.325	4.600E-01	10.0	XNLO79A
25	.350	6.600E-01	10.0	XNLO79A
25	.375	8.500E-01	10.0	XNLO79A
25	.400	1.110E+00	10.0	XNLO79A
25	.400	1.550E+00	12.0	XNBO75
25	.425	1.430E+00	10.0	XNLO79A
25	.450	1.730E+00	10.0	XNLO79A
25	.475	2.150E+00	10.0	XNLO79A
25	.500	2.600E+00	10.0	XNLO79A
25	.500	2.640E+00	10.0	XNLO78
25	.500	2.370E+00	10.0	XNLA79
25	.500	3.200E+00	12.0	XNBO75
25	.550	3.640E+00	10.0	XNLO78
25	.600	4.720E+00	10.0	XNLO78
25	.600	4.240E+00	10.0	XNLA79
25	.600	5.700E+00	12.0	XNBO75
25	.650	6.120E+00	10.0	XNLO78
25	.700	7.440E+00	10.0	XNLO78
25	.700	9.300E+00	12.0	XNBO75
25	.750	9.140E+00	10.0	XNLO78
25	.800	1.110E+01	10.0	XNLO78
25	.800	1.080E+01	10.0	XNLA79
25	.800	1.450E+01	12.0	XNBO75
25	.850	1.320E+01	10.0	XNLO78
25	.900	1.550E+01	10.0	XNLO78
25	.900	1.980E+01	12.0	XNBO75
25	.950	1.810E+01	10.0	XNLO78
25	1.000	2.100E+01	10.0	XNLO78
25	1.000	2.180E+01	10.0	XNLA79
25	1.000	2.600E+01	12.0	XNBO75
25	1.050	2.380E+01	10.0	XNLO78
25	1.100	2.680E+01	10.0	XNLO78
25	1.100	3.400E+01	12.0	XNBO75
25	1.150	3.000E+01	10.0	XNLO78
25	1.200	3.350E+01	10.0	XNLO78
25	1.200	3.520E+01	10.0	XNLA79
25	1.200	4.100E+01	12.0	XNBO75
25	1.250	3.750E+01	10.0	XNLO78
25	1.300	4.030E+01	10.0	XNLO78
25	1.300	5.100E+01	12.0	XNBO75
25	1.350	4.430E+01	10.0	XNLO78
25	1.400	4.830E+01	10.0	XNLO78
25	1.400	6.000E+01	12.0	XNBO75
25	1.450	5.370E+01	10.0	XNLO78
25	1.500	5.620E+01	10.0	XNLO78
25	1.500	6.900E+01	12.0	XNBO75
25	1.550	6.150E+01	10.0	XNLO78
25	1.600	6.580E+01	10.0	XNLO78
25	1.600	6.630E+01	10.0	XNLA79
25	1.600	8.200E+01	12.0	XNBO75
25	1.650	7.040E+01	10.0	XNLO78
25	1.700	7.290E+01	10.0	XNLO78
25	1.700	9.300E+01	12.0	XNBO75
25	1.750	7.930E+01	10.0	XNLO78

APPENDIX: TABLE II. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger-Electron Production, and Ionization

See page 112 for Explanation of Tables

Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
25	1.800	8.290E+01	10.0	XNLO78
25	1.800	1.060E+02	12.0	XNBO75
25	1.850	8.620E+01	10.0	XNLO78
25	1.900	9.130E+01	10.0	XNLO78
25	1.900	1.190E+02	12.0	XNBO75
25	1.950	9.920E+01	10.0	XNLO78
25	2.000	1.030E+02	10.0	XNLO78
25	2.000	1.030E+02	10.0	XNLA79
25	2.000	1.330E+02	12.0	XNBO75
25	2.100	1.400E+02	12.0	XNBO75
25	2.200	1.590E+02	12.0	XNBO75
25	2.300	1.710E+02	12.0	XNBO75
25	2.400	1.590E+02	10.0	XNLA79
25	2.500	1.560E+02	15.0	XNLI73
25	3.000	2.260E+02	15.0	XNLI73
25	3.000	2.180E+02	5.0	XNBS78
25	4.500	3.560E+02	15.0	XNLI73
25	6.000	4.460E+02	15.0	XNLI73
25	6.500	4.370E+02	15.0	XNLI73
25	7.000	5.190E+02	15.0	XNLI73
25	8.000	5.280E+02	15.0	XNLI73
25	8.500	5.970E+02	15.0	XNLI73
25	9.000	5.680E+02	15.0	XNLI73
25	10.000	5.470E+02	15.0	XNLI73
25	10.500	5.640E+02	15.0	XNLI73
25	11.000	6.200E+02	15.0	XNLI73
25	12.000	6.560E+02	15.0	XNLI73
26	.070	2.850E-05	30.0	XKZA79
26	.080	8.160E-05	30.0	XKZA79
26	.085	3.090E-04	22.0	XNBG80
26	.090	2.060E-04	30.0	XKZA79
26	.100	1.180E-03	29.4	XKDU72
26	.100	4.730E-04	30.0	XKZA79
26	.110	9.950E-04	30.0	XKZA79
26	.110	1.670E-03	22.0	XNBG80
26	.120	1.960E-03	30.0	XKZA79
26	.125	4.789E-03	29.7	XKDU72
26	.130	3.660E-03	30.0	XKZA79
26	.135	5.710E-03	21.0	XNBG80
26	.140	7.391E-03	15.0	XKME58
26	.140	6.500E-03	30.0	XKZA79
26	.150	1.214E-02	28.6	XKDU72
26	.150	1.048E-02	14.9	XKME58
26	.150	1.110E-02	30.0	XKZA79
26	.160	1.416E-02	15.0	XKME58
26	.160	1.440E-02	21.0	XNBG80
26	.180	2.533E-02	15.1	XKME58
26	.185	2.980E-02	20.0	XNBG80
26	.200	3.991E-02	14.8	XKME58
26	.200	5.780E-02	9.0	XNJL80
26	.220	5.864E-02	14.8	XKME58
26	.235	9.150E-02	20.0	XNBG80
26	.240	8.328E-02	15.0	XKME58
26	.250	1.020E-01	10.0	XNLO79A
26	.250	1.466E-01	8.0	XNJL80
26	.260	1.110E-01	15.0	XKME58
26	.275	1.930E-01	10.0	XNLO79A
26	.280	1.541E-01	15.1	XKME58
26	.285	2.130E-01	20.0	XNBG80
26	.300	1.971E-01	15.0	XKME58
26	.300	2.440E-01	10.0	XNLO79A
26	.300	3.092E-01	7.0	XNJL80
26	.320	2.533E-01	15.1	XKME58
26	.325	3.470E-01	10.0	XNLO79A
26	.340	2.984E-01	15.1	XKME58
26	.350	4.700E-01	10.0	XNLO79A
26	.360	3.852E-01	15.3	XKME58
26	.375	6.600E-01	10.0	XNLO79A
26	.390	7.210E-01	20.0	XNBG80
26	.400	5.691E-01	15.2	XKME58
26	.400	7.900E-01	10.0	XNLO79A
26	.400	1.240E+00	10.0	XNBO75
26	.400	8.960E-01	7.0	XNJL80
26	.425	1.030E+00	10.0	XNLO79A
26	.440	8.016E-01	15.2	XKME58
26	.441	5.517E-01	80.0	XKJO62
26	.450	1.260E+00	10.0	XNLO79A
26	.475	1.460E+00	10.0	XNLO79A
26	.490	1.560E+00	20.0	XNBG80
26	.500	1.400E+00	7.1	XNLE73
26	.500	1.830E+00	10.0	XNLO79A
26	.500	2.040E+00	10.0	XNLO78
26	.500	1.940E+00	10.0	XNLA79
26	.500	2.300E+00	10.0	XNBO75
26	.500	2.048E+00	7.0	XNJL80
26	.550	2.820E+00	10.0	XNLO78
26	.600	4.400E+00	6.8	XNLE73
26	.600	3.480E+00	10.0	XNLO78
26	.600	3.820E+00	10.0	XNLA79
26	.600	4.200E+00	10.0	XNBO75
26	.600	3.728E+00	6.0	XNJL80
26	.650	4.480E+00	10.0	XNLO78
26	.690	4.860E+00	20.0	XNBG80
26	.700	7.200E+00	8.3	XNLE73
26	.700	6.620E+00	10.0	XNLO78
26	.700	6.700E+00	10.0	XNBO75
26	.700	5.985E+00	6.0	XNJL80
26	.740	3.609E+00	15.4	XKME58
26	.750	6.830E+00	10.0	XNLO78
26	.790	7.310E+00	20.0	XNBG80
26	.800	9.500E+00	7.4	XNLE73
26	.800	8.470E+00	10.0	XNLO78
26	.800	9.230E+00	7.0	XNLA79
26	.800	1.050E+01	10.0	XNBO75
26	.800	9.097E+00	6.0	XNJL80
26	.830	8.100E+00	14.8	XNHO75B
26	.850	1.130E+01	10.0	XNLO78
26	.900	1.420E+01	7.7	XNLE73
26	.900	1.210E+01	10.0	XNLO78
26	.900	1.390E+01	10.0	XNBO75
26	.900	1.269E+01	6.0	XNJL80
26	.935	6.766E+00	14.9	XKME58
26	.950	9.300E+00	15.1	XNMI76
26	.950	1.400E+01	10.0	XNLO78
26	1.000	1.139E+01	20.0	XNBE73
26	1.000	1.490E+01	4.0	XKRS77
26	1.000	1.690E+01	7.7	XNLE73
26	1.000	1.250E+01	8.0	XNKN77
26	1.000	1.570E+01	10.0	XNLO78
26	1.000	1.680E+01	5.0	XNLA79
26	1.000	1.870E+01	10.0	XNBO75
26	1.040	1.079E+01	15.1	XKME58
26	1.050	1.810E+01	10.0	XNLO78
26	1.100	2.400E+01	7.9	XNLE73
26	1.100	2.190E+01	10.0	XNLO78
26	1.100	2.500E+01	10.0	XNBO75
26	1.150	2.490E+01	10.0	XNLO78
26	1.200	1.385E+01	15.0	XKME58
26	1.200	2.320E+01	3.9	XKRS77
26	1.200	2.960E+01	7.8	XNLE73
26	1.200	2.710E+01	10.0	XNLO78
26	1.200	2.740E+01	6.0	XNLA79
26	1.200	3.100E+01	10.0	XNBO75
26	1.250	3.020E+01	10.0	XNLO78
26	1.300	1.725E+01	15.1	XKME58
26	1.300	3.440E+01	7.8	XNLE73
26	1.300	3.230E+01	10.0	XNLO78
26	1.300	3.900E+01	10.0	XNBO75
26	1.350	3.670E+01	10.0	XNLO78
26	1.400	3.920E+01	7.9	XNLE73
26	1.400	4.100E+01	10.0	XNLO78
26	1.400	4.700E+01	10.0	XNBO75
26	1.450	4.360E+01	10.0	XNLO78
26	1.500	3.800E+01	80.0	XKOG64
26	1.500	4.670E+01	4.9	XNAK74
26	1.500	3.900E+01	4.1	XKRS77
26	1.500	5.020E+01	7.8	XNLE73
26	1.500	4.510E+01	10.0	XNLO78
26	1.500	5.400E+01	10.0	XNBO75
26	1.550	4.770E+01	10.0	XNLO78
26	1.580	5.900E+01	15.3	XNHO75B
26	1.600	5.690E+01	7.7	XNLE73
26	1.600	5.220E+01	10.0	XNLO78
26	1.600	5.000E+01	6.0	XNLA79
26	1.600	6.100E+01	10.0	XNBO75
26	1.650	5.530E+01	10.0	XNLO78
26	1.700	6.570E+01	7.8	XNLE73
26	1.700	6.020E+01	10.0	XNLO78
26	1.700	7.000E+01	10.0	XNBO75
26	1.750	6.260E+01	10.0	XNLO78
26	1.800	5.920E+01	4.1	XKRS77
26	1.800	6.920E+01	7.8	XNLE73
26	1.800	6.600E+01	10.0	XNLO78
26	1.800	8.200E+01	10.0	XNBO75
26	1.850	6.940E+01	10.0	XNLO78
26	1.900	7.270E+01	7.8	XNLE73
26	1.900	7.530E+01	10.0	XNLO78
26	1.900	9.300E+01	10.0	XNBO75
26	1.950	7.790E+01	10.0	XNLO78
26	2.000	8.660E+01	5.0	XNAK74
26	2.000	7.390E+01	4.1	XKRS77
26	2.000	7.650E+01	7.8	XNLE73
26	2.000	6.320E+01	8.2	XNKN77
26	2.000	8.310E+01	10.0	XNLO78
26	2.000	8.570E+01	5.0	XNLA79
26	2.000	9.900E+01	10.0	XNBO75
26	2.100	1.110E+02	10.0	XNBO75
26	2.200	1.200E+02	10.0	XNBO75
26	2.250	7.404E+01	15.4	XNBE73
26	2.300	9.710E+01	4.0	XKRS77
26	2.300	1.280E+02	10.0	XNBO75
26	2.400	1.130E+02	6.0	XNLA79
26	2.500	1.320E+02	5.3	XNAK74
26	2.500	1.120E+02	4.5	XKRS77
26	2.500	1.190E+02	15.0	XNLI73
26	2.560	1.340E+02	14.9	XNHO75B
26	2.800	1.290E+02	3.9	XKRS77
26	2.900	1.340E+02	3.7	XKRS77
26	3.000	1.253E+02	15.5	XNBE73
26	3.000	1.750E+02	5.1	XNAK74
26	3.000	1.770E+02	15.0	XNLI73
26	3.000	1.760E+02	3.4	XNBS78
26	3.280	2.270E+02	15.0	XNHO75B
26	4.000	2.670E+02	10.0	XNPE79
26	4.000	2.680E+02	15.0	XNLI73
26	4.500	2.750E+02	15.0	XNLI73
26	4.970	3.240E+02	4.9	XNAK74
26	5.000	2.350E+02	8.5	XNKN77
26	5.000	3.120E+02	15.0	XNLI73
26	5.960	3.880E+02	4.9	XNAK74
26	6.000	3.990E+02	10.0	XNPE79
26	6.000	4.050E+02	15.0	XNLI73
26	6.500	3.860E+02	15.0	XNLI73
26	6.940	4.390E+02	5.0	XNAK74
26	6.960	4.320E+02	5.1	XNAK74
26	7.000	4.290E+02	15.0	XNLI73
26	8.000	4.500E+02	10.0	XNPE79
26	8.000	4.190E+02	15.0	XNLI73
26	8.500	4.910E+02	15.0	XNLI73
26	8.940	4.950E+02	5.1	XNAK74
26	9.000	5.000E+02	15.0	XNLI73
26	10.000	5.480E+02	10.0	XNPE79
26	10.000	4.900E+02	15.0	XNLI73
26	10.500	4.860E+02	15.0	XNLI73
26	10.930	5.290E+02	1.9	XNAK74
26	11.000	4.990E+02	15.0	XNLI73
26	12.000	6.040E+02	10.0	XNPE79
26	12.000	4.940E+02	15.0	XNLI73
26	11.000	5.690E+02	10.0	XNPE79
26	16.000	5.860E+02	10.0	XNPE79
26	18.000	6.180E+02	10.0	XNPE79
26	20.000	5.830E+02	10.0	XNPE79
26	22.000	5.340E+02	10.0	XNPE79
26	160.000	2.287E+02	8.0	XNJA72
27	.100	4.762E-04	30.1	XKDU72
27	.125	2.248E-03	30.5	XKDU72
27	.150	6.591E-03	30.1	XKDU72
27	.200	3.570E-02	10.0	XNLO79B
27	.250	9.940E-02	10.0	XNLO79B
27	.300	2.060E-01	10.0	XNLO79B
27	.300	2.031E-01	5.5	XNBR82
27	.350	3.750E-01	10.0	XNLO79B
27	.400	6.290E-01	10.0	XNLO79B
27	.400	6.210E-01	5.4	XNBR82
27	.450	9.940E-01	10.0	XNLO79B
27	.500	1.600E+00	6.3	XNLE73
27	.500	1.350E+00	10.0	XNLO79B
27	.500	1.900E+00	10.0	XNBO75
27	.500	1.106E+00	5.3	XNBR82
27	.600	2.700E+00	7.1	XNLE73
27	.600	2.570E+00	10.0	XNLO79B
27	.600	3.500E+00	10.0	XNBO75
27	.600	2.637E+00	5.2	XNBR82
27	.600	2.477E+00	5.0	XNBR81
27	.700	4.200E+00	7.1	XNLE73
27	.700	4.040E+00	10.0	XNLO79B
27	.700	5.700E+00	10.0	XNBO75
27	.700	4.343E+00	5.1	XNBR82
27	.700	4.115E+00	5.0	XNBR81
27	.800	6.400E+00	10.9	XNLE73
27	.800	6.020E+00	10.0	XNLO79B
27	.800	9.200E+00	10.0	XNBO75
27	.800	6.477E+00	5.0	XNBR82
27	.800	6.248E+00	5.0	XNBR81
27	.900	9.500E+00	7.1	XNLE73
27	.900	8.570E+00	10.0	XNLO79B
27	.900	1.210E+01	10.0	XNBO75
27	1.000	1.270E+01	7.9	XNLE73
27	1.000	1.130E+01	10.0	XNLO79B
27	1.000	1.660E+01	10.0	XNBO75
27	1.000	1.253E+01	5.0	XNBR82
27	1.000	1.227E+01	5.0	XNBR81
27	1.100	1.650E+01	7.3	XNLE73
27	1.100	1.480E+01	10.0	XNLO79B
27	1.100	2.100E+01	10.0	XNBO75
27	1.200	1.990E+01	7.5	XNLE73
27	1.200	1.880E+01	10.0	XNLO79B
27	1.200	2.600E+01	10.0	XNBO75
27	1.200	2.080E+01	4.9	XNBR82
27	1.200	2.000E+01	5.0	XNBR81
27	1.300	2.470E+01	7.7	XNLE73
27	1.300	2.340E+01	10.0	XNLO79B
27	1.300	3.200E+01	10.0	XNBO75
27	1.400	2.790E+01	7.5	XNLE73
27	1.400	2.780E+01	10.0	XNLO79B
27	1.400	3.900E+01	10.0	XNBO75

APPENDIX: TABLE II. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger-Electron Production, and Ionization

See page 112 for Explanation of Tables

Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
27	1.400	3.052E+01	4.9	XNBR82
27	1.400	2.926E+01	5.0	XNBR84
27	1.500	3.720E+01	5.1	XNAK74
27	1.500	3.340E+01	7.5	XNLE73
27	1.500	3.220E+01	10.0	XNLO79B
27	1.500	4.500E+01	10.0	XNBO75
27	1.600	3.910E+01	7.7	XNLE73
27	1.600	3.680E+01	10.0	XNLO79B
27	1.600	5.200E+01	10.0	XNBO75
27	1.600	4.305E+01	4.8	XNBR82
27	1.600	4.077E+01	5.0	XNBR84
27	1.700	4.420E+01	7.5	XNLE73
27	1.700	4.270E+01	10.0	XNLO79B
27	1.700	5.900E+01	10.0	XNBO75
27	1.800	4.600E+01	7.6	XNLE73
27	1.800	4.840E+01	10.0	XNLO79B
27	1.800	6.700E+01	10.0	XNBO75
27	1.800	5.677E+01	4.8	XNBR82
27	1.800	5.296E+01	5.0	XNBR84
27	1.900	5.350E+01	7.5	XNLE73
27	1.900	5.520E+01	10.0	XNLO79B
27	1.900	8.000E+01	10.0	XNBO75
27	2.000	7.020E+01	5.0	XNAK74
27	2.000	5.930E+01	7.6	XNLE73
27	2.000	5.144E+01	8.5	XNKN77
27	2.000	6.250E+01	10.0	XNLO79B
27	2.000	8.600E+01	10.0	XNBO75
27	2.000	6.820E+01	4.7	XNBR82
27	2.000	6.591E+01	5.0	XNBR84
27	2.100	9.600E+01	10.0	XNBO75
27	2.200	1.030E+02	10.0	XNBO75
27	2.200	8.382E+01	4.7	XNBR82
27	2.200	8.001E+01	5.0	XNBR84
27	2.300	1.100E+02	10.0	XNBO75
27	2.300	8.992E+01	4.7	XNBR82
27	2.400	9.792E+01	4.7	XNBR82
27	2.500	1.090E+02	4.6	XNAK74
27	3.000	1.460E+02	4.8	XNAK74
27	3.000	1.350E+02	4.4	XNBS78
27	4.960	2.670E+02	4.9	XNAK74
27	5.000	1.993E+02	8.4	XNKN77
27	5.960	3.200E+02	5.0	XNAK74
27	6.940	3.660E+02	4.9	XNAK74
27	6.960	3.640E+02	4.9	XNAK74
27	8.940	4.250E+02	4.9	XNAK74
27	10.930	4.620E+02	5.0	XNAK74
28	.055	7.100E-07	50.0	XKLL80
28	.060	2.300E-06	50.0	XKLL80
28	.065	6.700E-06	25.0	XKLL80
28	.070	1.400E-05	25.0	XKLL80
28	.075	2.400E-05	25.0	XKLL80
28	.080	2.670E-05	30.0	XKZA79
28	.080	4.400E-05	25.0	XKLL80
28	.090	7.590E-05	30.0	XKZA79
28	.090	1.400E-04	25.0	XKLL80
28	.100	2.484E-04	30.0	XKDU72
28	.100	1.930E-04	30.0	XKZA79
28	.100	3.400E-04	25.0	XKLL80
28	.110	4.480E-04	30.0	XKZA79
28	.110	6.800E-04	25.0	XKLL80
28	.120	9.630E-04	30.0	XKZA79
28	.120	1.200E-03	25.0	XKLL80
28	.125	1.449E-03	31.4	XKDU72
28	.130	1.950E-03	30.0	XKZA79
28	.130	1.900E-03	25.0	XKLL80
28	.140	3.720E-03	30.0	XKZA79
28	.140	3.200E-03	25.0	XKLL80
28	.150	4.554E-03	30.0	XKDU72
28	.150	6.810E-03	30.0	XKZA79
28	.150	4.300E-03	25.0	XKLL80
28	.160	5.700E-03	25.0	XKLL80
28	.180	1.100E-02	25.0	XKLL80
28	.200	2.130E-02	10.0	XNLO79B
28	.200	1.700E-02	25.0	XKLL80
28	.220	2.900E-02	25.0	XKLL80
28	.240	4.300E-02	25.0	XKLL80
28	.250	6.090E-02	10.0	XNLO79B
28	.260	6.100E-02	25.0	XKLL80
28	.280	8.800E-02	25.0	XKLL80
28	.300	1.300E-01	10.0	XNLO79B
28	.300	1.300E-01	25.0	XKLL80
28	.300	1.437E-01	5.5	XNBR82
28	.350	2.500E-01	10.0	XNLO79B
28	.400	3.820E-01	10.0	XNLO79B
28	.400	4.430E-01	5.4	XNBR82
28	.450	6.290E-01	10.0	XNLO79B
28	.500	9.850E-01	10.0	XNLA79
28	.500	8.900E-01	10.0	XNLO79B
28	.500	1.500E+00	10.0	XNBO75
28	.500	1.006E+00	5.3	XNBR82
28	.600	1.810E+00	10.0	XNLA79
28	.600	1.690E+00	10.0	XNLO79B
28	.600	2.600E+00	10.0	XNBO75
28	.600	1.967E+00	5.2	XNBR82
28	.700	2.850E+00	10.0	XNLO79B
28	.700	4.300E+00	10.0	XNBO75
28	.700	3.279E+00	5.1	XNBR82
28	.800	4.350E+00	7.0	XNLA79
28	.800	4.310E+00	10.0	XNLO79B
28	.800	6.200E+00	10.0	XNBO75
28	.800	5.134E+00	5.0	XNBR82
28	.900	6.170E+00	10.0	XNLO79B
28	.900	8.300E+00	10.0	XNBO75
28	.950	5.300E+00	9.4	XNMI76
28	1.000	1.000E+01	5.0	XKRS76
28	1.000	7.182E+00	14.3	XNBE73
28	1.000	7.452E+00	8.3	XNKN77
28	1.000	9.030E+00	6.0	XNLA79
28	1.000	8.400E+00	10.0	XNLO79B
28	1.000	1.160E+01	10.0	XNBO75
28	1.000	9.688E+00	5.0	XNBR82
28	1.000	8.984E+00	5.0	XNBR84
28	1.100	1.280E+01	4.7	XKRS76
28	1.100	9.462E+00	14.5	XNBE73
28	1.100	1.110E+01	10.0	XNLO79B
28	1.100	1.530E+01	10.0	XNBO75
28	1.200	1.600E+01	5.0	XKRS76
28	1.200	1.254E+01	18.2	XNBE73
28	1.200	1.520E+01	6.0	XNLA79
28	1.200	1.420E+01	10.0	XNLO79B
28	1.200	1.820E+01	10.0	XNBO75
28	1.200	1.641E+01	4.9	XNBR82
28	1.200	1.490E+01	5.0	XNBR84
28	1.300	1.970E+01	5.1	XKRS76
28	1.300	1.482E+01	15.4	XNBE73
28	1.300	1.780E+01	10.0	XNLO79B
28	1.300	2.400E+01	10.0	XNBO75
28	1.400	2.380E+01	5.0	XKRS76
28	1.400	1.824E+01	12.5	XNBE73
28	1.400	2.180E+01	6.0	XNLA79
28	1.400	2.100E+01	10.0	XNLO79B
28	1.400	2.700E+01	10.0	XNBO75
28	1.400	2.541E+01	10.0	XNSN81
28	1.400	2.409E+01	4.9	XNBR82
28	1.400	2.256E+01	5.0	XNBR84
28	1.500	2.840E+01	4.9	XKRS76
28	1.500	2.166E+01	15.8	XNBE73
28	1.500	2.750E+01	5.1	XNAK74
28	1.500	2.480E+01	10.0	XNLO79B
28	1.500	3.500E+01	10.0	XNBO75
28	1.500	2.997E+01	10.0	XNSN81
28	1.600	3.350E+01	5.1	XKRS76
28	1.600	2.508E+01	13.6	XNBE73
28	1.600	3.050E+01	6.0	XNLA79
28	1.600	2.980E+01	10.0	XNLO79B
28	1.600	4.100E+01	10.0	XNBO75
28	1.600	3.305E+01	10.0	XNSN81
28	1.600	3.349E+01	4.8	XNBR82
28	1.600	3.130E+01	5.0	XNBR84
28	1.700	3.900E+01	5.1	XKRS76
28	1.700	2.964E+01	15.4	XNBE73
28	1.700	3.370E+01	10.0	XNLO79B
28	1.700	4.700E+01	10.0	XNBO75
28	1.700	3.989E+01	10.0	XNSN81
28	1.800	4.470E+01	4.9	XKRS76
28	1.800	3.420E+01	13.3	XNBE73
28	1.800	4.050E+01	6.0	XNLA79
28	1.800	3.810E+01	10.0	XNLO79B
28	1.800	5.300E+01	10.0	XNBO75
28	1.800	4.330E+01	10.0	XNSN81
28	1.800	4.388E+01	4.8	XNBR82
28	1.800	4.223E+01	5.0	XNBR84
28	1.900	5.090E+01	4.9	XKRS76
28	1.900	3.762E+01	15.2	XNBE73
28	1.900	4.430E+01	10.0	XNLO79B
28	1.900	5.800E+01	10.0	XNBO75
28	1.900	4.877E+01	10.0	XNSN81
28	2.000	5.710E+01	5.1	XKRS76
28	2.000	4.218E+01	13.5	XNBE73
28	2.000	5.530E+01	5.1	XNAK74
28	2.000	4.099E+01	8.1	XNKN77
28	2.000	5.010E+01	6.0	XNLA79
28	2.000	4.970E+01	10.0	XNLO79B
28	2.000	6.800E+01	10.0	XNBO75
28	2.000	5.402E+01	10.0	XNSN81
28	2.000	5.548E+01	4.7	XNBR82
28	2.000	5.216E+01	5.0	XNBR84
28	2.100	6.330E+01	5.1	XKRS76
28	2.100	4.674E+01	14.6	XNBE73
28	2.100	7.700E+01	10.0	XNBO75
28	2.200	7.000E+01	5.0	XKRS76
28	2.200	5.130E+01	15.6	XNBE73
28	2.200	6.170E+01	6.0	XNLA79
28	2.200	6.748E+01	4.7	XNBR82
28	2.200	6.600E+01	10.0	XNLO79B
28	2.200	8.500E+01	10.0	XNBO75
28	2.200	6.293E+01	5.0	XNBR84
28	2.250	5.358E+01	14.9	XNBE73
28	2.300	7.620E+01	5.0	XKRS76
28	2.300	5.586E+01	14.3	XNBE73
28	2.300	7.369E+01	4.7	XNBR82
28	2.400	8.240E+01	5.0	XKRS76
28	2.400	6.156E+01	14.8	XNBE73
28	2.400	7.330E+01	6.0	XNLA79
28	2.400	7.949E+01	4.7	XNBR82
28	2.500	8.820E+01	5.0	XKRS76
28	2.500	6.612E+01	13.8	XNBE73
28	2.500	8.640E+01	5.0	XNAK74
28	2.500	7.550E+01	10.0	XNBB78
28	2.600	9.360E+01	5.0	XKRS76
28	2.600	7.296E+01	15.6	XNBE73
28	2.700	9.810E+01	5.0	XKRS76
28	2.700	7.524E+01	15.2	XNBE73
28	2.800	1.030E+02	4.9	XKRS76
28	2.800	8.094E+01	15.5	XNBE73
28	2.900	1.060E+02	4.7	XKRS76
28	2.900	8.550E+01	14.7	XNBE73
28	2.980	1.140E+02	5.3	XNAK74
28	3.000	1.090E+02	5.5	XKRS76
28	3.000	9.120E+01	15.0	XNBE73
28	3.000	1.200E+02	5.0	XNAK74
28	3.000	1.050E+02	10.0	XNBB78
28	3.000	1.230E+02	15.0	XNLI73
28	3.000	1.190E+02	4.2	XNBS78
28	3.100	9.576E+01	16.5	XNBE73
28	3.200	1.015E+02	14.6	XNBE73
28	3.300	1.060E+02	15.1	XNBE73
28	3.400	1.117E+02	15.3	XNBE73
28	3.500	1.243E+02	15.0	XNBE73
28	3.600	1.231E+02	15.0	XNBE73
28	3.700	1.311E+02	15.0	XNBE73
28	4.000	2.130E+02	10.0	XNBD78
28	4.000	1.820E+02	15.0	XNLI73
28	4.500	2.120E+02	15.0	XNLI73
28	4.970	2.330E+02	5.2	XNAK74
28	5.000	2.944E+02	12.7	XNBI70
28	5.000	2.000E+02	8.5	XNKN77
28	5.000	2.120E+02	10.0	XNBB78
28	5.000	2.490E+02	15.0	XNLI73
28	5.960	2.820E+02	5.0	XNAK74
28	6.000	2.680E+02	15.0	XNLI73
28	6.500	2.560E+02	15.0	XNLI73
28	6.940	3.210E+02	5.0	XNAK74
28	7.000	3.530E+02	15.0	XNLI73
28	7.000	3.030E+02	13.0	XNDO81A
28	7.500	3.070E+02	10.0	XNBB78
28	8.000	4.099E+02	13.1	XNBI70
28	8.000	3.180E+02	15.0	XNLI73
28	8.500	4.180E+02	15.0	XNLI73
28	8.940	3.840E+02	4.9	XNAK74
28	9.000	3.430E+02	10.0	XNBB78
28	9.000	3.620E+02	15.0	XNLI73
28	9.750	3.480E+02	10.0	XNBB78
28	10.000	4.060E+02	15.0	XNLI73
28	10.500	4.550E+02	15.0	XNLI73
28	10.930	4.170E+02	5.0	XNAK74
28	11.000	4.761E+02	13.0	XNBI70
28	11.000	4.540E+02	15.0	XNLI73
28	12.000	4.930E+02	15.0	XNLI73
28	14.000	5.258E+02	12.6	XNBI70
28	17.000	5.175E+02	13.6	XNBI70
28	20.000	5.465E+02	14.4	XNBI70
28	24.000	5.092E+02	15.4	XNBI70
28	28.000	5.051E+02	15.6	XNBI70
29	.100	1.370E-04	17.0	XNRI81
29	.110	3.340E-04	22.0	XNBG80
29	.120	5.920E-04	17.0	XNRI81
29	.125	9.300E-04	30.1	XKDU72
29	.135	1.330E-03	22.0	XNBG80
29	.140	1.887E-03	15.1	XKME58
29	.140	1.690E-03	17.0	XNRI81
29	.150	2.982E-03	14.9	XKKH64
29	.150	3.070E-03	30.4	XKDU72
29	.150	2.830E-03	14.9	XKME58
29	.160	4.539E-03	14.7	XKME58
29	.160	4.020E-03	21.0	XNBG80
29	.160	3.820E-03	12.4	XNRI81
29	.180	8.188E-03	15.2	XKME58
29	.180	7.600E-03	12.4	XNRI81
29	.185	8.890E-03	20.0	XNBG80
29	.200	1.299E-02	15.1	XKKH64
29	.200	2.100E-02	80.0	XKJO62
29	.200	1.393E-02	15.0	XKME58
29	.200	4.672E-03	80.0	XKHA57
29	.200	1.650E-02	15.0	XNLO79B
29	.200	1.330E-02	12.4	XNRI81
29	.220	2.172E-02	15.0	XKME58

APPENDIX: TABLE II. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger-Electron Production, and Ionization

See page 112 for Explanation of Tables

Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
29	.235	2.980E-02	20.0	XNBG80	29	.830	4.300E+00	14.0	XNHO75B	29	1.700	3.300E+01	16.1	XNKL76
29	.240	3.249E-02	15.1	XKME58	29	.830	2.772E+00	15.0	XNBH82A	29	1.700	2.680E+01	7.5	XNLE73
29	.249	4.680E-02	25.0	XKBR78	29	.870	3.256E+00	15.0	XNBH82A	29	1.700	2.920E+01	10.0	XNLO79B
29	.250	1.015E-02	80.0	XKHA57	29	.875	3.950E+00	12.0	XNSE80	29	1.700	3.600E+01	10.0	XNBO75
29	.250	4.450E-02	15.0	XNLO79B	29	.900	5.050E+00	15.8	XNKL76	29	1.700	2.943E+01	10.0	XNSN81
29	.250	3.910E-02	12.1	XNRI81	29	.900	4.100E+00	7.3	XNLE73	29	1.710	2.376E+01	15.0	XNBH82A
29	.260	4.628E-02	15.4	XKME58	29	.900	5.120E+00	10.0	XNLO79B	29	1.720	2.610E+01	25.0	XKBR78
29	.280	6.542E-02	15.0	XKME58	29	.900	4.700E+00	25.0	XKBR78	29	1.750	2.920E+01	12.0	XNSE80
29	.280	9.240E-02	15.0	XNBH82A	29	.900	6.400E+00	10.0	XNBO75	29	1.750	3.109E+01	11.0	XNBA80
29	.285	7.300E-02	20.0	XNBG80	29	.900	7.700E+00	12.0	XNSI57	29	1.800	3.620E+01	5.0	XKRS76
29	.300	8.811E-02	15.2	XKKH64	29	.910	4.400E+00	15.0	XNBH82A	29	1.800	3.540E+01	16.1	XNKL76
29	.300	7.120E-02	80.0	XKJO62	29	.935	4.495E+00	14.9	XKME58	29	1.800	3.220E+01	7.5	XNLE73
29	.300	9.033E-02	14.8	XKME58	29	.950	4.200E+00	9.5	XNMI76	29	1.800	3.310E+01	5.0	XNLA79
29	.300	2.109E-02	80.0	XKHA57	29	1.000	8.589E+00	80.0	XKHA57	29	1.800	3.350E+01	10.0	XNLO79B
29	.300	9.520E-02	15.0	XNLO79B	29	1.000	7.300E+00	15.1	XNHO75A	29	1.800	4.100E+01	10.0	XNBO75
29	.300	1.080E-01	25.0	XKBR78	29	1.000	7.160E+00	5.0	XKRS76	29	1.800	3.400E+01	10.0	XNSN81
29	.300	8.480E-02	12.4	XNRI81	29	1.000	5.462E+00	14.6	XNBE73	29	1.800	3.502E+01	1.8	XNBR82
29	.300	9.434E-02	5.5	XNBR82	29	1.000	7.890E+00	15.2	XNKL76	29	1.800	3.249E+01	5.0	XNBR84
29	.320	1.188E-01	15.0	XKME58	29	1.000	6.000E+00	6.7	XNLE73	29	1.810	3.036E+01	15.0	XNBH82A
29	.330	1.760E-01	15.0	XNBH82A	29	1.000	7.120E+00	3.0	XNLA79	29	1.900	4.410E+01	16.0	XNKL76
29	.340	1.558E-01	15.1	XKME58	29	1.000	7.480E+00	10.0	XNLO79B	29	1.900	3.590E+01	7.5	XNLE73
29	.350	3.912E-02	80.0	XKHA57	29	1.000	6.010E+00	12.0	XNSE80	29	1.900	3.780E+01	10.0	XNLO79B
29	.350	1.780E-01	10.0	XNLO79B	29	1.000	8.700E+00	10.0	XNBO75	29	1.900	4.800E+01	10.0	XNBO75
29	.353	2.110E-01	25.0	XKBR78	29	1.000	7.253E+00	11.0	XNBA80	29	1.900	3.582E+01	10.0	XNSN81
29	.360	2.007E-01	15.1	XKME58	29	1.000	7.253E+00	5.0	XNBR82	29	1.910	3.400E+01	25.0	XKBR78
29	.370	2.508E-01	15.0	XNBH82A	29	1.000	1.110E+01	12.0	XNSI57	29	2.000	2.300E+01	13.0	XNRN76
29	.380	3.115E-01	14.3	XKKH64	29	1.000	6.898E+00	5.0	XNBR84	29	2.000	4.260E+01	15.0	XNHO75A
29	.380	1.900E-01	80.0	XKJO62	29	1.000	6.960E+00	9.5	XNGB86	29	2.000	4.630E+01	5.0	XKRS76
29	.380	2.541E-01	15.1	XKME58	29	1.010	5.720E+00	15.0	XNBH82A	29	2.000	4.270E+01	4.0	XNAK74
29	.380	3.070E-01	14.5	XKME58	29	1.040	6.274E+00	14.9	XKME58	29	2.000	4.630E+01	16.0	XNKL76
29	.390	2.680E-01	20.0	XNBG80	29	1.100	1.270E+01	15.7	XNKL76	29	2.000	3.920E+01	7.1	XNLE73
29	.400	8.589E-02	80.0	XKHA57	29	1.100	8.000E+00	7.5	XNLE73	29	2.000	4.280E+01	3.0	XNLA79
29	.400	3.030E-01	10.0	XNLO79B	29	1.100	9.790E+00	10.0	XNLO79B	29	2.000	4.240E+01	10.0	XNLO79B
29	.400	3.160E-01	5.4	XNBR82	29	1.100	8.180E+00	25.0	XKBR78	29	2.000	3.950E+01	12.0	XNSE80
29	.400	4.060E-01	12.0	XNSI57	29	1.100	1.160E+01	10.0	XNBO75	29	2.000	5.300E+01	10.0	XNBO75
29	.403	3.510E-01	25.0	XKBR78	29	1.110	7.920E+00	15.0	XNBH82A	29	2.000	4.494E+01	11.0	XNBA80
29	.420	3.652E-01	15.0	XNBH82A	29	1.135	7.787E+00	14.9	XKME58	29	2.000	4.153E+01	10.0	XNSN81
29	.440	3.871E-01	14.9	XKME58	29	1.200	9.434E+00	15.1	XKME58	29	2.000	4.263E+01	1.7	XNBR82
29	.450	4.810E-01	10.0	XNLO79B	29	1.200	1.210E+01	5.0	XKRS76	29	2.000	4.152E+01	5.0	XNBR84
29	.460	1.994E-01	80.0	XKHA57	29	1.200	1.750E+01	16.0	XNKL76	29	2.000	4.601E+01	5.3	XNAV84
29	.484	6.770E-01	25.0	XKBR78	29	1.200	1.070E+01	7.5	XNLE73	29	2.000	4.260E+01	9.5	XNGB86
29	.490	6.420E-01	20.0	XNBG80	29	1.200	1.160E+01	5.0	XNLA79	29	2.100	4.620E+01	16.0	XNKL76
29	.500	6.319E-01	14.8	XKKH64	29	1.200	1.210E+01	10.0	XNLO79B	29	2.100	5.700E+01	10.0	XNBO75
29	.500	4.717E-01	80.0	XKJO62	29	1.200	1.140E+01	10.0	XNBO75	29	2.200	5.610E+01	5.0	XKRS76
29	.500	5.000E-01	20.0	XNLE73	29	1.200	1.219E+01	4.9	XNBR82	29	2.200	5.380E+01	16.0	XNKL76
29	.500	7.650E-01	10.0	XNLA79	29	1.200	1.148E+01	5.0	XNBR84	29	2.200	5.030E+01	5.0	XNLA79
29	.500	6.810E-01	10.0	XNLO79B	29	1.210	9.680E+00	15.0	XNBH82A	29	2.200	6.000E+01	10.0	XNBO75
29	.500	6.990E-01	12.0	XNSE80	29	1.215	1.602E+01	80.0	XKHA57	29	2.200	5.310E+01	1.7	XNBR82
29	.500	9.000E-01	10.0	XNBO75	29	1.250	1.190E+01	12.0	XNSE80	29	2.200	5.073E+01	5.0	XNBR84
29	.500	7.565E-01	5.3	XNBR82	29	1.250	1.380E+01	11.0	XNBA80	29	2.250	4.211E+01	16.2	XNBE73
29	.500	8.400E-01	12.0	XNSI57	29	1.300	1.850E+01	16.2	XNKL76	29	2.250	5.170E+01	12.0	XNSE80
29	.500	6.600E-01	15.0	XNBH82A	29	1.300	1.340E+01	7.5	XNLE73	29	2.250	5.829E+01	11.0	XNBA80
29	.500	6.764E-01	5.0	XNBR84	29	1.300	1.420E+01	10.0	XNLO79B	29	2.250	5.625E+01	5.3	XNAV84
29	.552	1.060E+00	25.0	XKBR78	29	1.300	1.820E+01	10.0	XNBO75	29	2.300	6.100E+01	16.1	XNKL76
29	.560	8.800E-01	15.0	XNBH82A	29	1.310	1.300E+01	25.0	XKBR78	29	2.300	6.600E+01	10.0	XNBO75
29	.590	1.250E+00	20.0	XNBG80	29	1.310	1.276E+01	15.0	XNBH82A	29	2.300	5.740E+01	4.7	XNBR82
29	.600	1.188E+00	15.0	XKME58	29	1.333	1.510E+01	11.3	XNHO75A	29	2.300	5.720E+01	5.0	XNBR84
29	.600	1.000E+00	10.0	XNLE73	29	1.400	1.880E+01	4.8	XKRS76	29	2.333	5.450E+01	15.0	XNHO75A
29	.600	1.460E+00	7.0	XNLA79	29	1.400	2.080E+01	15.9	XNKL76	29	2.400	6.500E+01	5.1	XKRS76
29	.600	1.380E+00	10.0	XNLO79B	29	1.400	1.630E+01	7.4	XNLE73	29	2.400	7.420E+01	16.2	XNKL76
29	.600	1.800E+00	10.0	XNBO75	29	1.400	1.790E+01	5.0	XNLA79	29	2.400	6.190E+01	5.0	XNLA79
29	.600	1.442E+00	5.2	XNBR82	29	1.400	1.780E+01	10.0	XNLO79B	29	2.400	6.319E+01	4.7	XNBR82
29	.600	1.750E+00	12.0	XNSI57	29	1.400	2.200E+01	10.0	XNBO75	29	2.400	6.336E+01	5.0	XNBR84
29	.600	1.322E+00	5.0	XNBR84	29	1.400	1.985E+01	10.0	XNSN81	29	2.500	6.940E+01	5.0	XKRS76
29	.610	1.100E+00	15.0	XNBH82A	29	1.400	1.860E+01	1.9	XNBR82	29	2.500	6.870E+01	3.9	XNAK74
29	.625	1.350E+00	12.0	XNSE80	29	1.400	1.740E+01	5.0	XNBR84	29	2.500	6.950E+01	15.8	XNKL76
29	.635	1.660E+00	25.0	XKBR78	29	1.410	1.628E+01	15.0	XNBH82A	29	2.500	6.610E+01	15.0	XNLI73
29	.650	1.320E+00	15.0	XNBH82A	29	1.500	1.998E+01	80.0	XKOG64	29	2.500	6.850E+01	12.0	XNSE80
29	.667	2.000E+00	15.0	XNHO75A	29	1.500	2.270E+01	4.8	XKRS76	29	2.500	7.031E+01	11.0	XNBA80
29	.690	2.090E+00	20.0	XNBG80	29	1.500	2.150E+01	4.2	XNAK74	29	2.500	6.732E+01	5.0	XNBR84
29	.690	1.628E+00	15.0	XNBH82A	29	1.500	2.240E+01	16.1	XNKL76	29	2.500	7.062E+01	5.1	XNAV84
29	.700	1.994E+00	80.0	XKHA57	29	1.500	1.980E+01	7.6	XNLE73	29	2.560	6.800E+01	14.7	XNHO75B
29	.700	1.800E+00	5.6	XNLE73	29	1.500	2.150E+01	10.0	XNLO79B	29	2.600	7.300E+01	5.1	XKRS76
29	.700	2.300E+00	10.0	XNLO79B	29	1.500	1.930E+01	12.0	XNSE80	29	2.667	7.460E+01	15.0	XNHO75A
29	.700	2.900E+00	10.0	XNBO75	29	1.500	2.600E+01	10.0	XNBO75	29	2.750	8.410E+01	12.0	XNSE80
29	.700	2.305E+00	5.1	XNBR82	29	1.500	2.399E+01	11.0	XNBA80	29	2.750	8.678E+01	11.0	XNBA80
29	.700	2.730E+00	12.0	XNSI57	29	1.500	2.156E+01	10.0	XNSN81	29	2.750	8.108E+01	5.2	XNAV84
29	.700	2.305E+00	5.0	XNBR84	29	1.500	2.203E+01	4.6	XNAV84	29	2.800	7.970E+01	5.0	XKRS76
29	.720	2.440E+00	25.0	XKBR78	29	1.510	1.936E+01	15.0	XNBH82A	29	2.985	9.220E+01	12.0	XNSE80
29	.740	2.172E+00	15.0	XKME58	29	1.520	1.910E+01	25.0	XKBR78	29	3.000	9.050E+01	15.0	XNHO75A
29	.740	1.936E+00	15.0	XNBH82A	29	1.580	2.900E+01	13.8	XNHO75B	29	3.000	8.410E+01	5.0	XKRS76
29	.750	2.470E+00	12.0	XNSE80	29	1.600	2.690E+01	4.8	XKRS76	29	3.000	7.511E+01	15.2	XNBE73
29	.790	3.240E+00	20.0	XNBG80	29	1.600	2.800E+01	16.1	XNKL76	29	3.000	9.510E+01	4.0	XNAK74
29	.790	2.376E+00	15.0	XNBH82A	29	1.600	2.340E+01	7.3	XNLE73	29	3.000	9.810E+01	15.0	XNLI73
29	.800	2.800E+00	7.1	XNLE73	29	1.600	2.510E+01	5.0	XNLA79	29	3.000	9.700E+01	12.0	XNSE80
29	.800	3.800E+00	7.0	XNLA79	29	1.600	2.510E+01	10.0	XNLO79B	29	3.000	1.032E+02	11.0	XNBA80
29	.800	3.590E+00	10.0	XNLO79B	29	1.600	3.200E+01	10.0	XNBO75	29	3.000	8.700E+01	10.0	XNFB74
29	.800	4.700E+00	10.0	XNBO75	29	1.600	2.510E+01	10.0	XNSN81	29	3.000	9.720E+01	4.0	XNBS78
29	.800	3.671E+00	5.0	XNBR82	29	1.600	2.639E+01	4.8	XNBR82	29	3.000	9.612E+01	5.4	XNAV84
29	.800	4.370E+00	12.0	XNSI57	29	1.600	2.443E+01	5.0	XNBR84	29	3.280	1.230E+02	15.4	XNHO75B
29	.800	3.542E+00	5.0	XNBR84	29	1.610	2.200E+01	15.0	XNBH82A	29	3.333	1.160E+02	14.7	XNHO75A
29	.810	3.470E+00	25.0	XKBR78	29	1.667	2.790E+01	15.1	XNHO75A	29	3.667	1.340E+02	14.9	XNHO75A

APPENDIX: TABLE II. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger-Electron Production, and Ionization

See page 112 for Explanation of Tables

Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
29	3.940	1.450E+02	12.0	XNSE80
29	4.000	1.440E+02	15.3	XNHO75A
29	4.000	1.950E+02	10.8	XNBD78
29	4.000	1.560E+02	10.0	XNPE79
29	4.000	1.560E+02	15.0	XNLI73
29	4.333	1.850E+02	15.1	XNHO75A
29	4.500	1.800E+02	15.0	XNLI73
29	4.667	1.990E+02	15.1	XNHO75A
29	4.960	2.060E+02	3.9	XNAK74
29	5.000	2.130E+02	15.0	XNHO75A
29	5.000	1.860E+02	15.0	XNLI73
29	5.000	1.800E+02	10.0	XNFB74
29	5.333	2.310E+02	15.2	XNHO75A
29	5.667	2.470E+02	15.0	XNHO75A
29	5.960	2.520E+02	4.0	XNAK74
29	6.000	2.800E+02	10.0	XNPE79
29	6.000	2.530E+02	15.0	XNLI73
29	6.070	2.460E+02	12.0	XNSE80
29	6.500	2.460E+02	15.0	XNLI73
29	6.960	2.970E+02	4.0	XNAK74
29	7.000	2.990E+02	15.0	XNLI73
29	8.000	3.690E+02	10.0	XNPE79
29	8.000	2.900E+02	15.0	XNLI73
29	8.500	3.460E+02	15.0	XNLI73
29	8.940	3.550E+02	3.9	XNAK74
29	9.000	3.250E+02	15.0	XNLI73
29	9.130	3.500E+02	12.0	XNSE80
29	10.000	4.090E+02	10.0	XNPE79
29	10.000	3.800E+02	15.0	XNLI73
29	10.500	3.620E+02	15.0	XNLI73
29	10.930	3.980E+02	4.0	XNAK74
29	11.000	4.050E+02	15.0	XNLI73
29	12.000	4.490E+02	10.0	XNPE79
29	12.000	4.250E+02	15.0	XNLI73
29	12.210	4.060E+02	12.0	XNSE80
29	14.000	4.810E+02	10.0	XNPE79
29	16.000	4.720E+02	10.0	XNPE79
29	18.000	4.940E+02	10.0	XNPE79
29	18.290	4.230E+02	12.0	XNSE80
29	20.000	4.850E+02	10.0	XNPE79
29	22.000	4.760E+02	10.0	XNPE79
29	23.980	4.120E+02	12.0	XNSE80
29	30.130	3.960E+02	12.0	XNSE80
29	35.610	3.740E+02	12.0	XNSE80
29	39.560	3.560E+02	12.0	XNSE80
29	160.000	2.203E+02	8.1	XNJA72
30	.100	6.080E-05	30.0	XKZA79
30	.110	1.340E-04	30.0	XKZA79
30	.120	2.740E-04	30.0	XKZA79
30	.130	5.300E-04	30.0	XKZA79
30	.140	9.730E-04	30.0	XKZA79
30	.150	1.720E-03	30.0	XKZA79
30	.200	1.300E-02	15.0	XNLO79B
30	.250	3.410E-02	15.0	XNLO79B
30	.300	7.710E-02	15.0	XNLO79B
30	.300	6.371E-02	5.5	XNBR82
30	.350	1.110E-01	10.0	XNLO79B
30	.400	2.340E-01	10.0	XNLO79B
30	.400	2.232E-01	5.1	XNBR82
30	.450	3.000E-01	33.3	XNLE73
30	.450	3.830E-01	10.0	XNLO79B
30	.500	5.700E-01	10.0	XNLO79B
30	.500	5.221E-01	5.3	XNBR82
30	.500	4.838E-01	5.0	XNBR84
30	.560	5.000E-01	20.0	XNLE73
30	.600	9.630E-01	10.0	XNLO79B
30	.600	1.015E+00	5.2	XNBR82
30	.600	9.484E-01	5.0	XNBR84
30	.660	1.600E+00	12.5	XNLE73
30	.700	2.630E+00	14.8	XNTA74
30	.700	1.610E+00	10.0	XNLO79B
30	.700	1.691E+00	5.1	XNBR82
30	.700	1.619E+00	5.0	XNBR84
30	.770	2.600E+00	7.7	XNLE73
30	.800	2.490E+00	10.0	XNLO79B
30	.800	2.678E+00	5.0	XNBR82
30	.800	2.524E+00	5.0	XNBR84
30	.870	3.600E+00	8.3	XNLE73
30	.900	3.610E+00	10.0	XNLO79B
30	.970	5.300E+00	7.5	XNLE73
30	1.000	5.600E+00	3.9	XKRS77
30	1.000	6.420E+00	15.0	XNTA74
30	1.000	5.130E+00	10.0	XNLO79B
30	1.000	5.461E+00	5.0	XNBR82
30	1.000	5.269E+00	5.0	XNBR84
30	1.080	7.000E+00	8.6	XNLE73
30	1.100	7.500E+00	4.0	XKRS77
30	1.100	7.380E+00	10.0	XNLO79B
30	1.200	9.700E+00	4.0	XKRS77
30	1.200	9.100E+00	7.7	XNLE73
30	1.200	9.240E+00	10.0	XNLO79B
30	1.200	9.245E+00	4.9	XNBR82
30	1.200	8.766E+00	5.0	XNBR84
30	1.300	1.140E+01	7.9	XNLE73
30	1.300	1.200E+01	10.0	XNLO79B
30	1.400	1.500E+01	4.0	XKRS77
30	1.400	1.400E+01	7.9	XNLE73
30	1.400	1.570E+01	15.3	XNTA74
30	1.400	1.380E+01	10.0	XNLO79B
30	1.400	1.427E+01	4.9	XNBR82
30	1.400	1.351E+01	5.0	XNBR84
30	1.500	1.660E+01	7.8	XNLE73
30	1.500	1.600E+01	10.0	XNLO79B
30	1.600	2.160E+01	4.2	XKRS77
30	1.600	1.940E+01	8.2	XNLE73
30	1.600	2.000E+01	15.0	XNTA74
30	1.600	1.840E+01	10.0	XNLO79B
30	1.600	2.031E+01	4.8	XNBR82
30	1.600	1.926E+01	5.0	XNBR84
30	1.700	2.220E+01	8.1	XNLE73
30	1.700	2.210E+01	10.0	XNLO79B
30	1.800	2.920E+01	4.1	XKRS77
30	1.800	2.520E+01	7.9	XNLE73
30	1.800	2.670E+01	15.0	XNTA74
30	1.800	2.610E+01	10.0	XNLO79B
30	1.800	2.783E+01	4.8	XNBR82
30	1.800	2.553E+01	5.0	XNBR84
30	1.900	2.870E+01	8.0	XNLE73
30	1.900	2.950E+01	10.0	XNLO79B
30	2.000	3.780E+01	4.0	XKRS77
30	2.000	3.190E+01	7.8	XNLE73
30	2.000	3.310E+01	15.1	XNTA74
30	2.000	3.330E+01	10.0	XNLO79B
30	2.000	3.473E+01	4.7	XNBR82
30	2.000	3.267E+01	5.0	XNBR84
30	2.200	4.680E+01	4.1	XKRS77
30	2.200	4.250E+01	15.1	XNTA74
30	2.200	4.282E+01	4.7	XNBR82
30	2.200	4.062E+01	5.0	XNBR84
30	2.300	4.934E+01	4.7	XNBR82
30	2.400	5.560E+01	4.0	XKRS77
30	2.400	5.320E+01	15.0	XNTA74
30	2.400	5.173E+01	4.7	XNBR82
30	2.500	5.340E+01	15.0	XNLI73
30	2.600	6.420E+01	4.0	XKRS77
30	2.600	6.470E+01	15.0	XNTA74
30	2.800	7.140E+01	4.1	XKRS77
30	2.800	7.620E+01	15.0	XNTA74
30	3.000	7.760E+01	4.0	XKRS77
30	3.000	8.720E+01	15.0	XNTA74
30	3.000	7.800E+01	15.0	XNLI73
30	3.200	1.010E+02	14.9	XNTA74
30	3.400	1.110E+02	15.3	XNTA74
30	3.600	1.230E+02	14.6	XNTA74
30	3.800	1.360E+02	14.7	XNTA74
30	4.000	1.540E+02	14.9	XNTA74
30	4.000	1.340E+02	15.0	XNLI73
30	4.200	1.660E+02	15.1	XNTA74
30	4.100	1.780E+02	15.2	XNTA74
30	4.500	1.520E+02	15.0	XNLI73
30	5.000	1.710E+02	15.0	XNLI73
30	6.000	2.120E+02	15.0	XNLI73
30	6.500	2.170E+02	15.0	XNLI73
30	7.000	2.510E+02	15.0	XNLI73
30	7.000	2.060E+02	12.1	XNDO81A
30	8.000	2.810E+02	15.0	XNLI73
30	8.500	3.230E+02	15.0	XNLI73
30	9.000	3.030E+02	15.0	XNLI73
30	10.000	3.370E+02	15.0	XNLI73
30	10.500	3.390E+02	15.0	XNLI73
30	11.000	3.580E+02	15.0	XNLI73
30	12.000	3.820E+02	15.0	XNLI73
31	.500	3.000E-01	16.7	XNLE73
31	.600	6.000E-01	16.7	XNLE73
31	.700	1.100E+00	9.1	XNLE73
31	.800	1.800E+00	11.1	XNLE73
31	.900	2.700E+00	7.4	XNLE73
31	1.000	4.000E+00	7.0	XKRS77
31	1.000	4.100E+00	7.3	XNLE73
31	1.100	5.300E+00	7.5	XNLE73
31	1.200	6.400E+00	7.0	XKRS77
31	1.200	7.600E+00	7.9	XNLE73
31	1.300	8.900E+00	7.9	XNLE73
31	1.400	9.200E+00	7.0	XKRS77
31	1.400	1.110E+01	7.2	XNLE73
31	1.400	1.032E+01	10.0	XNSN81
31	1.500	1.360E+01	7.4	XNLE73
31	1.500	1.216E+01	10.0	XNSN81
31	1.600	1.280E+01	7.0	XKRS77
31	1.600	1.610E+01	7.5	XNLE73
31	1.600	1.503E+01	10.0	XNSN81
31	1.700	1.880E+01	7.4	XNLE73
31	1.700	1.744E+01	10.0	XNSN81
31	1.800	1.740E+01	6.9	XKRS77
31	1.800	2.190E+01	7.3	XNLE73
31	1.800	2.007E+01	10.0	XNSN81
31	1.900	2.480E+01	7.7	XNLE73
31	1.900	2.225E+01	10.0	XNSN81
31	2.000	2.290E+01	7.0	XKRS77
31	2.000	2.830E+01	7.4	XNLE73
31	2.000	2.478E+01	10.0	XNSN81
31	2.200	2.890E+01	6.9	XKRS77
31	2.400	3.470E+01	6.9	XKRS77
31	2.600	3.940E+01	7.1	XKRS77
31	2.800	4.210E+01	6.9	XKRS77
31	3.000	4.210E+01	6.9	XKRS77
31	4.000	1.470E+02	10.2	XNBD78
32	.200	3.660E-03	10.0	XNLO79B
32	.300	3.070E-02	10.0	XNLO79B
32	.330	8.025E-02	15.0	XNBH82A
32	.350	5.560E-02	10.0	XNLO79B
32	.370	1.177E-01	15.0	XNBH82A
32	.400	1.080E-01	10.0	XNLO79B
32	.420	1.658E-01	15.0	XNBH82A
32	.450	1.710E-01	10.0	XNLO79B
32	.460	2.140E-01	15.0	XNBH82A
32	.500	3.000E-01	16.7	XNLE73
32	.500	2.670E-01	10.0	XNLO79B
32	.520	3.370E-01	15.0	XNBH82A
32	.560	4.280E-01	15.0	XNBH82A
32	.600	5.000E-01	10.0	XNLE73
32	.600	5.450E-01	10.0	XNLO79B
32	.600	5.296E-01	15.0	XNBH82A
32	.650	6.420E-01	15.0	XNBH82A
32	.700	9.000E-01	11.1	XNLE73
32	.700	9.070E-01	10.0	XNLO79B
32	.710	9.095E-01	15.0	XNBH82A
32	.750	1.123E+00	15.0	XNBH82A
32	.790	1.337E+00	15.0	XNBH82A
32	.800	1.400E+00	7.1	XNLE73
32	.800	1.430E+00	10.0	XNLO79B
32	.900	2.000E+00	7.5	XNLE73
32	.900	2.100E+00	10.0	XNLO79B
32	.910	1.872E+00	15.0	XNBH82A
32	1.000	2.300E+00	15.0	XNBE73
32	1.000	2.800E+00	7.1	XNLE73
32	1.000	2.940E+00	10.0	XNLO79B
32	1.010	2.782E+00	15.0	XNBH82A
32	1.100	2.990E+00	15.4	XNBE73
32	1.100	3.700E+00	8.1	XNLE73
32	1.100	3.970E+00	10.0	XNLO79B
32	1.110	3.424E+00	15.0	XNBH82A
32	1.200	3.910E+00	14.7	XNBE73
32	1.200	4.800E+00	8.3	XNLE73
32	1.200	5.170E+00	10.0	XNLO79B
32	1.210	4.655E+00	15.0	XNBH82A
32	1.300	5.060E+00	15.9	XNBE73
32	1.300	6.000E+00	8.3	XNLE73
32	1.300	6.370E+00	10.0	XNLO79B
32	1.310	5.350E+00	15.0	XNBH82A
32	1.400	6.325E+00	14.5	XNBE73
32	1.400	7.200E+00	6.9	XNLE73
32	1.400	8.100E+00	10.0	XNLO79B
32	1.410	7.490E+00	15.0	XNBH82A
32	1.500	7.705E+00	14.9	XNBE73
32	1.500	8.700E+00	6.9	XNLE73
32	1.500	1.020E+01	10.0	XNLO79B
32	1.510	8.560E+00	15.0	XNBH82A
32	1.600	9.315E+00	14.8	XNBE73
32	1.600	1.100E+01	7.3	XNLE73
32	1.600	1.180E+01	10.0	XNLO79B
32	1.610	1.070E+01	15.0	XNBH82A
32	1.700	1.069E+01	15.1	XNBE73
32	1.700	1.220E+01	7.4	XNLE73
32	1.700	1.380E+01	10.0	XNLO79B
32	1.710	1.231E+01	15.0	XNBH82A
32	1.800	1.265E+01	18.2	XNBE73
32	1.800	1.410E+01	7.8	XNLE73
32	1.800	1.590E+01	10.0	XNLO79B
32	1.810	1.391E+01	15.0	XNBH82A
32	1.900	1.495E+01	15.4	XNBE73
32	1.900	1.600E+01	7.5	XNLE73
32	1.900	1.830E+01	10.0	XNLO79B
32	2.000	1.610E+01	14.3	XNBE73
32	2.000	1.850E+01	7.6	XNLE73
32	2.000	2.060E+01	10.0	XNLO79B
32	2.100	1.840E+01	12.5	XNBE73
32	2.200	2.070E+01	16.7	XNBE73
32	2.250	2.185E+01	15.8	XNBE73
32	2.300	2.300E+01	15.0	XNBE73
32	2.400	2.530E+01	13.6	XNBE73
32	2.500	2.760E+01	16.7	XNBE73
32	2.600	2.990E+01	15.4	XNBE73
32	2.700	3.220E+01	14.3	XNBE73
32	2.800	3.565E+01	16.1	XNBE73
32	2.900	3.680E+01	15.6	XNBE73

APPENDIX: TABLE II. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger-Electron Production, and Ionization

See page 112 for Explanation of Tables

Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
32	3.000	4.025E+01	14.3	XNBE73	35	1.200	2.710E+00	8.1	XNCR74	39	2.000	3.940E+00	8.1	XNCR71
32	3.000	5.440E+01	5.5	XNBS78	35	1.400	4.470E+00	8.1	XNCR74	39	2.000	5.190E+00	25.0	XKAN78
32	3.100	4.255E+01	16.2	XNBE73	35	1.500	4.043E+00	4.6	XNAV84	39	2.200	4.700E+00	10.0	XKRS77
32	3.200	4.600E+01	15.0	XNBE73	35	1.600	6.030E+00	8.0	XNCR74	39	2.400	5.800E+00	10.0	XKRS77
32	3.300	4.830E+01	14.3	XNBE73	35	1.800	8.310E+00	7.9	XNCR74	39	2.500	7.440E+00	15.0	XNLI73
32	3.400	5.060E+01	15.9	XNBE73	35	2.000	1.040E+01	7.7	XNCR74	39	2.500	9.240E+00	25.0	XKAN78
32	3.500	5.520E+01	14.6	XNBE73	35	2.000	9.952E+00	5.0	XNAV84	39	2.600	7.000E+00	10.0	XKRS77
32	3.600	5.750E+01	14.0	XNBE73	35	2.250	1.294E+01	5.3	XNAV84	39	2.800	8.100E+00	10.0	XKRS77
32	3.700	6.095E+01	15.1	XNBE73	35	2.500	1.735E+01	5.0	XNAV84	39	3.000	9.200E+00	10.0	XKRS77
32	3.800	6.440E+01	14.3	XNBE73	35	2.750	2.183E+01	5.1	XNAV84	39	3.000	1.250E+01	15.0	XNLI73
33	.470	1.000E-01	20.0	XNLE73	35	3.000	2.918E+01	5.3	XNAV84	39	3.000	9.300E+00	10.0	XNFB74
33	.570	3.000E-01	16.7	XNLE73	35	4.000	5.100E+01	10.0	XNBD78	39	4.000	2.100E+01	10.0	XNPE79
33	.680	5.000E-01	10.0	XNLE73	36	.500	6.600E-02	20.0	XNLN75	39	4.000	2.540E+01	15.0	XNLI73
33	.780	9.000E-01	11.1	XNLE73	36	.500	5.993E-02	10.0	XNLO80	39	4.500	3.200E+01	15.0	XNLI73
33	.890	1.400E+00	7.1	XNLE73	36	.600	1.135E-01	10.0	XNLO80	39	5.000	3.570E+01	15.0	XNLI73
33	.990	2.100E+00	9.5	XNLE73	36	.700	2.488E-01	10.0	XNLO80	39	5.000	2.800E+01	10.0	XNFB74
33	1.000	2.268E+00	30.0	XNLD73	36	.800	4.409E-01	10.0	XNLO80	39	6.000	5.700E+01	10.0	XNPE79
33	1.100	2.800E+00	7.1	XNLE73	36	.817	5.200E-01	20.0	XNLN75	39	6.000	5.220E+01	15.0	XNLI73
33	1.200	3.800E+00	7.9	XNLE73	36	.900	6.798E-01	10.0	XNLO80	39	6.500	6.090E+01	15.0	XNLI73
33	1.300	4.800E+00	8.3	XNLE73	36	.915	6.600E-01	20.0	XNLN75	39	7.000	5.840E+01	15.0	XNLI73
33	1.400	6.000E+00	8.3	XNLE73	36	.998	9.900E-01	20.0	XNLN75	39	8.000	8.500E+01	10.0	XNPE79
33	1.500	7.938E+00	30.0	XNLD73	36	1.000	9.570E-01	10.0	XNLO80	39	8.000	7.280E+01	15.0	XNLI73
33	1.500	7.400E+00	8.1	XNLE73	36	1.100	1.175E+00	10.0	XNLO80	39	8.500	9.600E+01	15.0	XNLI73
33	1.600	8.800E+00	8.0	XNLE73	36	1.200	1.604E+00	10.0	XNLO80	39	9.000	9.500E+01	15.0	XNLI73
33	1.700	1.040E+01	7.7	XNLE73	36	1.290	2.200E+00	20.0	XNLN75	39	10.000	1.140E+02	10.0	XNPE79
33	1.800	1.220E+01	8.2	XNLE73	36	1.300	2.125E+00	10.0	XNLO80	39	10.000	1.100E+02	15.0	XNLI73
33	1.900	1.410E+01	13.5	XNLE73	36	1.400	2.990E+00	10.0	XNLO80	39	10.500	1.020E+02	15.0	XNLI73
33	2.000	1.593E+01	29.9	XNLD73	36	1.500	3.386E+00	10.0	XNLO80	39	11.000	1.110E+02	15.0	XNLI73
33	2.000	1.590E+01	7.5	XNLE73	36	1.600	4.488E+00	10.0	XNLO80	39	12.000	1.490E+02	10.0	XNPE79
33	2.500	3.005E+01	30.2	XNLD73	36	1.630	4.300E+00	20.0	XNLN75	39	12.000	1.030E+02	15.0	XNLI73
33	3.000	4.366E+01	29.9	XNLD73	36	1.700	5.293E+00	10.0	XNLO80	39	11.000	1.560E+02	10.0	XNPE79
33	3.500	6.180E+01	30.3	XNLD73	36	1.800	6.125E+00	10.0	XNLO80	39	16.000	1.850E+02	10.0	XNPE79
33	4.000	7.201E+01	29.9	XNLD73	36	1.900	6.798E+00	10.0	XNLO80	39	18.000	1.920E+02	10.0	XNPE79
33	4.500	8.845E+01	30.1	XNLD73	36	2.000	6.600E+00	20.0	XNLN75	39	20.000	2.130E+02	10.0	XNPE79
33	5.000	1.066E+02	30.3	XNLD73	36	2.000	7.920E+00	10.0	XNLO80	39	22.000	2.060E+02	10.0	XNPE79
33	5.500	1.225E+02	30.1	XNLD73	37	.350	5.745E-03	11.6	XNMA79	40	.200	1.197E-04	80.0	XKJO62
34	.200	2.084E-03	10.0	XNLO79B	37	.400	2.770E-02	7.9	XNCR74	40	.260	5.690E-04	27.0	XNBG80
34	.250	6.201E-03	10.0	XNLO79B	37	.400	1.149E-02	11.6	XNMA79	40	.285	1.150E-03	26.0	XNBG80
34	.300	1.435E-02	10.0	XNLO79B	37	.450	2.325E-02	11.5	XNMA79	40	.300	1.000E-03	80.0	XKJO62
34	.350	3.593E-02	10.0	XNLO79B	37	.500	3.507E-02	11.4	XNMA79	40	.335	2.910E-03	25.0	XNBG80
34	.400	7.160E-02	8.0	XNCR74	37	.600	1.240E-01	8.1	XNCR74	40	.365	4.380E-03	25.0	XNBG80
34	.400	5.855E-02	10.0	XNLO79B	37	.600	8.016E-02	12.5	XNMA79	40	.390	6.340E-03	25.0	XNBG80
34	.450	9.123E-02	10.0	XNLO79B	37	.700	1.703E-01	11.8	XNMA79	40	.400	4.000E-03	80.0	XKJO62
34	.500	1.519E-01	10.0	XNLO79B	37	.800	4.080E-01	8.1	XNCR74	40	.400	3.511E-03	10.4	XNMA79
34	.600	2.910E-01	7.9	XNCR74	37	1.000	5.772E-01	14.3	XNBE73	40	.441	8.760E-03	80.0	XKJO62
34	.600	3.190E-01	10.0	XNLA79	37	1.000	9.270E-01	8.0	XNCR74	40	.450	5.694E-03	10.3	XNMA79
34	.600	3.059E-01	10.0	XNLO79B	37	1.200	1.320E+00	8.3	XNCR74	40	.490	1.880E-02	25.0	XNBG80
34	.700	5.457E-01	10.0	XNLO79B	37	1.400	2.370E+00	8.0	XNCR74	40	.500	1.197E-02	80.0	XKJO62
34	.800	8.490E-01	8.0	XNCR74	37	1.600	3.620E+00	8.0	XNCR74	40	.500	1.752E-02	12.5	XNMA79
34	.800	8.280E-01	10.0	XNLA79	37	1.800	4.470E+00	8.1	XNCR74	40	.590	4.260E-02	25.0	XNBG80
34	.800	8.704E-01	10.0	XNLO79B	37	2.000	5.430E+00	7.9	XNCR74	40	.690	8.000E-02	25.0	XNBG80
34	.900	1.278E+00	10.0	XNLO79B	37	2.250	7.539E+00	15.6	XNBE73	40	.700	6.278E-02	11.6	XNMA79
34	1.000	1.860E+00	8.1	XNCR74	37	3.000	1.531E+01	15.4	XNBE73	40	.790	1.350E-01	25.0	XNBG80
34	1.000	1.790E+00	3.0	XNLA79	38	.400	9.881E-03	10.5	XNMA79	40	.900	2.102E-01	10.4	XNMA79
34	1.000	1.791E+00	10.0	XNLO79B	38	.450	2.211E-02	12.5	XNMA79	40	1.000	3.097E-01	15.4	XNBE73
34	1.100	2.388E+00	10.0	XNLO79B	38	.500	3.420E-02	12.1	XNMA79	40	1.000	3.770E-01	3.0	XNLA79
34	1.200	2.810E+00	7.8	XNCR74	38	.600	7.970E-02	8.0	XNCR74	40	1.000	3.211E-01	11.3	XNMA79
34	1.200	3.180E+00	5.0	XNLA79	38	.600	7.394E-02	9.3	XNMA79	40	1.100	4.555E-01	11.2	XNMA79
34	1.200	3.153E+00	10.0	XNLO79B	38	.700	1.444E-01	12.0	XNMA79	40	1.200	5.881E-01	11.2	XNMA79
34	1.300	4.022E+00	10.0	XNLO79B	38	.800	3.640E-01	8.0	XNCR74	40	1.300	7.738E-01	9.4	XNMA79
34	1.400	4.410E+00	7.9	XNCR74	38	.800	2.059E-01	11.7	XNMA79	40	1.400	9.490E-01	11.5	XNMA79
34	1.400	4.890E+00	5.0	XNLA79	38	.900	3.317E-01	10.4	XNMA79	40	1.500	1.278E+00	11.4	XNMA79
34	1.400	5.049E+00	10.0	XNLO79B	38	1.000	7.020E-01	8.0	XNCR74	40	2.000	3.340E+00	3.0	XNLA79
34	1.500	5.887E+00	10.0	XNLO79B	38	1.000	5.189E-01	12.0	XNMA79	40	2.250	3.573E+00	16.7	XNBE73
34	1.600	6.540E+00	8.0	XNCR74	38	1.100	7.255E-01	9.5	XNMA79	40	3.000	7.503E+00	14.3	XNBE73
34	1.600	7.210E+00	5.0	XNLA79	38	1.200	1.180E+00	7.6	XNCR74	40	7.000	6.351E+01	9.7	XNDO81A
34	1.600	7.395E+00	10.0	XNLO79B	38	1.200	1.030E+00	10.1	XNMA79	40	160.000	1.351E+02	7.0	XNJA72
34	1.700	8.463E+00	10.0	XNLO79B	38	1.400	1.810E+00	7.7	XNCR74	41	.200	7.070E-05	12.4	XNRI81
34	1.800	8.570E+00	8.1	XNCR74	38	1.400	1.520E+00	9.1	XNMA79	41	.240	3.150E-04	12.4	XNRI81
34	1.800	1.040E+01	5.0	XNLA79	38	1.500	1.810E+00	11.5	XNMA79	41	.280	9.610E-04	12.4	XNRI81
34	1.800	9.836E+00	10.0	XNLO79B	38	1.600	2.690E+00	8.2	XNCR74	41	.320	2.140E-03	12.4	XNRI81
34	1.900	1.131E+01	10.0	XNLO79B	38	1.800	3.940E+00	8.1	XNCR74	41	.360	4.280E-03	12.4	XNRI81
34	2.000	1.150E+01	7.8	XNCR74	38	2.000	5.280E+00	8.0	XNCR74	41	.400	7.650E-03	12.4	XNRI81
34	2.000	1.340E+01	3.0	XNLA79	38	3.000	1.230E+01	4.9	XNBS78	41	.450	1.390E-02	12.4	XNRI81
34	2.000	1.278E+01	10.0	XNLO79B	39	.500	3.630E-02	25.0	XKAN78	41	.500	2.320E-02	12.4	XNRI81
34	2.200	1.620E+01	5.0	XNLA79	39	.600	7.200E-02	8.1	XNCR74	41	1.130	3.600E-01	11.1	XNWL76
34	2.400	1.900E+01	5.0	XNLA79	39	.750	1.850E-01	25.0	XKAN78	41	1.340	6.500E-01	9.2	XNWL76
34	3.000	3.420E+01	5.0	XNBS78	39	.800	2.200E-01	8.2	XNCR74	41	1.550	1.000E+00	10.0	XNWL76
34	4.000	6.260E+01	10.0	XNPE79	39	1.000	4.100E-01	10.0	XKRS77	41	1.760	1.600E+00	6.3	XNWL76
34	6.000	1.252E+02	10.0	XNPE79	39	1.000	4.800E-01	7.9	XNCR74	41	1.970	2.100E+00	9.5	XNWL76
34	7.000	1.526E+02	12.1	XNDO81A	39	1.000	5.470E-01	25.0	XKAN78	41	2.180	3.000E+00	10.0	XNWL76
34	8.000	1.910E+02	10.0	XNPE79	39	1.200	7.700E-01	10.0	XKRS77	41	2.390	3.600E+00	8.3	XNWL76
34	10.000	2.270E+02	10.0	XNPE79	39	1.200	8.810E-01	7.9	XNCR74	41	2.600	4.500E+00	8.9	XNWL76
34	12.000	2.740E+02	10.0	XNPE79	39	1.250	1.210E+00	25.0	XKAN78	41	2.700	4.800E+00	8.3	XNWL76
34	14.000	2.900E+02	10.0	XNPE79	39	1.400	1.300E+00	10.0	XKRS77	41	3.000	6.800E+00	4.4	XNBS78
34	16.000	2.980E+02	10.0	XNPE79	39	1.400	1.430E+00	7.7	XNCR74	42	.240	5.501E-04	15.3	XKME58
34	18.000	3.220E+02	10.0	XNPE79	39	1.500	2.200E+00	25.0	XKAN78	42	.250	1.925E-04	80.0	XKHA57
34	20.000	3.280E+02	10.0	XNPE79	39	1.600	1.900E+00	10.0	XKRS77	42	.260	8.251E-04	14.8	XKME58
34	22.000	3.100E+02	10.0	XNPE79	39	1.600	2.110E+00	8.1	XNCR74	42	.280	1.215E-03	15.1	XKME58
35	.600	3.340E-01	8.1	XNCR74	39	1.800	2.800E+00	10.0	XKRS77	42	.300	1.750E-03	14.8	XKME58
35	.800	7.390E-01	8.0	XNCR74	39	1.800	3.090E+00	8.1	XNCR74	42	.300	5.501E-04	80.0	XKHA57
35	1.000	1.560E+00	7.7	XNCR74	39	2.000	3.700E+00	10.0	XKRS77	42	.320	2.483E-03	15.1	XKME58

APPENDIX: TABLE II. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger-Electron Production, and Ionization

See page 112 for Explanation of Tables

Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
42	.340	3.576E-03	15.0	XKME58	42	5.000	2.165E+01	10.6	XNDI84	47	.400	8.170E-04	7.0	XNLA79
42	.350	1.574E-03	80.0	XKHA57	42	5.500	2.517E+01	10.6	XNDI84	47	.400	7.006E-04	14.0	XNJL80
42	.360	4.676E-03	15.0	XKME58	42	6.000	2.260E+01	15.0	XKLI73	47	.400	1.000E-03	12.4	XNRI81
42	.380	6.265E-03	14.6	XKME58	42	6.000	3.014E+01	10.9	XNDI84	47	.403	9.340E-04	34.0	XKBR78
42	.390	4.131E-03	15.0	XNBH82A	42	6.500	2.170E+01	15.0	XKLI73	47	.420	1.579E-03	15.0	XNBH82A
42	.400	8.098E-03	15.1	XKME58	42	7.000	2.210E+01	15.0	XKLI73	47	.430	9.900E-04	25.0	XNBG80
42	.400	3.285E-03	80.0	XKHA57	42	7.000	3.660E+01	16.1	XNDO81A	47	.450	2.010E-03	12.4	XNRI81
42	.400	4.720E-03	8.1	XNCR74	42	8.000	3.800E+01	15.0	XKLI73	47	.460	2.825E-03	15.0	XNBH82A
42	.430	6.808E-03	15.0	XNBH82A	42	8.500	4.120E+01	15.0	XKLI73	47	.480	1.870E-03	25.0	XNBG80
42	.440	1.261E-02	15.2	XKME58	42	9.000	3.730E+01	15.0	XKLI73	47	.484	2.730E-03	34.0	XKBR78
42	.441	5.654E-03	80.0	XKJO62	42	10.000	5.350E+01	15.0	XKLI73	47	.500	2.980E-03	6.0	XNLA79
42	.480	1.147E-02	15.0	XNBH82A	42	10.500	4.670E+01	15.0	XKLI73	47	.500	2.820E-03	25.0	XKAN78
42	.500	1.380E-02	25.0	XKAN78	42	11.000	6.350E+01	15.0	XKLI73	47	.500	2.464E-03	12.0	XNJL80
42	.500	1.308E-02	13.4	XNDI84	42	12.000	7.110E+01	15.0	XKLI73	47	.500	3.640E-03	12.4	XNRI81
42	.550	1.912E-02	15.0	XNBH82A	42	160.000	1.184E+02	3.9	XNJA72	47	.510	1.321E-03	15.0	XNBH82A
42	.600	3.430E-02	14.9	XKME58	44	7.000	2.641E+01	15.9	XNDO81A	47	.550	6.080E-03	12.4	XNRI81
42	.600	2.940E-02	8.2	XNCR74	45	1.030	1.000E-01	10.0	XNWL76	47	.550	6.482E-03	15.0	XNBH82A
42	.600	2.907E-02	15.0	XNBH82A	45	1.240	2.100E-01	9.5	XNWL76	47	.552	4.510E-03	34.0	XKBR78
42	.600	3.175E-02	11.6	XNDI84	45	1.450	3.400E-01	8.8	XNWL76	47	.580	5.100E-03	25.0	XNBG80
42	.660	3.748E-02	15.0	XNBH82A	45	1.660	5.200E-01	7.7	XNWL76	47	.600	1.112E-02	14.9	XKME58
42	.700	6.494E-02	80.0	XKHA57	45	1.870	7.700E-01	7.8	XNWL76	47	.600	9.400E-03	8.5	XNWL76
42	.700	6.013E-02	11.3	XNDI84	45	2.070	1.100E+00	9.1	XNWL76	47	.600	6.700E-03	10.4	XNKH75
42	.710	4.972E-02	15.0	XNBH82A	45	2.290	1.500E+00	6.7	XNWL76	47	.600	7.490E-03	6.0	XNLA79
42	.740	8.480E-02	15.3	XKME58	45	2.500	1.900E+00	10.5	XNWL76	47	.600	6.620E-03	10.0	XNJL80
42	.750	7.490E-02	25.0	XKAN78	45	2.700	2.200E+00	9.1	XNWL76	47	.600	9.570E-03	12.4	XNRI81
42	.770	6.885E-02	15.0	XNBH82A	45	160.000	1.162E+02	50.0	XNJA72	47	.600	4.160E-03	12.0	XNSI57
42	.800	9.680E-02	8.0	XNCR74	46	.215	8.500E-06	50.0	XKLL80	47	.600	9.141E-03	15.0	XNBH82A
42	.800	9.945E-02	15.0	XNBH82A	46	.230	1.900E-05	50.0	XKLL80	47	.635	8.930E-03	34.0	XKBR78
42	.800	1.040E-01	11.0	XNDI84	46	.245	4.200E-05	25.0	XKLL80	47	.610	1.217E-02	15.0	XNBH82A
42	.850	1.147E-01	15.0	XNBH82A	46	.260	6.500E-05	25.0	XKLL80	47	.680	1.050E-02	25.0	XNBG80
42	.900	1.377E-01	15.0	XNBH82A	46	.275	9.700E-05	25.0	XKLL80	47	.690	1.745E-02	15.0	XNBH82A
42	.900	1.561E-01	11.8	XNDI84	46	.300	1.700E-04	25.0	XKLL80	47	.700	1.358E-02	10.0	XNJL80
42	.935	2.116E-01	15.2	XKME58	46	.500	4.116E-03	13.4	XNDI84	47	.700	1.018E-02	12.0	XNSI57
42	1.000	2.284E-01	80.0	XKHA57	46	.600	8.120E-03	8.0	XNCR74	47	.720	1.540E-02	34.0	XKBR78
42	1.000	2.350E-01	5.1	XKRS76	46	.600	1.066E-02	11.5	XNDI84	47	.740	2.382E-02	15.0	XKME58
42	1.000	2.180E-01	7.8	XNCR74	46	.700	2.124E-02	11.2	XNDI84	47	.740	2.244E-02	15.0	XNBH82A
42	1.000	2.290E-01	25.0	XKAN78	46	.800	3.280E-02	7.9	XNCR74	47	.750	1.830E-02	25.0	XKAN78
42	1.000	2.387E-01	10.9	XNDI84	46	.800	3.854E-02	10.9	XNDI84	47	.780	1.930E-02	25.0	XNBG80
42	1.010	2.218E-01	15.0	XNBH82A	46	.900	6.101E-02	10.9	XNDI84	47	.800	3.300E-02	9.1	XNWL76
42	1.040	2.850E-01	15.0	XKME58	46	1.000	7.140E-02	8.0	XNCR74	47	.800	2.700E-02	11.1	XNKH75
42	1.110	3.060E-01	15.0	XNBH82A	46	1.000	9.266E-02	11.5	XNDI84	47	.800	2.770E-02	5.0	XNLA79
42	1.200	5.348E-01	14.3	XKME58	46	1.200	1.520E-01	7.9	XNCR74	47	.800	2.405E-02	10.0	XNJL80
42	1.200	4.210E-01	5.0	XKRS76	46	1.200	1.837E-01	10.7	XNDI84	47	.800	1.760E-02	12.0	XNSI57
42	1.200	4.020E-01	8.0	XNCR74	46	1.400	2.490E-01	8.0	XNCR74	47	.810	2.580E-02	34.0	XKBR78
42	1.200	4.536E-01	10.5	XNDI84	46	1.400	3.141E-01	10.7	XNDI84	47	.820	3.241E-02	15.0	XNBH82A
42	1.210	4.284E-01	15.0	XNBH82A	46	1.600	3.860E-01	8.0	XNCR74	47	.880	4.404E-02	15.0	XNBH82A
42	1.215	4.424E-01	80.0	XKHA57	46	1.600	4.863E-01	10.1	XNDI84	47	.900	4.020E-02	34.0	XKBR78
42	1.250	5.200E-01	25.0	XKAN78	46	1.800	5.410E-01	7.9	XNCR74	47	.900	4.047E-02	10.0	XNJL80
42	1.310	5.355E-01	15.0	XNBH82A	46	1.800	7.060E-01	9.8	XNDI84	47	.900	3.357E-02	12.0	XNSI57
42	1.400	6.950E-01	5.0	XKRS76	46	2.000	7.550E-01	7.9	XNCR74	47	.935	6.557E-02	15.2	XKME58
42	1.400	6.790E-01	8.0	XNCR74	46	2.000	9.758E-01	10.1	XNDI84	47	1.000	7.060E-02	6.1	XKRS76
42	1.400	7.520E-01	10.1	XNDI84	46	2.200	1.312E+00	10.0	XNDI84	47	1.000	6.429E-02	15.1	XNBE73
42	1.410	6.809E-01	15.0	XNBH82A	46	2.400	1.689E+00	10.2	XNDI84	47	1.000	7.900E-02	7.6	XNWL76
42	1.500	8.710E-01	5.1	XKRS76	46	2.600	2.099E+00	10.2	XNDI84	47	1.000	6.400E-02	10.9	XNKH75
42	1.500	9.930E-01	25.0	XKAN78	46	2.800	2.534E+00	10.0	XNDI84	47	1.000	7.080E-02	3.0	XNLA79
42	1.510	7.650E-01	15.0	XNBH82A	46	3.000	3.091E+00	10.1	XNDI84	47	1.000	6.310E-02	25.0	XKAN78
42	1.600	1.080E+00	4.6	XKRS76	46	3.200	3.551E+00	10.1	XNDI84	47	1.000	6.181E-02	10.0	XNJL80
42	1.600	1.130E+00	8.0	XNCR74	46	3.500	4.535E+00	10.5	XNDI84	47	1.000	4.650E-02	12.0	XNSI57
42	1.600	1.155E+00	10.6	XNDI84	46	4.000	6.371E+00	10.6	XNDI84	47	1.010	7.615E-02	15.0	XNBH82A
42	1.610	1.215E+00	80.0	XKHA57	46	4.500	8.364E+00	10.8	XNDI84	47	1.040	9.960E-02	15.0	XKME58
42	1.610	1.071E+00	15.0	XNBH82A	46	5.000	1.058E+01	11.6	XNDI84	47	1.100	9.880E-02	6.0	XKRS76
42	1.710	1.224E+00	15.0	XNBH82A	46	5.500	1.304E+01	11.3	XNDI84	47	1.100	9.098E-02	14.7	XNBE73
42	1.800	1.580E+00	5.1	XKRS76	46	6.000	1.525E+01	10.8	XNDI84	47	1.100	8.600E-02	34.0	XKBR78
42	1.800	1.480E+00	8.1	XNCR74	47	.200	2.800E-06	50.0	XKLL80	47	1.110	1.163E-01	15.0	XNBH82A
42	1.800	1.645E+00	10.2	XNDI84	47	.215	5.500E-06	50.0	XKLL80	47	1.200	1.726E-01	14.9	XKME58
42	1.810	1.530E+00	15.0	XNBH82A	47	.230	1.000E-05	25.0	XKLL80	47	1.200	1.340E-01	6.0	XKRS76
42	2.000	2.200E+00	5.0	XKRS76	47	.245	1.900E-05	25.0	XKLL80	47	1.200	1.213E-01	20.0	XNBE73
42	2.000	2.050E+00	7.8	XNCR74	47	.249	1.760E-05	34.0	XKBR78	47	1.200	1.400E-01	7.1	XNWL76
42	2.000	2.440E+00	25.0	XKAN78	47	.250	1.885E-05	20.0	XNJL80	47	1.200	1.300E-01	7.7	XNKH75
42	2.000	2.241E+00	9.9	XNDI84	47	.250	1.960E-05	12.4	XNRI81	47	1.200	1.380E-01	3.0	XNLA79
42	2.200	2.920E+00	5.1	XKRS76	47	.260	1.270E-04	15.0	XKME58	47	1.210	1.196E-01	15.0	XNBH82A
42	2.200	2.984E+00	10.0	XNDI84	47	.260	3.500E-05	25.0	XKLL80	47	1.250	1.490E-01	25.0	XKAN78
42	2.400	7.869E+00	25.2	XKLE53	47	.275	5.900E-05	25.0	XKLL80	47	1.300	1.780E-01	6.2	XKRS76
42	2.400	3.680E+00	4.9	XKRS76	47	.280	2.050E-04	15.0	XKME58	47	1.300	1.698E-01	14.3	XNBE73
42	2.400	3.733E+00	10.0	XNDI84	47	.300	3.287E-04	14.9	XKME58	47	1.310	1.911E-01	15.0	XNBH82A
42	2.500	4.060E+00	4.9	XKRS76	47	.300	9.840E-05	34.0	XKBR78	47	1.400	2.320E-01	6.0	XKRS76
42	2.500	3.130E+00	15.0	XKLI73	47	.300	1.400E-04	25.0	XKLL80	47	1.400	2.183E-01	16.7	XNBE73
42	2.500	4.510E+00	25.0	XKAN78	47	.300	1.047E-04	17.0	XNJL80	47	1.400	2.300E-01	8.7	XNWL76
42	2.600	4.420E+00	5.0	XKRS76	47	.300	1.160E-04	12.4	XNRI81	47	1.400	2.200E-01	9.1	XNKH75
42	2.600	4.628E+00	9.9	XNDI84	47	.320	4.922E-04	15.0	XKME58	47	1.400	2.400E-01	3.0	XNLA79
42	2.800	5.070E+00	4.9	XKRS76	47	.325	1.450E-04	27.0	XNBG80	47	1.410	2.576E-01	15.0	XNBH82A
42	2.800	5.615E+00	9.9	XNDI84	47	.340	7.387E-04	14.6	XKME58	47	1.500	2.960E-01	6.1	XKRS76
42	3.000	5.520E+00	5.1	XKRS76	47	.350	3.300E-04	10.0	XNLA79	47	1.500	2.790E-01	13.0	XNBE73
42	3.000	5.380E+00	15.0	XKLI73	47	.350	2.520E-04	26.0	XNBG80	47	1.500	2.850E-01	3.9	XNAK74
42	3.000	6.663E+00	10.0	XNDI84	47	.350	3.260E-04	15.0	XNJL80	47	1.500	2.990E-01	25.0	XKAN78
42	3.200	7.803E+00	10.8	XNDI84	47	.350	3.750E-04	12.4	XNRI81	47	1.510	3.241E-01	15.0	XNBH82A
42	3.500	9.639E+00	10.3	XNDI84	47	.353	4.090E-04	34.0	XKBR78	47	1.520	2.640E-01	34.0	XKBR78
42	4.000	1.000E+01	15.0	XKLI73	47	.360	1.062E-03	14.8	XKME58	47	1.600	3.700E-01	5.9	XKRS76
42	4.000	1.316E+01	10.5	XNDI84	47	.375	4.420E-04	25.0	XNBG80	47	1.600	3.396E-01	11.3	XNBE73
42	4.500	1.310E+01	15.0	XKLI73	47	.380	1.544E-03	15.1	XKME58	47	1.600	3.600E-01	8.3	XNWL76
42	4.500	1.683E+01	10.4	XNDI84	47	.400	2.133E-03	15.2	XKME58	47	1.600	3.100E-01	11.8	XNKH75
42	5.000	1.710E+01	15.0	XKLI73	47	.400	9.000E-04	11.1	XNWL76	47	1.600	3.700E-01	3.0	XNLA79

APPENDIX: TABLE II. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger-Electron Production, and Ionization

See page 112 for Explanation of Tables

Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
47	1.610	3.989E-01	15.0	XNBH82A	47	9.000	2.140E+01	15.0	XNLI73	49	.700	4.250E-03	12.0	XNMA79
47	1.700	5.312E-01	25.0	XKLE53	47	9.000	2.240E+01	6.0	XNBE78	49	.800	1.900E-02	10.5	XNWL76
47	1.700	4.570E-01	5.9	XKRS76	47	10.000	3.602E+01	15.0	XNBI72	49	.800	1.505E-02	10.0	XNJL82
47	1.700	4.124E-01	14.7	XNBE73	47	10.000	2.600E+01	15.0	XNLI73	49	.800	8.500E-03	10.0	XNMA79
47	1.710	4.654E-01	15.0	XNBH82A	47	10.000	2.620E+01	6.0	XNBE78	49	.900	3.700E-02	10.8	XNKL76
47	1.720	4.150E-01	34.0	XKBR78	47	10.500	2.450E+01	15.0	XNLI73	49	.900	2.482E-02	10.0	XNJL82
47	1.800	5.540E-01	6.0	XKRS76	47	10.930	3.220E+01	4.0	XNAK74	49	.900	1.878E-02	11.3	XNMA79
47	1.800	5.095E-01	14.3	XNBE73	47	11.000	3.901E+01	15.1	XNBI72	49	1.000	3.600E-02	6.1	XKRS77
47	1.800	5.200E-01	7.7	XNWL76	47	11.000	3.090E+01	15.0	XNLI73	49	1.000	6.100E-02	13.1	XNKL76
47	1.800	4.700E-01	10.6	XNKH75	47	11.000	3.240E+01	6.0	XNBE78	49	1.000	5.000E-02	8.0	XNWL76
47	1.800	5.230E-01	3.0	XNLA79	47	12.000	4.399E+01	15.1	XNBI72	49	1.000	3.765E-02	10.0	XNJL82
47	1.810	5.900E-01	15.0	XNBH82A	47	12.000	3.690E+01	15.0	XNLI73	49	1.000	2.635E-02	11.3	XNMA79
47	1.900	6.630E-01	6.0	XKRS76	47	12.000	3.580E+01	6.0	XNBE78	49	1.100	9.400E-02	9.6	XNKL76
47	1.900	5.944E-01	14.3	XNBE73	47	13.000	4.997E+01	15.0	XNBI72	49	1.100	3.901E-02	12.0	XNMA79
47	1.910	5.800E-01	34.0	XKBR78	47	13.000	3.930E+01	6.0	XNBE78	49	1.200	7.300E-02	6.0	XKRS77
47	1.920	9.960E-01	25.0	XKLE53	47	14.000	5.727E+01	14.5	XNBI72	49	1.200	1.300E-01	15.4	XNKL76
47	2.000	8.134E-01	15.3	XNBI72	47	14.000	4.290E+01	6.0	XNBE78	49	1.200	8.500E-02	8.2	XNWL76
47	2.000	7.820E-01	6.0	XKRS76	47	15.000	5.893E+01	15.5	XNBI72	49	1.200	5.703E-02	11.9	XNMA79
47	2.000	7.035E-01	15.5	XNBE73	47	15.000	4.860E+01	6.0	XNBE78	49	1.300	1.700E-01	17.6	XNKL76
47	2.000	7.520E-01	4.0	XNAK74	47	17.000	7.387E+01	14.6	XNBI72	49	1.300	7.225E-02	11.8	XNMA79
47	2.000	7.500E-01	8.0	XNWL76	47	18.000	8.217E+01	15.2	XNBI72	49	1.400	1.300E-01	6.2	XKRS77
47	2.000	6.700E-01	10.4	XNKH75	47	19.000	7.885E+01	14.7	XNBI72	49	1.400	2.300E-01	13.0	XNKL76
47	2.000	7.520E-01	3.0	XNLA79	47	20.000	7.885E+01	14.7	XNBI72	49	1.400	1.500E-01	6.7	XNWL76
47	2.000	7.890E-01	25.0	XKAN78	47	21.000	8.632E+01	15.4	XNBI72	49	1.400	9.265E-02	9.2	XNMA79
47	2.100	9.130E-01	6.0	XKRS76	47	22.000	9.794E+01	15.3	XNBI72	49	1.500	2.600E-01	11.5	XNKL76
47	2.100	8.127E-01	14.9	XNBE73	47	23.000	9.794E+01	15.3	XNBI72	49	1.500	1.224E-01	10.4	XNMA79
47	2.170	1.618E+00	25.1	XKLE53	47	24.000	8.383E+01	14.9	XNBI72	49	1.600	2.000E-01	6.0	XKRS77
47	2.200	1.050E+00	5.7	XKRS76	47	25.000	8.466E+01	14.7	XNBI72	49	1.600	3.400E-01	11.8	XNKL76
47	2.200	9.340E-01	15.6	XNBE73	47	26.000	9.462E+01	14.9	XNBI72	49	1.600	2.300E-01	8.7	XNWL76
47	2.200	9.900E-01	8.1	XNWL76	47	27.000	9.628E+01	14.7	XNBI72	49	1.600	1.488E-01	11.4	XNMA79
47	2.200	1.000E+00	5.0	XNLA79	48	.600	4.800E-03	10.4	XNKH75	49	1.700	3.900E-01	12.8	XNKL76
47	2.250	1.031E+00	15.3	XNBE73	48	.600	2.268E-03	11.1	XNMA79	49	1.700	1.981E-01	10.7	XNMA79
47	2.300	1.200E+00	5.8	XKRS76	48	.700	6.140E-03	11.6	XNMA79	49	1.800	3.100E-01	6.1	XKRS77
47	2.300	1.067E+00	14.8	XNBE73	48	.800	1.800E-02	11.1	XNKH75	49	1.800	4.100E-01	12.2	XNKL76
47	2.400	2.307E+00	25.2	XKLE53	48	.800	1.571E-02	10.7	XNMA79	49	1.800	3.500E-01	8.6	XNWL76
47	2.400	1.340E+00	6.0	XKRS76	48	.900	2.520E-02	11.7	XNMA79	49	1.800	2.388E-01	10.7	XNMA79
47	2.400	1.213E+00	20.0	XNBE73	48	1.000	4.600E-02	6.1	XKRS77	49	1.900	5.100E-01	11.8	XNKL76
47	2.400	1.300E+00	7.7	XNWL76	48	1.000	4.900E-02	10.2	XNKH75	49	1.900	2.533E-01	11.7	XNMA79
47	2.500	1.490E+00	6.0	XKRS76	48	1.000	3.461E-02	12.1	XNMA79	49	2.000	4.300E-01	6.0	XKRS77
47	2.500	1.334E+00	18.2	XNBE73	48	1.100	5.460E-02	12.3	XNMA79	49	2.000	4.300E-01	11.6	XNKL76
47	2.500	1.440E+00	4.2	XNAK74	48	1.200	9.000E-02	6.0	XKRS77	49	2.000	5.000E-01	8.0	XNWL76
47	2.500	1.080E+00	15.0	XNLI73	48	1.200	1.000E-01	10.0	XNKH75	49	2.000	3.034E-01	11.2	XNMA79
47	2.500	1.660E+00	25.0	XKAN78	48	1.200	7.896E-02	11.7	XNMA79	49	2.100	6.900E-01	13.0	XNKL76
47	2.600	1.630E+00	6.1	XKRS76	48	1.300	1.075E-01	11.7	XNMA79	49	2.100	3.715E-01	11.4	XNMA79
47	2.600	1.577E+00	15.4	XNBE73	48	1.400	1.600E-01	6.3	XKRS77	49	2.200	5.900E-01	5.9	XKRS77
47	2.640	3.320E+00	25.0	XKLE53	48	1.400	1.700E-01	11.8	XNKH75	49	2.200	7.100E-01	12.7	XNKL76
47	2.700	1.770E+00	6.2	XKRS76	48	1.400	1.403E-01	12.0	XNMA79	49	2.200	6.600E-01	7.6	XNWL76
47	2.700	1.698E+00	14.3	XNBE73	48	1.500	1.697E-01	12.4	XNMA79	49	2.300	9.900E-01	12.1	XNKL76
47	2.800	1.890E+00	5.8	XKRS76	48	1.600	2.500E-01	6.0	XKRS77	49	2.400	7.600E-01	6.1	XKRS77
47	2.800	1.941E+00	12.5	XNBE73	48	1.600	2.700E-01	11.1	XNKH75	49	2.400	1.310E+00	11.5	XNKL76
47	2.880	6.308E+00	25.0	XKLE53	48	1.600	2.251E-01	11.2	XNMA79	49	2.400	8.500E-01	7.1	XNWL76
47	2.900	2.010E+00	6.0	XKRS76	48	1.700	2.730E-01	12.3	XNMA79	49	2.500	1.490E+00	12.1	XNKL76
47	2.900	2.062E+00	17.6	XNBE73	48	1.800	3.800E-01	6.1	XKRS77	49	2.600	9.000E-01	6.0	XKRS77
47	2.980	2.190E+00	4.1	XNAK74	48	1.800	3.900E-01	10.3	XNKH75	49	2.800	1.100E+00	6.4	XKRS77
47	3.000	2.598E+00	15.0	XNBI72	48	1.800	3.184E-01	11.9	XNMA79	49	3.000	1.300E+00	6.2	XKRS77
47	3.000	2.100E+00	6.2	XKRS76	48	1.900	3.973E-01	10.6	XNMA79	49	3.000	1.510E+00	10.0	XNFB74
47	3.000	2.305E+00	15.8	XNBE73	48	2.000	5.300E-01	6.0	XKRS77	49	3.000	1.120E+00	5.4	XNBS78
47	3.000	2.420E+00	4.1	XNAK74	48	2.000	5.400E-01	9.3	XNKH75	49	5.000	5.200E+00	10.0	XNFB74
47	3.000	1.880E+00	15.0	XNLI73	48	2.000	4.738E-01	12.4	XNMA79	50	.260	5.755E-05	14.9	XKME58
47	3.000	1.930E+00	5.2	XNBS78	48	2.100	5.250E-01	12.0	XNMA79	50	.280	9.707E-05	15.0	XKME58
47	3.100	2.547E+00	14.3	XNBE73	48	2.200	7.100E-01	6.1	XKRS77	50	.320	2.268E-04	14.0	XKME58
47	3.200	2.790E+00	17.4	XNBE73	48	2.200	6.325E-01	12.0	XNMA79	50	.360	5.248E-04	15.1	XKME58
47	3.300	3.033E+00	16.0	XNBE73	48	2.300	6.619E-01	12.1	XNMA79	50	.380	7.731E-04	14.4	XKME58
47	3.400	3.275E+00	14.8	XNBE73	48	2.400	9.000E-01	6.0	XKRS77	50	.400	1.065E-03	15.3	XKME58
47	3.500	3.518E+00	13.8	XNBE73	48	2.500	9.070E-01	15.0	XNLI73	50	.420	8.620E-04	15.0	XNBH82A
47	3.600	3.760E+00	16.1	XNBE73	48	2.600	1.100E+00	6.4	XKRS77	50	.440	2.027E-03	14.8	XKME58
47	3.700	4.003E+00	15.2	XNBE73	48	2.800	1.300E+00	6.2	XKRS77	50	.450	2.285E-03	15.0	XKME58
47	3.800	4.367E+00	13.9	XNBE73	48	3.000	1.600E+00	6.3	XKRS77	50	.500	1.030E-03	25.0	XKAN78
47	4.000	6.142E+00	14.9	XNBI72	48	3.000	1.650E+00	15.0	XNLI73	50	.500	1.362E-03	13.9	XNDI84
47	4.000	3.700E+00	10.8	XNBD78	48	3.000	1.470E+00	4.8	XNBS78	50	.510	1.810E-03	15.0	XNBH82A
47	4.000	4.180E+00	15.0	XNLI73	48	4.000	3.860E+00	15.0	XNLI73	50	.600	5.584E-03	15.1	XKME58
47	4.000	5.000E+00	6.0	XNBE78	48	4.500	4.310E+00	15.0	XNLI73	50	.600	3.200E-03	9.4	XNKH75
47	4.500	5.490E+00	15.0	XNLI73	48	5.000	5.500E+00	15.0	XNLI73	50	.600	3.103E-03	15.0	XNBH82A
47	4.970	7.680E+00	1.0	XNAK74	48	6.000	8.710E+00	15.0	XNLI73	50	.600	3.595E-03	10.3	XNDI84
47	5.000	1.004E+01	14.9	XNBI72	48	6.500	8.530E+00	15.0	XNLI73	50	.700	5.689E-03	15.0	XNBH82A
47	5.000	6.810E+00	15.0	XNLI73	48	7.000	1.170E+01	15.0	XNLI73	50	.700	7.629E-03	9.5	XNDI84
47	5.960	1.120E+01	3.6	XNAK74	48	7.000	1.319E+01	15.9	XNDO81A	50	.740	1.409E-02	15.2	XKME58
47	6.000	1.403E+01	14.8	XNBI72	48	8.000	1.500E+01	15.0	XNLI73	50	.750	7.990E-03	25.0	XKAN78
47	6.000	8.880E+00	15.0	XNLI73	48	8.500	1.820E+01	15.0	XNLI73	50	.750	7.758E-03	15.0	XNBH82A
47	6.500	1.250E+01	15.0	XNLI73	48	9.000	1.770E+01	15.0	XNLI73	50	.790	1.034E-02	15.0	XNBH82A
47	6.940	1.540E+01	3.9	XNAK74	48	10.000	2.200E+01	15.0	XNLI73	50	.800	8.200E-03	9.8	XNKH75
47	6.960	1.480E+01	4.1	XNAK74	48	10.500	2.120E+01	15.0	XNLI73	50	.800	1.457E-02	10.1	XNDI84
47	7.000	1.901E+01	14.8	XNBI72	48	11.000	2.700E+01	15.0	XNLI73	50	.830	1.465E-02	15.0	XNBH82A
47	7.000	1.410E+01	15.0	XNLI73	48	12.000	2.990E+01	15.0	XNLI73	50	.880	2.069E-02	15.0	XNBH82A
47	7.000	1.701E+01	16.1	XNDO81A	49	.300	3.358E-05	18.0	XNJL82	50	.900	2.431E-02	9.6	XNDI84
47	7.500	1.690E+01	6.0	XNBE78	49	.350	1.317E-04	16.0	XNJL82	50	.935	3.805E-02	14.9	XKME58
47	8.000	2.399E+01	14.9	XNBI72	49	.400	3.468E-04	14.0	XNJL82	50	1.000	3.180E-02	15.4	XNBE73
47	8.000	1.880E+01	15.0	XNLI73	49	.500	1.462E-03	12.0	XNJL82	50	1.000	2.600E-02	6.2	XKRS77
47	8.000	1.830E+01	6.0	XNBE78	49	.600	5.500E-03	7.3	XNWL76	50	1.000	3.200E-02	9.4	XNKH75
47	8.500	2.050E+01	15.0	XNLI73	49	.600	3.774E-03	10.0	XNJL82	50	1.000	2.920E-02	25.0	XKAN78
47	8.940	2.360E+01	3.8	XNAK74	49	.600	1.734E-03	9.8	XNMA79	50	1.000	3.698E-02	9.6	XNDI84
47	9.000	2.897E+01	14.9	XNBI72	49	.700	8.262E-03	10.0	XNJL82	50	1.010	3.362E-02	15.0	XNBH82A

APPENDIX: TABLE II. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger-Electron Production, and Ionization

See page 112 for Explanation of Tables

Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
50	1.040	5.927E-02	18.8	XKME58
50	1.100	5.439E-02	9.2	XNDI81
50	1.110	4.827E-02	15.0	XNBH82A
50	1.200	5.600E-02	6.1	XKRS77
50	1.200	5.800E-02	10.3	XNKH75
50	1.200	7.499E-02	9.2	XNDI84
50	1.210	6.810E-02	15.0	XNBH82A
50	1.250	7.300E-02	25.0	XKAN78
50	1.300	9.999E-02	9.5	XNDI84
50	1.310	9.482E-02	15.0	XNBH82A
50	1.400	1.000E-01	6.0	XKRS77
50	1.400	1.000E-01	10.0	XNKH75
50	1.400	1.319E-01	9.1	XNDI84
50	1.410	1.293E-01	15.0	XNBH82A
50	1.500	1.435E-01	29.9	XNLD73
50	1.500	1.460E-01	25.0	XKAN78
50	1.510	1.638E-01	15.0	XNBH82A
50	1.600	1.600E-01	6.3	XKRS77
50	1.600	1.600E-01	12.5	XNKH75
50	1.600	2.121E-01	8.9	XNDI84
50	1.610	1.896E-01	15.0	XNBH82A
50	1.710	2.069E-01	15.0	XNBH82A
50	1.800	2.400E-01	5.8	XKRS77
50	1.800	2.400E-01	12.5	XNKH75
50	1.800	3.130E-01	15.0	XNIS74
50	1.800	3.138E-01	9.3	XNDI84
50	1.810	2.672E-01	15.0	XNBH82A
50	2.000	4.037E-01	29.8	XNLD73
50	2.000	3.400E-01	5.9	XKRS77
50	2.000	3.200E-01	9.4	XNKH75
50	2.000	4.040E-01	25.0	XKAN78
50	2.000	4.370E-01	8.9	XNDI84
50	2.200	4.500E-01	6.0	XKRS77
50	2.200	5.820E-01	14.9	XNIS74
50	2.200	5.922E-01	8.9	XNDI84
50	2.250	5.504E-01	15.6	XNBE73
50	2.333	6.100E-01	15.1	XNHO75A
50	2.400	5.800E-01	6.0	XKRS77
50	2.400	7.620E-01	8.9	XNDI84
50	2.500	7.903E-01	30.4	XNLD73
50	2.500	7.650E-01	25.0	XKAN78
50	2.600	7.300E-01	6.0	XKRS77
50	2.600	9.190E-01	15.0	XNIS74
50	2.600	9.741E-01	8.9	XNDI84
50	2.667	9.410E-01	15.0	XNHO75A
50	2.800	9.000E-01	6.0	XKRS77
50	2.800	1.190E+00	9.4	XNDI84
50	2.900	1.000E+00	6.0	XKRS77
50	3.000	1.366E+00	30.2	XNLD73
50	3.000	1.380E+00	15.2	XNHO75A
50	3.000	1.345E+00	18.2	XNBE73
50	3.000	1.200E+00	5.8	XKRS77
50	3.000	1.320E+00	15.2	XNIS74
50	3.000	9.400E-01	5.3	XNBS78
50	3.000	1.457E+00	8.9	XNDI84
50	3.200	1.710E+00	15.2	XNIS74
50	3.200	1.724E+00	10.0	XNDI84
50	3.333	1.760E+00	14.8	XNHO75A
50	3.500	2.079E+00	30.2	XNLD73
50	3.500	2.224E+00	9.7	XNDI84
50	3.600	2.250E+00	15.1	XNIS74
50	3.667	2.390E+00	15.1	XNHO75A
50	4.000	3.007E+00	31.4	XNLD73
50	4.000	2.960E+00	14.9	XNHO75A
50	4.000	3.310E+00	15.1	XNIS74
50	4.000	3.112E+00	9.7	XNDI84
50	4.333	3.690E+00	14.9	XNHO75A
50	4.400	4.030E+00	14.9	XNIS74
50	4.500	4.037E+00	29.8	XNLD73
50	4.500	4.172E+00	9.7	XNDI84
50	4.667	4.400E+00	15.0	XNHO75A
50	5.000	4.982E+00	29.3	XNLD73
50	5.000	5.250E+00	15.0	XNHO75A
50	5.000	5.258E+00	9.7	XNDI84
50	5.333	6.050E+00	15.0	XNHO75A
50	5.500	5.927E+00	30.4	XNLD73
50	5.500	6.637E+00	9.7	XNDI84
50	5.667	7.030E+00	15.1	XNHO75A
50	6.000	6.700E+00	29.5	XNLD73
50	6.000	7.930E+00	9.7	XNDI84
50	7.000	1.065E+01	12.0	XNDO81A
50	160.000	9.449E+01	5.5	XNJA72
51	.300	1.860E-05	12.4	XNRI81
51	.350	7.500E-05	23.0	XNJL82
51	.350	7.250E-05	12.4	XNRI81
51	.400	1.916E-04	14.0	XNJL82
51	.400	2.030E-04	12.4	XNRI81
51	.450	5.490E-04	12.4	XNRI81
51	.500	8.670E-04	12.0	XNJL82
51	.500	1.120E-03	12.4	XNRI81
51	.550	1.820E-03	12.4	XNRI81
51	.600	1.400E-03	14.3	XNKH75
51	.600	2.280E-03	10.0	XNJL82
51	.600	3.000E-03	12.4	XNRI81
51	.650	4.770E-03	12.4	XNRI81
51	.700	5.124E-03	10.0	XNJL82
51	.800	1.000E-02	10.0	XNKH75
51	.800	8.670E-03	10.0	XNJL82
51	.900	1.483E-02	10.0	XNJL82
51	1.000	2.577E-02	11.3	XNBE73
51	1.000	2.600E-02	11.5	XNKH75
51	1.000	2.566E-02	10.0	XNJL82
51	1.200	5.500E-02	10.9	XNKH75
51	1.400	1.000E-01	10.0	XNKH75
51	1.500	1.925E-01	30.2	XNLD73
51	1.600	1.600E-01	12.5	XNKH75
51	1.800	2.300E-01	8.7	XNKH75
51	2.000	3.988E-01	30.4	XNLD73
51	2.000	3.400E-01	8.8	XNKH75
51	2.250	5.399E-01	15.9	XNBE73
51	2.500	7.543E-01	29.9	XNLD73
51	3.000	1.318E+00	30.3	XNLD73
51	3.000	1.227E+00	20.0	XNBE73
51	3.500	1.985E+00	30.1	XNLD73
51	4.000	2.852E+00	30.1	XNLD73
51	4.500	3.988E+00	30.4	XNLD73
51	5.000	4.855E+00	30.4	XNLD73
51	5.500	6.156E+00	29.6	XNLD73
52	.600	2.200E-03	9.1	XNKH75
52	.800	9.000E-03	10.0	XNKH75
52	1.000	1.900E-02	10.5	XNKH75
52	1.200	3.300E-02	9.1	XNKH75
52	1.400	5.900E-02	10.2	XNKH75
52	1.600	9.200E-02	9.8	XNKH75
52	1.600	6.143E-02	6.0	XNAV83
52	1.800	1.200E-01	8.3	XNKH75
52	1.800	8.225E-02	6.4	XNAV83
52	2.000	1.800E-01	11.1	XNKH75
52	2.000	1.251E-01	5.6	XNAV83
52	2.200	1.794E-01	5.9	XNAV83
52	2.400	2.415E-01	4.7	XNAV83
52	2.600	2.817E-01	5.0	XNAV83
52	2.800	4.165E-01	5.0	XNAV83
52	3.000	5.189E-01	4.9	XNAV83
52	7.000	7.017E+00	16.2	XNDO81A
53	.600	1.500E-03	13.3	XNWL76
53	.800	6.200E-03	8.1	XNWL76
53	1.000	1.400E-02	7.1	XNWL76
53	1.200	3.200E-02	9.4	XNWL76
53	1.400	5.900E-02	8.5	XNWL76
53	1.500	8.224E-02	6.0	XNAV84
53	1.600	9.200E-02	7.6	XNWL76
53	1.800	1.500E-01	6.7	XNWL76
53	2.000	2.100E-01	9.5	XNWL76
53	2.000	2.059E-01	6.0	XNAV84
53	2.200	3.000E-01	10.0	XNWL76
53	2.250	3.009E-01	4.1	XNAV84
53	2.400	3.600E-01	8.3	XNWL76
53	2.500	4.648E-01	5.3	XNAV84
53	2.750	4.907E-01	5.5	XNAV84
53	3.000	5.500E-01	5.5	XNBS78
53	3.000	8.027E-01	6.1	XNAV84
55	1.130	1.600E-02	12.5	XNWL76
55	1.340	2.800E-02	7.1	XNWL76
55	1.500	5.915E-02	6.3	XNAV84
55	1.550	5.000E-02	8.0	XNWL76
55	1.760	7.800E-02	7.7	XNWL76
55	1.970	1.100E-01	9.1	XNWL76
55	2.000	1.649E-01	7.5	XNAV84
55	2.180	1.700E-01	5.9	XNWL76
55	2.250	2.244E-01	5.5	XNAV84
55	2.390	2.100E-01	9.5	XNWL76
55	2.500	3.534E-01	7.0	XNAV84
55	2.600	2.900E-01	6.9	XNWL76
55	2.700	3.000E-01	6.7	XNWL76
55	2.750	4.303E-01	5.8	XNAV84
55	3.000	6.014E-01	6.2	XNAV84
55	4.000	6.200E-01	22.6	XNBD78
56	.600	7.640E-04	13.3	XNDI84
56	.700	1.723E-03	12.6	XNDI84
56	.800	3.500E-03	11.1	XNDI84
56	.900	6.278E-03	10.5	XNDI84
56	1.000	8.400E-03	15.5	XNKH75
56	1.000	1.028E-02	10.5	XNDI84
56	1.100	1.515E-02	10.1	XNDI84
56	1.200	1.400E-02	14.3	XNKH75
56	1.200	2.183E-02	9.9	XNDI84
56	1.300	2.941E-02	9.8	XNDI84
56	1.400	2.000E-02	30.0	XNKH75
56	1.400	3.933E-02	9.6	XNDI84
56	1.500	5.060E-02	9.6	XNDI84
56	1.500	4.512E-02	8.3	XNAV84
56	1.600	3.000E-02	23.3	XNKH75
56	1.600	6.675E-02	9.6	XNDI84
56	1.700	8.217E-02	9.6	XNDI84
56	1.800	4.100E-02	22.7	XNKH75
56	1.800	9.651E-02	10.3	XNDI84
56	2.000	4.900E-02	22.4	XNKH75
56	2.000	1.116E-01	9.6	XNDI84
56	2.000	1.038E-01	7.2	XNAV84
56	2.200	1.903E-01	9.0	XNDI84
56	2.250	1.989E-01	6.3	XNAV84
56	2.100	2.571E-01	9.1	XNDI84
56	2.500	3.170E-01	5.9	XNAV84
56	2.600	3.274E-01	8.5	XNDI84
56	2.750	3.928E-01	6.3	XNAV84
56	2.800	4.104E-01	9.0	XNDI84
56	3.000	4.979E-01	9.1	XNDI84
56	3.000	5.511E-01	6.7	XNAV84
56	3.200	6.089E-01	9.6	XNDI84
56	3.500	7.956E-01	9.6	XNDI84
56	4.000	1.146E+00	10.2	XNDI84
56	4.500	1.533E+00	10.0	XNDI84
56	5.000	2.002E+00	9.9	XNDI84
56	5.500	2.562E+00	9.9	XNDI84
56	6.000	3.067E+00	9.7	XNDI84
56	7.000	3.757E+00	16.1	XNDO81A
56	160.000	4.226E+01	20.9	XNJA72
57	.800	2.700E-03	11.8	XNKH75
57	1.000	7.400E-03	14.9	XNKH75
57	1.200	1.100E-02	14.3	XNKH75
57	1.100	2.500E-02	16.0	XNKH75
57	1.600	3.100E-02	16.1	XNKH75
57	1.800	3.900E-02	15.4	XNKH75
57	2.000	5.200E-02	15.4	XNKH75
57	2.000	1.158E-01	6.5	XNAV84
57	2.250	1.880E-01	6.6	XNAV84
57	2.500	2.366E-01	5.3	XNAV84
57	2.750	3.424E-01	7.3	XNAV84
57	3.000	4.719E-01	5.3	XNAV84
58	.500	9.565E-05	13.0	XNJL82
58	.600	3.200E-04	9.4	XNWL76
58	.600	3.434E-04	11.0	XNJL82
58	.600	4.378E-04	12.3	XNDI84
58	.650	6.886E-04	11.5	XNDI84
58	.700	8.818E-04	10.0	XNJL82
58	.700	1.058E-03	12.1	XNDI84
58	.800	1.700E-03	5.9	XNWL76
58	.800	1.968E-03	10.0	XNJL82
58	.800	2.225E-03	12.3	XNDI84
58	.900	3.872E-03	10.0	XNJL82
58	.900	4.013E-03	10.9	XNDI84
58	1.000	5.100E-03	7.8	XNWL76
58	1.000	5.691E-03	10.0	XNJL82
58	1.000	6.512E-03	10.6	XNDI84
58	1.100	1.003E-02	10.9	XNDI84
58	1.200	1.100E-02	9.1	XNWL76
58	1.200	1.459E-02	10.6	XNDI84
58	1.300	2.016E-02	10.4	XNDI84
58	1.400	2.100E-02	9.5	XNWL76
58	1.100	2.700E-02	10.5	XNDI84
58	1.500	3.484E-02	10.7	XNDI84
58	1.600	3.100E-02	6.5	XNWL76
58	1.600	4.478E-02	10.6	XNDI84
58	1.700	5.618E-02	10.6	XNDI84
58	1.800	5.600E-02	7.1	XNWL76
58	1.800	6.867E-02	10.5	XNDI84
58	2.000	7.400E-02	8.1	XNWL76
58	2.000	9.941E-02	11.0	XNDI84
58	2.200	1.100E-01	9.1	XNWL76
58	2.200	1.341E-01	10.9	XNDI84
58	2.400	1.400E-01	7.1	XNWL76
58	2.400	1.821E-01	10.5	XNDI84
58	2.600	2.344E-01	9.3	XNDI84
58	2.800	2.928E-01	9.3	XNDI84
58	3.000	3.621E-01	9.3	XNDI84
58	3.200	4.396E-01	9.1	XNDI84
58	3.500	5.727E-01	9.7	XNDI84
58	4.000	8.299E-01	9.6	XNDI84
58	4.500	1.140E+00	9.6	XNDI84
58	5.000	1.496E+00	9.8	XNDI84
58	5.500	1.897E+00	9.6	XNDI84
58	6.000	2.314E+00	9.7	XNDI84
58	7.000	2.642E+00	15.9	XNDO81A
59	.500	3.840E-05	25.0	XKAN78
59	.600	2.900E-04	10.3	XNWL76
59	.750	6.220E-04	25.0	XKAN78
59	.800	1.300E-03	7.7	XNWL76
59	1.000	3.700E-03	8.1	XNWL76
59	1.000	3.290E-03	25.0	XKAN78
59	1.200	8.300E-03	8.4	XNWL76
59	1.250	9.150E-03	25.0	XKAN78
59	1.400	1.600E-02	6.3	XNWL76
59	1.500	1.920E-02	25.0	XKAN78
59	1.600	2.700E-02	7.4	XNWL76
59	1.800	3.900E-02	7.7	XNWL76

APPENDIX: TABLE II. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger-Electron Production, and Ionization

See page 112 for Explanation of Tables

Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
59	2.000	5.700E-02	7.0	XNWL76
59	2.000	6.110E-02	25.0	XKAN78
59	2.200	8.100E-02	7.4	XNWL76
59	2.400	1.000E-01	10.0	XNWL76
59	2.500	1.460E-01	25.0	XKAN78
60	.600	2.100E-04	14.3	XNWL76
60	.700	6.834E-04	13.3	XNDI84
60	.800	1.100E-03	9.1	XNWL76
60	.800	1.501E-03	12.3	XNDI84
60	.880	2.020E-03	7.9	INZE86
60	.900	2.662E-03	10.4	XNDI84
60	1.000	3.400E-03	8.8	XNWL76
60	1.000	4.430E-03	10.0	XNDI84
60	1.100	6.861E-03	9.7	XNDI84
60	1.100	6.100E-03	8.2	INZE86
60	1.200	8.100E-03	8.6	XNWL76
60	1.200	1.022E-02	9.9	XNDI84
60	1.300	1.109E-02	9.8	XNDI84
60	1.310	1.380E-02	6.5	INZE86
60	1.400	1.600E-02	6.3	XNWL76
60	1.400	1.906E-02	9.7	XNDI84
60	1.500	2.487E-02	9.6	XNDI84
60	1.510	2.540E-02	6.3	INZE86
60	1.600	2.600E-02	7.7	XNWL76
60	1.600	3.187E-02	9.8	XNDI84
60	1.720	1.300E-02	7.0	INZE86
60	1.800	3.900E-02	7.7	XNWL76
60	1.800	4.918E-02	9.6	XNDI84
60	1.930	6.100E-02	6.6	INZE86
60	2.000	5.300E-02	7.5	XNWL76
60	2.000	7.064E-02	9.3	XNDI84
60	2.130	8.900E-02	6.7	INZE86
60	2.200	7.500E-02	8.0	XNWL76
60	2.200	9.947E-02	9.3	XNDI84
60	2.330	1.200E-01	6.7	INZE86
60	2.400	9.700E-02	8.2	XNWL76
60	2.400	1.299E-01	9.2	XNDI84
60	2.540	1.540E-01	6.5	INZE86
60	2.550	1.680E-01	13.1	INEA87
60	2.600	1.685E-01	9.3	XNDI84
60	2.660	1.470E-01	12.9	INSZ87
60	2.750	1.940E-01	8.2	INEA87
60	2.800	2.109E-01	9.2	XNDI84
60	2.950	2.310E-01	8.2	INSZ87
60	2.950	2.640E-01	8.3	INEA87
60	3.000	2.100E-01	10.0	XNFB74
60	3.000	2.634E-01	9.4	XNDI84
60	3.150	3.300E-01	8.8	INEA87
60	3.200	3.233E-01	10.5	XNDI84
60	3.350	4.050E-01	10.9	INEA87
60	3.500	4.209E-01	10.5	XNDI84
60	3.550	4.590E-01	10.9	INEA87
60	3.960	5.450E-01	8.8	INSZ87
60	4.000	6.060E-01	10.8	XNDI84
60	4.430	8.630E-01	9.5	INSZ87
60	4.500	8.317E-01	10.5	XNDI84
60	4.830	1.135E+00	8.6	INSZ87
60	5.000	8.400E-01	10.0	XNFB74
60	5.000	1.087E+00	11.0	XNDI84
60	5.340	1.370E+00	8.8	INSZ87
60	5.500	1.409E+00	11.1	XNDI84
60	5.950	2.000E+00	8.5	INSZ87
60	6.000	1.731E+00	10.6	XNDI84
60	6.450	2.320E+00	9.1	INSZ87
62	.700	3.967E-04	14.8	XNDI84
62	.710	6.100E-04	7.0	INZE86
62	.750	4.270E-04	25.0	XKAN78
62	.800	9.253E-04	12.8	XNDI84
62	.900	1.737E-03	14.4	XNDI84
62	.950	2.390E-03	6.3	INZE86
62	1.000	2.130E-03	25.0	XKAN78
62	1.000	2.964E-03	12.2	XNDI84
62	1.100	4.691E-03	12.1	XNDI84
62	1.160	5.650E-03	6.2	INZE86
62	1.200	7.060E-03	12.0	XNDI84
62	1.250	6.310E-03	25.0	XKAN78
62	1.300	9.847E-03	12.3	XNDI84
62	1.370	1.130E-02	6.2	INZE86
62	1.400	1.356E-02	11.6	XNDI84
62	1.500	1.390E-02	25.0	XKAN78
62	1.500	1.765E-02	12.1	XNDI84
62	1.560	1.890E-02	5.8	INZE86
62	1.600	2.304E-02	11.7	XNDI84
62	1.700	2.889E-02	11.6	XNDI84
62	1.760	3.100E-02	6.5	INZE86
62	1.800	3.567E-02	11.2	XNDI84
62	1.970	4.700E-02	6.4	INZE86
62	2.000	4.270E-02	25.0	XKAN78
62	2.000	5.082E-02	11.3	XNDI84
62	2.170	6.600E-02	6.1	INZE86
62	2.200	7.116E-02	11.1	XNDI84
62	2.370	9.000E-02	6.7	INZE86
62	2.400	9.569E-02	11.6	XNDI84
62	2.500	8.910E-02	25.0	XKAN78
62	2.560	1.330E-01	10.5	INEA87
62	2.570	1.180E-01	6.8	INZE86
62	2.600	1.245E-01	11.2	XNDI84
62	2.660	1.230E-01	11.1	INSZ87
62	2.760	1.660E-01	10.8	INEA87
62	2.800	1.570E-01	10.6	XNDI84
62	2.960	1.840E-01	9.2	INSZ87
62	2.960	1.960E-01	10.7	INEA87
62	3.000	1.951E-01	10.5	XNDI84
62	3.160	2.570E-01	10.9	INEA87
62	3.200	2.341E-01	10.3	XNDI84
62	3.360	3.030E-01	10.9	INEA87
62	3.500	3.112E-01	9.9	XNDI84
62	3.560	3.710E-01	11.1	INEA87
62	3.770	4.170E-01	14.1	INEA87
62	3.970	4.570E-01	9.0	INSZ87
62	4.000	4.515E-01	10.1	XNDI84
62	4.430	6.110E-01	9.5	INSZ87
62	4.500	6.271E-01	10.2	XNDI84
62	4.920	8.050E-01	9.4	INSZ87
62	5.000	8.222E-01	10.2	XNDI84
62	5.440	1.090E+00	10.1	INSZ87
62	5.500	1.050E+00	10.6	XNDI84
62	5.940	1.370E+00	10.2	INSZ87
62	6.000	1.319E+00	10.6	XNDI84
62	6.390	1.640E+00	9.8	INSZ87
62	6.870	2.040E+00	9.8	INSZ87
62	7.000	1.754E+00	12.2	XNDO81A
62	160.000	4.547E+01	5.9	XNJA72
63	.800	8.500E-04	9.4	XNWL76
63	1.000	2.000E-03	10.0	XNWL76
63	1.200	4.500E-03	8.9	XNWL76
63	1.400	9.100E-03	8.8	XNWL76
63	1.600	1.400E-02	7.1	XNWL76
63	1.800	2.300E-02	8.7	XNWL76
63	2.000	3.300E-02	9.1	XNWL76
63	2.200	4.400E-02	9.1	XNWL76
63	2.400	5.900E-02	8.5	XNWL76
64	.800	7.100E-04	9.9	XNWL76
64	.800	5.604E-04	8.3	XNGO83
64	1.000	2.200E-03	9.1	XNWL76
64	1.110	3.176E-03	6.5	XNGO83
64	1.200	4.900E-03	8.2	XNWL76
64	1.320	7.005E-03	6.7	XNGO83
64	1.400	8.600E-03	8.1	XNWL76
64	1.530	1.308E-02	5.7	XNGO83
64	1.600	1.500E-02	6.7	XNWL76
64	1.730	2.055E-02	5.5	XNGO83
64	1.800	2.100E-02	9.5	XNWL76
64	1.940	3.148E-02	5.3	XNGO83
64	2.000	3.300E-02	9.1	XNWL76
64	2.140	4.558E-02	5.5	XNGO83
64	2.200	4.400E-02	6.8	XNWL76
64	2.340	6.239E-02	5.5	XNGO83
64	2.100	5.600E-02	8.9	XNWL76
64	2.540	8.004E-02	5.5	XNGO83
64	4.620	2.880E-01	7.1	XNCK79
64	7.000	9.900E-01	8.0	XNBE78
64	8.000	1.480E+00	8.0	XNBE78
64	9.000	1.850E+00	8.0	XNBE78
64	10.000	2.480E+00	8.0	XNBE78
64	11.000	3.060E+00	8.0	XNBE78
64	12.000	3.750E+00	8.0	XNBE78
64	13.000	4.100E+00	8.0	XNBE78
64	14.000	4.940E+00	8.0	XNBE78
64	15.000	5.680E+00	8.0	XNBE78
65	.730	1.970E-04	10.2	INEA87
65	.940	1.060E-03	11.3	INEA87
65	1.150	3.210E-03	8.4	INEA87
65	1.360	6.900E-03	8.0	INEA87
65	1.560	1.250E-02	6.4	INEA87
65	1.760	1.850E-02	9.2	INEA87
65	1.960	2.950E-02	6.4	INEA87
65	2.160	4.160E-02	6.3	INEA87
65	2.360	5.610E-02	5.9	INEA87
65	2.560	7.260E-02	5.9	INEA87
65	2.560	8.300E-02	10.8	INEA87
65	2.650	9.700E-02	11.3	INSZ87
65	2.760	1.020E-01	10.8	INEA87
65	2.950	1.070E-01	9.3	INSZ87
65	2.960	1.210E-01	10.7	INEA87
65	3.160	1.510E-01	10.6	INEA87
65	3.360	1.710E-01	10.5	INEA87
65	3.560	2.090E-01	10.5	INEA87
65	3.960	2.940E-01	8.5	INSZ87
65	4.380	3.740E-01	9.1	INSZ87
65	4.910	5.080E-01	9.1	INSZ87
65	5.390	7.110E-01	9.4	INSZ87
65	5.910	1.110E+00	10.8	INSZ87
65	6.430	1.130E+00	11.5	INSZ87
65	6.950	1.660E+00	14.5	INSZ87
65	160.000	2.942E+01	19.1	XNJA72
67	.750	1.130E-04	25.0	XKAN78
67	.790	2.260E-04	10.2	INSZ87
67	.800	3.620E-04	14.9	XNKA77
67	.900	7.320E-04	15.0	XNKA77
67	.900	6.450E-04	9.6	INSZ87
67	1.000	7.070E-04	25.0	XKAN78
67	1.050	1.750E-03	14.9	XNKA77
67	1.110	1.960E-03	7.7	INSZ87
67	1.200	3.190E-03	15.0	XNKA77
67	1.250	2.450E-03	25.0	XKAN78
67	1.310	4.290E-03	5.8	INSZ87
67	1.350	5.500E-03	14.9	XNKA77
67	1.500	8.460E-03	15.0	XNKA77
67	1.500	6.040E-03	25.0	XKAN78
67	1.510	8.070E-03	5.7	INSZ87
67	1.600	1.090E-02	14.7	XNKA77
67	1.730	1.414E-02	5.5	INSZ87
67	1.800	1.780E-02	15.2	XNKA77
67	1.940	2.130E-02	5.6	INSZ87
67	2.000	2.440E-02	15.2	XNKA77
67	2.000	1.890E-02	25.0	XKAN78
67	2.100	2.810E-02	14.9	XNKA77
67	2.140	3.060E-02	5.6	INSZ87
67	2.200	3.350E-02	10.1	XNKA77
67	2.340	4.240E-02	5.4	INSZ87
67	2.400	4.330E-02	15.0	XNKA77
67	2.500	4.150E-02	25.0	XKAN78
67	2.540	5.780E-02	5.5	INSZ87
67	2.600	5.570E-02	15.1	XNKA77
67	2.670	6.470E-02	10.4	INSZ87
67	2.800	7.230E-02	14.9	XNKA77
67	2.970	9.040E-02	8.4	INSZ87
67	3.000	9.020E-02	15.0	XNKA77
67	3.200	1.190E-01	15.1	XNKA77
67	3.400	1.380E-01	15.2	XNKA77
67	3.600	1.580E-01	15.2	XNKA77
67	3.800	1.850E-01	15.1	XNKA77
67	3.970	2.160E-01	8.8	INSZ87
67	4.000	2.180E-01	15.1	XNKA77
67	4.400	3.010E-01	8.6	INSZ87
67	4.900	4.100E-01	8.8	INSZ87
67	5.410	5.440E-01	8.8	INSZ87
67	5.920	6.850E-01	8.9	INSZ87
67	6.420	9.490E-01	9.0	INSZ87
67	6.920	1.046E+00	8.9	INSZ87
67	7.000	9.996E-01	12.3	XNDO81A
67	7.430	1.290E+00	9.3	INSZ87
67	7.930	1.390E+00	9.4	INSZ87
67	8.430	1.750E+00	9.7	INSZ87
67	8.940	2.080E+00	10.1	INSZ87
67	9.440	2.130E+00	10.8	INSZ87
67	9.940	2.620E+00	11.4	INSZ87
69	.940	4.880E-04	9.4	INZE86
69	1.140	1.540E-03	8.4	INZE86
69	1.360	3.280E-03	7.9	INZE86
69	1.560	5.950E-03	8.4	INZE86
69	1.760	1.010E-02	7.9	INZE86
69	1.960	1.480E-02	8.1	INZE86
69	2.160	2.390E-02	7.9	INZE86
69	2.360	3.010E-02	7.6	INZE86
69	2.560	4.830E-02	8.5	INZE86
69	3.000	5.300E-02	10.0	XNFB74
69	5.000	2.400E-01	10.0	XNFB74
69	7.000	7.309E-01	12.1	XNDO81A
70	4.000	1.190E-01	10.0	XNBE78
70	7.500	5.640E-01	10.0	XNBE78
70	8.000	7.100E-01	10.0	XNBE78
70	9.000	9.060E-01	10.0	XNBE78
70	10.000	1.100E+00	10.0	XNBE78
70	11.000	1.390E+00	10.0	XNBE78
70	12.000	1.740E+00	10.0	XNBE78
70	13.000	2.040E+00	10.0	XNBE78
70	14.000	2.350E+00	10.0	XNBE78
70	15.000	2.740E+00	10.0	XNBE78
72	.750	5.060E-05	25.0	XKAN78
72	1.000	3.720E-04	25.0	XKAN78
72	1.250	1.340E-03	25.0	XKAN78
72	1.500	2.960E-03	25.0	XKAN78
72	2.000	9.250E-03	25.0	XKAN78
72	2.500	2.100E-02	25.0	XKAN78
73	.750	3.350E-05	25.0	XKAN78
73	1.000	8.891E-04	15.1	XKME58
73	1.000	3.060E-04	25.0	XKAN78
73	1.030	4.590E-04	13.1	INEA87
73	1.120	1.463E-03	15.0	XKME58
73	1.190	1.060E-03	12.3	INEA87
73	1.250	2.514E-03	14.8	XKME58
73	1.250	1.630E-03	25.0	XKAN78
73	1.400	2.340E-03	12.0	INEA87
73	1.500	2.770E-03	25.0	XKAN78

APPENDIX: TABLE II. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger-Electron Production, and Ionization

See page 112 for Explanation of Tables

Z2	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
73	1.620	4.480E-03	12.1	INEA87
73	1.830	7.150E-03	11.7	INEA87
73	1.920	1.271E-02	24.8	XKLE53
73	2.000	8.700E-03	25.0	XKAN78
73	2.020	1.130E-02	12.4	INEA87
73	2.170	2.275E-02	25.2	XKLE53
73	2.180	1.460E-02	12.3	INEA87
73	2.380	2.070E-02	12.1	INEA87
73	2.400	3.270E-02	25.1	XKLE53
73	2.500	2.010E-02	25.0	XKAN78
73	2.550	2.670E-02	10.9	INEA87
73	2.580	2.660E-02	13.2	INEA87
73	2.640	6.023E-02	25.4	XKLE53
73	2.650	2.930E-02	12.6	INSZ87
73	2.750	3.630E-02	10.7	INEA87
73	2.880	7.744E-02	24.7	XKLE53
73	2.950	4.100E-02	12.4	INSZ87
73	2.950	4.330E-02	10.9	INEA87
73	3.150	1.090E-01	25.4	XKLE53
73	3.150	5.370E-02	10.8	INEA87
73	3.250	5.480E-02	12.4	INSZ87
73	3.350	6.820E-02	10.9	INEA87
73	3.560	7.430E-02	12.2	INSZ87
73	3.560	7.840E-02	9.4	INEA87
73	3.760	8.880E-02	10.8	INEA87
73	3.960	1.020E-01	12.7	INSZ87
73	4.450	1.440E-01	12.5	INSZ87
73	4.960	2.050E-01	12.7	INSZ87
73	5.470	2.840E-01	13.0	INSZ87
73	5.970	3.320E-01	12.3	INSZ87
73	6.470	3.980E-01	12.6	INSZ87
73	6.970	5.150E-01	13.8	INSZ87
73	7.000	3.270E-01	14.0	XKBE78
73	7.000	4.551E-01	12.0	XNDO81A
73	7.470	6.250E-01	13.8	INSZ87
73	7.970	6.720E-01	13.8	INSZ87
73	8.000	5.000E-01	14.0	XKBE78
73	8.470	8.500E-01	20.0	INSZ87
73	9.000	6.620E-01	14.0	XKBE78
73	10.000	8.170E-01	14.0	XKBE78
73	11.000	9.940E-01	14.0	XKBE78
73	12.000	1.410E+00	14.0	XKBE78
73	13.000	1.530E+00	14.0	XKBE78
73	14.000	1.630E+00	14.0	XKBE78
73	15.000	1.870E+00	14.0	XKBE78
73	160.000	2.285E+01	7.1	XNJA72
74	.650	9.015E-06	11.0	XKCF84
74	.720	2.507E-05	11.0	XKCF84
74	.800	6.106E-05	11.0	XKCF84
74	.900	1.436E-04	11.0	XKCF84
74	1.000	2.794E-04	11.0	XKCF84
74	1.140	5.838E-04	11.5	XNGO83
74	1.250	8.986E-04	11.0	XKCF84
74	1.383	1.636E-03	5.8	XNGO83
74	1.500	2.010E-03	11.0	XKCF84
74	1.530	2.680E-03	7.1	XNGO83
74	1.750	3.799E-03	11.0	XKCF84
74	1.930	6.603E-03	7.2	XNGO83
74	2.000	6.565E-03	11.0	XKCF84
74	2.215	1.101E-02	6.1	XNGO83
74	2.250	1.024E-02	11.0	XKCF84
74	2.340	1.589E-02	7.8	XNGO83
74	2.500	1.560E-02	11.0	XKCF84
74	2.593	1.914E-02	5.0	XNGO83
74	2.750	2.440E-02	10.6	XNGO83
74	2.750	2.249E-02	11.0	XKCF84
74	3.000	3.187E-02	11.0	XKCF84
74	3.194	3.895E-02	4.9	XNGO83
74	3.250	4.268E-02	11.0	XKCF84
74	3.500	5.292E-02	11.0	XKCF84
74	3.750	7.350E-02	11.0	XKCF84
74	3.795	6.594E-02	5.1	XNGO83
74	4.495	1.081E-01	5.3	XNGO83
74	4.560	1.050E-01	6.4	XNCK79
74	5.195	1.617E-01	5.3	XNGO83
74	5.690	2.144E-01	5.8	XNGO83
74	6.620	3.034E-01	6.0	XNGO83
74	7.740	4.326E-01	6.0	XNGO83
75	.750	2.210E-05	25.0	XKAN78
75	1.000	1.630E-04	25.0	XKAN78
75	1.250	7.380E-04	25.0	XKAN78
75	1.500	2.110E-03	25.0	XKAN78
75	2.000	7.100E-03	25.0	XKAN78
75	2.500	9.590E-03	25.0	XKAN78
78	.750	1.160E-05	25.0	XKAN78
78	1.000	1.250E-04	25.0	XKAN78
78	1.250	5.100E-04	25.0	XKAN78
78	1.500	1.250E-03	25.0	XKAN78
78	2.000	4.040E-03	25.0	XKAN78
78	2.500	9.240E-03	25.0	XKAN78
78	7.000	2.725E-01	12.0	XNDO81A
78	160.000	1.820E+01	6.9	XNJA72
79	.710	4.319E-06	10.0	XKCF84
79	.750	7.710E-06	25.0	XKAN78
79	.760	8.772E-06	10.0	XKCF84
79	.820	1.754E-05	10.0	XKCF84
79	.900	3.895E-05	10.0	XKCF84
79	1.000	9.700E-05	14.9	XNKA77
79	1.000	9.350E-05	25.0	XKAN78
79	1.000	8.936E-05	10.0	XKCF81
79	1.100	1.880E-04	14.9	XNKA77
79	1.200	3.070E-04	15.0	XNKA77
79	1.250	3.860E-04	25.0	XKAN78
79	1.250	3.856E-04	10.0	XKCF84
79	1.300	4.960E-04	14.9	XNKA77
79	1.400	7.610E-04	15.0	XNKA77
79	1.500	1.030E-03	11.6	XNKA77
79	1.500	9.640E-04	25.0	XKAN78
79	1.500	1.051E-03	10.0	XKCF81
79	1.574	1.128E-03	9.4	XNPF84
79	1.600	1.410E-03	14.9	XNKA77
79	1.700	1.960E-03	14.8	XNKA77
79	1.750	2.208E-03	10.0	XKCF84
79	1.800	2.500E-03	14.8	XNKA77
79	1.976	3.740E-03	9.0	XNPF84
79	2.000	4.040E-03	15.1	XNKA77
79	2.000	3.180E-03	25.0	XKAN78
79	2.000	3.952E-03	10.0	XKCF84
79	2.183	5.495E-03	8.8	XNPF84
79	2.200	5.750E-03	15.0	XNKA77
79	2.250	6.305E-03	10.0	XKCF84
79	2.377	7.134E-03	8.1	XNPF84
79	2.400	1.600E-02	25.3	XKLE53
79	2.400	8.330E-03	15.0	XNKA77
79	2.500	7.520E-03	25.0	XKAN78
79	2.500	9.245E-03	10.0	XKCF81
79	2.580	1.118E-02	10.3	XNPF84
79	2.600	1.120E-02	15.2	XNKA77
79	2.780	1.359E-02	9.2	XNPF84
79	2.800	1.470E-02	15.0	XNKA77
79	3.000	1.890E-02	14.8	XNKA77
79	3.000	1.687E-02	10.0	XKCF84
79	3.200	2.360E-02	14.8	XNKA77
79	3.400	2.960E-02	14.9	XNKA77
79	3.500	2.699E-02	10.0	XKCF84
79	3.600	3.700E-02	14.9	XNKA77
79	3.800	4.350E-02	11.9	XNKA77
79	4.000	4.830E-02	14.9	XNKA77
79	7.000	2.130E-01	12.2	XNDO81A
79	7.500	2.250E-01	8.0	XNBE78
79	8.000	2.460E-01	8.0	XNBE78
79	9.000	3.230E-01	8.0	XNBE78
79	10.000	4.120E-01	8.0	XNBE78
79	11.000	5.500E-01	8.0	XNBE78
79	12.000	6.750E-01	8.0	XNBE78
79	13.000	8.030E-01	8.0	XNBE78
79	14.000	9.410E-01	8.0	XNBE78
79	15.000	1.160E+00	8.0	XNBE78
79	160.000	1.677E+01	5.7	XNJA72
82	.750	2.610E-06	25.0	XKAN78
82	1.000	4.180E-05	21.0	XNLA79
82	1.000	6.100E-05	25.0	XKAN78
82	1.250	2.440E-04	16.0	XNLA79
82	1.250	2.420E-04	25.0	XKAN78
82	1.500	6.380E-04	12.0	XNLA79
82	1.500	6.580E-04	25.0	XKAN78
82	1.750	1.220E-03	11.0	XNLA79
82	1.920	3.601E-03	25.0	XKLE53
82	2.000	2.450E-03	10.0	XNLA79
82	2.000	2.320E-03	25.0	XKAN78
82	2.170	5.905E-03	24.6	XKLE53
82	2.250	3.690E-03	10.0	XNLA79
82	2.400	1.045E-02	25.0	XKLE53
82	2.500	5.130E-03	25.0	XKAN78
82	2.880	3.049E-02	25.1	XKLE53
82	3.000	9.870E-03	11.0	XNLA79
82	3.000	1.280E-02	10.0	XNFB74
82	4.000	2.610E-02	10.0	XNLA79
82	4.000	4.100E-02	6.0	XNBE78
82	5.000	5.370E-02	10.0	XNLA79
82	5.000	5.300E-02	10.0	XNFB71
82	6.000	9.130E-02	10.0	XNLA79
82	7.000	1.350E-01	10.0	XNLA79
82	7.000	1.781E-01	13.0	XNDO81A
82	7.500	1.570E-01	6.0	XNBE78
82	8.000	1.850E-01	10.0	XNLA79
82	8.000	1.860E-01	6.0	XNBE78
82	9.000	2.410E-01	6.0	XNBE78
82	10.000	2.980E-01	6.0	XNBE78
82	11.000	3.990E-01	6.0	XNBE78
82	12.000	4.780E-01	6.0	XNBE78
82	13.000	5.680E-01	6.0	XNBE78
82	14.000	6.750E-01	6.0	XNBE78
82	15.000	8.010E-01	6.0	XNBE78
82	160.000	1.839E+01	8.9	XNJA72
83	.750	1.900E-06	25.0	XKAN78
83	1.000	5.600E-05	25.0	XKAN78
83	1.250	2.500E-04	25.0	XKAN78
83	1.500	6.200E-04	25.0	XKAN78
83	2.000	2.000E-03	25.0	XKAN78
83	2.500	5.100E-03	25.0	XKAN78
83	7.000	1.550E-01	11.0	XNDO81A
90	1.130	2.250E-05	14.7	INHO81
90	1.510	2.480E-04	10.1	INHO81
90	1.750	5.610E-04	8.4	INHO81
90	1.940	9.230E-04	11.1	INHO81
90	2.230	1.910E-03	7.9	INHO81
90	2.350	2.360E-03	10.6	INHO81
90	2.740	4.530E-03	8.4	INHO81
90	2.750	4.090E-03	11.0	INHO81
90	3.250	9.300E-03	8.9	INHO81
90	3.740	1.470E-02	8.8	INHO81
90	4.670	3.000E-02	10.0	INHO81
90	4.760	3.420E-02	9.9	INHO81
90	5.960	6.180E-02	10.4	INHO81
90	6.730	5.800E-02	13.8	INHO81
90	7.000	7.000E-02	12.0	XNBE78
90	7.170	7.700E-02	14.3	INHO81
90	8.000	1.060E-01	12.0	XNBE78
90	8.790	1.070E-01	13.1	INHO81
90	9.000	1.280E-01	12.0	XNBE78
90	10.000	1.610E-01	12.0	XNBE78
90	10.050	1.940E-01	25.3	INHO81
90	11.000	2.020E-01	12.0	XNBE78
90	12.000	2.380E-01	12.0	XNBE78
90	13.000	2.750E-01	12.0	XNBE78
90	14.000	2.940E-01	12.0	XNBE78
90	15.000	3.720E-01	12.0	XNBE78
92	1.100	1.669E-05	15.0	XKCF84
92	1.200	3.123E-05	15.0	XKCF81
92	1.250	3.610E-05	25.0	XKAN78
92	1.300	5.348E-05	15.0	XKCF84
92	1.400	1.090E-04	18.3	XNKA77
92	1.400	8.589E-05	15.0	XKCF81
92	1.500	1.270E-04	25.0	XKAN78
92	1.600	2.530E-04	18.2	XNKA77
92	1.600	1.932E-04	15.0	XKCF84
92	1.800	5.010E-04	18.0	XNKA77
92	1.800	3.797E-04	15.0	XKCF84
92	2.000	9.190E-04	18.0	XNKA77
92	2.000	5.860E-04	25.0	XKAN78
92	2.000	6.812E-04	15.0	XKCF84
92	2.200	1.430E-03	18.2	XNKA77
92	2.250	1.288E-03	15.0	XKCF84
92	2.400	2.060E-03	18.0	XNKA77
92	2.500	2.050E-03	25.0	XKAN78
92	2.500	2.225E-03	15.0	XKCF84
92	2.600	2.840E-03	18.0	XNKA77
92	2.750	3.601E-03	15.0	XKCF84
92	2.800	4.310E-03	18.1	XNKA77
92	3.000	5.320E-03	18.0	XNKA77
92	3.200	6.710E-03	18.0	XNKA77
92	3.400	7.950E-03	18.0	XNKA77
92	3.600	8.530E-03	18.1	XNKA77
92	3.800	1.150E-02	18.3	XNKA77
92	4.000	1.410E-02	17.7	XNKA77
92	4.230	1.460E-02	20.0	XNCK79
92	160.000	1.005E+01	18.4	XNJA72

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